# Praxia Update

Daily orthopaedic literature digest · Friday, 06 November 2026 · 497 new articles across 10 journals

## American Journal of Sports Medicine

Volume 54, Issue 13 · November 2026 · 31 new articles

### The Changing Climate of Musculoskeletal Research Funding.

*Fleming BC*

PubMed · DOI

### Lateral Extra-articular Tenodesis Does Not Alter Early Gait Biomechanics After ACL Reconstruction in a Pediatric Population: A Motion Analysis Study.

*Manitiu A, Mazy D, Bayol A, Blouin V, Saunders P, Hagemeister N, Nault ML*

BACKGROUND: Lateral extra-articular tenodesis (LET) is increasingly added to anterior cruciate ligament reconstruction (ACLR) to reduce graft failure and improve rotatory stability. However, concerns remain that LET may overconstrain the lateral compartment, alter gait biomechanics, and contribute to degenerative changes, especially in pediatric patients.

HYPOTHESIS/PURPOSE: The purpose of this study was to compare 3-dimensional (3D) knee kinematics during gait between pediatric patients who underwent ACLR with LET (ACLR+LET) and those who underwent isolated ACLR (iACLR). It was hypothesized that patients with ACLR+LET would show modified postoperative gait biomechanics compared to those with iACLR, namely decreased tibial internal rotation and increased valgus during the unloading phase.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: This retrospective study included pediatric patients (≤18 years of age) who underwent ACLR with or without LET between January 2023 and January 2026. All participants underwent standardized gait assessment with knee kinesiography pre- and postoperatively. Primary outcomes were 3D knee kinematic waveforms in the sagittal, frontal, and transverse planes, analyzed using 1-dimensional statistical parametric mapping.

RESULTS: In total, 70 pediatric patients (70 knees; mean age 15.8 ± 1.3 years) were included: 35 in the iACLR group and 3 […]

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### ACL Graft Type Effects on Outcomes Were Not Significantly Modified by Medial Collateral Ligament Status in Combined ACL-MCL Injuries: A Multicenter Registry Study From the French Arthroscopic Society.

*Bouguennec N, Neri T, Carrozzo A, Freychet B, Hardy A, Herce C, Kajetanek C, Ollivier M et al.*

BACKGROUND: Combined anterior cruciate ligament (ACL) and medial collateral ligament (MCL) injuries are common in pivoting sport athletes and may affect postoperative stability and recovery. Although hamstring autografts are widely used, concerns persist that harvesting the semitendinosus-gracilis complex could impair medial dynamic stabilization in knees with concomitant MCL injury.

HYPOTHESIS/PURPOSE: The purpose was to determine whether MCL status modifies the association between ACL graft type and postoperative patient-reported outcomes. It was hypothesized that MCL status would not significantly modify this relationship.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: This prospective multicenter registry study included patients undergoing primary ACL reconstruction between January 2020 and June 2022 in French Arthroscopy Society centers. Patients were categorized by medial

injury status (isolated ACL, combined ACL-MCL treated nonoperatively, or operatively) and graft type (hamstring vs nonhamstring). The primary outcome was the International Knee Documentation Committee (IKDC) subjective score. Secondary outcomes included Anterior Cruciate Ligament-Return to Sport after Injury (ACL-RSI), Subjective Knee Value (SKV), and Tegner activity level. Inverse probability weighting based on propensity scores was used to reduce allocation bias. Weighted multivariable r […]

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### Changes in Coronal and Sagittal Plane Alignment After Transphyseal ACL Reconstruction Performed Within Two Years of Skeletal Maturity: Results From the PLUTO Study Group.

*Bram JT, Saville GH, Whalen MM, Farah S, Sugimoto D, Green DW, PLUTO Study Group, Ganley TJ et al.*

BACKGROUND: For patients with open physes approaching skeletal maturity, transphyseal anterior cruciate ligament reconstruction (ACLR) with soft tissue autograft and metaphyseal fixation is often preferred. Although iatrogenic physeal injury is rare, growth disturbances have been reported. This study sought to evaluate postoperative changes in lower extremity alignment 1 year after transphyseal ACLR in skeletally immature patients within 2 years of skeletal maturity.

HYPOTHESIS: Transphyseal ACLR would not result in clinically significant changes in alignment.

STUDY DESIGN: Cohort study; Level of evidence, 2.

METHODS: This was a prospective multicenter study of skeletally immature patients with ≤2 years of remaining growth who underwent primary transphyseal soft tissue autograft ACLR. The change in the difference in alignment parameters between involved and uninvolved extremities from preoperative assessment to 1 year postoperative was calculated (difference of differences = $\Delta 2$) to account for naturally occurring alignment changes during growth. Radiographic measurements included mechanical axis deviation (MAD), anatomic lateral distal femoral angle (aLDFA), mechanical medial proximal tibial angle (mMPTA), posterior tibial slope (PTS), and leg length discrepancy (LLD).

RESULTS: A total of 91 patients (48 males) merited inclusion with a median skeletal age of 14.0 years. Rec […]

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### Association Between Early Postoperative Limb Symmetry Index Scores and 2-Year Patient-Reported Outcomes After Quadriceps Tendon Autograft ACL Reconstruction.

*Kang LS, Ina JG, Johnson A, Iyer S, Duchman KR, Hevesi M, Camp CL, Saris DBF et al.*

BACKGROUND: The limb symmetry index (LSI) is a measure comparing strength and functional movement between the surgical and nonsurgical limbs, used after anterior cruciate ligament reconstruction (ACLR) to guide rehabilitation and assess return-to-sport readiness. With increasing use of the quadriceps tendon autograft (QT-ACLR), limited data exist on whether the LSI score correlates with patient-reported outcome measures (PROMs).

HYPOTHESIS/PURPOSE: To evaluate the predictive value of postoperative strength and functional LSI tests at 3, 6, and 9 months for short-term PROMs, determine Patient Acceptable Symptom State (PASS) thresholds, and assess differences based on PASS achievement. It was hypothesized that quadriceps strength LSI score would more strongly predict 2-year PROMs than hamstring strength LSI score, and that higher 9-month isokinetic quadriceps LSI scores would be associated with higher International Knee Documentation Committee (IKDC) and Lysholm scores.

STUDY DESIGN: Case series; Level of evidence, 4.

METHODS: Patients undergoing primary QT-ACLR from 2021 to 2024 with LSI testing during rehabilitation (isometric and isokinetic lower extremity strength, Y-Balance, hop testing) and no revision or additional knee surgery were included. The LSI score was assessed at 3, 6, and 9 months, and PROMs, including Lysholm and IKDC scores, were collected at a mean follow-u […]

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### Lateral Tibial Posterior Slope Induces Anterior Tibial Translation While Lateral-Medial Slope Difference Induces Internal Rotation: A Cadaveric Biomechanical Study.

*Nejima S, Holt J, van Arkel R, Metcalfe A, Smith N, Amis A*

BACKGROUND: Posterior tibial slope (PTS) influences tibiofemoral kinematics and anterior cruciate ligament (ACL) function. While previous studies have evaluated symmetrical slope modifications, the isolated influences of the medial or lateral PTS remain unclear. This study investigated the influence of medial and lateral PTS changes on anterior tibial translation (ATT) and internal rotation (IR) under axial loading, using unicondylar slope-changing osteotomies.

HYPOTHESIS: Increased lateral PTS increases ATT, while a greater delta PTS (lateral minus medial PTS) increases IR.

STUDY DESIGN: Controlled laboratory study.

METHODS: Eight fresh-frozen knee specimens were mounted in a custom-built fixture in a compression/torsion loading machine. Tibiofemoral translation/rotation kinematics were measured under a 500-N axial compression load at 0° and 20° of knee flexion using optical motion tracking. Unicondylar tibial slope-changing osteotomies were performed on the medial and lateral tibial plateaus with a custom cutting guide, preserving key ligament and meniscus root attachments. Medial and lateral PTS were independently adjusted to -5°, 0°, +5°, and +10°, resulting in medial and lateral PTS ranging from -5° to +10°, using 3-dimensionally printed wedges. Repeated-measures analyses of variance, post hoc t tests, and Pearson correlations were used for analysis.

RESULTS: An increa […]

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## Age-Dependent Variations in Knee Synovial Fluid Biomarker Type and Concentration Predict 5- to 10-Year Outcomes After Anterior Cruciate Ligament Reconstruction.

*Ehlers M, Kurtz JL, Montgomery SR, Kaplan DJ, Strauss EJ*

BACKGROUND: Synovial fluid biomarkers reflect the intra-articular response to an anterior cruciate ligament (ACL) rupture and may provide insight into biological processes associated with posttraumatic joint degeneration. While increasing age has been associated with inferior clinical outcomes after ACL reconstruction, age-dependent differences in biomarker expression and their relationship to longer term outcomes remain unclear.

PURPOSE/HYPOTHESIS: The purpose of this study was to evaluate the association between patient age and preoperative synovial biomarker concentrations after an ACL injury to determine whether age modifies biomarker-outcome relationships at longer term follow-up. The authors hypothesized that increasing age would be associated with an elevated inflammatory response and modify biomarker-outcome associations.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: Patients undergoing ACL reconstruction between July 2011 and December 2020 were prospectively enrolled. Synovial fluid was aspirated before incision, and concentrations of proinflammatory and anti-inflammatory biomarkers were quantified. Patients completed patient-reported outcome (PRO) measures, including the visual analog scale for pain, Lysholm scale, Tegner Activity Scale, and Knee injury and Osteoarthritis Outcome Score-Physical Function Short Form, at a minimum of 5 years postoperative […]

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## High-Grade Percutaneous Medial Collateral Ligament Pie-Crusting Is Not Associated With Clinically Meaningful Valgus Instability.

*Nishioka T, Thamrongskulsiri N, Casanova F, Ackermann J, Morgan JT, Moews LD, Vega TF, Spaan J et al.*

BACKGROUND: Medial collateral ligament (MCL) pie-crusting is used during medial meniscal surgery to improve visualization and reduce the risk of chondral injuries. However, some surgeons extend the release site beyond routine incremental perforation, and the effect of this more extensive release on postoperative medial laxity remains unclear.

PURPOSE: To determine whether high-grade percutaneous MCL pie-crusting during medial meniscus posterior root tear (MMPRT) repair results in clinically meaningful valgus instability postoperatively as assessed by valgus stress radiography.

STUDY DESIGN: Case series; Level of evidence, 4.

METHODS: A retrospective review was performed of patients who underwent isolated MMPRT repair with high-grade MCL pie-crusting and no prior knee ligament surgery. High-grade pie-crusting was defined intraoperatively by an audible and palpable release and a substantial medial compartment opening (≥6 mm) under arthroscopic visualization. Postoperative bilateral valgus stress radiographs were obtained at a mean of 5.9 ± 2.0 months with the knee positioned at 20° of flexion. Gapping was measured in millimeters, and the side-to-side difference was calculated using the contralateral knee as an internal control. An a priori power analysis based on a 3.2-mm threshold for clinically meaningful medial-sided laxity demonstrated a required sample size of 15 patients […]

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### A Retensionable Inlay/Onlay Hybrid Double-Bundle Medial Collateral Ligament Reconstruction Produces Superior Restoration of the Uninjured State When Compared to a Single-Bundle Inlay Reconstruction.

*Anz AW, Waller G, Carroll J, Colbrunn G, Magoolaghan AC, Truett J, Cohen A, Kulkarni M et al.*

BACKGROUND: Traditional superficial medial collateral ligament (sMCL) reconstruction techniques have not consistently restored native laxity.

PURPOSE/HYPOTHESIS: The purpose of this study was to develop a retensionable inlay/onlay hybrid medial collateral ligament (MCL) reconstruction through comparison to a biomechanically and clinically validated reconstruction. It was hypothesized that increasing the distal footprint on the tibia through a double-bundle construct and a retensionable construct would better restore native laxity.

STUDY DESIGN: Controlled laboratory study.

METHODS: A total of 27 cadaveric knees were tested on a 6 degrees of freedom robot system. Knees were tested with valgus and external rotation (ER). Each knee was tested in a native state and an injured state after sectioning of the sMCL. One of 3 reconstructions using an anterior tibialis allograft was then performed: (1) single-bundle inlay reconstruction, (2) double-bundle inlay/onlay hybrid reconstruction with tibial fixation at both the proximal and distal (PD) tibial insertions of the sMCL footprint, or (3) double-bundle inlay/onlay hybrid reconstruction with tibial fixation at the anterior and posterior (AP) aspects of the distal sMCL tibial insertion. The constructs used retensionable variable-loop devices, allowing for retensioning after knee cycling. The constructs were tested after cycling and r […]

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### Concomitant Centralization Procedure With Medial Meniscus Posterior Root Repair May Have Zone-Specific Effects on the Medial Femoral Condyle and Enhance Repair Healing Compared With Isolated Root Repair: An In Vivo Experimental Study in Rabbit Knees.

*Dzidzishvili L, Nelson B, Santangelo K, Sikes KJ, Burton L, Del Águila López-Pintor Ó, Fernández-Valle ME, Goodrich L et al.*

BACKGROUND: Short-term clinical outcomes of meniscal centralization are promising, but questions remain whether it can protect articular cartilage and delay osteoarthritis progression.

PURPOSE: To assess cartilage status and healing potential after isolated medial meniscus posterior root (MMPR) repair (RR) versus MMPR repair augmented with a centralization procedure (RR-CP) in a rabbit stifle model.

STUDY DESIGN: Controlled laboratory study.

METHODS: MMPR tears were surgically induced in 16 New Zealand White rabbits. Eight underwent RR and 8 underwent RR-CP using interlocking knotless anchors; contralateral limbs served as controls. After 4 months, animals were euthanized and articular cartilage was harvested for macro- and microscopic evaluation, including repair tissue assessment for healing potential. Microscopic assessment was performed using the Osteoarthritis Research Society International (OARSI) scoring system. Cartilage thickness and meniscal extrusion (ME) were assessed using high-resolution magnetic resonance imaging (MRI) at 9.4 T. The medial femoral condyle was divided into 3 regions of interest (anterior, central, and posterior), and zone-specific changes were evaluated.

RESULTS: The mean ± standard deviation gross pathology joint degeneration score was significantly higher in the RR group compared with the RR-CP group (12.3 ± 2.2 vs 9.1 ± 2.0; P = .01). The m […]

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## Can Psychological Readiness Predict Successful Return to Sports After Patellar Stabilization Surgery in Adolescent Patients? A Retrospective Cohort Study.

*Kim CL, Turlip RW, Thothadri M, Redler LH, Greenberg E, Williams BA*

BACKGROUND: While many metrics are used to subjectively assess a patient's knee function with patellofemoral instability (PFI), psychological readiness to return to sport (RTS) after patellar stabilization surgery is not well established in this population. The Anterior Cruciate Ligament Return to Sport After Injury (ACL-RSI) scale assesses psychological readiness to RTS after anterior cruciate ligament reconstruction and may be similarly beneficial in understanding this domain in those with PFI.

HYPOTHESIS: Higher ACL-RSI scores in patients with PFI are associated with an increased likelihood of RTS after patellar stabilization surgery.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A retrospective review was conducted of adolescent patients (10-19 years old) treated surgically for PFI (medial patellofemoral ligament reconstruction [MPFLR] with or without tibial tubercle osteotomy) at a single institution from January 2020 to October 2024. Patients actively participating in sports at their first orthopaedic visit who completed the ACL-RSI during postoperative rehabilitation were included. Patient demographics, injury characteristics, surgical procedure, ACL-RSI scores, and RTS status were recorded. Descriptive statistics and bivariate analysis were performed. Based on the euclidean distance, the optimal ACL-RSI cutoff score for predicting successful RTS was dete […]

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## Clinical Pathway Management for Platelet-Rich Plasma Therapy in Kellgren-Lawrence Grade I and II Knee Osteoarthritis: A Prospective, Multicenter, Double-Blind Randomized Controlled Trial.

*Yang S, Wang H, Liu Y, Fu F, Xu P, Li Q, Li J, Tian X et al.*

BACKGROUND: Clinical outcomes after platelet-rich plasma (PRP) treatment for knee osteoarthritis are inconsistent, and some well-designed trials show no benefit. Whether a standardized multicomponent pathway improves outcomes relative to conventional PRP care is uncertain.

PURPOSE: To compare the efficacy, safety, satisfaction, and costs of a standardized PRP pathway versus physician-directed conventional PRP care in Kellgren-Lawrence grade I and II knee osteoarthritis.

STUDY DESIGN: Randomized controlled trial; Level of evidence, 1.

METHODS: At 3 hospitals, 240 patients were randomized 1:1 to a standardized PRP clinical pathway (PRP-CP; n = 120) or physician-directed conventional PRP care (PRP-C; n = 120). PRP-CP combined standardized selection, quality-controlled leukocyte-poor PRP, ultrasound-guided injections, staged rehabilitation, and systematic follow-up. Primary 12-month outcomes were visual analog scale (VAS) pain, Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) total, and International Knee Documentation Committee (IKDC) scores; secondary outcomes included achievement of the minimal clinically important difference, satisfaction, adverse events, and costs.

RESULTS: At 12 months, PRP-CP produced greater changes than PRP-C in VAS (mean difference [MD], -2.98 [95% CI, -3.64 to -2.32]), WOMAC total (MD, -8.63 [95% CI, -11.35 to -5.90]), and IKDC […]

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## Autologous Platelet-Rich Plasma-Enriched Chondral Matrix Implantation Improves Cartilage Repair in an Ovine Model: A Magnetic Resonance Assessment Using the MOCART 2.0 Scoring System.

*Peláez P, Damiá Giménez E, Torres-Torrillas M, Hernández-Guerra ÁM, Rubio M, Sopena JJ, Colomer-Selva R, Rodríguez-Altónaga JA et al.*

BACKGROUND: Articular cartilage defects exhibit a limited intrinsic regenerative capacity and continue to represent a significant challenge in sports medicine, frequently progressing to degenerative joint disease if left

untreated.

PURPOSE: To assess cartilage repair after autologous chondral platelet-rich plasma matrix implantation (PACI) in an ovine model using magnetic resonance imaging (MRI) and the MOCART (magnetic resonance observation of cartilage repair tissue) 2.0 scoring system.

STUDY DESIGN: Controlled laboratory study.

METHODS: Full-thickness chondral defects (8 mm in diameter) were surgically induced in the medial femoral condyles of both stifles in 7 skeletally mature Merino sheep (n = 14 stifles). Right stifles were treated with PACI combined with intra-articular plasma rich in growth factors (PRGF), whereas left stifles received a single intra-articular injection of either Ringer lactate or hyaluronic acid. MRI evaluation was performed 18 months postoperatively to assess cartilage repair using the MOCART 2.0 knee scoring system.

RESULTS: PACI-treated stifles demonstrated significantly higher MOCART scores compared with both control groups. Enhanced defect filling, improved integration with adjacent cartilage, and superior repair tissue architecture were observed in the PACI group. Furthermore, PACI-treated stifles exhibited signal intensity values more close […]

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### Indications for Isolated Arthroscopic Bankart Repair in the Adolescent Shoulder May Be Limited: Bipolar Bone Loss and Low Distance to Dislocation Found Commonly in an Adolescent Population With Glenohumeral Instability.

*Arvesen JE, Ellis HB, Do DH, Todesco C, Dinulescu AL, Van Pelt RL, Wyatt CW, Wilson PL*

BACKGROUND: Bone loss is an established risk factor for recurrent anterior shoulder instability and failure of isolated arthroscopic Bankart repair; however, its prevalence and clinical significance at initial presentation in adolescents remains poorly characterized.

PURPOSE/HYPOTHESIS: The purpose of this study was to evaluate the prevalence of bone loss and the factors that correlate with bipolar bone loss in adolescent patients evaluated with anterior shoulder instability. It was hypothesized that male sex, older age, higher number of dislocations, and contact sports participation would be associated with bipolar bone loss at initial presentation.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A retrospective institutional review board-approved study of 181 patients aged 10 to 19 years diagnosed with anterior shoulder instability from January 2022 to March 2024 was performed at a single tertiary referral center. Glenoid bone loss, Hill-Sachs interval, on-track versus off-track status, and distance to dislocation (DTD) were measured on magnetic resonance imaging. The association between bipolar bone loss and patient characteristics, sports participation, and prior dislocation history was evaluated using chi-square and Mann-Whitney U tests with univariate and multivariate logistic regression.

RESULTS: Overall, 74.6% of patients exhibited bone loss; 34.3% had bi […]

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### Five-Year Outcomes and Return to Sport After Arthroscopic Bankart Repair in Young Patients With Minimal (<10%) or No Glenoid Bone Loss.

*Fritsch L, Mitchell BC, Horan MP, Nocek M, Featherall J, Rupp MC, Korber SS, Rogers TC et al.*

BACKGROUND: Arthroscopic Bankart repair is commonly used to address anterior instability in adolescents and young adults with no or minimal bone loss. However, failure rates remain high, particularly in high-risk populations.

PURPOSE/HYPOTHESIS: The purpose of this study was to evaluate clinical outcomes, return-to-sport (RTS) rates, and predictors of failure after arthroscopic Bankart repair in patients ≤24 years of age with minimal glenoid bone loss. It was hypothesized that patients would demonstrate favorable outcomes, low recurrence rates, and high RTS rates, and smaller distance-to-dislocation (DTD) measurements, and contact sports participation would be associated with failure.

STUDY DESIGN: Case series; Level of evidence, 4.

METHODS: The data of patients with anterior shoulder instability and minimal bone loss (<10%) who underwent primary arthroscopic Bankart repair between 2006 and 2020 were retrospectively analyzed. Remplissage was not performed. Data collected included preoperative instability events, patient-reported outcome measures, satisfaction, radiographic measurements (glenoid bone loss, glenoid track, and DTD), and RTS outcomes. Failure was defined as re-dislocation or recurrent instability requiring surgery.

RESULTS: A total of 130 patients were evaluated; 18 were excluded, leaving 112. Of these, 96 patients (85.7%) were available for the final follow-up […]

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### Forearm Flexor Strain and Subsequent Ulnar Collateral Ligament Surgery Risk in Major League Baseball Pitchers: A Covariate-Matched Survival Analysis.

*Dillon MR, Mastroianni MA, Frappa N, Manzi J, Fuller SI, Luzzi AJ, Alexander FJ, Ablove R et al.*

BACKGROUND: Ulnar collateral ligament (UCL) injury remains a significant source of time loss for baseball pitchers despite work identifying biomechanical, workload, and velocity risk factors. Because medial elbow stability also depends on the contribution of the forearm flexor-pronator complex, forearm flexor strains may be a marker of downstream UCL risk.

PURPOSE: To quantify subsequent UCL surgery risk among Major League Baseball (MLB) pitchers with a nonoperatively managed forearm flexor strain who returned to play, and to evaluate whether return timing and pitching characteristics predicted post-return UCL surgery.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: A retrospective cohort study (2015-2024) identified nonoperative forearm flexor strains using publicly available MLB Injury List (IL) data corroborated by official/team sources. Pitchers were included if they returned to MLB within 1 year and were excluded if they had any history of UCL surgery before the index flexor strain. Each case was matched 1 to 2 to controls without a recorded forearm flexor strain who were active at the time of the case's return, using a multivariable distance algorithm across age, body mass index, MLB experience, preindex workload (pitches/game; total pitches), fastball velocity, and prior season elbow/forearm IL days as a marker of baseline elbow/forearm morbidity. Time-to-e […]

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### Comparable Survivorship and Clinically Significant Outcome Achievement With Both Interportal and T-Type Capsulotomy in Contemporary Hip Arthroscopy: A Propensity Score-Matched Analysis.

*Brinkman JC, Monty TL, Sparks CA, Kunze KN, Nho SJ*

BACKGROUND: Interportal and T-type capsulotomy are the most common capsulotomy techniques in contemporary hip arthroscopy. Given the critical role of the hip capsule in maintaining joint stability, the extent and configuration of capsulotomy may have meaningful implications for postoperative outcomes. However, investigations directly comparing clinical outcomes of capsulotomy types in a contemporary study population are lacking.

PURPOSE: To compare patient-reported outcomes (PROs), achievement of clinically significant outcomes, and reoperation-free survival between interportal and T-capsulotomy in patients undergoing hip arthroscopy at a minimum 2-year follow-up.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: Patients who underwent hip arthroscopy between October 2020 and November 2022 with ≥2-year follow-up were included. Patients with interportal capsulotomy were propensity score matched to patients with T-capsulotomy by age, sex, and body mass index. All patients underwent capsular closure. PROs included the Hip Outcome Score-Activities of Daily Living (HOS-ADL), Hip Outcome Score-Sports Specific Subscale (HOS-SSS), 12-item International Hip Outcome Tool (iHOT-12) score, Patient-Reported Outcomes Measurement Information System-Pain Interference (PROMIS-PI) score, and Patient-Reported Outcomes Measurement Information System-Physical Function (PROMIS-PF) score. […]

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## The Incidence and Breadth of Postoperative Pudendal Nerve Palsy and Its Relationship to Table Traction During Hip Arthroscopy With a Perineal Post.

*Hoit G, Dwyer T, Chahal J, Burkhart TA, Curd E, Docter S, Farag J, Theodoropoulos J et al.*

BACKGROUND: Pudendal nerve palsy (PNP) is a poorly characterized and underreported postoperative complication of hip arthroscopy (HA) with a perineal post. PNP may result in external genital numbness and sexual dysfunction in both men and women.

PURPOSE: To describe the incidence, breadth, and duration of postoperative PNP symptoms and to evaluate the relationship between PNP and intraoperative limb traction.

STUDY DESIGN: Cohort study; Level of evidence, 3.

METHODS: All patients underwent primary unilateral HA in a supine position with a perineal post. Patients with preexisting PNP symptoms, including sexual dysfunction, were excluded. Intraoperative traction duration and force were recorded using a sensor applied to the perineal post. A PNP screening questionnaire and validated sexual function questionnaires were administered confidentially at 2 weeks, 6 weeks, 3 months, 6 months, and 12 months postoperatively. Patient-reported hip outcomes (International Hip Outcome Tool-33 [iHOT-33]) were completed preoperatively and at 6 and 12 months.

RESULTS: A total of 93 patients were included (mean age, 36.4 years [SD, 9.2 years]; 55% male). The mean traction duration was 80 minutes (SD, 19 minutes). In the acute postoperative period, 77 (83%) patients experienced symptoms consistent with PNP. The most common symptoms were perineal numbness (75%), paresthesia (26%), and pain (18%) […]

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## Association Between Autograft Choice and Risk of Contralateral ACL Injury in Patients ≤25 Years of Age: A Systematic Review and Meta-analysis.

*Petit CB, Hussain ZB, Lloyd RS, Kember LS, McPherson AL, Pradip K, Petushek EJ, White MS et al.*

BACKGROUND: Understanding the risk of secondary contralateral (CL) anterior cruciate ligament (ACL) injury in young patients after ACL reconstruction (ACLR) may help identify targetable risk factors to optimize rehabilitation and return to sport. It is not clear if the autograft type used in the index procedure affects the risk of having a CL injury. Additionally, the aggregated CL injury rates across studies in young patients have not been well described.

PURPOSE: To compare CL ACL injury rates by graft type in patients ≤25 years of age at a minimum follow-up of 24 months after ACLR, and to stratify CL injury rates by potential modifiers, including sex.

STUDY DESIGN: Systematic review and meta-analysis; Level of evidence, 4.

METHODS: A systematic review of the literature for studies reporting ACL autograft reinjury rates after ACLR using hamstring tendon (HT), bone-patellar tendon-bone (BPTB), or quadriceps tendon (QT) autograft in patients ≤25 years of age with a minimum follow-up of 24 months was conducted. From this data set, CL ACL injury rates from studies in which they were reported were evaluated.

RESULTS: The meta-analysis included 22 studies pooling 4435 total patients who received BPTB (n = 2349), HT (n = 1561), or QT (n = 525) autografts for their ALCR. The pooled CL ACL injury rate was 10.8% (95% CI, 8.2%-14.0%) for HT, 10.5% (95% CI, 7.4%-14.8%) for BPTB, 12.2 […]

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## A Comparison of Arthroscopic Bankart Repair and the Latarjet Procedure in Contact Athletes With Anterior Shoulder Instability: A Systematic Review and Meta-analysis.

*Peckston DC, Deng D, Farshidfar SS, Jenkins M, De Franceschi F, Baba M*

BACKGROUND: Anterior shoulder instability is a frequent and often career-limiting problem in contact sports. Surgical intervention is frequently required to restore stability and reduce the risk of recurrence. Arthroscopic Bankart repair (ABR) and the Latarjet procedure are widely used, but direct comparative data in contact athletes remain limited.

PURPOSE: To compare clinical outcomes, recurrence, complications, and return-to-sport (RTS) timelines between ABR and the Latarjet procedure in contact athletes.

STUDY DESIGN: Systematic review and meta-analysis; Level of evidence, 4.

METHODS: A systematic review and meta-analysis was conducted in accordance with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. The PubMed, Ovid MEDLINE, Embase and Cochrane databases were searched for studies reporting recurrence, revision, complications, RTS, and functional outcomes in contact athletes undergoing primary ABR or the Latarjet procedure for anterior shoulder instability.

RESULTS: In total, 29 studies involving 2103 patients (1443 ABR, 660 Latarjet) were included in this analysis. The Latarjet procedure demonstrated a lower recurrence rate (7% vs 15%; P = .014) but similar revision rates (6% for the Latarjet procedure vs 7% for ABR; P = .402). ABR was associated with fewer complications (2% vs 8%; P < .0001). Both procedures achieved high RTS r […]

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## Comparative Outcomes of Circumferential Versus Segmental Labral Reconstruction in Hip Arthroscopy: A Systematic Review and Meta-analysis.

*Kurapatti M, Yuro M, Tao BS, Ramey MD, Swaminathan S, Chahla J, Parisien RL*

BACKGROUND: Labral reconstruction has become an essential technique for treating irreparable acetabular labral pathology. While both segmental and circumferential reconstruction techniques are utilized, it remains unclear whether one provides superior clinical outcomes.

PURPOSE: To compare patient-reported outcomes, complications, and reoperation rates between segmental and circumferential labral reconstruction.

STUDY DESIGN: Meta-analysis.

METHODS: A systematic review of the PubMed, Embase, and Scopus databases was performed in accordance with PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines on February 2, 2025. Studies were included if they reported clinical outcomes after arthroscopic hip labral reconstruction using either segmental or circumferential techniques. Data were extracted on patient characteristics, surgical details, patient-reported outcome measures, complications, revision arthroscopy, and conversion to total hip arthroplasty (THA). Meta-analyses were performed using random-effects models, and heterogeneity was assessed using the I2 statistic.

RESULTS: A total of 28 studies consisting of 1817 hips were included, with 1086 undergoing segmental and 731 undergoing circumferential reconstruction. Patient characteristics were similar between groups, except for a lower proportion of women in the segmental cohort than circumfer […]

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## Early Versus Delayed Repair of Proximal Hamstring Tendon Ruptures: A Systematic Review and Meta-analysis.

*Akhtar M, Ramanis M, Pasko K, Wang D*

BACKGROUND: Early recognition and repair of proximal hamstring tendon ruptures is critical, as delays can complicate surgical repair. However, the influence of surgical timing on patient outcomes remains unclear, with previous studies reporting variable results.

PURPOSE: To evaluate the effect of early versus delayed repair on clinical outcomes after primary proximal hamstring tendon repair.

STUDY DESIGN: Meta-analysis.

METHODS: A search was performed in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines in the PubMed, Embase, and Scopus databases. Studies were included if they (1) reported outcomes of primary proximal hamstring tendon repair, (2) had ≥12 months of follow-up, (3) included patients ≥18 years, and (4) compared early versus delayed treatment. Studies were excluded if they examined outcomes of proximal hamstring tendon repair/reconstruction using an allograft. Early repair was defined as surgery within 4 to 6 weeks of injury. Risk ratios (RRs) were calculated using a random-effects model, with significance set at P < .05.

RESULTS: Twelve studies were included, encompassing 992 early and 728 delayed repairs. The mean patient age ranged from 29.7 to 53.8 years in the early group and from 29.2 to 51.8 years in the delayed group. The mean follow-up ranged from 24 to 58.8 months. Nine studies reported patient-r […]

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### Return to Work Outcomes After Common Sports-Related Orthopaedic Procedures: A Systematic Review Stratified by Occupational Intensity and Workers' Compensation Status.

*Nishioka T, Lalvani S, Thamrongskulsiri N, Hummel A, Morgan JT, Moews LD, Vega TF, Casanova F et al.*

BACKGROUND: Return to work (RTW), particularly return to full-duty occupational function, is a clinically meaningful outcome after sports-related orthopaedic surgery. However, reporting across procedures is heterogeneous and often lacks occupational context.

PURPOSE: To synthesize full-duty RTW outcomes after common sports-related orthopaedic procedures in civilian populations. Secondary objectives were to describe the time to RTW and to evaluate outcomes stratified by occupational intensity and workers' compensation (WC) status.

STUDY DESIGN: Systematic review; Level of evidence, 4.

METHODS: A systematic review was conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. Civilian studies reporting RTW outcomes after meniscectomy, meniscal repair, anterior cruciate ligament (ACL) reconstruction (ACLR), rotator cuff repair (RCR), distal biceps repair, biceps tenodesis, or hip arthroscopy were included. Data were extracted at the cohort level. Full-duty RTW rates were calculated using sample-size weighted pooled estimates. Time to RTW was summarized using the median of reported cohort-level mean RTW times. Occupational intensity-specific and WC-specific outcomes were analyzed descriptively when reported separately at the cohort level.

RESULTS: A total of 87 unique studies met the inclusion criteria. Pooled full-d […]

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### Looking Beyond the Score: Response.

*Graham GD, Testa EJ, Riboh J*

PubMed · DOI

### Looking Beyond the Score: Letter to the Editor.

*Magosch A, Mouton C, Seil R*

PubMed · DOI

### Artificial Intelligence Cannot Replace Peer Reviewers But May Help Editors Triage: Response.

*Patel R, Shultz C, Ollivier M, Wascher DC*

PubMed · DOI

### Artificial Intelligence Cannot Replace Peer Reviewers But May Help Editors Triage: Letter to the Editor.

*Matsubara S*

PubMed · DOI

### Is Deltoid Osteomuscular Transfer Biomechanically Effective? A Comment on Sample Size and Statistical Power: Response.

*Iio R, Fleet CT, Johnson JA, Athwal GS*

PubMed · DOI

### Is Deltoid Osteomuscular Transfer Biomechanically Effective? A Comment on Sample Size and Statistical Power: Letter to the Editor.

*Wu Z, Liu C, Wu W*

PubMed · DOI

### Two Instability Episodes: A Structural-Risk Marker Rather Than a Stand-Alone Surgical Threshold: Letter to the Editor.

*Tang X, Zhang J, Yu B*

PubMed · DOI

### Two Instability Episodes: A Structural-Risk Marker Rather Than a Stand-Alone Surgical Threshold: Response.

*Dadoo S, Lin A, Lesniak BP*

PubMed · DOI

BJJ

## The Bone & Joint Journal

Volume 108-B, Issue 11 · November 2026 · 14 new articles

### Novel and long-term research: where are we?

*Haddad FS*

PubMed · DOI

### The robotic arm-assisted versus conventional total knee arthroplasty (RACER-Knee) trial : the question of value and where we go next.

*Clement ND, Metcalfe A, Davis ET, Haddad FS*

PubMed · DOI

### A comparison of the efficacy and safety of direct oral anticoagulants with low molecular weight heparin for venous thromboembolism prophylaxis after total hip arthroplasty : a systematic review and meta-analysis.

*Mohammad HR, Rahman A, Muhammad IM, Gill SS, Selim A, Thomas G, Murray DW, Griffin XL*

AIMS: To evaluate differences in the efficacy and safety of low molecular weight heparin (LMWH) and direct oral anticoagulants (DOACs) used for venous thromboembolism (VTE) prophylaxis after elective total hip arthroplasty (THA).

METHODS: This systematic review and meta-analysis reviewed MEDLINE, Embase, and CENTRAL from conception to 13 December 2025, for any randomized controlled trials (RCTs) comparing the efficacy (VTE risk) and safety (bleeding risk) of LMWH and DOACs after elective THA. Pooled odds ratios (ORs) of VTE, bleeding, reoperation, and mortality risks were assessed.

RESULTS: A total of 22 RCTs involving 28,112 participants were included. The risk of VTE in the DOAC group was lower compared with the LMWH group (OR 0.62 (95% CI 0.47 to 0.82); p = 0.002), driven by a reduction in deep vein thrombosis (DVT) risk (OR 0.60 (95% CI 0.42 to 0.86); p = 0.008). The findings remained consistent in the subgroup analysis restricted to anticoagulants and dosages licensed for use in the UK. Overall bleeding risk was higher in the DOAC group (OR 1.19 (95% CI 1.03 to 1.38); p = 0.023), but no difference was seen when comparing only UK-licensed anticoagulants and doses (OR 1.06 (95% CI 0.95 to 1.18); p = 0.279). Reoperation risk (OR 1.21 (95% CI 0.75 to 1.93); p = 0.351) and mortality (OR 1.03 (95% CI 0.46 to 2.33); p = 0.937) did not differ between groups.

CONCLUSION: DOACs r […]

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### Clinical outcomes of transposition osteotomy of the acetabulum in patients aged 50 years or older with hip dysplasia : a propensity score-matched comparative study.

*Fujii M, Mochizuki H, Tanaka S, Kawano S, Ueno M, Oba Y, Morimoto T*

AIMS: To clarify whether transposition osteotomy of the acetabulum (TOA) in patients aged ≥ 50 years with hip dysplasia provides clinical, radiological, and patient-reported outcomes comparable with those in younger patients.

METHODS: Among 615 patients undergoing TOA between November 1998 and July 2019, 131 patients aged ≥ 50 years were propensity score-matched 1:1 to 131 patients aged < 50 years using sex, BMI, preoperative modified Harris Hip Score (mHHS), Tönnis grade, and lateral centre-edge angle. Matching was repeated with controls aged < 45 years. Clinical and radiological outcomes, joint survival, and patient-reported outcome measures (PROMs) were compared. Failure was defined as progression to Tönnis grade 3 osteoarthritis or conversion to total hip arthroplasty. Predictors of failure in patients aged ≥ 50 years were examined using Cox regression.

RESULTS: In patients aged ≥ 50 years, the median mHHS improved from 62 (IQR 54 to 75) to 92 (IQR 84 to 96) ($p < 0.001$), with scores comparable with those in patients aged < 50 years ($p = 0.685$). Failure occurred in seven (5.3%) and ten hips (7.6%), respectively ($p = 0.607$), with similar ten- and 20-year joint survival (98% (95% CI 94 to 100) and 79% (95% CI 55 to 91) vs 97% (95% CI 89 to 99) and 72% (95% CI 51 to 86); $p = 0.655$). PROMs did not differ significantly between groups. When compared with patients aged < 45 years […]

PubMed · DOI

### Total hip arthroplasty improves global sagittal and coronal spinal imbalance and spinopelvic mobility : a prospective cohort study.

*Leung TKC, Chan PK, Cheung A, Luk MH, Lau LCM, Cheung WK, Chiu KY, Fu H*

AIMS: The impact of total hip arthroplasty (THA) on global spinal alignment is under-reported. This study prospectively evaluates the effect of THA on coronal and sagittal global spinal balance and spinopelvic kinematics using dynamic whole-body EOS radiographs.

METHODS: We prospectively studied 163 patients (62 male, 101 female; mean age 68.4 years (SD 9.1)) undergoing primary unilateral THA for end-stage osteoarthritis. Full-body EOS images were acquired preoperatively and at a six-month follow-up in standing, relaxed-seated, and deep-seated postures. The primary outcome was change in sagittal vertical axis (SVA). Secondary outcomes included changes in pelvic tilt (PT), sacral slope (SS), proximal femoral angle (PFA), and coronal parameters (C7-central sacral vertical line (C7-CSVL), limb length discrepancy (LLD), and pelvic obliquity (PO)). Preoperative SVA > 50 mm defined sagittal imbalance; C7-CSVL > 20 mm defined coronal imbalance. Statistical analysis employed paired non-parametric tests and multivariate regression.

RESULTS: In the 58 patients with preoperative sagittal imbalance (SVA > 50 mm), mean SVA improved significantly from 77.9 mm (SD 24.4) to 59.9 mm (SD 36.1; $p < 0.001$), with 27.6% (16 out of 58) moving to the sagittal balance group postoperatively. Preoperative SVA was the sole independent predictor of SVA correction ($R^2$ 0.25; $p = 0.009$). In the 41 coronally […]

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### Periprosthetic femoral fracture following primary hip arthroplasty : should the choice of femoral component design and fixation method vary according to the sex and age of the patient?

*Børsheim S, Kristensen TB, Hallan G, Gjertsen JE, Furnes O, Dybvik E, Lie SA, Dale H*

AIMS: Periprosthetic femoral fracture is a severe complication following hip arthroplasty, influenced by the design of the femoral component and the method of fixation. The aim of this study was to investigate the influence of sex and age on the risk of reoperation for periprosthetic femoral fracture.

METHODS: We analyzed 204,041 primary hip arthroplasties from the Norwegian Arthroplasty Register and the Norwegian Hip Fracture Register (1 January 2005 to 31 December 2025). Femoral components were categorized as cemented composite beam (CB), cemented polished taper-slip (PTS), uncemented collarless (UCC-), and uncemented collared (UCC+). Cox regression analysis was used to compare the risk (adjusted hazard rate ratio

(aHRR)) of reoperation for periprosthetic femoral fracture stratified by sex and age.

RESULTS: In males, PTS and UCC- had a nine-times higher risk of reoperation for periprosthetic femoral fracture than CB. For UCC+, the risk was four times higher than CB. The risk of reoperation for periprosthetic femoral fracture was increased in males aged > 65 years for UCC-; for UCC+ the risk was increased in males aged > 75 years, whereas PTS had higher risk for periprosthetic femoral fracture at all ages, compared with CB. In females, the risk of reoperation for periprosthetic femoral fracture was increased in those aged > 70 years for PTS and UCC+ compared with CB, whereas […]

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### Health-related quality of life following periprosthetic femoral fracture after total hip arthroplasty : a propensity score-matched cohort study with trajectory analysis.

*Ablett AD, Powell-Bowns MFR, Super JT, Neo C, Yapp LZ, Clement ND, Scott CEH*

AIMS: The aim of this study was to quantify the impact of a postoperative periprosthetic femoral fracture following a primary total hip arthroplasty (THA) on patient-reported outcome measures (PROMs) compared with normal values in the general population, and to assess the agreement between recalled pre-fracture and prospectively collected PROMs.

METHODS: This single-centre retrospective cohort study involved 140 operatively managed patients with a periprosthetic femoral fracture which was sustained around a primary THA, between 2007 and 2025, with a minimum follow-up of one year. The EQ-5D-3L, EQ-VAS, and Oxford Hip Score (OHS) were collected at a median of 2.9 years (IQR 1.9 to 5.5) after the fracture. Complete case propensity score matching (1:5) balanced patients who had a periprosthetic fracture with Health Survey for England general population controls (n = 700) by sex, age, and deprivation. An agreement cohort with data from a linked arthroplasty database enabled comparison between prospectively collected PROMs one year after THA and recalled pre-fracture PROMs, as well as the analysis of PROMs at four timepoints (before THA, one year after THA, pre-fracture recall, and after the periprosthetic femoral fracture).

RESULTS: The mean EQ-5D-3L scores after periprosthetic femoral fracture were significantly lower than those in the matched general population (mean difference […]

PubMed · DOI

### Outcomes of a long modular uncemented femoral stem for reconstruction in metastatic femoral disease.

*Downie S, Kennedy M, Farhan-Alanie OM, Kennedy IW, McKean G, Kelly MP, Young PS*

AIMS: Cemented arthroplasty has previously been recommended for treating metastatic bone disease (MBD) around the hip. However, cementing has disadvantages, including restricted femoral component length, loosening due to disease progression, and bone cement implantation syndrome (BCIS). A long uncemented implant spanning the whole femur could negate these disadvantages; however, there is a paucity of data regarding their use. Our aim was to assess the outcomes of a long uncemented femoral component in MBD.

METHODS: A total of 205 patients with MBD underwent femoral reconstruction using the Restoration Modular Revision Hip System Plasma stem between January 2008 and December 2025. They were identified from the prospectively collected database of a tertiary MBD service. Any patient with symptomatic bone metastases involving the acetabulum or femur, with completed or impending pathological fracture, were included. The primary endpoint was reintervention for femoral implant failure, with exclusion of reintervention for deep infection and/or instability.

RESULTS: Indications for surgery included pathological fracture (46 of 205; 22%) or lesions at risk of fracture (78%; n = 159). The mean follow-up was 45 months (12 to 131). A total of 140 patients (68%) died during the study period (mean 20 months (0.1 to 128) postoperatively). The mean distance from the centre of the femoral hea […]

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### Supervised physiotherapeutic scoliosis-specific exercises provide limited additional benefit at 24 months in braced adolescents with idiopathic scoliosis: a randomized trial.

*Zhang X, Liang M, Huang Y, Lin B, Lin Y, Yang Y, Wang X, Huang X et al.*

AIMS: Physiotherapeutic scoliosis-specific exercises (PSSE) are used as an adjunct to bracing in adolescent idiopathic scoliosis (AIS), although evidence for their long-term benefit remains limited. This randomized trial assessed whether supervised PSSE provide clinically important radiological and patient-reported benefits compared with self-directed exercise in patients with AIS undergoing brace treatment.

METHODS: Patients with AIS (major Cobb angle 25° to 45°; Risser grade 0 to 2) were recruited from a tertiary scoliosis centre between December 2018 and December 2022. Patients were randomized to supervised Schroth-based PSSE, consisting of physiotherapist-led sessions plus daily home exercises, or to self-directed general exercise. The co-primary outcomes were change in major Cobb angle and change in health-related quality of life measured using the Chinese Brace Questionnaire-34. A 5° between-group difference in Cobb angle was prespecified as the minimal clinically important difference (MCID). To provide a clinically anchored interpretation, 24-month radiological outcomes were classified as improved (> 5° decrease), stable (-5° to +5°), or worsened (> 5° increase), and compared using adjusted ordinal logistic regression in intention-to-treat (ITT) and per-protocol (PP) analyses.

RESULTS: Supervised PSSE resulted in a greater reduction in major Cobb angle than self-direct […]

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## Marginal excision of dedifferentiated liposarcoma of the limbs without radiation may be safe in select patients.

*Barriga J, Magnan MC, Arafah MA, Griffin AM, Tsoi KM, Wunder JS, Ferguson PC*

AIMS: Dedifferentiated liposarcoma (DDLPS) is a heterogeneous tumour consisting of low-grade atypical lipomatous tumour (ALT) and high-grade sarcomatous components. This study evaluates local recurrence and metastasis rates after resection of a limb DDLPS, focusing on the impact of tumour grade at surgical margins and the role of adjuvant radiation therapy.

METHODS: A retrospective analysis was conducted of 137 patients with a pathologically confirmed limb DDLPS who underwent surgical resection between July 1995 and September 2024. Outcomes assessed included local recurrence-free survival, metastasis-free survival, and overall survival. Statistical analyses used chi-squared tests for categorical variables, independent-samples t-tests for continuous variables, and Kaplan-Meier survival estimates.

RESULTS: Of the 137 patients (mean age 67 years (28 to 96); mean tumour size 16 cm (2.4 to 39)), 54.7% (n = 75) had negative surgical margins, and 45.3% (n = 62) had positive margins. Of those with positive margins, 48.3% involved only the ALT component. Notably, patients with positive margins confined to the ALT component who did not receive radiation therapy 40% (n = 12) exhibited no local recurrences or metastases. Those patients with negative margins who were treated without radiation therapy (n = 21; 28%) had low rates of recurrence (n = 1; 4.7%) and metastasis (n = 5; 23%).

CON […]

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## Barriers to recruitment in paediatric orthopaedic surgical trials : a qualitative study embedded in the Big BOSS trial.

*Kirton JA, Achten J, Young B, Perry DC, Cherry MG*

AIMS: Recruitment to randomized controlled trials (RCTs) remains a widely recognized challenge, particularly in surgical and paediatric contexts. The British Orthopaedic Surgery Surveillance (Big BOSS) trial aimed to compare surgical approaches for treating severe stable slipped capital femoral epiphysis (SCFE), a relatively rare but serious adolescent hip condition. Despite national estimates suggesting feasibility, recruitment fell significantly short of targets, and Big BOSS was discontinued. This embedded qualitative study explored the perspectives of consultant surgeons to understand the barriers that contributed to the low recruitment for the trial.

METHODS: Semistructured interviews were carried out with 24 consultant surgeons from 21 NHS sites, including recruiting, non-recruiting, and setup sites. Participants were purposely sampled to ensure variation in experience and involvement in trials. Data were analyzed inductively and thematically, guided by constant comparative methods.

RESULTS: Five overarching themes were identified: 1) limited availability of eligible patients, with many cases of SCFE either unstable, too mild or managed outside trial centres; 2) challenges in interpreting and standardizing the eligibility criteria; 3) difficulty maintaining clinical equipoise, as surgeons often relied on personal experience and preferred treatment pathways; 4) systemic […]

PubMed · DOI

## Hip-related tenotomy and progression to severe displacement of the hip in nonambulant children with cerebral palsy : a nationwide cohort study.

*Bakhtiyar M, Iljazi A, Wong C, Petersen MM, Balslev-Clausen A*

AIMS: The aim of this study was to investigate whether a hip-related tenotomy is associated with a lower risk of severe hip outcomes in nonambulant children with cerebral palsy and established displacement of the hip.

METHODS: This nationwide register-based cohort study included children classified as Gross Motor Function Classification System (GMFCS) level IV or V who were enrolled in the Danish Cerebral Palsy Follow-Up Program (CPOP) between 1 January 2008 and 31 December 2023. Hips with a baseline migration percentage (MP) of between 30% and 49% were included and followed until the first occurrence of a severe hip outcome, defined as a MP of ≥ 50% or osteotomy of the hip. Death was treated as a competing event. The cumulative incidence was estimated using the Aalen-Johansen estimator. Associations were analyzed using Fine-Gray competing-risk regression with cluster-robust standard errors.

RESULTS: A total of 259 children, contributing 406 hips, were included, of which 110 underwent a tenotomy. At ten years, the cumulative incidence of a severe hip outcome was 34.7% (95% CI 28.8 to 40.7) without tenotomy, and 20.4% (95% CI 12.4 to 28.5) with tenotomy. In adjusted competing-risk regression, tenotomy was associated with a significantly lower risk of a severe hip outcome (sub-distribution hazard ratio (sHR) 0.53 (95% CI 0.34 to 0.84); p = 0.006). GMFCS-specific Fine-Gray analy […]

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## Consensus guidelines in the management of paediatric flexible flatfoot : a Delphi study of the British Society of Children's Orthopaedic Surgery.

*Morrow EM, Bennet S, Morrison SC, Bridgens A, Mavrotas J, Brousil J, Knight D, Kokkinakis M et al.*

AIMS: The assessment and management of flexible flatfoot in children is inconsistent, reflecting a limited and heterogeneous evidence base. This study aimed to develop a British Society for Children's Orthopaedic Surgery (BSCOS)-led consensus guideline for the multidisciplinary assessment and management of flexible flatfoot in children.

METHODS: A modified Delphi consensus opinion study was conducted among the membership of the BSCOS. A multidisciplinary steering committee developed an initial set of statements based on clinical expertise, current evidence, and recognized variation in practice. A total of 33 statements addressing assessment, conservative, and surgical management were scored over two Delphi rounds using a nine-point Likert scale. Participant responses were anonymized. Consensus was defined prospectively as ≥ 75%. Statements that did not meet the consensus threshold were revised and rescored, or discussed and voted on by the steering committee.

RESULTS: A total of 169 clinicians participated in Round One and 167 in Round Two. Most respondents were consultant orthopaedic surgeons (80%; n = 136), with further representation from registrars/fellows (5%; n = 9), and allied health professionals (AHPs) and advanced nurse practitioners (14%; n = 24). Overall, 24 of 33 statements met the consensus threshold. A further two statements were included following steering com […]

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## Corrigendum.

*Mancino F, Haddad FS*

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# Clinical Orthopaedics & Related Research

Volume 484, Issue 11 · November 2026 · 35 new articles

## A Conversation With … Kent R. Anderson MBA, Veteran of Medical Publishing, on What Readers Can Trust.

*Leopold SS*

PubMed · DOI

## Editor's Spotlight/Take 5: Is Undiagnosed Dysglycemia Common in Orthopaedic Surgery and Is it Associated With 30-day Outcomes? A Large-database Study.

*Leopold SS*

PubMed · DOI

## Is Undiagnosed Dysglycemia Common in Orthopaedic Surgery and Is it Associated With 30-day Outcomes? A Large-database Study.

*Schaschinger T, Munzinger R, Niederegger T, Brandt J, Marcela FO, Knoedler L, Rendenbach C, Heiland M et al.*

BACKGROUND: Given that patients undergoing orthopaedic surgery frequently share risk factors that are associated with metabolic syndrome, the patients orthopaedic surgeons treat may be especially likely to have undiagnosed dysglycemia. Although its broader impact is documented in other specialties, its specific prevalence and association with postoperative outcomes after orthopaedic procedures remains incompletely defined. Quantifying this hidden burden is important to developing individual risk profiles, refining perioperative decision-making, and establishing targeted preoperative medical management strategies for patients considering orthopaedic surgery.

QUESTIONS/PURPOSES: In a large-database study, we asked: (1) In adults who underwent orthopaedic surgery, what proportion of patients had previously undiagnosed dysglycemia? (2) Was undiagnosed dysglycemia associated with increased odds of 30-day readmission and reoperation compared with patients with normoglycemia? (3) What was the relationship between preoperative HbA1c levels and the 30-day risk of complications in patients who underwent orthopaedic surgery?

METHODS: Among the 2,990,063 procedures recorded in the American College of Surgeons National Surgical Quality Improvement Program database (2021-2023), 25% (735,259 of 2,990,063) were orthopaedic. After excluding 75% (2,254,804 of 2,990,063) of nonorthopaedic proce […]

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## Value-based Healthcare: Scaling Value-Based Care Through Surgeon Leadership.

*Balu AR, Makhni EC, Bozic KJ*

PubMed · DOI

## Residency Diary: The Dark Places.

*Friedman LGM*

PubMed · DOI

## CORR® Curriculum-Orthopaedic Education: Perspective of a Medical Student at an LCME-accredited School in Puerto Rico.

*Flanagan JP*

PubMed · DOI

## ArtiFacts: The Surgeon Who Hated Rust and Dirt Like the Black Death.

*Drost A*

PubMed · DOI

### The Intersection of Gender and Orthopaedic Culture: American Orthopaedic Foremothers, Part I.

*Zillmer DA*

PubMed · DOI

### Art in Science: Horus Has Been Bitten.

*Green SA*

PubMed · DOI

### Is Neighborhood Socioeconomic Deprivation Associated With Outcomes Following Primary THA and TKA? A Systematic Review Utilizing the Area Deprivation Index.

*Furyes AR, Elmenawi KA, Hackett L, Deren ME, Piuzzi NS*

BACKGROUND: Socioeconomic disadvantage is increasingly recognized as an important factor associated with outcomes after total joint arthroplasty (TJA). The Area Deprivation Index (ADI) has emerged as a validated measure of socioeconomic disadvantage that specifically captures individual neighborhood levels of structural and material deprivation based on income, education, employment, and housing characteristics. Prior research has demonstrated ADI to be associated with surgical outcomes; however, its relationship to postoperative outcomes after THA and TKA, procedures in which outcomes depend highly on institutional and structural support, home environment, and accessible postacute care resources, has not been well synthesized. Prior studies provide a broad overview of heterogeneous measures of socioeconomic disadvantage that are not directly comparable.

QUESTIONS/PURPOSES: (1) Is a higher ADI associated with increased postoperative healthcare utilization after primary THA and TKA? (2) Is a higher ADI associated with differences in patient-reported outcome measures (PROMs)? (3) Is a higher ADI associated with increased postoperative complications? (4) Is ADI associated with differences in implant survivorship after TJA?

METHODS: A systematic review of four databases-Ovid Medline, Embase, Cochrane Library, and Web of Science Core Collection-was performed from their inception t […]

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### CORR Insights®: Is Neighborhood Socioeconomic Deprivation Associated With Outcomes Following Primary THA and TKA? A Systematic Review Utilizing the Area Deprivation Index.

*Blumenfeld TJ*

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### Is Robotic-assisted THA Associated With Decreased Risk of All-cause Revision or Revision for Dislocation? An Analysis of the American Joint Replacement Registry.

*Kirchner GJ, Riley C, Jimenez E, Stambough JB, Nikkel LE*

BACKGROUND: Robot-assisted THA is becoming more common, and evidence suggests that robot assistance improves the accuracy of acetabular cup positioning. By means of reducing implant malpositioning, robot-assisted THA has been proposed as a method to reduce the risk of revision surgery and particularly revision due to dislocation. However, this has not yet been explored using an arthroplasty registry.

QUESTIONS/PURPOSES: (1) In a dataset drawn from the American Joint Replacement Registry (AJRR) and its Medicare claims data linkage, after controlling for confounding variables, is robot-assisted THA associated with a decreased risk of all-cause revision at minimum 2-year follow-up? (2) After controlling for confounding variables, is robot-assisted THA associated with a decreased risk of revision for dislocation at minimum 2-year follow-up?

METHODS: The AJRR and its Medicare claims data linkage were used to create a retrospective comparative study of patients aged 65 years or older who underwent primary elective robot-assisted THA (n = 10,988) or conventional THA (n = 113,477) between 2017 and 2021. All patients within the AJRR who had undergone primary THA and

had complete records with respect to robot data were included. The AJRR was selected for this study because of its Medicare linkage, and therefore follow-up data are considered near complete. Both cohorts had the same mean […]

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### Is a Resorbable Citrate-based Bioceramic Device Associated With Osseous Integration? An Early Retrospective MRI Analysis.

*Berberian WS, Day J, Schon L*

BACKGROUND: Bioresorbable fixation devices may be advantageous because they eliminate the question of whether they ought to be removed, have biomimetic qualities, and minimize artifacts seen on MRI. Most bioresorbable devices utilized during orthopaedic surgery consist of thermoplastic polymers derived from lactic and glycolic acids. These devices are linked to implant-related osteolysis, inflammation, and bone tunnel enlargement, and their integration with surrounding bone is inconsistent. Citregen®, a resorbable composite of a citrate-based thermoset polymer and hydroxyapatite, recently has been approved for use in the United States; yet to date, there are no published studies documenting device-related MRI findings.

QUESTIONS/PURPOSES: (1) Was there observable bone tunnel enlargement on MRI at a minimum of 2 years after implantation of a resorbable citrate-based bioceramic device? (2) Were there other observable adverse findings (such as bone marrow edema or device breakage) on MRI at a minimum of 2 years after implantation of a resorbable citrate-based bioceramic device? (3) Did citrate-based bioceramic devices demonstrate integration with surrounding bone on MRI at a minimum of 2 years after implantation? (4) What was the frequency of early clinical complications after soft tissue fixation with citrate-based bioceramic interference devices?

METHODS: All 15 patients who u […]

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### CORR Insights®: Is a Resorbable Citrate-based Bioceramic Device Associated With Osseous Integration? An Early Retrospective MRI Analysis.

*DeCoster TA*

PubMed · DOI

### Preoperative Shoulder Stiffness Was Accompanied by Distinct Supraspinatus Tendon Inflammatory Signatures but Not Inferior Long-term Outcomes After Rotator Cuff Repair.

*Liu M, Cai H, Tang C, Luo J, Wang X, Liu H, Ruan D, Wang C et al.*

BACKGROUND: Rotator cuff tears with concomitant preoperative shoulder stiffness present a clinical challenge because the rehabilitation needed to restore motion may conflict with the protection typically used to support tendon healing after rotator cuff repair. Although recent studies suggest that preoperative stiffness may not be associated with inferior long-term outcomes, whether it is associated with a different pattern of postoperative improvement and distinct inflammatory features in the torn tendon remains unclear.

QUESTIONS/PURPOSES: (1) Do a greater proportion of patients with preoperative shoulder stiffness achieve improvements in functional outcomes that exceed the minimum clinically important difference (MCID) at 3, 6, 12, and 24 months? (2) Do supraspinatus tendons from patients with shoulder stiffness show molecular and histologic features consistent with greater inflammation than tendons from patients without stiffness?

METHODS: This prospective, observational cohort study included patients scheduled to undergo arthroscopic rotator cuff repair for unilateral rotator cuff tears with medium-sized, full-thickness supraspinatus tears, with or without concomitant preoperative shoulder stiffness. A medium-sized tear was defined as a supraspinatus tendon tear measuring 3 cm or less on preoperative coronal MRI. Preoperative shoulder stiffness was defined as passive sho […]

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### CORR Insights®: Preoperative Shoulder Stiffness Was Accompanied by Distinct Supraspinatus Tendon Inflammatory Signatures but Not Inferior Long-term Outcomes After Rotator Cuff Repair.

*Rojas Lievano J*

PubMed · DOI

### Radial Head Fractures Cluster in the Anterolateral and Anteromedial Quadrants and Do Not Correlate With Coronoid Fracture Types.

*Azib N, de Klerk HH, Doornberg JN, Nettuno N, van den Bekerom MPJ, Chen NC, Bhashyam AR*

BACKGROUND: Combined radial head and coronoid fractures, including terrible triad injuries, are challenging patterns associated with elbow instability, poor clinical outcomes, and other complications. While both radial head and coronoid fractures contribute to joint stability, the relationship between their fracture morphologies remains poorly understood. Clarifying these associations may aid in preoperative planning and guide surgical decision-making for restoring elbow stability.

QUESTIONS/PURPOSES: (1) In patients with combined coronoid and radial head fractures, what radial head fracture patterns are most common? (2) Are radial head fracture patterns associated with coronoid fracture types?

METHODS: This study was a secondary data analysis of a retrospective, multicenter descriptive study of patients with combined coronoid and radial head fractures identified on elbow CT scans. The cohort was derived from two urban tertiary level 1 trauma centers with specialized upper extremity and orthopaedic trauma services. Of 10,016 patients with elbow or forearm injuries who underwent CT imaging, 2% (175) met eligibility criteria for combined coronoid and radial head fractures; eight patients were excluded because of prior pathophysiologic conditions of the elbow, previous surgery, or inadequate imaging quality, leaving 167 patients for analysis, 54% (90) of whom were female, with a […]

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### CORR Insights®: Radial Head Fractures Cluster in the Anterolateral and Anteromedial Quadrants and Do Not Correlate With Coronoid Fracture Types.

*Itamura JM*

PubMed · DOI

### Long-term Oncologic Outcomes After Unplanned Soft Tissue Sarcoma Surgery: Implications for Surveillance Strategies.

*Jahn J, Dean KK, Travis LM, Daftari M, Costello JP, Shloush M, Brutti J, Shae J et al.*

BACKGROUND: Unplanned excision refers to the resection of soft tissue sarcomas without appropriate preoperative imaging, histologic confirmation, or oncologic planning. Although the short-term oncologic risks of unplanned excisions are well described, data on long-term outcomes remain limited.

QUESTIONS/PURPOSES: After controlling for tumor grade, clinical stage, Charlson comorbidity index (CCI), and surgical margin status, is unplanned excision independently associated with (1) worse overall survival, (2) local recurrence-free survival, and (3) metastatic disease-free survival compared with planned excision?

METHODS: In this study of 1100 patient records initially identified by Current Procedural Terminology code query, 403 were identified as duplicate records, 253 were excluded for insufficient baseline data or lack of cancer center entry, and 48 were excluded for insufficient follow-up or not undergoing surgery at the institution, yielding 396 patients, 279 who underwent planned excision and 117 who underwent unplanned excision. We defined an unplanned excision as resection of a soft tissue mass performed without appropriate preoperative advanced imaging (such as MRI) or histologic tissue diagnosis (such as core needle biopsy), or when imaging and biopsy findings were misinterpreted as benign, leading to resection without oncologic planning. The mean ± SD age was similar b […]

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### CORR Insights®: Long-term Oncologic Outcomes After Unplanned Soft Tissue Sarcoma Surgery: Implications for Surveillance Strategies.

*Kelly S*

PubMed · DOI

## What Are the Clinical Characteristics and Outcomes of Brucellar Spondylitis in a Nonendemic Region of Southern China?

*Lin Z, Tang J, Chen Q, Mi H, Zhang W, Zhang J, Sun H, Le S et al.*

BACKGROUND: Brucellosis, historically endemic to northern China, has recently shown a southward spread into previously nonendemic regions. Although brucellar spondylitis represents one of the most disabling complications of the disease, its clinical and epidemiologic characteristics in these emerging southern regions remain poorly defined, warranting further investigation.

QUESTIONS/PURPOSES: Among patients diagnosed with brucellar spondylitis from nonendemic regions of southern China, we asked: (1) What are the predominant clinical and laboratory findings? (2) For focal lesion aspiration cultures and blood cultures, what proportion of patients had positive cultures and what factors were associated with having positive cultures? (3) What are the characteristic imaging findings? (4) At a minimum 1-year follow-up, what were the proportions of patients with mild sequelae (exertional pain without functional impairment), moderate sequelae (neurologic deficits or work interference), treatment failure, and infection recurrence?

METHODS: Between January 2015 and April 2025, we treated 47 patients with brucellar spondylitis at a tertiary medical center in a nonendemic region of southern China. Two patients were excluded because of incomplete clinical records, leaving 45 patients for the baseline analyses. An additional seven patients were excluded from the evaluation of clinical outco […]

PubMed · DOI

## Diabetes Is Associated With Higher Odds of Junctional Complications and More Reoperations After Adult Spinal Deformity Surgery: A Propensity-matched Study.

*Yamada T, Yamato Y, Hasegawa T, Yoshida G, Banno T, Arima H, Oe S, Ushirozako H et al.*

BACKGROUND: Mechanical complications remain a major concern after adult spinal deformity surgery. Diabetes has been associated with impaired bone quality and delayed healing in orthopaedic surgery, but its relationship with mechanical complications and patient-reported outcomes after adult spinal deformity surgery remains incompletely understood.

QUESTIONS/PURPOSES: (1) Is diabetes associated with a higher odds of mechanical complications after adult spinal deformity surgery? (2) Is revision-free survivorship for proximal junctional failure lower in patients with diabetes than in those without diabetes? (3) Do radiographic alignment parameters differ between groups after surgery? (4) Are patient-reported outcomes lower in patients with diabetes at 2 years?

METHODS: Between January 2010 and December 2023, a total of 585 patients underwent corrective fusion for adult spinal deformity at our institution. Of these, 11% (65) had diabetes and 89% (520) did not. Among patients with diabetes, none died before 2 years, but 3% (2 of 65) were lost to follow-up for reasons other than death. Among patients without diabetes, 0.6% (3 of 520) died, and 4% (23) were lost to follow-up before 2 years. This analysis therefore consisted of 557 patients, of whom 11% (63) had diabetes and 89% (494) did not; their data were analyzed at least 2 years after the index fusion. Diabetes was defined as a […]

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## CORR Insights®: Diabetes Is Associated With Higher Odds of Junctional Complications and More Reoperations After Adult Spinal Deformity Surgery: A Propensity-matched Study.

*Patralekh MK*

PubMed · DOI

## Do Perineural Dexamethasone and Dexmedetomidine Improve Analgesia and Inflammatory Response After Lumbar Spine Surgery?

*Reysner T, Ciftci B, Alver S, Wieczorowska-Tobis K, Neumann-Podczaska A, Reysner M*

BACKGROUND: Postoperative pain after lumbar spine surgery often requires substantial opioid use. Perineural adjuvants such as dexamethasone and dexmedetomidine are increasingly used to prolong the duration of regional blocks and improve analgesic quality; however, their comparative effects remain uncertain. Ultrasound-guided erector spinae plane block (ESPB) is widely used as part of multimodal analgesia, but the selection of perineural

adjuvants remains uncertain.

QUESTIONS/PURPOSES: (1) Does adding adjuvants (dexamethasone or dexmedetomidine) to the ESPB result in longer time to first rescue opioid? (2) Do patients who received adjuvants have lower total opioid consumption in the first 48 hours after surgery, and do they have lower pain scores? (3) Does dexamethasone or dexmedetomidine result in lower postoperative systemic inflammatory indices?

METHODS: This study was conducted at a tertiary academic referral center specializing in spine surgery. In this prospective, randomized, triple-blinded controlled trial, between March 2024 and December 2025, a total of 103 patients were assessed for eligibility. Of these, 94 were randomized. Four patients were excluded after randomization because of intraoperative changes in surgical plan or hemodynamic instability, leaving 96% (90) of adult patients undergoing posterior lumbar decompression with instrumented fusion. These patients […]

PubMed · DOI

### Impact Microindentation Evaluates Bone Strength, Bone Quality, and Fracture Susceptibility Across Skeletal Sites: A Cadaver Study.

*Vaidya RS, Jahani B, Jin JM, Gonzalez CC, Pujari B, Krothapalli S, Tang SY*

BACKGROUND: Impact microindentation (IMI) is a minimally invasive technique for assessing bone material properties that can be performed clinically. In this study, we aimed to determine how measurements obtained by IMI (bone material strength index [BMSi]) relate to bone strength, fracture resistance, and susceptibility to fracture.

QUESTIONS/PURPOSES: (1) Is BMSi associated with bone strength of the femur, radius, and vertebrae, and does incorporating BMSi alongside bone mineral density (BMD) improve the estimation of strength? (2) Does BMSi evaluate fracture toughness and mechanical properties of the bone tissue? (3) Can BMSi discriminate fracture susceptibility of the hip and wrist?

METHODS: This single-site laboratory study included the tibiae, femurs, radii, and L4 vertebrae from fresh-frozen cadavers (n = 67; mean ± SD age 71 ± 13 years). To answer the first question, IMI was performed on the tibia midshaft, dual-energy x-ray absorptiometry (DXA) was used to assess hip and spine BMD, and bone strength (ultimate fracture force) was mechanically determined for each site. To address the second question, stress intensity factor (K C ) was quantified from notched cortical bone beam specimens from tibiae using the fracture toughness test. Bone mechanical behavior (ultimate stress, elastic modulus, work-to-fracture), microstructure (cortical porosity, tissue mineral density), […]

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### CORR Insights®: Impact Microindentation Evaluates Bone Strength, Bone Quality, and Fracture Susceptibility Across Skeletal Sites: A Cadaver Study.

*Wright TM*

PubMed · DOI

### Letter to the Editor: What Is the Association Between Radiographic Measurements of Initial and Residual Fracture Displacement, Tibial Alignment, and Recovery in Terms of Patient-reported Outcome After Surgical Treatment of Tibial Plateau Fractures?

*Qin K, Chen D, Sun P, Liang W*

PubMed · DOI

### Reply to the Letter to the Editor: What Is the Association Between Radiographic Measurements of Initial and Residual Fracture Displacement, Tibial Alignment, and Recovery in Terms of Patient-reported Outcome After Surgical Treatment of Tibial Plateau Fractures?

*Vaartjes TP, Assink N, IJpma FFA*

PubMed · DOI

### Letter to the Editor: Isolated Proximal Femoral Valgus Osteotomy for Hinge Abduction Improves Femoroacetabular Relationship in Patients With Perthes Disease.

*Emara K, Eissa M*

PubMed · DOI

### Reply to the Letter to the Editor: Isolated Proximal Femoral Valgus Osteotomy for Hinge Abduction Improves Femoroacetabular Relationship in Patients With Perthes Disease.

*Park BK, Kim HW*

PubMed · DOI

### Letter to the Editor: Does Aspiration With Culture Augmented With Percutaneous Core Biopsy Analysis Have Better Diagnostic Value Than Culture Alone for PJI in Large Joints? A Prospective Enhancement and Feasibility Study.

*Yang X*

PubMed · DOI

### Letter to the Editor: Most Contemporary TKA Implant Systems Meet International Benchmark Survivorship Standards: Evidence From a US Registry.

*Wera GD*

PubMed · DOI

### Reply to the Letter to the Editor: Most Contemporary TKA Implant Systems Meet International Benchmark Survivorship Standards: Evidence From a US Registry.

*Prentice HA, Kelly MP, Reddy NC, Paxton EW*

PubMed · DOI

### Letter to the Editor: What Is the Relationship Between Clinician-reported and Patient-reported Outcomes in Orthopaedic Surgery?

*Lee TC, Fu YC*

PubMed · DOI

### Reply to the Letter to the Editor: What Is the Relationship Between Clinician-reported and Patient-reported Outcomes in Orthopaedic Surgery?

*Aepala MR, Kiani S, Yeramosu T, Sabharwal S*

PubMed · DOI

## JBJS (American)

Volume 108, Issue 21 · November 2026 · 18 new articles

### Transparency and Accountability in the Age of AI: Updated Policy for JBJS Authors.

*Strelzow J, Ghert M, Bini SA*

PubMed · DOI

### Acing the Base in Adult Spinal Deformity: Commentary on an article by Benjamin D. Elder, MD, PhD, et al.: " Placement of Triangular Titanium Implants Suprajacent to Pelvic Fixation Screws Reduces Pelvic Fixation Failure in Multilevel Spinal Fusion Surgery. Results of a Randomized Controlled Trial ".

*Soh RCC*

PubMed · DOI

### Design Matters: Rethinking Femoral Stem Choices: Commentary on an article by P.A. Bouché, MD, PhD, et al.: " Impact of Stem Design on Periprosthetic Femoral Fracture Risk. Findings from 182,118 Primary Total Hip Arthroplasties in the Swiss National Joint Registry ".

*Gjertsen JE*

PubMed · DOI

### Beyond the Echo Chamber: Upholding Clinical Objectivity in the Era of Sycophantic Large Language Models: Commentary on an article by Arthur J. Perry, BS, et al.: " Current Artificial Intelligence Large Language Models Exhibit Sycophantic Behavior in Orthopaedic Contexts ".

*Shea GKH, Wang H*

PubMed · DOI

### Early Periprosthetic Femoral Fracture After Total Hip Arthroplasty Is Not a Complication; It Is a Crisis: Commentary on an article by Alexander J. Acuña, MD, et al.: "Alarmingly High Rates of Complications and Refracture Among Patients with Early Periprosthetic Femoral Fracture Within 30 Days After THA".

*Linke P*

PubMed · DOI

### Between-Domain Interactions in Multi-Domain Platform Trials: Commentary on an article by Nathan N. O'Hara, PhD, MHA, et al.: " Reimagining Orthopaedic Research: The Musculoskeletal Adaptive Platform Trial (MAPT) Design Rationale ".

*Greenstreet P*

PubMed · DOI

### What's New in Musculoskeletal Basic Science.

*Goodman SB, Ahn J, Ebramzadeh E*

PubMed · DOI

### The Variable Axis Knee Prosthesis: A Legacy of Dr. David G. Murray.

*Shaw JA*

An important and often overlooked legacy and orthopaedic innovation of Dr. David G. Murray is the Variable Axis knee prosthesis. Conceptualized by Dr. Murray in 1972 and engineered over the ensuing 2 years, the unique "ball-in-socket" design of the prosthesis represented the first unconstrained cruciate-substituting knee prosthesis. Its metal-backed tibial component with modular polyethylene inserts of varying thicknesses and angular designs was also an innovative first and is a design feature that was incorporated into virtually all subsequent prosthetic knees. Released for clinical application in 1974, the Variable Axis knee prosthesis enjoyed over a decade of successful clinical use but was ultimately supplanted by the more anatomically configured Total Condylar-type designs.

PubMed · DOI

### Radiation Safety in Orthopaedic Surgery.

*Florentino SA*

PubMed · DOI

### The 2025 North American Traveling Fellowship.

*Puvanesarajah V, Chrea B, Parisien RL, Stepan JG, Ekhtiari S*

PubMed · DOI

### Postoperative Rehabilitation Milestones and Functional Recovery Among Geriatric Patients After Total Knee Arthroplasty.

*Kamal H*

PubMed · DOI

### Reimplantation in 2-Stage Exchange: Breaking the Dogma of the 2-Week Antibiotic Holiday.

*Ascione T, Balato G, Pagliano P, Parvizi J*

➢ The optimal timing for reimplantation in patients with periprosthetic joint infection undergoing 2-stage exchange revision is challenging. The metrics used to determine reimplantation timing, such as serum inflammatory markers and culture results, often have low predictive value and can be misleading or discordant. ➢ A combination of clinical signs indicating infection resolution together with declining serological and synovial markers may be used to guide the timing of reimplantation. ➢ The rationale for an antibiotic holiday is to allow for a persistent infection to "resurface" in that period, prior to reimplantation. ➢ Available data have shown no clear evidence of improved outcomes when an antibiotic holiday is implemented. ➢ However, continuous antibiotic therapy offers important advantages, including a shorter overall treatment duration and higher cure rates, particularly in immunocompromised patients. There is a dire need to develop a tool to guide the optimal timing of reimplantation.

PubMed · DOI

### Lexicon on the Pathomechanics of the Native Hip Joint at Risk for Arthritis.

*Beaulé PE, Beck M, Wagner M, Witt J, Grammatopoulos G, Zaltz I, Sugano N, Anderson AE et al.*

BACKGROUND: Degenerative osteoarthritis (OA) of the hip often arises from structural abnormalities that alter load distribution, which leads to progressive chondrolabral damage. Although substantial advances have been made, the existing terminology in the field of hip preservation remains heterogeneous and inconsistently applied, limiting scientific communication and treatment standardization. The aim of this article is to link evidence from existing literature to develop a framework in the form of a structured lexicon, one that defines the morphological features and pathomechanical mechanisms that are relevant to the symptomatic, pre-arthritic hip.

METHODS: This descriptive framework was developed through the synthesis of current biomechanical, radiographic, and clinical evidence, combined with expert consensus from an international, multidisciplinary panel. Foundational concepts were derived from basic science research, including tribology, mechanobiology, and contact mechanics, to establish a unified terminology.

RESULTS: Three principal wear modes-instability, impingement, and mixed mechanisms-are proposed, corresponding to distinct pathological loading environments. Wear mechanisms include dysplasia with rim overload and stress-related cartilage failure, impingement with cam-induced shear or pincer rim conflict, and secondary inflammatory degeneration. Wear damage is cla […]

PubMed · DOI

### Placement of Triangular Titanium Implants Suprajacent to Pelvic Fixation Screws Reduces Pelvic Fixation Failure in Multilevel Spinal Fusion: Results of a Randomized Controlled Trial.

*Elder BD, Mundis G, Eastlack R, Arrigoni P, Franke J, Lee R, Polly D, the SILVIA Investigator Group*

BACKGROUND: The purpose of this study was to determine whether placement of triangular titanium implants (TTI) suprajacent to S2-alar-iliac (S2AI) pelvic fixation screws during multilevel spinal fusion surgery for adult spine deformity reduces pelvic fixation failures.

METHODS: This prospective, multicenter, international, partially blinded randomized clinical trial included 222 patients scheduled for multilevel spinal fusion surgery with S2AI-based pelvic fixation. In the S2AI+TTI group, the mean age was 64.9 years, 54.1% were female, 3.7% were Hispanic, 5.5% were African American, and 1.8% were Asian. In the S2AI-alone group, the mean age was 64.8 years, 58.4% were female, 3.5% were Hispanic, 7.1%

were African American, and 0.9% were Asian. Subjects were randomized to pelvic fixation using S2AI screws alone versus S2AI+TTI. Scheduled follow-up included a 24-month full-spine radiograph and a high-resolution computed tomography (CT) scan, which was independently interpreted.

RESULTS: The primary composite radiographic end point (S2AI screw breakage, lucency, or pullout, or posterior spinal rod breakage distal to S1) occurred less frequently in the S2AI+TTI group than in the S2AI-alone group (11.0% versus 20.4%, p = 0.0415). Except for rod-screw dissociation and rod fracture distal to S1, all components of the primary composite end point occurred less frequently in the S2AI+TT […]

PubMed · DOI

## Impact of Stem Design on Periprosthetic Femoral Fracture Risk: Findings from 182,118 Primary Total Hip Arthroplasties in the Swiss National Joint Registry.

*Bouché PA, Lübbeke A, Brand C, Gonzalez AI, Beck M, Hannouche D*

BACKGROUND: Periprosthetic femoral fractures (PFFs) are an increasingly common indication for revision total hip arthroplasty (THA). While patient-related risk factors are well documented, the influence of femoral stem design on PFF risk remains poorly characterized. In this study using nationwide data, we assessed the association between stem design and PFF risk.

METHODS: We analyzed 182,118 primary THAs (performed from 2015 to 2023) from the Swiss National Joint Registry (SIRIS). Cementless stems were categorized according to the Kheir classification, and cemented stems were categorized as double-tapered polished, triple-tapered polished, composite-beam, or custom. A multivariable Cox regression model, including variables such as age, sex, American Society of Anesthesiologists (ASA) class, body mass index (BMI), surgical indication, prior ipsilateral hip surgery, stem design, collar, dual-mobility cup, bearing, and head size, was analyzed. Hazard ratios (HRs) with 95% confidence intervals (CIs) are reported.

RESULTS: Among 182,118 THAs (mean patient age, 68.9 ± 11.5 years; female sex in 53.1% of cases), 1,226 (0.7%) were complicated by PFF. The cumulative incidence of PFF reached 0.7% at 5 years and 1.3% at 10 years. Higher PFF risk was associated with an age of 75 to 84 years (HR = 1.68 [95% CI = 1.44 to 1.96]) and ≥85 years (HR = 1.86 [95% CI = 1.47 to 2.35]), ASA class o […]

PubMed · DOI

## Current Artificial Intelligence Large Language Models Exhibit Sycophantic Behavior in Orthopaedic Contexts.

*Perry AJ, Kalva S, Fucich D, Muppidi S, Aggarwal M, Virk MS, Zuckerman JD, Yao JJ*

BACKGROUND: The use of large language models (LLMs) is increasingly common. However, LLMs may exhibit sycophancy, echoing users' beliefs while avoiding contradiction. In the present study, we describe sycophancy in general-purpose LLMs when applied to orthopaedic contexts.

METHODS: We investigated sycophancy in 2 general-purpose LLMs. We evaluated performance on 3 tasks: (1) accuracy on benchmark answering: LLMs were tested on validated benchmark orthopaedic questions, with correct and incorrect cues, and the change in accuracy and sycophancy error rate were determined; (2) user belief agreement: LLMs were provided with ambiguous statements and a user belief, and LLM agreement, contradiction, and uncertainty were described; and (3) false information detection: false information was placed within a task prompt to measure noncontradiction and propagation rates.

RESULTS: Baseline factual accuracy on benchmark questioning was 78%, decreasing with correct hints (71%) (p = 0.49). With incorrect hints, LLM accuracy declined significantly (48%) (p < 0.001), with a sycophancy error rate of 52%. Presented with user beliefs about an indefinite, controversial statement, models echoed user beliefs in 56%, expressed uncertainty in 12%, and contradicted users in 32% of statements. In noncontradiction tasks, models perpetuated incorrect attributions 99% of the time yet reliably corrected sta […]

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## Alarmingly High Rates of Complications and Refracture Among Patients with Early Periprosthetic Femoral Fracture Within 30 Days After THA.

*Acuña AJ, Forlenza EM, Jones CM, DeBenedetti A, Terhune EB, Della Valle CJ*

BACKGROUND: The purpose of this study was to compare the outcomes of patients who underwent a total hip arthroplasty (THA) and sustained a postoperative periprosthetic femoral fracture either within the early postoperative period, within 30 days, or at later time points.

METHODS: A large, all-payer national database was queried to identify patients who underwent a primary THA between 2010 and 2022. Patients who underwent a primary THA and sustained a periprosthetic femoral fracture within 30 days postoperatively were propensity-score matched on the basis of age, sex, and comorbidity burden to 3 separate cohorts based on timing: fracture between 31 and 90 days postoperatively (n = 1,725), fracture between 91 days and 1 year (n = 1,887), and no fracture within 1 year (n = 4,465). The 30-day fracture group had a mean age of 66.63 ± 10.15 years, and 75% of patients in this group were female. Regression analyses were utilized to evaluate outcomes at 90 days and 2 years. Survivorship analyses utilizing death as a competing end-point were conducted for rates of infection, refracture, and reoperation at up to 2 years.

RESULTS: Patients who sustained a periprosthetic femoral fracture within 30 days after primary THA demonstrated significantly greater rates of complications within 90 days relative to each comparison cohort. The 2-year survivorships free of refracture and reoperation we […]

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## Reimagining Orthopaedic Research: The Musculoskeletal Adaptive Platform Trial (MAPT) Design Rationale.

*O'Hara NN, Sprague SA, Patterson JP, Druker V, Connor JT, Marvel D, Slobogean GP*

UNLABELLED: Musculoskeletal disorders are common and can profoundly affect the survival, mobility, and independence of patients. The overall objective of the Musculoskeletal Adaptive Platform Trial (MAPT) is to incrementally compare the effectiveness of available interventions to improve outcomes for adult patients with health conditions affecting physical mobility. The MAPT framework is flexible, allowing the detection of varying levels of meaningful benefit or harm (or equivalence) on the basis of the burden of disease and the available treatments. Treatment comparisons will be evaluated as intervention domains on the platform in order to achieve the overarching objective. Specific objectives for each intervention domain will be included in intervention domain-specific protocol appendices. The primary outcome is a hierarchically assessed composite of death within 120 days, ambulation status at 120 days, and days alive and out of the hospital within 120 days of randomization. Secondary outcomes will include the components of the composite outcome and patient-reported outcomes. Participants will be contacted at 6 weeks, 4 months, 8 months, and 12 months after randomization to ascertain trial outcomes. The MAPT will enable researchers to efficiently evaluate multiple treatment options for patients with musculoskeletal impairments, which may improve patient care and outcomes glob […]

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# The Journal of Arthroplasty

Volume 41, Issue 12 · December 2026 · 48 new articles

## The Aggregation of Marginal Gains in Total Joint Arthroplasty: A Practical Framework for Scalable, High-Quality Care.

*Jenkins S, Smitterberg CW, Scuderi GR, Mont MA*

Demand for total hip and knee arthroplasties continues to increase faster than many programs can expand operating room capacity, clinic access, and workforce. Although major technological investments may improve care, small recurring inefficiencies throughout the episodepisode of care can also produce substantial cumulative losses. The aggregation of marginal gains offers a disciplined approach: identify a defined constraint, establish a baseline, introduce a targeted change, measure process, outcome and balancing measures, and retain only changes that improve the system. This perspective applies that approach across the preoperative, intraoperative, and postoperative phases of care, using operative efficiency, wound management, blood management, discharge, follow-up, and workforce design as illustrative domains. A practice-level example demonstrates how reductions in closure time and routine postoperative visits can be translated into theoretical annual capacity. These calculations are arithmetic estimates rather than causal or product-specific effects. The framework emphasizes sequential implementation, explicit ownership, standardized definitions, prospective exposure capture, and predetermined decision criteria. Marginal gains should not be viewed as a collection of preferred products or isolated efficiency measures but as a learning system designed to improve access while […]

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## Artificial Intelligence in Arthroplasty: From Workflow Efficiency to Clinical Decision Support.

*Scuderi GR, Taunton MJ, Smitterberg CW, Mont MA*

PubMed · DOI

## Evaluating the Impact of Artificial Intelligence Scribes on Clinical Workflow and Documentation Quality: A Randomized Controlled Trial.

*Etagiuri A, Chaudhry D, van der Merwe J, Nickol M, van der Merwe JM*

BACKGROUND: The adoption of artificial intelligence (AI) scribes has grown rapidly in recent years, aiming to improve clinical workflow, increase efficiency, and reduce administrative burden by transcribing doctor-patient interactions. These systems capture in-office conversations and generate summaries using either standard or surgeon-specific templates. However, to date, no randomized controlled trials have rigorously assessed the proposed benefits of AI scribes. This randomized controlled trial focused on assessing AI scribe efficiency, error rate, accuracy, and patients' perceptions in comparison to traditional clinical dictation-transcription methods.

METHODS: A randomized controlled trial was conducted involving 304 patients in a single-surgeon adult reconstruction practice. Patients were allocated into the folowing two groups: a conventional documentation group (using dictation transcribed by the office administrator or preconstructed templates: 140 patients) and an AI scribe group (139 patients). Timing data were recorded from when the surgeon entered a consultation room to when they began with the next patient. Additional data included administrative processing times, such as transcription and faxing of medical letters. All documentation was reviewed for medical errors by the surgeon in both groups. Grammar, spelling, and wrong wording were assessed by unblinded resea […]

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## Machine Learning Using Preoperative Patient Factors Can Predict the Severity of Pain Following Primary Total Hip Arthroplasty.

*Ade-Conde AM, Ahmed I, Shah A, Hardisty M, Whyne C, Ravi B, Chaudhry H*

BACKGROUND: Persistent postoperative pain is a challenge in total hip arthroplasty (THA), yet a lack of accurate predictive tools exists for this outcome. This study aimed to develop and validate machine learning (ML) models to

predict long-term pain following THA and to identify key predictors.

METHODS: A secondary analysis of 513 patients who underwent primary unilateral THA from 2000 to 2011 was performed. Hip pain was measured on a 0 to 10 visual analog scale at 3-years (T3) and 5-years (T5) follow-up and served as the target variable. The dataset was split into 80% training and 20% testing sets. Various ML models (linear, tree-based ensemble, and neural network approaches) were evaluated using (1) mean squared error (where lower values indicate higher accuracy), (2) buffer accuracy within ± 1 and ± 2 points of the observed pain score, and (3) accuracy in classifying pain severity as low (0 to 3), moderate (4 to 6), or high (7 to 10).

RESULTS: Across 13 models, performance was the strongest for nonlinear ML approaches. The best-performing model, CatBoost, achieved low prediction error (mean squared error 2.70 at T3 and 4.11 at T5) and classified pain severity with 90.1% accuracy. Age, body mass index, history of back/neck problems, and specific items from the Western Ontario and McMaster Universities Osteoarthritis Index and Intermittent and Constant Osteoarthritis Pain t […]

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### Utility of Machine Learning in Predicting Catastrophic Cardiac Complications Following Hip and Knee Periprosthetic Fracture Surgery Using Preoperative Parameters.

*Sampson WT, Freeman IA, Shimizu MR, Kwon OJ, Xiao P, Kwon YM*

BACKGROUND: With the rising prevalence of primary total hip (THA) and knee arthroplasties (TKAs), the number of periprosthetic fractures (PPFs) is expected to increase. Myocardial infarction (MI) and cardiac arrest (CA) are devastating complications following PPF surgery. There is a paucity of literature demonstrating the utility of machine learning (ML) in predicting rare clinical outcomes in revision arthroplasty. This study aimed to develop ML models to predict cardiac complications following PPF surgery.

METHODS: Patients who underwent revision arthroplasty or open reduction internal fixation (ORIF) due to PPF were extracted from a national quality improvement database. There were four ML models that were developed to predict MI and CA: artificial neural network, random forest, histogram-based gradient boosting, and k-nearest neighbor. Area under the curve (AUC) was used to assess discriminatory ability. Validation was performed with calibration slopes, intercepts, and Brier scores.

RESULTS: The final population included 3,805 PPF surgery patients, consisting of 1,869 revision arthroplasties (1,378 THAs, 491 TKAs) and 1,936 ORIF (811 THAs, 1,125 TKAs) procedures. Among revisions, 1.97% suffered MI (27 THAs, 10 TKAs) and 0.92% suffered CA (13 THAs, five TKAs). In ORIF patients, 1.81 and 0.67% experienced MI (14 THAs, 21 TKAs) and CA (one THA, 12 TKAs), respectively. All MI […]

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### Nontraumatic Osteonecrosis of the Femoral Head: An International Evidence-Based Clinical Practice Guideline.

*Cheng EY, Mirzaei A, Goodman SB, Cui Q, Mont MA, Jones LC, ARCO Clinical Practice Guideline workgroup*

BACKGROUND: Nontraumatic osteonecrosis of the femoral head (ONFH) can lead to major disability in patients of all ages. It presents at various levels of severity and can be either symptomatic or asymptomatic. There is a vast array of management strategies. Treatment is often subject to physician bias. Clinical practice guidelines that are broad-based, internationally developed, consensus-driven, and strictly evidence-based are needed. The aim of this guideline by the Association Research Circulation Osseous (ARCO) was to develop international evidence-based recommendations to assist physicians and patients in managing ONFH.

METHODS: ARCO convened an international, multidisciplinary guideline panel that was balanced to minimize potential bias from conflicts of interest. The Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach was utilized, including GRADE Evidence-to-Decision frameworks. The panel prioritized clinical questions, defined criteria for the systematic review of evidence, evaluated the statistical analysis, and, by consensus, approved recommendation statements, which were then subject to external review by content experts and stakeholders (a health policy-maker and a patient).

RESULTS: The Panel Agreed on 12 recommendations for the diagnosis, evaluation, and management of ONFH.

CONCLUSIONS: Key recommendations of these guidelines requ […]

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### Closer and Wealthier, Not Healthier: Proximity and Socioeconomic Status Drive Ambulatory Surgery Center Selection More than Comorbidity.

*Oleisky ER, Kayali Z, Wong BW, Chandrashekar AS, Vesperman RI, Fox JA, Polkowski GG, Martin JR*

BACKGROUND: Ambulatory surgery centers (ASCs) are increasingly used for total joint arthroplasty (TJA), offering lower costs and shorter stays than hospitals. However, access to ASC-based arthroplasty remains uneven, with socioeconomic and geographic disparities suspected to influence which patients receive outpatient care. This study evaluated how community-level socioeconomic status (SES) and travel distance affect the likelihood of undergoing TJA at an ASC versus a hospital.

METHODS: A retrospective cohort analysis was conducted of 5,560 patients who underwent primary total hip or knee arthroplasty at a single academic center between 2018 and 2024. Of the cohort, 2,113 patients underwent surgery at the ASC and 3,447 at the hospital. Patient demographics, comorbidities, distressed communities index scores, and travel distance to the surgical facility were collected. A multivariable logistic regression was used to identify independent predictors of ASC utilization.

RESULTS: The ASC patients were significantly younger, traveled shorter distances, and lived in communities with lower socioeconomic distress. In adjusted models, increasing age (OR [odds ratio] = 0.96), greater travel distance (OR = 0.999), and higher distressed communities index scores (OR = 0.995) were associated with lower odds of ASC surgery. Black patients had significantly reduced ASC utilization compared wi […]

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### Racial Differences in Patient-Reported Outcomes After Total Joint Arthroplasty.

*Prado IP, Jahan M, Wu M, O'Donnell JA, Pean CA, Bolognesi MP, Wellman SS, Ryan SP*

BACKGROUND: Racial differences in total joint arthroplasty (TJA) remain poorly defined. The purpose of this study was to investigate racial/ethnic differences in postoperative clinical and patient-reported functional outcomes between non-Hispanic White, African American, Hispanic, Asian, and Native American populations using the American Joint Replacement Registry.

METHODS: All primary TJAs from 2012 to 2022 with reported race/ethnicity and a minimum of 1-year follow-up data were retrospectively reviewed. Patient-reported outcome measures were evaluated based on identified racial differences. Descriptive statistics were performed using a Chi-square, Fisher exact tests, or analyses of variance (ANOVA). Multivariate regression models were used to assess for predictors of functional outcomes. Statistical significance was set at $P < 0.05$. There were 896,154 total hip arthroplasties (THAs) and 1,401,359 total knee arthroplasties (TKAs) evaluated. The population of THAs was 88% White and 55% women, and their mean age was 66 years. The population of TKAs was 85% White and 61% women, and their mean age was 68 years.

RESULTS: In the multivariate logistic regression model, Asian patients had a significantly decreased odds of achieving minimal clinically important difference after TJA compared to White patients ($P = 0.02$, THA; $P < 0.01$, TKA). Overall postoperative functional scores were […]

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### Comparing Medicare and Non-Medicare Inpatient Total Hip Arthroplasty: Is the New Centers for Medicare & Medicaid Services Patient-Reported Outcomes Performance Measure Mandate Applicable to All?

*Elmenawi KA, Kwas C, Molloy LG, Zhang C, Pasqualini I, Deren ME, Cleveland Clinic Adult Reconstruction Research (CCARR), Piuzzi NS*

BACKGROUND: The Centers for Medicare & Medicaid Services implemented the Patient Reported Outcomes Performance Measure policy, mandating reporting of patient-reported outcome measures for total hip arthroplasty. A 22-point improvement in Hip disability and Osteoarthritis Outcome Score for Joint Replacement defines substantial clinical benefit (SCB). However, it remains unclear whether SCB achievement differs across insurance groups.

METHODS: We analyzed a prospectively collected cohort of 3,240 total hip arthroplasty patients (2016 to 2023) stratified by insurance: Medicare (n = 2,108), commercial (n = 921), and Medicaid (n = 211). Baseline

demographics, comorbidities, and patient-reported outcome measures were compared between Medicare and non-Medicare patients. Multivariable regressions identified predictors of SCB for Hip disability and Osteoarthritis Outcome Score for Joint Replacement. Medicare patients were older (median 72 versus 60 to 61 years, P < 0.001), had lower body mass index (median 29 versus 31 to 31.1, P < 0.001), and greater comorbidity, with over 40% having a Charlson Comorbidity Index ≥ 2 compared to less than 23% in non-Medicare patients (P < 0.001).

RESULTS: Medicare patients demonstrated less baseline disability than Medicaid, but lower mental health than commercial patients. Independent predictors of SCB failure included non-White race (odds ratio (OR) […]

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### Comparable Outcomes of Total Joint Arthroplasty Performed at Ambulatory Surgery Centers and Hospital Outpatient Departments.

*Potluri AS, Yadav AS, Acuña AJ, Majji I, Forlenza EM, Gerlinger TL, Della Valle CJ*

BACKGROUND: As the volume of outpatient total joint arthroplasty (TJA) continues to rise, evaluating outcomes across surgical settings has become increasingly important. Our study compared rates of 90-day complications between patients undergoing outpatient TJA at an ambulatory surgery center (ASC) and a hospital outpatient department (HOPD).

METHODS: Patients undergoing primary TJA at a single institution between June 2016 and April 2024 with a minimum of 90-day follow-up were retrospectively identified (ASC: n = 615, 49.6%; HOPD: n = 624; 50.4%). Demographics and 90-day outcomes, including complications, readmissions, and reoperations, were documented. Univariate and multivariate logistic regressions, including Firth's correction for scarce events, were performed for all evaluated outcomes while adjusting for sex, body mass index, Charlson Comorbidity Index, and procedure type.

RESULTS: The incidence of medical (1.1 versus 1.9%; odds ratio (OR) 0.55; P = 0.230) and surgical complications (3.3 versus 2.9%; OR 1.01; P = 0.977) were comparable between the ASC and HOPD cohorts, respectively. Specifically, the risk of periprosthetic joint infection (ASC: 0.8% versus HOPD: 0.5%; OR 1.25; P = 0.770), wound complications (1.8 versus 2.0%, OR 0.72; P = 0.458), and 90-day readmission (2.8 versus 4.0%; OR 0.66; P = 0.356) were not significantly different. Similarly, there were no diff […]

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### Negative Influence of Joint Line Change on 10.8-Year Outcomes and Survival of Medial Unicompartmental Knee Arthroplasty.

*Yang HY, Song EK, Zheng Y, Kang SJ, Na HS, Seon JK*

BACKGROUND: This study aimed to investigate the effect of changes in joint line height on clinical outcomes and survival rates of unicompartmental knee arthroplasty (UKA) for medial compartmental knee osteoarthritis at a mean follow-up of 10.8 years METHODS: We reviewed 205 patients who underwent medial UKA between March 2005 and December 2017 and had a minimum follow-up of five years. The mean overall follow-up duration was 10.8 years (range, 5.1 to 18.2). Patients were divided into two groups based on the degree of joint line restitution following surgery: the restored joint line group (joint line change less than 2 mm; n = 125) and the joint line lowering group (joint line change ≥ 2 mm; n = 80). Radiographic evaluations included the mechanical hip-knee-ankle angle, medial proximal tibial angle, lateral distal femoral angle, joint line convergence angle, and posterior tibial slope. Clinical outcomes were assessed using the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) and the Forgotten Joint Score. Kaplan-Meier survival analysis was performed using conversion to total knee arthroplasty as the endpoint.

RESULTS: There were significant differences in postoperative WOMAC function score, WOMAC total score, and Forgotten Joint Score between the restored joint line group and the joint line lowering group (all P < 0.05). During the follow-up period, 13 imp […]

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### Do Patients Undergoing Unicompartmental Knee Arthroplasty Who Fail to Achieve Minimal Clinically Important Difference Have Higher Rates of Conversion to Total Knee Arthroplasty?

*Singh R, Narravula R, Williams DL, Sauder N, Booth M, Lim PL, Melnic CM, Bedair HS*

BACKGROUND: With an increase in unicompartmental knee arthroplasty (UKA) being performed in the United States, arthroplasty surgeons may benefit from strategies to identify the UKA patients at high risk for conversion to total knee arthroplasty (TKA). These patients may benefit from closer monitoring, targeted physical therapy, and timely interventions. A strategy for monitoring at-risk patients may be the utilization of the Minimal Clinically Important Difference (MCID) threshold. The UKA patients who experience unresolved pain, suboptimal biomechanics, or are unable to actively participate in their recovery may not achieve MCID, which may contribute to higher conversion rates. This study investigated the association between MCID achievement following UKA and rate of conversion.

METHODS: This was a retrospective analysis of a prospectively maintained multi-institutional arthroplasty registry. A total of 760 UKAs with a mean 5-year follow-up were identified. Patient demographics, baseline characteristics, and the Patient-Reported Outcomes Measurement Information System Physical Function Short-Form 10a were collected. The Patient-Reported Outcomes Measurement Information System Physical Function Short-Form 10a scores were categorized according to a literature-derived MCID threshold of +3.0. Conversion-free survival through 5-year follow-up was compared between patients who achi […]

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## Comparison of Functional Recovery Between Unicompartmental Knee Arthroplasty and Medial Meniscus Posterior Root Tear Repair in Patients Aged over 50 Years: A One-Year Propensity Score-Weighted Study.

*Boontanapibul K, Nitayavardhana S, Tanariyakul Y, Pinsornsak P*

BACKGROUND: Medial meniscus posterior root tear (MMPRT) accelerates knee osteoarthritis progression. Meniscal root repair (MRR) in patients older than 50 years is controversial due to unpredictable healing and mild to moderate cartilage degeneration, which may lead surgeons to consider unicompartmental knee arthroplasty (UKA). This study compared clinical outcomes of UKA versus MRR in patients older than 50 years who had MMPRT and Kellgren-Lawrence (KL) grade ≤ 3 arthritis.

METHODS: We retrospectively reviewed patients aged 50 to 79 years who had magnetic resonance image-confirmed MMPRT who underwent either MRR or mobile-bearing UKA. Patients who had severe medial compartment osteoarthritis (KL-4), varus deformity greater than 5°, or body mass index greater than 35 were excluded. Propensity score weighting balanced baseline demographics. Clinical outcomes-including the Knee Injury and Osteoarthritis Outcome Score (KOOS), Tegner activity score, University of California at Los Angeles activity score, and visual analog scale for pain-were assessed at six weeks, six months, and one year postoperatively.

RESULTS: Unicompartmental knee arthroplasty demonstrated superior outcomes in total KOOS, KOOS symptoms/stiffness, KOOS sport and recreation, and KOOS quality of life from six weeks to one year, as well as visual analog scale pain scores from six months to one year. In the UKA gro […]

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## Does Prior Arthroscopy Delay Clinically Relevant Improvement Timelines Following Unicompartmental Knee Arthroplasty and Primary Total Knee Arthroplasty?

*Williams DL, Sauder N, Singh R, Murrietta AJ, Lim PL, Bedair HS, Melnic CM*

BACKGROUND: Prior knee arthroscopy before total knee arthroplasty (TKA) may lead to worse patient-reported outcomes. However, the impact of prior arthroscopy on unicompartmental knee arthroplasty (UKA) is unclear. Moreover, to our knowledge, no previous investigation has examined how prior arthroscopy is associated with the time to achieve the minimal clinically important difference (MCID). Thus, this study investigated whether prior arthroscopy is associated with the time to MCID achievement after both UKA and TKA.

METHODS: We conducted a retrospective analysis of a prospectively maintained multi-institutional arthroplasty registry. A total of 55 UKA and 136 TKA patients underwent prior arthroscopy within 1 year, while 51 UKA and 83 TKA patients underwent prior arthroscopy between 1 and 2 years. A one-to-four propensity-score match identified UKA and primary TKA patients who had similar characteristics, but no prior arthroscopy. Patients completed the Knee Injury and Osteoarthritis Outcome Score Physical Function Short Form preoperatively and at multiple time points postoperatively.

RESULTS: Arthroscopy was not associated with a delayed time to MCID for UKA, whether performed within 1 year (3.03 to 3.04 versus 3.03 to 3.04 months; P = 0.82) or 1 to 2 years prior (3.31 to 3.37 versus 1.67 to 1.84 months; P = 0.77). In contrast, arthroscopy within 1 year before TKA was associa […]

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### Patients Who Have a Preoperative Flexion Contracture Demonstrate Inferior Outcomes Following Unicompartmental Knee Arthroplasty Compared to Total Knee Arthroplasty.

*Acuña AJ, Potluri AS, Yadav AS, Forlenza EM, Alfonsi S, Della Valle CJ*

BACKGROUND: The traditional criteria for unicompartmental knee arthroplasty (UKA) excluded patients who have a flexion contracture greater than 5°. However, the outcomes of patients undergoing UKA who have a flexion contracture remain poorly studied.

METHODS: Patients undergoing primary UKA and total knee arthroplasty (TKA) between 2010 and 2022 who had a minimum of 2-years follow-up were retrospectively identified. Patients undergoing UKA who had a flexion contracture ≥ 10° were matched in a 1:1 manner based on age, sex, body mass index, smoking status, Charlson Comorbidity Index, and diabetes prevalence to two cohorts: TKA patients who had a flexion contracture ≥ 10° (TKA controls) and UKA patients, who did not have a flexion contracture (UKA controls). Changes in the range of motion as well as Knee Society Score (KSS) were compared. A total of 138 patients were included in the final analysis (n = 46 per cohort).

RESULTS: There were comparable changes in the range of mation and KSS values for UKA patients who had a flexion contracture and UKA controls. Patients undergoing UKA who had a flexion contracture had a smaller change in their contracture postoperatively relative to TKA controls (8.4 versus 12.0°; P < 0.001). Despite similar preoperative KSS scores, patients undergoing UKA who had a flexion contracture had lower KSS objective scores at final follow-up relative to TK […]

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### Kinematic Impact of Patella Resurfacing in Bicruciate-Stabilized Total Knee Arthroplasty: A Comparative Analysis of Femoro-Tibial Motion and Patellar Classification.

*LaCour MT, Smith LA, Dessinger GM, Chesney CD, Komistek RD*

BACKGROUND: Understanding patello-femoral and femoro-tibial kinematics is essential for optimizing outcomes following total knee arthroplasty (TKA). Registry data indicate higher revision rates in TKA patients who have unresurfaced patellae, yet the kinematic implications of patellar resurfacing remain underexplored. This study investigated the influence of patellar resurfacing on patellar height classification and knee kinematics.

METHODS: A retrospective observational study of 167 bicruciate-stabilized TKA patients who were from six surgeons was evaluated-128 with resurfaced and 39 with unresurfaced patellae. Subjects underwent fluoroscopic analysis during weight-bearing deep knee bends. Femoro-tibial kinematics, including antero-posterior condylar motion and axial rotation, were assessed via three-dimensional model-fitting. Patellar height was classified using the Blackburne-Peel ratio at 30° flexion.

RESULTS: The resurfaced cohort exhibited a lower incidence of patella baja (22%) and alta (2%) compared to the unresurfaced group (38 and 5%, respectively), with patella baja significantly more common in the unresurfaced group (P = 0.033). The resurfaced group also demonstrated more lateral condylar rollback (-14.4 ± 5.4 versus -11.8 ± 5.6 mm, P = 0.012) and greater overall axial rotation during flexion (9.4 ± 5.4 versus 7.4 ± 5.8°, P = 0.043), with fewer instances of reverse […]

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### Building Bridges, Not Dependencies: Empowering Local Surgeons as the Future of Global Arthroplasty Outreach.

*Alcerro JC, Christie MJ*

BACKGROUND: Short-term medical missions have provided episodic access to total knee arthroplasty (TKA) in low- and middle-income countries, but concerns persist regarding sustainability and continuity of care. We hypothesized that a Honduran surgeon-led TKA program would achieve clinically meaningful improvements in patient-reported outcomes with complication rates within national registry-reported benchmarks.

METHODS: We retrospectively analyzed consecutive primary TKAs performed from October 2023 to June 2025. Of 120 knee arthroplasties, 20 were excluded (two revisions, three without research consent, seven with incomplete baseline evaluation, and eight without complete follow-up), leaving 100 primary TKAs with paired data at 9 to 15 months (mean age 68 years; 80.0% women). Comorbidity burden was assessed using the Goldman cardiac risk index (99.0% class I to II). The primary outcome was change in the Knee Injury and Osteoarthritis Outcome Score for Joint Replacement (KOOS Jr.). Secondary outcomes included the proportion achieving the minimal clinically important difference (≥ 14.0 points), the proportion reaching KOOS Jr ≥ 71.0 (national registry-reported benchmark), and 90-day complications. Multivariable linear regression evaluated predictors of KOOS Jr improvement.

RESULTS: The mean KOOS Jr. score improved from 33.2 ± 12.6 to 88.6 ± 9.6 (mean improvement, 55.3 points; P […]

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## Conversion Total Knee Arthroplasty in Patients Who Have Periarticular Hardware: One Operation May Be Enough.

*Molloy IB, Abbaszadeh A, Yilmaz MK, Sherman MB, Restrepo C, Parvizi J*

BACKGROUND: Patients undergoing total knee arthroplasty (TKA) not infrequently have periarticular hardware in place from prior surgical intervention(s). The optimal management of hardware around the knee is not known, with some surgeons preferring simultaneous removal while others stage hardware removal and TKA. This study compared outcomes in patients with hardware in place who underwent simultaneous or staged management of the hardware and TKA.

METHODS: This retrospective study included 266 patients undergoing conversion TKA between 2000 and 2023. Simultaneous hardware removal at the time of TKA was performed in 170 patients, while 96 patients underwent staged procedures. Primary outcomes included 90-day postoperative infection and postoperative complications. Hardware removal was categorized as major hardware (plate and intramedullary nail) versus minor hardware (screw, pin, staple, and button). There were no statistically significant differences in patient age, sex, body mass index, or comorbidities between the two cohorts.

RESULTS: Patients with previous infections were more likely to undergo a staged procedure (P = 0.002), with no significant difference in infection rate between the staged cohort (6.25%) and simultaneous cohort (2.35%) (P = 0.176). The overall rate of TKA complications was also not significantly different between groups (13.5% simultaneous, 20.8% staged […]

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## Effects of Ankylosing Spondylitis on Total Knee Arthroplasty Outcomes and Postoperative Manipulation Under Anesthesia Rates.

*Gadda MN, Norton JB, Mittal MM, Collett GA, Huo MH, Weinschenk RC*

BACKGROUND: Ankylosing spondylitis (AS) is a chronic autoimmune spondyloarthropathy primarily affecting the spine and sacroiliac joints, with up to 70% of patients developing asymmetric oligoarthritis in lower-limb joints. With the rise in total knee arthroplasty (TKA) among the elderly, more AS patients are undergoing this procedure. This study aimed to assess whether AS patients undergoing TKA have a higher rate of manipulation under anesthesia (MUA) compared to non-AS patients, hypothesizing that AS patients' increased inflammatory state would lead to more synovial hypertrophy, scarring, and stiffness, resulting in a higher need for MUA.

METHODS: We conducted a retrospective analysis using a national insurance claims database, identifying patients who underwent primary TKA from 2010 to 2020. Patients were divided into AS-TKA (who had a preoperative AS diagnosis) and non-AS-TKA groups. Propensity score matching (1:5 ratio) was used to compare outcomes between the groups. Postoperative complications, including MUA, were analyzed using Chi-square tests. Out of 2,055,785 TKAs, 4,131 involved AS patients. After exclusions and matching, the AS-TKA group had 3,512 patients, and the non-AS-TKA group had 17,560.

RESULTS: There were no differences in MUA rates postoperatively at 90 days (2.36 versus 2.90%, P = 0.08), one year (3.42 versus 3.95%, P = 0.14), and two years (3.62 versus […]

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## Intraosseous Vancomycin for Primary Total Knee Arthroplasty Is Associated With a Lower Rate of Postoperative Acute Kidney Injury Than Intravenous Vancomycin.

*Aflatooni JO, Wininger AE, Sullivan TC, Clyburn TA, Incavo SJ, Brown TS, Park KJ*

BACKGROUND: Perioperative vancomycin administration can be difficult to time and can lead to adverse systemic effects. Intraosseous vancomycin (IOV) is becoming increasingly utilized for total knee arthroplasty (TKA) prophylaxis. The primary purpose was to compare the rate of acute kidney injury (AKI) following primary TKA that utilized IOV or intravenous vancomycin (IVV).

METHODS: A single-institution retrospective review was conducted on 4,184 primary TKAs from July 2016 to November 2024 that received IOV (1,297 cases) or IVV (2,887 cases) as part of antibiotic prophylaxis. All included patients had a basic metabolic panel on postoperative day 1. Patients were excluded if they had a solid organ transplant, were on dialysis, or received a blood transfusion following TKA. An AKI was defined as a creatinine increase of 0.3 mg/dL from baseline. Data were analyzed using independent samples t-tests and Chi-square analyses. Logistic regressions were used to identify significant risk factors for AKI.

RESULTS: The rate of AKI was lower in the IOV than the IVV group (1.9 versus 3.3%, P = 0.017). Patients who developed AKI had longer hospitalizations (59.9 ± 39.1 hours versus 42.1 ± 33.3, P < 0.001) and a higher likelihood of discharge to skilled nursing facilities (10.1 versus 3.0%, P < 0.001) than non-AKI patients. Logistic regression demonstrated that IVV (odds ratio [OR]: 1.87; 95 […]

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## Patient-Reported Outcomes and Satisfaction Following Total Knee Arthroplasty in Rheumatoid Arthritis: An Observational Cohort Study.

*Kumagai K, Kubo M, Nosaka Y, Amano Y, Mimura T, Yayama T, Imai S*

BACKGROUND: Advances in pharmacologic therapy for rheumatoid arthritis (RA), including biologics and Janus kinase inhibitors, have reduced joint damage and the need for surgery. However, total knee arthroplasty (TKA) remains essential in patients who have RA and are generally favorable; most studies emphasize physician-reported outcomes, with limited focus on patient-reported outcome measures (PROMs).

METHODS: We retrospectively reviewed 61 patients who have RA (61 knees) who underwent primary TKA between June 2018 and June 2023. Clinical outcomes were assessed using range of motion, Japanese Orthopaedic Association score, and Knee Society Score. The PROMs included the Knee Injury and Osteoarthritis Outcome Score (KOOS) and a five-grade satisfaction scale. Patients were categorized as satisfied or dissatisfied for subgroup analysis.

RESULTS: The range of motion improved significantly from -10.1 to 112.9° preoperatively to -1.8 to 118.3° postoperatively. The Japanese Orthopaedic Association scores increased from 50.0 to 65.6, and both Knee Society Score knee and functional scores improved significantly (P < 0.01). All KOOS domains improved, with 85.2% of patients reporting satisfaction. In subgroup analysis, physician-reported scores did not differ significantly between satisfied and dissatisfied groups, whereas KOOS activities of daily living and quality of life subscales sho […]

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## Impact of Chronic Obstructive Pulmonary Disease on Complications and Readmissions Following Total Knee Arthroplasty: A Retrospective Matched Cohort Study.

*Reynolds C, Sontag-Milobsky IL, Riccobono G, Manning DW*

BACKGROUND: Chronic obstructive pulmonary disease (COPD) is a common comorbidity among patients undergoing total knee arthroplasty (TKA). We compared preoperative characteristics, perioperative management methods, postoperative complications, and readmissions after TKA in patients who did and did not have COPD.

METHODS: We performed a retrospective matched cohort study of primary, elective TKAs at an urban academic center (August 2012 to December 2020). We identified 785 patients who had COPD and matched them 1:3 to 2,274 patients who did not have COPD by age, sex, and body mass index. Groups were compared using univariate analyses. Multivariable regressions evaluated the independent associations of COPD with complications and 30-day readmissions. Temporal trends in complications and readmissions were assessed across the study

period.

RESULTS: Compared with non-COPD patients, those who had COPD had modest but consistently higher prevalence of several cardiopulmonary and medical comorbidities, reflecting a higher overall comorbidity burden at baseline. Intraoperatively, COPD patients more often received general anesthesia (10.1 versus 7.6%, P = 0.037) and carried higher American Society of Anesthesiologists classifications (class III or IV: 51.1 versus 30.3%, P < 0.001). Univariate analyses showed higher rates of postoperative anemia (3.6 versus 2.1%, P = 0.033), any complicat […]

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### Dual Mobility versus Conventional Total Hip Arthroplasty for Femoral Neck Fractures: A Multicenter Propensity-Matched Study on Dislocation Risk.

*Koh CK, Lee J, Young S, Monk P, Munro J, Lynch J, Beaule P, Grammatopoulos G*

BACKGROUND: Total hip arthroplasty (THA) is a common treatment for displaced femoral neck fractures (FNFs). Dual mobility (DM) THA has been advocated to reduce dislocation risk in high-risk patients, such as those who have FNF. However, comparative studies remain limited. The purpose of our study was to answer the following questions: (1) What is the dislocation incidence between DM-THA and conventional THA (c-THA) in patients who have FNF? (2) What is the revision incidence between DM-THA and c-THA in patients who have FNF? and (3) What are the risk factors for dislocation in patients who have THA for FNF?

METHODS: A multicenter retrospective cohort study involving 822 patients who underwent THA for displaced FNF at four public hospitals between 2012 and 2020 was conducted. Propensity matching (2:1) balanced age, sex, American Society of Anesthesiologists score, surgical approach, and surgeon experience to analyze dislocation and revision incidence between DM-THA and c-THA. Kaplan-Meier survival analysis and multivariable logistic regressions were performed.

RESULTS: Following propensity matching (209 DM-THA versus 418 c-THA), DM-THA demonstrated lower dislocation incidence up to five years (0.5% [95% confidence interval 0 to 2.5%] versus 4.8% [95% confidence interval 3.0 to 7.2%], P = 0.005). However, there was no difference in the revision for instability (DM THA: 0.5% ver […]

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### Clinically Relevant Threshold Achievement Evolves Over Time Following Primary Total Hip Arthroplasty.

*Cohen MF, Walsh EG, Patel DH, Quesada-Jimenez R, Kuhns BD, Domb BG*

BACKGROUND: Patient-reported outcome thresholds add clinical relevance to functional outcome scores. This study aimed to define the patient acceptable symptom state (PASS) and minimal clinically important difference (MCID) for the modified Harris Hip Score (mHHS), visual analog scale (VAS) for pain, Hip Osteoarthritis Outcome Score for Joint Replacement (HOOS-JR), and Forgotten Joint Score (FJS) and identify differences in threshold achievement over time.

METHODS: Data were retrospectively reviewed for patients who underwent primary total hip arthroplasty between December 2016 and February 2020. A total of 221 hips were included and had complete 2- and 5-year outcomes with anchor questions for the mHHS, VAS, and HOOS-JR, as well as FJS postoperatively. Receiver operating characteristic curves determined PASS threshold values. The MCID was determined using the distribution method for the mHHS and VAS. Rates of threshold achievement were compared between timepoints.

RESULTS: The PASS thresholds and achievement rates at the 2-year mark for the mHHS, HOOS-JR, FJS, and VAS were as follows: 88.0 (76%); 71.9 (85.5%); 53.1 (89.1%); and 2.5 (84.6%). The PASS thresholds and rates of achievement at the 5-year mark for the mHHS, HOOS-JR, FJS, and VAS were as follows: 75.0 (86.4%), 60.4 (91.0%), 51.0 (86.9%), and 2.3 (83.3%). More patients achieved the PASS at the 5-year compared to the 2 […]

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### The 10- to 23-Year Outcomes of Cemented Total Hip Arthroplasty Utilizing Impaction Bone Grafting for Severe Acetabular Bone Defects in Osteoarthritis and Rheumatoid Arthritis.

*Hashimoto K, Nakamura Y, Otsuka H, Takahashi N, Morishima T*

BACKGROUND: Long-term outcomes of impaction bone grafting (IBG) with a cemented cup in primary total hip arthroplasty (THA), particularly in Asian populations who have prevalent developmental dysplasia of the hip, remain limited. We evaluated outcomes at ≥ 10 years after primary cemented THA using IBG for severe acetabular defects in osteoarthritis or rheumatoid arthritis (RA).

METHODS: We retrospectively reviewed consecutive primary cemented THAs with IBG and metal-mesh containment (2000 to 2015). The cohort comprised 102 hips (99 patients). Clinical outcomes used the Japanese Orthopaedic Association hip score. Radiographs assessed component position, maximum acetabular defect distance, horizontal graft coverage, graft incorporation, and loosening. Kaplan-Meier survivorship used aseptic loosening as the endpoint. The mean follow-up was 12.8 years (range, zero to 23); 66 hips had a follow-up of ≥ 10 years, 33 had ≥ 15 years, and 14 had ≥ 20 years.

RESULTS: In all assessed hips, the Japanese Orthopaedic Association hip score improved by ≥ 20 points. No acetabular component showed aseptic loosening or migration ≥ four mm. The hip center was restored to a mean of 19.1 mm above the teardrop line, and trabecular continuity between host and graft bone was present in ≥ two zones in 96% at final follow-up. Survivorship was 100% at 10 and 15 years. There was one hip that underwent ace […]

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## Iliopsoas Impingement Risk in Collared Femoral Implants: Beware of Small Femora in the Setting of a Constant Length Collar.

*Kelly JJ, Yanamala SR, Taunton MJ, Pagnano MW, Sierra RJ, Hevesi M*

BACKGROUND: Collared femoral stems have seen a resurgence in total hip arthroplasty (THA) due to improved axial and rotational mechanical stability and lower early complication rates, including periprosthetic fracture. However, the presence of a collar over the femoral calcar may cause pain due to iliopsoas impingement (IPI). The objective of this article was to analyze temporal trends in collared stem use and determine risk factors for associated IPI.

METHODS: A retrospective institutional review was conducted to assess collared stem utilization and the incidence of IPI resulting in arthroscopic iliopsoas fractional lengthening (IPFL) between 2002 and 2025. Patient demographics, collar design, and implant sizing were analyzed as risk factors for IPFL. A total of 108 IPFLs and 29,070 primary THAs were performed during the study period.

RESULTS: While collared stem use remained relatively stable between 2002 and 2019, there was a significant increase (P = 0.003) in collar usage from 2020 onward, averaging 9.4% annual growth. A rise in IPFL procedures was observed over the same period, with a notable increase in cases involving collar overhang greater than two mm. Namely, collar overhang was observed in none of the IPFL cases in 2015, 16.7% of cases in 2020, and 31.6% of all cases in 2025 (P = 0.031). Constant length collar stems, though comprising only 19.6% of all collared st […]

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## Outcomes Following Total Hip Arthroplasty in Patients Who Have Cutaneous Psoriasis and Psoriatic Arthritis.

*Durbin JW, Cui E, Kesick A, Anderson B, Parel PM, Das A, Gu A, Sterling R et al.*

BACKGROUND: Cutaneous psoriasis (PsC) and psoriatic arthritis (PsA) are associated with higher risks for soft-tissue infection. However, there is sparse literature examining psoriasis and infectious complications following total hip arthroplasty (THA). Therefore, the purpose of this study was to observe and compare the rates of (1) superficial surgical site infections (superficial SSIs), (2) deep SSIs (deep SSI), and (3) 90-day sepsis rates, and 2-year revision rates following THA in patients who have PsC and PsA compared to patients who did not have psoriasis.

METHODS: A retrospective database analysis was performed utilizing a large national database. Patients who underwent THA were identified using Current Procedural Terminology codes. In total, 593,013 patients were included, of whom 10,230 patients (1.7%) had PsC and 1,445 patients (0.2%) had PsA. Patients were then classified into three cohorts: (1) PsA, (2) PsC, and (3) patients who do not have either condition (control). Multivariate logistic regressions were utilized to compare the outcomes between study cohorts.

RESULTS: Patients who have PsC had higher 90-day superficial SSI (odds ratio (OR): 2.3; confidence interval (CI): 2.2 to 2.5; P < 0.001), deep SSI (OR: 1.4; 1.2 to 1.5; P < 0.001), wound complications (OR: 1.1; 1.1 to 1.3; P < 0.001), sepsis (OR: 2.8; 2.6 to 3.0; P < 0.001), and aseptic revision (OR: 1.2; CI […]

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## Polymerase Chain Reaction as an Adjunct to Conventional Culture for Periprosthetic Joint Infection Diagnosis.

*Sarfraz A, Ruff G, DePillis R, Wang GH, Otto C, Pham V, Schwarzkopf R, Aggarwal VK*

BACKGROUND: Periprosthetic joint infection (PJI) remains one of the most challenging complications after total joint arthroplasty. While conventional culture remains the standard for identification of causative pathogens in PJI, rapid identification techniques such as polymerase chain reaction (PCR) are becoming increasingly popular, with little comparative data.

METHODS: A retrospective review was performed on a database of consecutive patients who underwent workup for PJI at a tertiary referral center from January 1, 2023, to May 31, 2025. Patients who underwent both standard synovial fluid culture and a PCR panel on the same day were included. Synovial fluid culture was used as the reference standard, and test result concordance and turnaround time were assessed.

RESULTS: Results from the PCR panel showed strong agreement with synovial fluid culture in the 225 cases analyzed, with 88% culture-positive agreement. The PCR false negatives were due to organisms not included in the PCR panel (n = 6), a false-positive culture from contamination (n = 1), and a patient who had a PJI due to Candida parapsilosis (n = 2). When only considering true positives with organisms included in the panel, the PCR displayed a 97.1% culture-positive agreement. The panel also detected organisms in eight culture-negative cases (94.7% culture-negative agreement). Those cases each met the 2018 Inter […]

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## High Mortality in Patients Presenting With Periprosthetic Joint Infection and Concomitant Sepsis.

*Tran DH, Yang KJ, Ha JH, Hand R, Schmitt DR, Brown NM*

BACKGROUND: Periprosthetic joint infection (PJI) has a 5-year mortality rate comparable to many cancers and can progress to a life-threatening sepsis. The aim of this study was to investigate the mortality of patients presenting with simultaneous PJI and sepsis after hip and knee arthroplasty.

METHODS: A retrospective chart review was performed on patients from an academic center between 2005 and 2022. Patients who underwent total joint arthroplasty, developed PJI, and presented for revision were identified. Septic patients met two or more systemic inflammatory response syndrome criteria and had suspected or confirmed bacteremia. The study included 306 admitted patients who had PJI. On admission, 43 patients (14%) had sepsis, while 263 did not (85.9%). The analysis was divided by knee or hip procedure, with 121 hip patients and 185 knee patients. Patient demographics, comorbidities, complications, and follow-up data were collected. Descriptive and univariate statistical analyses were conducted.

RESULTS: The all-cause mortality rate during follow-up was 35 of 43 (81.4%) in patients who presented with sepsis versus 133 of 263 (50.6%) in patients who presented without sepsis. The estimated probability of survival up to one year for septic patients was 77% (95% confidence interval (CI): 63 to 94) versus 92% for nonseptic patients (95% CI: 87 to 96) for knee patients and 42% (95% […]

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## Failure Incidence and Predictors Following Manipulation Under Anesthesia for the Stiff Total Knee Arthroplasty.

*Hurn MT, Balbuena JR, Heironimus EE, Henningsen JD, Smith LS, Tillett ED, Malkani AL*

BACKGROUND: Manipulation under anesthesia (MUA) is a common treatment for stiffness following total knee arthroplasty (TKA). The purpose of this study was to evaluate the outcomes following MUA and identify risk factors for additional surgical intervention.

METHODS: This study analyzed 227 knees that underwent MUA following primary TKA with 1-year follow-up. The 39 patients (17.2%) who required additional intervention were compared to 174 patients who underwent MUA without further interventions. There were no differences in body mass index, American Society of Anesthesiologists class, and prior knee surgeries. The mean follow-up was 38.5 months (range, 12 to 119.5) in the failure group and 39.2 (range, 12 to 124.2) in the control cohort.

RESULTS: Of the 39 patients requiring additional procedures, 29 (74%) underwent arthroscopic lysis of adhesions and 15 (38%) revision TKA. Range of motion (ROM) two weeks following MUA was lower in patients who had additional interventions (96.4 versus 107.4°, P < 0.001), as was ROM at one year (103.1 versus 112.9°, P < 0.001). Multivariate analyses found age, cruciate-retaining implants, and ROM two weeks after MUA as independently associated with failure. There were no differences in preoperative Knee Society Score (KSS) Function, KSS Knee, and Knee Injury and Osteoarthritis Outcome Score for Joint Replacement scores. At the latest follow-u […]

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## Effectiveness of Telerehabilitation After Total Knee Arthroplasty: An Umbrella Review.

*Xu H, Lou VWQ, Guo Z, Zhang X, Liu Y*

BACKGROUND: The use of telerehabilitation is increasing worldwide among patients who have undergone total knee arthroplasty (TKA). However, its efficacy and credibility in clinical settings remain unclear. An umbrella review (CRD420251070340) of the available scientific evidence was performed to determine whether telerehabilitation is an effective alternative to conventional rehabilitation among patients after TKA.

METHODS: We searched multiple databases through July 30, 2025, for relevant systematic reviews or meta-analyses. Effect sizes were recalculated as standardized mean differences (SMDs) with 95% confidence intervals (CIs). The initial search retrieved 1,192 articles.

RESULTS: The trials showed a significant effect of telerehabilitation interventions on patients undergoing TKA (SMD = -0.06, 95% CI [-0.16 to 0.03], P < 0.001, I2 = 69.9%). The bias estimate was -0.423 (standard error (SE) = 0.809, P = 0.604). In subgroup analyses, telerehabilitation interventions all had a significant effect on patients who underwent TKA (SMD = -0.14, 95% CI [-0.28 to 0.01], P < 0.001, I2 = 73.9%; SMD = 0.02, 95% CI [-0.09 to 0.14], P = 0.005, I2 = 47.7%; SMD = -0.15, 95% CI [-8.53 to 8.23], P < 0.001, I2 = 95.9%, respectively). Range of motion (passive flexion) (SMD = 0.26, 95% CI [-0.06 to 0.58], P = 0.041, I2 = 59.8%), Western Ontario and McMaster Universities Osteoarthritis Index (S […]

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## Total Knee Arthroplasty in Postpolio Residual Paralysis: A Systematic Review and Pooled Analysis of Its Outcomes and Complications.

*Regmi A, Baral S, Saini R, Jain VK, Sami A, Iyengar KP*

BACKGROUND: Total knee arthroplasty (TKA) in patients who have postpolio residual paralysis (PPRP) is technically challenging due to bone deformities, ligamentous laxity, and variable quadriceps muscle strength. Despite these challenges, TKA is increasingly being performed in this population, yet outcomes remain variably reported. This review aimed to systematically assess postoperative complications and revision rates following TKA in patients who had PPRP through a pooled analysis of available studies.

METHODS: A systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. Databases including PubMed, Embase, and Scopus were searched using the strategy ((total knee arthroplasty) OR (total knee replacement)) AND ((polio) OR (poliomyelitis)). A total of 12 studies comprising 272 knees were included. Data on demographics, knee characteristics, complications, and revision rates were extracted. Pooled proportions with 95% confidence intervals (CIs) were calculated using a random-effects model with the Freeman-Tukey double arcsine transformation, and heterogeneity was assessed using the I2 statistic.

RESULTS: The overall pooled complication rate was 16% (95% CI, 9 to 26), with instability or residual recurvatum (5%) and periprosthetic fractures (2%) being the most common. Other complications, including compon […]

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## Patello-femoral Tracking Optimization in Robotic-Assisted Total Knee Arthroplasty.

*Reddy H, Di Gangi C, DeGuzman G, Schaffer O, Rozell JC, Hepinstall MS, Meftah M*

BACKGROUND: Robotic-assisted total knee arthroplasty (RA-TKA) allows for intraoperative component positioning to personalize alignment and gaps. However, traditional trochlear designs not optimized for patellar tracking in kinematic alignment result in femoral internal rotation (IR) relative to the surgical transepicondylar axis. We sought to determine the femoral component alignment's effect on patellar tracking in RA-TKA.

METHODS: We retrospectively reviewed 932 RA-TKA cases performed from January 2023 to August 2024 using a computed tomography-based robotic platform with a single radius femoral component with a 6° trochlear sulcus angle. Femoral rotation was defined as IR and external rotation (ER) relative to transepicondylar axis. Femoral coronal alignment was defined as varus (Var) or valgus (Val) relative to the mechanical axis. Patient-reported outcomes were collected at six weeks, three months, and one year postoperatively. Outcomes were analyzed using 1-way analyses of variance and Chi-square tests.

RESULTS: There were 445 (48%) Var-ER, 242 (26%) Val-ER, 105 (11%) Var-IR, and 141 (15%) Val-IR cases. Mean femoral component rotation (° external) was Var-ER: 2.8° (range, 0.2 to 6.5); Val-ER: 2.4° (range, 0.1 to 5.0); Var-IR: -0.6° (range, -3.4 to 0); and Val-IR: -1.0° (range, -4.2 to 0). There were three patella-related complications, two of which had further reoperati […]

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## An Increase in the Treatment of High-Risk Total Joint Arthroplasty Patients Who Have Medicare at Teaching Hospitals.

*Kistler NM, Iyer A, O'Brien D, Culler M, Aron A, Lieberman JR, Heckmann ND*

BACKGROUND: The Comprehensive Care for Joint Replacement Medicare model may not adequately compensate for increased resource utilization in complex total joint arthroplasty (TJA) cases, raising concern for potential biased selection of healthier patients and avoidance of complex or high-comorbidity patients. This study quantified national trends among Medicare beneficiaries undergoing TJA, comparing comorbidity characteristics at teaching versus nonteaching hospitals.

METHODS: A comprehensive health care database was reviewed for adult patients (≥ 65 years) who underwent total knee or total hip arthroplasty from 2016 to 2023. Patients who had Medicare insurance reimbursement were examined at teaching and nonteaching hospitals. The Elixhauser Comorbidity Index (EI) was determined using International Classification of Disease, Tenth Revision, and Current Procedural Terminology codes. High-risk patients were defined as EI above the 95th and 99th percentiles. Average EI was examined by year. Univariate analyses were performed to assess the impact of hospital type on the proportion of high-risk patients in each cohort.

RESULTS: Of the 686,229 patients identified with Medicare, "high-risk" patients comprised 6.67% (95th percentile) and 1.23% of their respective cohorts (99th percentile). The percentage of high-risk patients was significantly greater at teaching hospitals compared t […]

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## Attention Should Be Paid to the Bone Mineral Density Status of Patients Undergoing Unicompartmental Knee Arthroplasty: A 1,164-Case Follow-Up Study.

*Liu D, Wang X, Dong X, Liu M, Ji G, Liu G*

BACKGROUND: Osteoporosis is a systemic skeletal disorder, which can compromise implant fixation strength and increase the risk of postoperative complications. Currently, the relationship between the abnormal bone mass and clinical outcomes following unicompartmental knee arthroplasty (UKA) remains unclear.

METHODS: A retrospective review was conducted on follow-up data from 1,164 patients who underwent UKA, who had a mean follow-up period of 32 months (range, 28 to 41). Bone mineral density (BMD) was measured and osteoporosis was diagnosed according to World Health Organization criteria. Among participants, 734 patients (63.1%) had normal BMD, 314 (27.0%) had osteopenia, and 116 (9.9%) had osteoporosis. Demographic and clinical data, including age, sex, body mass index, and length of hospital stay, were recorded. Postoperative functional outcomes were assessed using validated scoring systems: the Knee Society Score (KSS), Oxford Knee Score, visual analog scale, and Forgotten Joint Score (FJS).

RESULTS: The prevalence of osteoporosis was significantly higher in women than in men (P = 0.005). The BMD was positively correlated with body mass index (P < 0.001) and negatively correlated with age (P < 0.001). Postoperatively, all groups showed significant improvement in knee function (P < 0.001). However, the osteoporosis group exhibited significantly lower KSS clinical scores (P < […]

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## Untreated Onychomycosis Is Associated With Increased Risk of Periprosthetic Joint Infection Following Total Joint Arthroplasty.

*Shen V, Tan D, Pydi R, Ranson RA, Sterling RS, Villa JC*

BACKGROUND: Onychomycosis, a fungal infection of the nail, is often perceived as a benign condition. The purpose of this study was to evaluate whether onychomycosis is associated with an increased risk of periprosthetic joint infection (PJI) in patients undergoing total joint arthroplasty.

METHODS: Patients who underwent total knee arthroplasty (TKA) and total hip arthroplasty (THA) were identified using a large national database. Control cohorts of patients who did not have a history of onychomycosis were 1:1 propensity score matched to comparison cohorts of patients who had onychomycosis diagnosed within the year prior to their surgery. The comparison cohorts were further stratified by receipt of antifungal treatment. The 90-day surgical site infection (SSI) and 1-year PJI and reoperation rates were assessed. There was a total of 7,530 patients who had a history of onychomycosis, of which 4,620 patients underwent TKA, and 2,910 patients underwent THA.

RESULTS: Onychomycosis was associated with increased 1-year reoperations [odds ratio (OR): 1.43 (95% confidence interval (CI): 1.04 to 1.97), P = 0.029], but not with SSI or PJI (P > 0.05) after TKA. There were no significant differences in SSI, PJI, or reoperations (P > 0.05) after THA. Subgroup analyses revealed that patients who have onychomycosis and did not receive treatment had an elevated risk of 1-year PJI (both TKA an […]

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## Baseline Characteristics Explain Sex Differences in Primary Total Knee Arthroplasty Outcomes: A Prospective Multicenter Study With 4-Year Follow-Up.

*Bayram JM, Chopra S, Clement ND, Deehan DJ, London NJ, Pandit HG, Holloway NJ, Clarke JV*

BACKGROUND: Women report worse outcomes following total knee arthroplasty (TKA), but it remains unclear whether sex independently predicts outcomes or whether observed differences reflect baseline characteristics. This study aimed to characterize sex differences across a 4-year recovery period and determine whether they persist after adjusting for confounding factors.

METHODS: This prospective multicenter cohort study included 229 patients (114 women, 115 men; mean age 64 years) who underwent primary TKA. Patient-reported outcome measures (PROMs), including the Oxford Knee Score (OKS), Knee Injury and Osteoarthritis Outcome Score (KOOS) subscales, Forgotten Joint Score, and the EuroQol 5-Dimension, along with physical measures (Objective Knee Society Score and range of motion), were collected preoperatively, at six weeks, six months, and annually up to four years postoperatively. Linear mixed-effects models assessed sex differences in postoperative trajectories with three levels of adjustment: unadjusted; demographic-adjusted; and baseline-adjusted (including preoperative scores).

RESULTS: Women demonstrated lower-scoring postoperative trajectories than men across nearly all PROMs in unadjusted models, with the largest differences in functional outcomes (OKS: $\beta$ = -2.7, P < 0.001; KOOS Activities of Daily Living: $\beta$ = -5.7, P < 0.001). Women also had significantly worse preoper […]

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## Agreement of Patient Responses Between Electronic and Paper Patient-Reported Outcome Measures in Total Hip Arthroplasty.

*Swamy A, Dufresne M, Chambers K, McIlquham K, Lapner P, Beaule PE, Grammatopoulos G*

BACKGROUND: Total hip arthroplasty (THA) outcomes are commonly assessed using patient-reported outcome measures (PROMs). As electronic PROMs (ePROMs) gain popularity, ensuring equivalence to paper PROMs (pPROMs) is essential. This study aimed to evaluate the interchangeability of ePROMs and pPROMs in patients undergoing THA.

METHODS: A prospective cohort analysis of 96 THA patients who completed preoperative ePROMs and pPROMs was conducted. Electronic Oxford Hip Score (OHS) and EuroQol 5-Dimension 5-Level (EQ-5D-5L) were completed zero to 28 weeks prior to surgery, and pPROMs were completed on the day of surgery. Correlations were assessed with Spearman's rank correlation, and agreement and bias were evaluated using intraclass correlation coefficients (ICCs) and Bland-Altman analysis. The effect of time between completion of ePROM and pPROM on score difference was assessed.

RESULTS: Electronic scores were lower than paper format (OHS: median 19.0 [interquartile range, 13.8 to 25.3] versus 19.0 [14.0 to 25.0]; difference -1.5; P = 0.005, EQ-5D-5L: median 0.575 [0.455 to 0.681] versus 0.597 [0.472 to 0.704]; difference -0.028; P = 0.05). Differences were below Minimal Clinically Important Differences. Agreement was good for OHS (ICC: 0.873, 95% confidence interval: 0.805 to 0.917) and moderate for EQ-5D-5L (ICC: 0.677, 95% confidence interval: 0.551 to 0.773). Minimal bias (OHS […]

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## Asymptomatic Bacteriuria and Periprosthetic Joint Infection Risk: A Systematic Review and Meta-Analysis.

*Roy V, Van Brenk A, Benaroch LR, Alatassi R, Howard JL, Vasarhelyi EM, Lanting BA*

BACKGROUND: Periprosthetic joint infection (PJI) is a serious complication of hip and knee arthroplasty. Although hematogenous seeding is a recognized cause, the contribution of asymptomatic bacteriuria (ASB) remains uncertain. Therefore, we conducted a systematic review and meta-analysis to evaluate this association.

METHODS: A systematic search was conducted according to predefined eligibility criteria. There were 16 studies (14 cohorts [n = 48,562 joints] and two randomized controlled trials [n = 1,065 joints] that met inclusion criteria. The primary outcome was PJI rates after hip and knee arthroplasties in patients who had ASB. The secondary outcomes were subgroup analyses by procedure type and culture isolates in urinary and intraoperative cultures. Random-effects meta-analyses used Mantel-Haenszel pooling with Hartung-Knapp-Sidik-Jonkman adjustments.

RESULTS: The ASB cohort showed higher pooled odds of PJI (odds ratio 2.41, 95% confidence interval 1.20 to 4.83; P = 0.02), but subgroup analyses by follow-up and procedure type were not significant. Antibiotic treatment offered no benefit in elective arthroplasty, and ASB in hip fracture hemiarthroplasty showed no increased risk, with or without antibiotics.

CONCLUSIONS: The ASB cohort demonstrated a statistically significant, yet modest, increased odds of PJI limited to pooled undifferentiated, arthroplasty. In contrast […]

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## High Utilization of Neuraxial Anesthesia in Revision Total Joint Arthroplasty Is Safe and Effective.

*Playter KP, McCormick BP, Ly SM, Niu R, Hollenbeck BL, Erdman JS, Talmo CT, Smith EL*

BACKGROUND: Evidence in support of neuraxial anesthesia in revision total joint arthroplasty (rTJA) is still emerging. The purpose of this study was to demonstrate whether most rTJAs performed at an orthopaedic specialty hospital can be safely completed under neuraxial anesthesia.

METHODS: A retrospective cohort study including all patients who underwent revision total knee arthroplasty (rTKA) and revision total hip arthroplasty (rTHA) between June 3, 2024, and September 17, 2025, at a single institution was performed. Patients were divided by method of anesthesia. The primary outcome was the rate of neuraxial anesthesia utilization. Secondary outcomes included differences in baseline demographics and outcomes between groups. Propensity score matching for neuraxial versus general anesthesia was performed. Univariate analyses were utilized to compare groups. A total of 500 patients underwent rTJA (273 rTKA, 227 rTHA) during the study period. For rTKA and rTHA patients, 219 (80.2%) and 133 (58.6%) received neuraxial anesthesia, respectively.

RESULTS: On matched analysis for rTKA, general anesthesia was associated with higher intraoperative (57.2 versus 21.6 mL, P < 0.001) and total (175.9 versus 119.6 mL, P < 0.001) morphine equivalents administered and readmissions (15.7 versus 1.1%, P = 0.002). For rTHA, general anesthesia was associated with a longer length of stay (83.0 ver […]

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## Defining the Patient Acceptable Symptom State for the 12-Item Hip Disability and Osteoarthritis Outcome Score One Year After Total Hip Arthroplasty for Hip Joint Osteoarthritis.

*Matt AR, Kemp JL, Mosler AB, Semciw AI, Gomes DA, Gooden BR, O'Sullivan MD, Lyons MC et al.*

BACKGROUND: Patient-reported outcome measures (PROMs) are valuable for evaluating treatment success from the patient's perspective. The 12-item Hip disability and Osteoarthritis Outcome Score (HOOS-12) is a PROM scored from 0 to 100, worst to best, that assesses pain, function, and quality of life (QOL). Interpreting PROMs to determine patient satisfaction with treatment can be challenging. The patient acceptable symptom state (PASS) defines the PROM value beyond which patients consider their condition satisfactory. This study aimed to establish PASS values for the HOOS-12 one year after primary elective total hip arthroplasty for osteoarthritis.

METHODS: Data were drawn from a private Australian joint registry, including adults who underwent total hip arthroplasty between January 2020 and December 2022. A single-item satisfaction measure served as the anchor, and participants were categorized as satisfied or dissatisfied. PASS values were estimated using three methods: two predictive modeling methods (one with adjustment for unequal proportions of satisfied/dissatisfied participants and one without) and the receiver operating characteristic method. Nonparametric bootstrapping of confidence intervals was performed.

RESULTS: Among 746 participants, 96% reported satisfaction with the surgical outcome. The HOOS-12 PASS values derived from the three methods were lowest for QOL (r […]

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## Equivalent Functional Outcomes in Cruciate-Retaining and Posterior-Stabilized Total Knee Arthroplasty for Knee Osteoarthritis With Severe Fixed Flexion Deformity: A Propensity Matched-Pair Analysis.

*Soong J, Chen YJ, Youheng O, Lo NN, Yeo SJ, Liow MHL*

BACKGROUND: Cruciate-retaining (CR) total knee arthroplasty (TKA) is generally avoided in knees with severe fixed flexion deformity (FFD). The need for additional distal femoral resection, posterior soft-tissue releases, and gap balancing typically favors posterior stabilized (PS) implants. This study compared CR- and PS-TKA in severe FFD by evaluating clinical and patient-reported outcomes at two years: (1) range of motion (ROM), (2) Knee Society Score, (3) Oxford Knee Score, (4) Short-form 36 physical and (5) mental component scores.

METHODS: Patients who had severe preoperative FFD ≥ 15° from our institution's TKA registry were propensity-matched according to age, sex, body mass index, and preoperative scores. A total of 294 knees (147 CR-TKAs and 147 PS-TKAs) were matched and analyzed. Patients were evaluated preoperatively, at six months, and at two years. Postoperative manipulation under anesthesia was recorded.

RESULTS: Both groups achieved significant correction of FFD at two years (CR-TKA: 20.8° [95% confidence interval (CI), 19.7 to 21.9] to 4.0° [95% CI, 3.1 to 4.9]; PS-TKA: 19.9° [95% CI, 19.0 to 20.8] to 3.8° [95% CI, 3.0 to 4.6]; both $P < 0.001$) with no intergroup differences ($P = 0.534$). All patient-reported outcomes improved significantly from baseline ($P < 0.001$) and were comparable between groups. The PS-TKA achieved greater knee flexion and overall ROM (bot […]

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## Increased Risk of Postoperative Complications Following Total Knee Arthroplasty in Patients Who Have Metabolic Syndrome.

*Smitterberg CW, Misch M, Katanbaf RM, Nace J, Mont MA, Delanois RE*

BACKGROUND: Postoperative outcomes after total knee arthroplasty (TKA) are influenced by patient comorbidities. Metabolic syndrome (MetS) combines hypertension, diabetes, obesity, and dyslipidemia, which have been associated with increased short-term surgical complications. However, its independent impact on TKA outcomes beyond the perioperative period (> 90 days) remains unclear. This study compared outcomes following TKA between patients who had and did not have MetS, including (1) mechanical complications (mechanical loosening and periprosthetic fracture), (2) infectious complications (surgical site infection and periprosthetic joint infection), and (3) reoperations (all-cause revision, septic revision, and manipulation under anesthesia), at 90 days, 1 year, and 2 years.

METHODS: A national all-payer database was reviewed using International Classification of Diseases, Ninth and Tenth Revision codes to identify 1,255,079 patients who underwent TKA from 2010 to 2023. Among these, 386,813 (30.8%) had MetS. Outcomes were assessed at 90 days, 1 year, and 2 years. Student's t-tests and Pearson's Chi-square tests were used for unadjusted comparisons, and multivariable logistic regressions estimated odds ratios (ORs) with 95% confidence intervals (CIs).

RESULTS: At 2 years, MetS was associated with increased odds of mechanical loosening (OR 1.62, 95% CI 1.43 to 1.83, P < 0.001), […]

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### Impact of a Prior Debridement, Antibiotics, and Implant Retention on Outcomes of Two-Stage Exchange Knee Arthroplasty in Periprosthetic Joint Infection.

*Cisneros CL, Shannon MF, Heimroth J, Wong VR, Frear AJ, Gordon AC, Williams AA, Carlos NBT et al.*

BACKGROUND: In periprosthetic joint infection (PJI), prior studies evaluating failed debridement, antibiotics, and implant retention (DAIR) before two-stage exchange arthroplasty report conflicting outcomes and exclude initially successful DAIR cases. This study (1) compared two-stage exchange arthroplasty success with and without a preceding failed DAIR and (2) evaluated outcomes with initially successful DAIR cases included.

METHODS: This multihospital retrospective cohort study evaluated patients with PJI ≥ one year after primary total knee arthroplasty treated with initial DAIR or two-stage between 2015 and 2022. All patients met Musculoskeletal Infection Society (MSIS) or 2018 International Consensus Meeting criteria. We excluded patients who had prior PJI in the same joint or those who underwent more than one DAIR or two-stage exchange prior. Patients were categorized based on initial treatment: DAIR (n = 94) or two-stage exchange (n = 99). A subset of the DAIR cohort requiring later two-stage exchange (n = 22) was analyzed separately. Treatment success was defined as no reoperation and a MSIS Tier 1 or 2 outcome. Statistical analyses included t-tests, Chi-squares, Kaplan-Meier curves, and Cox models adjusted for key covariates (P < 0.05).

RESULTS: Prior DAIR for acute PJI was not significantly associated with reduced success rates when comparing primary two-stage excha […]

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### Radiographic Measures of Patellar Tracking After Total Knee Arthroplasty: Comparing the Native Patellar Midpoint Versus Median Ridge as Reference Landmarks.

*Sellig MT, Leipman JH, Bash HD, Sherman MB, Restrepo C, Lonner JH*

BACKGROUND: Accurate radiographic assessment of patellar tracking in total knee arthroplasty (TKA) is important. A commonly used method references the midpoint of the native patella (MPNP), which assumes central patella implant positioning and overlooks asymmetric patellar morphology in TKAs in which the patella is not resurfaced. Alternative measurement techniques that reference the patellar median ridge (MR) or center of the patellar prosthesis (CPP) may more accurately reflect true patellar position. This study compared MPNP with MR/CPP methods to determine whether landmark selection meaningfully alters the measured magnitude or direction of patellar translation.

METHODS: A retrospective review of 413 primary TKAs, including 213 unresurfaced and 200 resurfaced patellae, was conducted. Using preoperative and postoperative Merchant view radiographs, patellar translation relative to the trochlear sulcus was measured using (1) the MPNP method and (2) MR for unresurfaced patellae or CPP for resurfaced patellae. Magnitudes of shift and directional classifications were compared, with subgroup analyses performed by resurfacing status.

RESULTS: Across the full cohort, MPNP consistently measured greater lateral translation compared to MR/CPP, both preoperatively (5.01 ± 4.72 versus -0.12 ± 3.01 mm, P < 0.001) and postoperatively (2.92 ± 3.47 versus 0.48 ± 0.50 mm, P < 0.001). Differ […]

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### Is Low-Dose Aspirin the Safest Prophylaxis for Venous Thromboembolism Following Primary Total Hip Arthroplasty?

*Tummala S, Areti AS, Liu Y, Chen AF, Sambandam SN*

BACKGROUND: Venous thromboembolism (VTE) remains a serious complication following primary total hip arthroplasty (THA), prompting routine use of perioperative chemoprophylaxis. In 2022, the International Consensus Meeting (ICM) endorsed low-dose aspirin (LDA) as the preferred VTE prophylactic agent for both hip and knee arthroplasty patients. Prior THA studies lack stratification by VTE risk and direct comparisons between aspirin monotherapy, combination regimens, and alternative anticoagulants. This study aimed to evaluate the safety and efficacy of LDA monotherapy compared to other anticoagulants across high- and low-risk THA cohorts.

METHODS: A large national database identified 141,334 patients who underwent primary THA from 2012 to 2025. Patients were stratified into high- and low-risk cohorts based on ICM-VTE criteria. Propensity score matching (1:1) was performed to compare 90-day outcomes between patients receiving LDA monotherapy and those receiving alternative anticoagulants (low-molecular-weight heparin, direct oral anticoagulants, or warfarin). Primary outcomes included VTE, defined as deep vein thrombosis (DVT) or pulmonary embolism (PE). Secondary outcomes included bleeding complications, infections, mortalities, and health care utilization. Subgroup analyses compared LDA with combination prophylaxis (LDA plus another anticoagulant) and rivaroxaban monotherapy. […]

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## A Combined Multi-Route Regimen of Multiple Low-Dose Intravenous and Periarticular Dexamethasone Improves Perioperative Outcomes After Total Knee Arthroplasty: A Prospective, Double-Blind, Randomized Controlled Trial.

*Xia Y, Song H, Jin J, Zhou H, Li L, Xu W*

BACKGROUND: The objective of our study was to compare five low-dose perioperative dexamethasone (DEX) regimens on early recovery after total knee arthroplasty and identify an optimal route combination.

METHODS: In a prospective, double-blind, randomized controlled trial, patients were randomized to five groups. The COMB (combination) group received a combined regimen of pre and postoperative intravenous (IV) doses with intraoperative periarticular injection (PAI), the IV2 group received pre and postoperative IV doses, the IV1 group received a single preoperative IV dose, the PAI group received intraoperative PAI, and the CON (control) group received no DEX. Outcomes within 72 hours were visual analog scale pain (rest, passive flexion, and ambulation); postoperative nausea and vomiting (PONV); opioid consumption; range of motion (ROM); C-reactive protein; interleukin-6; white blood cell count and neutrophil percentage; Knee Society Score; knee swelling; Quality of Recovery-15; lengths of stay; and complications.

RESULTS: Versus the CON group, the COMB group showed lower rest pain on postoperative days (PODs) 1 to 3, and the PAI group showed lower pain on POD one to two. Both groups had lower passive flexion pain on POD one to two and lower ambulation pain on POD three. Compared with IV2 and IV1 groups, the COMB group yielded lower rest, passive-flexion, and ambulation pain acr […]

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## Aspirin Is Associated With Improved Thromboprophylaxis Versus Low-Molecular-Weight Heparin Following Revision Knee Arthroplasty for Periprosthetic Joint Infection.

*Wier J, Kumaran P, Telang SS, Culler MW, Buchanan C, Lieberman JR, Heckmann ND*

BACKGROUND: There is no consensus regarding the optimal venous thromboembolism (VTE) chemoprophylaxis agent following surgery for periprosthetic joint infection (PJI). Aspirin (ASA) and low-molecular-weight heparin (LMWH) are two commonly used agents; however, efficacy data in this population is lacking. This study compared ASA to LMWH among patients undergoing first-stage revision total knee arthroplasty for PJI.

METHODS: A large national database was queried from 2016 to 2023. Patients undergoing antibiotic spacer placement for a knee PJI were identified and divided into those who received either ASA or LMWH on postoperative day one. Patients taking other prophylactic agents or those who had a VTE history were excluded. Propensity score matching was used to balance cohorts. There were 10,472 patients 1:1 matched (5,236:5,236) based on VTE prophylaxis type and were well balanced for covariates (standardized mean difference < 0.10). Both cohorts had an average age of 66 years and an average Charlson comorbidity index of 3.2. Tranexamic acid was used in 61.2% of the ASA cohort and 60.3% of the LMWH cohort. To account for residual confounding, multivariable logistic regression models were used to assess our outcomes of VTE and postoperative transfusion.

RESULTS: The ASA patients had lower rates of VTE (1.0 versus 1.6%, P = 0.002; adjusted odds ratio = 0.6, 95% confidence interv […]

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# Journal of Orthopaedics & Traumatology

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## Fractures of the coronoid process: state of the art.

*Marinelli A, Riva M, Coliva F, Minerba M, Carbone G, Guerra E*

Coronoid fractures are rarely isolated and are much more frequently associated with other osseous or ligamentous structures injuries. On the basis of the coronoid fracture patterns, described by the O'Driscoll classification, it is possible to recognize three main patterns of injury that differ on traumatic mechanism and on associated lesions: posterolateral rotatory instability, posteromedial rotatory instability, and axial load injuries. The management of coronoid fractures is challenging and varies according to characteristics of the fracture, associated lesions, and amount of elbow instability. In general, operative treatment is indicated in every case the fracture is at least 50% of the whole coronoid, whether the sublime tubercle is involved, and whether the ulno-humeral joint is not perfectly reduced. In conclusion, the correct management of the coronoid, especially in the setting of complex elbow instability, represents a predictive factor for patient outcomes and functional results. The stability of the elbow, rather than the size of the coronoid fragment, is the main parameter for surgical indication, aimed to fix the coronoid and/or repair the associated lesions.

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## Total hip arthroplasty in Italy: an observational, population-based study on surgical volume growth from 2001 to 2023 and forecasts until 2050 with six different statistical models.

*Ciminello E, Cuccu A, Romanini E, Venosa M, Cazzato G, Tucci G, Boniforti F, Carpanese L et al.*

BACKGROUND: The number of total hip arthroplasty (THA) procedures has been steadily increasing worldwide, driven by aging population, improvements in surgical techniques and implant design. This study aimed to analyze the temporal trends of elective THA in Italy since 2001-2023 and forecast THA volumes up to 2050 to provide insights for healthcare planning.

MATERIALS AND METHODS: International Classification of Diseases, 9th Revision, Clinical Modification (ICD9-CM) coding system was used to extract records of interest (elective THA) from the Italian National Hospital Discharge Record database. Six statistical models were applied to forecast future THA volumes: logistic regression; Poisson regression; logarithmic regression; inverse/power regression; Poisson log-normal regression; and hierarchical Poisson regression with temporal effects (HPTE). Model performances were assessed by using error metrics and internal validation on the basis of a rolling-origin approach. An out-of-sample validation was conducted to ensure a robust assessment of forecasting reliability. THA volume forecasts were provided with 95% prediction intervals.

RESULTS: A total of 1,318,400 records for primary elective THAs performed in Italy since 2001-2023 were analyzed. The number of THAs increased by approximately 80%, rising from 68.270 in 2001 to 122.777 in 2023. Among the tested models, HPTE generally […]

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## Clinical relevance of patient-reported outcome measures in the surgical management of focal chondral defects of the knee: a systematic review.

*Migliorini F, Maffulli N, Memminger MK, Hofmann UK*

INTRODUCTION: To evaluate clinical outcome following surgical management of focal chondral defects in the knee, patient-reported outcome measures (PROMs) are used. To give these measures meaning, parameters such as the minimal clinically important difference (MCID), patient-acceptable symptom state (PASS), minimally detectable change (MDC), clinically important difference (CID) and substantial clinical benefit (SCB) have been

introduced. This systematic review investigated the MCID, SCB, CID, PASS and MDC of the most commonly used PROMs for assessing patients following surgical repair of focal chondral defects of the knee.

METHODS: This systematic review was conducted in accordance with the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and the recommendations of the Cochrane Handbook for Systematic Reviews of Interventions. All clinical studies investigating tools to assess the clinical relevance of PROMs in the surgical repair of focal chondral defects of the knee were reviewed. In April 2025, the following databases were accessed: PubMed, Web of Science and Embase. The PROMs of interest included: the International Knee Documentation Committee (IKDC) questionnaire, the Knee Injury and Osteoarthritis Outcome Score (KOOS) and its related subscales activities of daily living (ADL), pain, quality of life (QoL), sports and recreational […]

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### Simultaneous versus staged bilateral total hip arthroplasty: a meta-analysis.

*Soler F, Hernández J, Lamo-Espinosa JM, Benlloch M, Mariscal G*

BACKGROUND: There is debate regarding the optimal timing of bilateral total hip arthroplasty (THA), with simultaneous or staged approaches considered. Cost-effectiveness is an important factor that influences resource allocation. The main aim of this study was to assess the cost implications of bilateral simultaneous (sim-THA) versus staged (st-THA) total hip arthroplasty. The secondary objective was to evaluate the efficacy and safety of sim-THA compared those with of st-THA.

METHODS: A systematic review and meta-analysis were conducted according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines. Studies comparing simultaneous and staged bilateral THA in terms of cost, complications, length of hospital stay, and patient outcomes were eligible. Odds ratios (ORs), mean differences (MD), and standard mean differences (SMD) with 95% confidence intervals (CIs) were calculated. The risk of bias was assessed using the Methodological Index for Non-randomised Studies (MINORS). Meta-analyses were performed using Review Manager version 5.4 (Cochrane, Oxford, United Kingdom).

RESULTS: A total of 20 observational studies including 13,984 patients were included. Sim-THA was associated with significantly lower total costs (SMD -0.54, 95% CI -0.92 to -0.16; p = 0.005) and shorter hospital stay (MD -2.90, 95% CI -4.38 to -1.42; p = 0.0001) than tho […]

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### Long-term outcomes of the modified Dunn procedure in moderate and severe slipped capital femoral epiphysis: a prospective case series with 7-year follow-up.

*Fahmy M, Abdelazeem AH, Shawky MA*

INTRODUCTION: Slipped capital femoral epiphysis (SCFE) is the most common hip disorder in adolescents and may lead to femoroacetabular impingement, early osteoarthritis, and long-term functional disability if inadequately treated. While in situ pinning remains the standard treatment for mild slips, it fails to correct the deformity in moderate and severe cases, potentially predisposing to degenerative changes. The modified Dunn procedure (MDP) was developed to restore proximal femoral anatomy through surgical hip dislocation while preserving vascular supply. The aim of the study is to evaluate the long-term radiological and functional outcomes of the MDP in patients with moderate (14 cases) and severe (10 cases) SCFE, and to assess the incidence of avascular necrosis (AVN), osteoarthritis, and other complications.

METHODS: A prospective case series was conducted between August 2015 and January 2019 at a single tertiary institution. A total of 24 hips with moderate-to-severe SCFE and open physis were treated using the MDP via surgical hip dislocation. Mild and acute-only slips were excluded. MDP was used as a primary procedure, performed early in severe slips and in selected moderate slips after clinical assessment. Patients were followed clinically and radiologically for a mean duration of 84 ± 2.6 months (range 80-88 months). Functional outcomes were assessed using the Harris […]

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### The presence of a positive blood culture should be regarded a risk factor for DAIR failure in cases of haematogenous total knee arthroplasty infection.

*Pedemonte-Parramón G, Reynaga E, Molinos S, de Los Ríos JD, Vivero A, García-Oltra E, López-Pérez V, Paredes R et al.*

PURPOSE: Late acute haematogenous total knee arthroplasty (LAH TKA) infections represent approximately 20% of all prosthetic infections. Debridement, antibiotics, and implant retention (DAIR) is a widely accepted treatment; however, it carries a significant risk of failure. The study aims to identify risk factors for DAIR failure in LAH TKA infections, focusing on blood cultures (BC) as a potential contributor.

METHODS: A retrospective cohort study was conducted on 37 cases of LAH TKA infections from 2015 to 2023. Patients were divided into two groups: 20 in the success group (SG) and 17 in the failure group (FG). The study analyzes various factors, including demographics, comorbidities, serum and joint fluid biomarkers, blood and intraoperative cultures, prior infection or antibiotic use, and surgical and post-surgical variables.

RESULTS: Positive BC were more frequent in the FG compared with the SG (P = 0.03), and were also more common in patients with a history of prior infection (P = 0.03). Logistic regression identified positive BC as the only significant predictor of DAIR failure (odds ratio (OR) 12, 95% confidence interval (CI) 1.1-18, P = 0.04), even after adjusting for other variables. Positive intraoperative cultures were more frequent in the FG, particularly in those with a prior infection history (P = 0.08), even if they were receiving antibiotics (P = 0.05).

CON […]

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## Iatrogenic femoral neck fractures or separation of the proximal femoral epiphysis during closed reduction of irreducible femoral head fracture-dislocations in children: a review of 12 cases.

*Lu Y, Xue C, Canavese F, De Salvo S, Yao H, Huang D, Lin R, Zheng J et al.*

OBJECTIVES: This study aimed to analyze the radiologic characteristics of irreducible pediatric femoral head fracture-dislocations to prevent potential iatrogenic femoral neck fractures (FNF) or separation of the proximal femoral epiphysis (SPFE).

METHODS: This is a retrospective review of patients who were skeletally immature and diagnosed with traumatic hip dislocations combined with femoral head fractures. The collected data included patient demographics, fracture classification, fragment ratio, combined injuries, urgent reductions, treatment strategies, complications, and final outcomes.

RESULTS: We treated 12 patients with femoral head fractures and dislocations; 11 out of 12 patients underwent urgent closed reduction (91.7%). Six of the patients failed closed reduction and experienced FNF (n = 2, 33.3%) or SPFE (n = 4, 66.7%). Five patients presented with avascular necrosis of the femoral head after open reduction with internal fixation via a surgical hip dislocation approach, and two patients required further surgical treatment. Analysis of radiographs and computed tomography (CT) scans of irreducible femoral head fracture-dislocations revealed that the fractured femoral head was perched on the sharp angle of the posterior wall of the acetabulum, with a fragment ratio of 18-27%. After recognizing the irreducibility, one case with a fragment ratio of 26% underwent immed […]

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## Robotic-assisted total knee arthroplasty demonstrates superior early functional outcomes compared with conventional total knee arthroplasty: a quasi-randomised controlled trial.

*Migliorini F, Schäfer L, Nobili A, Kämmer D, Schneider J, Bell A*

BACKGROUND: Robotic-assisted total knee arthroplasty (TKA) has gained popularity over the last decade, with claimed advantages in implant positioning accuracy and soft-tissue balance. Whether these technical advantages translate into better clinical outcomes for patients remains debated. This investigation compared 1-year patient-reported outcomes between robotic-assisted and conventional TKA performed within the same institution but following structurally different surgical workflows.

METHODS: This quasi-randomised controlled trial enrolled patients who underwent primary TKA at the Department of Orthopaedic and Trauam Surgery of the Eifelklink St. Brigida (Simmerath, Germany) between 2022 and 2025. Group allocation was determined by surgeon assignment through the institutional digital scheduling system. All patients received the same implant (Smith & Nephew Legion Genesis II, posterior-stabilised).

Robotic-assisted procedures were performed using the CORI Robotic System (Smith & Nephew, UK) following a four-dimensional personalised alignment workflow (coronal, sagittal, transversal and balancing) guided intraoperatively by the Knee Balancing Score (KBS). Conventional procedures followed a femur-first technique with adjusted mechanical alignment. Outcomes included WOMAC total and subscores, visual analogue scale (VAS), Tegner activity score and a dedicated functional questionn […]

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### Advances in physiotherapy for chronic low-back pain: a level I systematic review.

*Migliorini F, Maffuli N, Schäfer L, Manocchio N, Bossa M, Foti C*

BACKGROUND: Non-pharmacologic management of chronic low back pain (cLBP) should be considered the first-line treatment. However, consensus guidelines and internationally accepted recommendations on the therapeutic algorithm are missing. The present level I systematic review presents a qualitative analysis of advancements in physiotherapy over the past decades to discuss the role of physical and rehabilitation medicine in the management of cLBP.

METHODS: In August 2025, the following databases were accessed: PubMed, Web of Science and Embase. All randomised controlled trials evaluating the efficacy of physiotherapy programs in patients with cLBP were assessed using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) criteria. The data extraction and risk-of-bias evaluation were performed independently by two authors.

RESULTS: Data from 150 randomised controlled trials (RCTs) (12,773 patients) were collected. The mean symptom duration was 61.2 ± 51.0 months, and the mean follow-up was 4.3 ± 5.9 months. The mean age was 44.5 ± 9.4 years, and the mean body mass index (BMI) was 25.8 ± 2.9 kg/m2.

CONCLUSIONS: Managing cLBP is challenging. While individual investigations highlight the efficacy of a specific intervention, most cLBP management strategies share common elements. Combining key elements from various approaches may prove crucial. Although this […]

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### Clinical examination findings were not associated with isometric strength or functional performance in adults with groin pain.

*Branco MO, Martins TB, da Costa GV, de Sousa VMM, Immich A, Dutra KB, Migliorini F, Maffulli N et al.*

OBJECTIVES: The aim of this study was to investigate whether positive clinical examination findings were associated with selected measures of lower-limb isometric strength or functional performance in physically active adults with groin pain.

METHODS: This was a cross-sectional, observational and exploratory study with blinded outcome assessment. A total of 37 physically active adults (18-45 years) with clinically diagnosed groin pain (≥ 6 weeks) underwent standardised orthopaedic testing (flexion, abduction and external rotation [FABER]; flexion, adduction and internal rotation [FADIR]; scour; squeeze), targeted palpation (adductors, pubic symphysis), isometric hip and knee strength testing using a handheld dynamometer and functional performance tests (modified Star Excursion Balance Test [mSEBT], cross-over hop). Associations between clinical findings and strength/performance measures were analysed using adjusted binary logistic regression models controlling for age, body mass index and height, with false discovery rate correction.

RESULTS: Participants demonstrated broadly similar right- and left-side performance in the mSEBT and cross-over hop tests, with strength ratios close to 1.0. Positive clinical examination findings were common, particularly for left adductor-region palpation (74.3% of valid observations) and the left FABER test (70.3%). After adjustment for multip […]

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### Increasing mobility while in hospital after hip fracture: even the best evidence may not translate to all national health services.

*Maffulli N, Rivera F, Truda G, Migliorini F, Oliva F*

Hip fracture rehabilitation is strongly influenced by the healthcare system in which postoperative care is delivered. The WHiTE 15: INITIATE trial represents an important pragmatic attempt to determine whether increasing the frequency and duration of in-hospital physiotherapy after hip fracture improves days at home within 30 days of admission. However, the transferability of such evidence to the Italian National Health System is limited by structural, regulatory and sociocultural factors. Italy has one of the oldest populations in Europe, a high annual burden of proximal femoral fractures, a limited public physiotherapy workforce capacity and a care model in which transfer to inpatient rehabilitation facilities is frequently shaped by patient and family preferences rather than clinical indication alone. Current evidence supports early mobilisation and structured rehabilitation, but it does not establish that a single discharge pathway is universally superior across all healthcare systems. In Italy, direct discharge home is often constrained by insufficient community-based rehabilitation services and strong expectations for post-acute institutional care. Therefore, intensifying in-hospital physiotherapy alone may not increase the number of days at home unless discharge preferences, family support and community rehabilitation capacity are addressed simultaneously. Future researc […]

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### Antibiofilm-coated intramedullary nailing for tibial and femoral diaphyseal fractures at high-risk infection: outcomes of 103 consecutive patients.

*Vicenti G, Migliorini F, Loiodice M, Scarpino A, D'attis P, Schäfer L, Colasuonno G, Solarino G*

BACKGROUND: Fracture-related infection (FRI) is a devastating complication of intramedullary nailing of tibial and femoral diaphyseal fractures. The present study evaluates the clinical efficacy of an antibacterially coated intramedullary nail in patients with high-risk tibial and femoral diaphyseal fractures. Open fractures, fractures requiring fasciotomy or closed fractures with Tscherne grade II or III soft-tissue damage were considered fractures at high risk of infection.

METHODS: A single-centre observational study was conducted between January 2022 and December 2024, in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines. Patients who underwent definitive intramedullary nailing with Bactiguard (Bactiguard AB, Tullinge, Sweden)-coated Natural Nail (Zimmer Biomet, Warsaw, IN, USA) for tibial or femoral diaphyseal fractures at high risk of infection were included. The primary outcome was the incidence of FRI, defined according to the internationally validated consensus criteria. Secondary outcomes included time to union, rates of nonunion, malunion, unplanned secondary procedures and functional outcomes at a minimum follow-up of 12 months.

RESULTS: A total of 103 patients were enrolled: 55 with tibial fractures and 48 with femoral fractures. The combined cohort comprised 90 males and 13 females, with a median age of […]

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### The efficacy of transcutaneous electrical nerve stimulation as an adjunctive therapy for perioperative pain management after total knee arthroplasty: an umbrella review.

*Huang M, Tang P, Wen T, Li Y, Lu B, Yang X, Xiao W, Zhang K*

BACKGROUND: Total knee arthroplasty (TKA) is a cornerstone intervention for end-stage knee pathologies, yet acute postoperative pain remains a critical barrier to rehabilitation. Transcutaneous electrical nerve stimulation (TENS), a non-pharmacological adjunct, has potential for pain mitigation, but its efficacy lacks robust evidence from umbrella reviews.

METHODS: This umbrella review systematically evaluated meta-analyses (MAs) published through September 2025 in PubMed/MEDLINE, Embase, the Cochrane Library, and Web of Science. Overlap among primary studies was assessed using the GROOVE tool, and evidence quality was graded via the AMSTAR 2 and GRADE frameworks. This umbrella review is registered on the Prospective Register of Systematic Reviews (PROSPERO) under accession no. CRD420251126630.

RESULTS: Eleven MAs were included in this review. The results showed that TENS significantly reduced acute postoperative pain at 24 h (MD = - 0.42, 95% CI - 0.59, - 0.25, P < 0.00001, I2 = 0%) and 72 h (MD = - 0.84, 95% CI - 1.17, - 0.52, P < 0.00001, I2 = 0%), but no significant difference was observed at 2 weeks (MD = 0.97, 95% CI - 0.66, 2.60, P = 0.24, I2 = 91%). Opioid consumption was lower in the TENS group (SMD = - 0.58, 95% CI - 1.06, - 0.11, P = 0.02, I2 = 53%). Improvements were also seen in range of motion (SMD = 0.40, 95% CI 0.11, 0.68, P = 0.006, I2 = 0%), Hospital for Spe […]

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### Orthopaedic publications receive fewer citations than general medicine articles: a scientometric analysis of temporal trends.

*Migliorini F, Vaishya R, Maffulli N, Eschweiler J, Kobbe P, Betsch M, Schäfer L, Oliva F et al.*

BACKGROUND: Orthopaedics, as a surgical speciality with a structurally narrower citing audience and lower citation density than general medicine, is systematically disadvantaged in any cross-disciplinary comparison relying on absolute citation counts. Whether this disadvantage is stable or has changed over time has not been previously quantified.

METHODS: The 100 most-cited articles indexed in the Web of Science Core Collection were identified for medicine and orthopaedics in 2014 and 2024. Citation data were extracted from a single snapshot taken on 19 January 2026. Total citations and citation velocity, defined as citations per year since publication, were compared between fields and across time points using Mann-Whitney U tests. Ratio-based analyses and distributional characteristics were also assessed.

RESULTS: Median citation counts were substantially higher in medicine than in orthopaedics in both years, reaching 5055 versus 288 in 2014 and 720.5 versus 34.0 in 2024, corresponding to 17.6- and 21.2-fold differences, respectively. Citation velocity followed the same pattern, with medicine maintaining an approximately 17- to 21-fold advantage at both time points. Within-field temporal analysis showed that citation velocity remained broadly stable in medicine between cohorts, whilst orthopaedics demonstrated reduced early citation accrual in the more recent cohort. Citatio […]

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### Conclusiveness of biopsy techniques for diagnosis of bone and soft tissue tumours: a retrospective, single-centre observational study.

*Smolle MA, Rauscher J, Wenzl FA, Igrec J, Szkandera J, Liegl-Atzwanger B, Leithner A, Andreou D*

BACKGROUND: Given the differences between biopsy techniques, especially regarding the representativeness and amount of tissue obtained upon diagnostic workup of suspicious bone and soft tissue lesions, this study aimed at evaluating the diagnostic conclusiveness of different biopsy techniques at a tertiary sarcoma centre.

MATERIALS AND METHODS: Overall, 222 consecutive patients (50.9% males, median age 63.3 [IQR 47.4-71.8] years) undergoing biopsy for a suspected bone or soft tissue lesion at a single centre between December 2021 and December 2023 were included. Biopsy techniques included core needle biopsy in the outpatient setting (without image guidance), sonography-guided and CT-guided and open biopsy. The conclusiveness rate of the first biopsy was defined as a histologically plausible result that enabled subsequent treatment planning.

RESULTS: Soft tissue and bone lesions were present in 114 (51.4%) and 108 (48.6%) patients, respectively. Core needle (n = 85; 38.3%) and CT-guided biopsies (n = 61; 27.5%) were most commonly employed. Median interval from first presentation to biopsy was 14 (IQR 3-29) days for bone and 0 (IQR 0-17) days for soft tissue lesions, attributable to the long delay of CT-guided biopsies (bone: median 28 [IQR 15-47] days; soft tissue: median 18 [14-45] days; $p < 0.001$). Complications were highest following CT-guided biopsies for soft tissue lesio […]

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### Clinical predictors of pathological femoral fractures in children: a retrospective comparative study.

*Kraus T, Clar C, Rünk J, Reinbacher P, Sackl V, Singer G, Rusch M, Leithner A*

BACKGROUND: Femoral fractures in children are uncommon and are usually trauma-related; however, some occur secondary to underlying bone pathology. Early differentiation between traumatic and pathological fractures is essential to avoid delayed diagnosis. This study aimed to identify clinical predictors associated with pathological femoral fractures in children.

MATERIAL AND METHODS: In this retrospective single-center comparative study, 432 pediatric patients (0-17 years) with femoral fractures were included. These fractures were classified as traumatic (n = 386, 89.4%; treated between 2011 and 2023) or pathological (n = 46, 10.6%; treated between 2002 and 2021) on the basis of radiological findings, advanced imaging, or histopathological confirmation. Demographic, anatomical, and

injury-related variables were compared between the two groups using Mann-Whitney U or chi square tests.

RESULTS: Patients with pathological fractures tended to be older (10.1 ± 4.3 years versus 8.3 ± 5.9 years, p = 0.090) and age-group distribution differed significantly between the two groups (p < 0.001). Traumatic fractures showed a bimodal age distribution, whereas pathological fractures followed a unimodal pattern. Both groups most commonly sustained diaphyseal fractures (58.7% pathological versus 59.8% traumatic). Low-energy trauma mechanisms were significantly more common in pathological fract […]

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## Development and validation of a vision transformer model for radiographic assessment of pediatric humeroradial joint alignment.

*Wang W, He S, Rao Y, Canavese F, Tang S, Zhao C, Liu H, Andreacchio A et al.*

BACKGROUND: Whether the radiographic assessments of pediatric anatomical (A-HRJA) and nonanatomical humeroradial joint alignment (NA-HRJA) among pediatric orthopedic surgeons (POS) relies on clinical experience remains unknown. No vision transformer (ViT) model for automated and accurate radiographic pediatric HRJAs categorization has been developed and validated. This study aims to evaluate the radiographic assessment outcomes of pediatric A-HRJAs and NA-HRJAs among POS with varying levels of experience, and to develop and validate a model for automated and accurate categorization of pediatric HRJAs on radiographs.

MATERIALS AND METHODS: A total of 294 pediatric radiographs were independently recorded by four junior-level, four mid-level, and three senior-level POS. Then, the performance of the POS with various levels of experience was evaluated. A total of 5118 pediatric radiographs (2932 A-HRJAs and 2186 NA-HRJAs) were prepared to construct the vision transfer (ViT) model. The ViT model's clinical efficacy was evaluated using an external testing dataset (84 A-HRJAs and 132 NA-HRJAs). A total of 11 POS independently recorded the HRJA types of the external testing dataset twice with a 1-month interval: first without ViT model assistance, and second with assistance. The rating performances of the POS without and with ViT model assistance were compared.

RESULTS: The accuracy, […]

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## Sleep quality and nocturnal pain in patients with elbow disorders: a prospective multicenter study.

*Schneider MM, Nietschke R, Migliorini F, Schoch C, Burkhart K, Hollinger B, Kimmeyer M, Lehmann L et al.*

BACKGROUND: Mental health disorders have become a growing global concern. Sleep quality is known to have a crucial impact on mental well-being. Poor sleep contributes to increasing economic costs owing to reduced productivity as well as the development of mental disorders. Conversely, anxiety and depression often lead to sleep disturbances. Pain is a key factor affecting sleep quality. Studies have shown that musculoskeletal disorders, especially in the shoulder, may cause sleep disturbances. Research on the relationship between elbow pain and sleep quality remains limited. This study aims to evaluate the impact of common elbow pathologies on sleep quality.

METHODS: This prospective multicenter cohort study used self-reported outcome measures in patients with elbow disorders, including the Pittsburgh Sleep Quality Index (PSQI), to assess sleep quality. The PSQI was compared with other outcome measures.

RESULTS: A total of 664 patients (284 female, 380 male) were included, of whom 81% reported nocturnal pain. The mean age was 47.9 ± 14.2 years, body mass index (BMI) 26.9 ± 6.7 kg/m2 and symptom duration 15.8 ± 7.3 months. The most common diagnoses were lateral epicondylar pain (66%), medial epicondylar pain, arthritis, ulnar nerve entrapment and biceps/triceps tendinopathies. The overall PSQI was 7.3 ± 3.8, disabilities of arm, shoulder and hand questionnaire (DASH) 35.0 ± 19. […]

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## A novel figure-of-eight polymeric cerclage wiring technique for greater trochanter fixation in revision arthroplasty and conversion of failed proximal femur fracture fixations to total hip arthroplasty.

*Leggieri F, Martín Cocilova FN, Secci G, Stimolo D, Zanna L, Civinini R, Innocenti M*

INTRODUCTION: Periprosthetic femoral fractures occur in 0.8-10% of primary total hip arthroplasties (THA) and up to 17.5% for revision THA (revTHA), with isolated greater trochanter (GT) fractures being the most predominant pattern. This study aimed to evaluate a novel figure-of-eight cerclage wiring technique for greater trochanter fixation in hip replacement cases.

METHODS: These retrospectively collected data included 36 hips (36 patients) undergoing revision THA or conversion of failed intertrochanteric fracture to THA with figure-of-eight cerclage wiring between April 2021 and June 2023. Patients under 18 years, with incomplete records, or lost to follow-up before 24 months were excluded. Primary outcomes included Trendelenburg sign, periprosthetic infections, and hip dislocation rates. Secondary outcomes evaluated THA stem subsidence and trochanteric migration using calibrated digital imaging. Statistical analysis included descriptive statistics, independent t-tests, linear regression, and correlation analyses with significance set at $p < 0.05$.

RESULTS: Mean follow-up was 404.76 ± 22.51 days. Patient demographics showed mean age 76.5 ± 12.3 years, body mass index (BMI) 25.8 ± 4.5 kg/m2, with 73.8% female patients. At final follow-up, one patient (2.8%) experienced post-surgical infection, one patient (2.8%) had dislocation, and three patients (8.3%) exhibited positive T […]

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### Minimal clinically important difference (MCID), patient-acceptable state (PASS), minimally detectable change (MDC), and substantial clinical benefit (SCB) in patients who have undergone arthroscopic surgery for femoroacetabular impingement: a systematic review.

*Migliorini F, Maffulli N, Schäfer L, Jeyaraman M, Betsch M, Mahmoud M*

INTRODUCTION: Hip arthroscopy has been established as a successful management for femoroacetabular impingement (FAI) syndrome. Patient-reported outcome measures (PROMs) are frequently used to assess treatment results in patients. However, clinically important outcome values have recently been calculated to better understand and interpret PROMs and define successful treatment. This systematic review aimed to investigate the minimal clinically important difference (MCID), patient-acceptable state (PASS), substantial clinical benefit (SCB), and minimally detectable change (MIC) of the most commonly used PROMs for assessing patients who have undergone arthroscopic surgery for FAI.

MATERIAL AND METHODS: This study was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses: the 2020 PRISMA statement. In October 2025, the following databases were accessed: PubMed, Web of Science, and Embase. All clinical studies investigating tools to assess the clinical relevance of PROMs used to evaluate patients who have undergone arthroscopic surgery for FAI were accessed. Only studies that assessed the minimal MCID, PASS, SCB, and/or MIC were considered.

RESULTS: The methodological quality of the 21 included studies was assessed, and the overall risk of bias was found to be low-to-moderate. The main characteristics of the included studies were analyzed and […]

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### Mattress impact on lifestyle and back pain: a 25-year orthopedic literature review guideline.

*Gianfilippo C, Filippo P, Stefano P, Alessandro A, Andrea F, Corrado C, Leonardo P, Giulia D et al.*

BACKGROUND: Sleep quality is influenced by several modifiable environmental factors, including the sleep surface. Low back pain, morning stiffness, shoulder discomfort, and muscle tightness on awakening may indicate inadequate bedding support. This systematic review evaluated whether mattress characteristics influence spinal alignment, sleep quality, and low back pain.

METHODS: A systematic literature search and qualitative synthesis were conducted for studies published from 1999 to 2023. PubMed, Cochrane Library, Web of Science, Scopus, EMBASE, PsycINFO, ScienceDirect, and Google Scholar were searched using terms related to mattress ergonomics, pain, vertebral-column alignment, sleep, and quality of life. Eligible studies evaluated adult or young-adult populations and reported outcomes related to low back pain, spinal alignment, pressure distribution, sleep comfort, or sleep quality. Methodological quality and risk of bias were appraised using literature evaluation and grading of evidence (LEGEND) and Critical Appraisal Skills Programme (CASP) tools.

RESULTS: A total of 47 studies were included. The evidence most consistently favored medium-firm mattresses over very firm or very soft surfaces. Medium-firm support was associated with better pressure distribution, improved sleep quality, reduced morning stiffness, and low back pain intensity. Zonal support, air-adjustable syst […]

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### Development and validation of a deep learning model for radiographic classification of pediatric femoral neck fractures.

*Wang W, He S, Ou L, Canavese F, Andreacchio A, Arrigoni C, Catena N, Corradin M et al.*

BACKGROUND: The Delbet-Colonna (DC) classification guides treatment of pediatric femoral neck fractures (PFNFs) but relies on clinical experience. No deep learning (DL) model has been developed and validated to differentiate between PFNFs and proximal femoral growth plates (PFGPs) and classify PFNFs via DC classification, in order to overcome this limitation.

MATERIALS AND METHODS: X-ray data including the annotations of 5555 PFGPs, 1306 PFNFs with various DC types, and 257 pediatric subtrochanteric femoral fractures (PSFFs), were prepared to construct a DL model based on the you-only-look-once (YOLO) model with wavelet transform (WT) and attention mechanism (AM) architectures. Two senior-level pediatric orthopedic surgeons (POS) performed the annotations independently by referring to the postoperative X-rays. The annotations were finalized if there were no differences. Otherwise, the two POS discussed and determined the final annotation. Thirty-one POS with different experience assessed the external testing dataset twice, without (first) and with (second) YOLO-WTAM model assistance. The rating performances of the YOLO-WTAM model and POS were evaluated. The kappa value reflecting reliability was obtained using a Fleiss kappa analysis.

RESULTS: According to the internal testing dataset, the area under the curve for different annotations ranged from 0.94 to 0.99. According to t […]

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### Association between Kaplan fiber injury and anterolateral tibial subluxation in patients with anterior cruciate ligament injury.

*Pan B, Chen S, Lin Y, Lin D, Zhou Z, Lin B, Zhou L, Ma J et al.*

BACKGROUND: Kaplan fibers (KFs) are critical stabilizers of the knee, yet their role in anterior tibial subluxation of the lateral compartment (ATSLC) in anterior cruciate ligament (ACL) injury remains unclear. This study is to determine whether KF injury is independently associated with increased ATSLC on magnetic resonance imaging (MRI) in patients with an injured ACL, and to identify risk factors for high-grade ATSLC (≥ 4.5 mm).

PATIENTS AND METHODS: This retrospective study enrolled 132 patients with ACL injury who underwent MRI within 12 weeks of trauma. MRI was used to evaluate the integrity of the ACL, KFs, anterolateral ligament (ALL), bony morphology, and ATSLC. Two trained observers independently assessed all images using standardized criteria. Agreement between observers was calculated using standard reliability metrics. Multivariable linear regression was performed to evaluate the independent association between KF injury and ATSLC. Logistic regression and receiver operating characteristic analysis were used to identify predictors of high-grade anterior tibial subluxation (≥ 4.5 mm) in patients with combined ACL and anterolateral complex injuries.

RESULTS: Of the 132 patients, 35.6% had KF injuries. These patients exhibited higher rates of ALL injury and distinct bony morphologic features. KF injury independently explained 15.1% of anterior tibial subluxation vari […]

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### Fibrous dysplasia: clinical and pathologic characteristics, surgical approaches, and outcomes in a large Italian cohort.

*Mordenti M, Trisolino G, Moroni A, Parisi SC, Nanni S, Staals EL, Righi A, Gambarotti M et al.*

Fibrous dysplasia (FD) is a rare benign bone disease caused by postzygotic activating mutations in the GNAS gene, leading to replacement of normal bone with fibrous tissue. It may be monostotic or polyostotic, often asymptomatic but sometimes associated with pain, deformity, or fractures. Management is mainly surgical in symptomatic or complicated cases. This retrospective study analyzes a large FD cohort to identify determinants of

treatment and outcomes, with a focus on pain and its evolution. The primary aim was to assess factors influencing surgical indication, while the secondary aim was to evaluate predictors of outcome in surgically treated patients, particularly symptom persistence at follow-up. A total of 227 patients were included. Mean age at diagnosis was 35.5 years (range 2-86 years); 79.3% had monostotic and 20.7% polyostotic FD. Pain was the most common presentation (62.6% at diagnosis; 78.4% during disease course), followed by incidental findings (21.6%), fractures (11.5%), and mass (4.4%). Surgery was performed in 51.6% of patients (127 lesions). Surgical patients were younger and had larger lesions and higher Mirels' scores (all $p < 0.001$), although Mirels' score showed only moderate predictive accuracy for surgery (area under the curve (AUC) 0.71). At a mean follow-up of 4.5 years, persistent pain was present in 24.1% of patients, with no significant differen […]

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### Glycemic variability improves risk prediction and reveals a mortality threshold in orthopedic trauma patients with sepsis.

*Long Z, Wang W, Wang S, Han T, Lin F, Li F, Qin Y, Lei M*

BACKGROUND: Sepsis is a life-threatening complication in patients with orthopedic trauma, associated with significant mortality. Dysregulated glucose metabolism is a hallmark of critical illness, but the specific role of glycemic variability (GV), beyond hyperglycemia, in this high-risk population remains inadequately explored.

METHODS: This retrospective cohort study utilized 1946 patients from the Beth Israel Deaconess Medical Center between 2008 and 2022. We included adult intensive care unit (ICU) patients with a primary diagnosis of orthopedic trauma and concurrent sepsis. GV was quantified as the coefficient of variation of all blood glucose measurements during the ICU stay. The primary outcome was all-cause mortality at in-hospital, 30-day, 90-day, and 365-day. Restricted cubic spline (RCS) analysis was used to determine the detailed relationship between GV and mortality and the optimal GV threshold for dichotomization. Propensity score matching and multivariable logistic regression were employed to control for confounders.

RESULTS: The study cohort experienced substantial mortality, with rates of 16.0% (in-hospital), 19.8% (30-day), 26.4% (90-day), and 33.9% (365-day). RCS analysis revealed a significant nonlinear, J-shaped relationship between GV and all mortality end points ($p < 0.001$), characterized by an initial modest increase in risk followed by a sharp escalati […]

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### Fibular allograft augmentation shows potential benefits in treating medial column comminuted proximal humeral fractures with osteoporosis: a secondary analysis of a randomized controlled trial.

*Chai Y, Shi T, Jiang H, Chen Y, Xu Z, Zhu Y, Ruan Z, Chen Y et al.*

BACKGROUND: Fibular allograft (FA) augmentation is proposed to address the lack of medial column support in plate-fixed proximal humeral fractures; however, the role of FA augmentation in these fractures remains inadequately elucidated. Our previous clinical trial did not show additional benefit in inserting an FA when treating adults with medial column comminuted proximal humeral fractures. Nonetheless, it is possible that some subgroups may still derive benefits from FA. This study aimed to investigate whether there were any subgroups benefit from FA.

MATERIALS AND METHODS: Our randomized controlled trial (RCT) recruited adults with a medial column comminuted proximal humeral fracture and randomly allocated them to receive a surgery fixed with a locking plate (LP group) or with an LP augmented with an FA (FA group). This study was a subsequent subgroup analysis of the previous randomized controlled trial, in which we included 72 participants who completed follow-up at the time of 12 months for the current subgroup analysis. The primary outcome was the Disabilities of the Arm, Shoulder and Hand (DASH) score at 12 months postoperatively. The secondary outcomes included the DASH score at other time points, changes in neck-shaft angle and humeral head height. We used generalized linear models for subgroup analysis to evaluate the effect of intervention (FA or LP) on 12-month DAS […]

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### Thicker polyethylene inserts for varus deformity versus thinner inserts for flexion contracture in total knee arthroplasty: a retrospective comparative study.

*Lee SH, Kwon HM, Lee WS, Park KK*

BACKGROUND: The success of total knee arthroplasty (TKA) depends on achieving precise mechanical alignment and stable soft tissue balance, with the thickness of the final polyethylene (PE) insert serving as a critical component. However, the impact of preoperative factors on PE thickness remains unclear. In this study, we investigated the effects of preoperative knee alignment and clinical deformities on PE thickness.

MATERIALS AND METHODS: The medical records of patients who underwent primary TKA between August 2024 and July 2025 were retrospectively evaluated. Preoperative radiographic parameters, including the hip-knee-ankle (HKA) angle, medial proximal tibial angle, lateral distal femoral angle (LDFA), joint-line convergence angle, weight-bearing line ratio (WBLR), and posterior tibial slope, were measured, along with the flexion contracture (FC) as a clinical factor. Intra-observer reliability was assessed using intraclass correlation coefficients (ICC). Linear regression analyses were performed to evaluate the effects of these preoperative factors on PE thickness.

RESULTS: This study included 233 knees (158 patients). The intra-observer reliability of the radiographic parameters was excellent (ICC > 0.9). Multivariable linear regression revealed that preoperative HKA angle ($\beta$ = 0.033, $p < 0.001$), WBLR ($\beta$ = -0.008, $p < 0.001$), and LDFA ($\beta$ = 0.065, $p = 0.003$) were signifi […]

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## PediLoc® blade plate yields faster consolidation rates than LCP pediatric condylar plate in distal femoral osteotomies: a retrospective comparison and description of surgical technique.

*Tschudi S, Delay C, Krautwurst B, Grisch D, Dreher T*

BACKGROUND: The PediLoc® Locking Cannulated Blade Plate System was originally designed for proximal femoral osteotomies but has also been used for multidimensional correction of distal femoral deformities. The present study delineates the surgical technique and undertakes a retrospective analysis of the consolidation rates of distal femoral osteotomies performed using the aforementioned system, comparing them with those performed using the DePuy Synthes® 90° LCP Pediatric Condylar Plate. We hypothesized that the semi-rigid design of the blade plate results in faster osteotomy healing rates compared with other locking screw plate systems.

MATERIALS AND METHODS: This retrospective single-centre study included patients who underwent distal femoral osteotomy with the PediLoc® Blade Plate (Blade Plate) and the DePuy Synthes® 90° LCP Pediatric Condylar Plate (90° LCP-PCP) at the University Children's Hospital Zurich (2020-2024). Consolidation rates were assessed using "Regenerating Bone Defects Using New Biomedical Engineering" (REBORNE), "Radiographic Union Score for Tibial Fractures" (RUST) and modified RUST (mRUST) scores at 6 weeks and 3 months postoperatively and compared using using Wilcoxon signed-rank test. For the comparison between both intervention groups, the unpaired t-test was used for normally distributed data and the Mann-Whitney U test for non-normally distributed a […]

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## Thin cortical translucent area of the iliac wing: morphometric analysis and implications for surgical safety.

*Yüceda■ Gündo■du H, Pasl■ B, Erçakmak Güne■ B*

BACKGROUND: The iliac wing is a frequently used fixation site in the surgical management of pelvic and acetabular fractures. Achieving safe implant placement in this region necessitates a comprehensive understanding of local morphology. However, the morphometric characteristics and surgical safety margins of the thin cortical translucent area (PLA) located in the central iliac wing have not been sufficiently defined. This may pose a significant challenge in evaluating the risk of cortical perforation, particularly during implant placement. This study aimed to evaluate the morphometric characteristics of the PLA and its relationship to fixation corridors.

METHODS: Fifty-one unpaired dry hip bones were examined. The thin cortical translucent region was identified using photoluminescence. Photograph-based measurements were conducted utilising ImageJ software on standardised photographs. Dimensions, area, distances to reference landmarks and relationships with the iliac pillar and supraacetabular corridors were assessed. The safe angular deviation range for anterior inferior iliac spine (AIIS)-origin screw fixation was calculated. Cortical thickness was measured using a digital thickness gauge.

RESULTS: The vertical and horizontal diameters and area of the PLA were 56.24 ± 16.56 mm, 59.92 ± 24.83 mm and 28.10 ± 16.47 cm2, respectively. Mean distances to the anterior superior ilia […]

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### Efficacy and safety of MIPO versus ORIF in distal tibial fractures: a systematic review and meta-analysis.

*Hassaan AM, Hassan AA, Abdelkhalik WS, Saad AM, Hosny AS, Elfeky BAEH, Mandour I, Ragheb MG et al.*

BACKGROUND: Distal tibial fractures are challenging to manage owing to limited soft-tissue coverage and compromised blood supply. Minimally invasive plate osteosynthesis (MIPO) and open reduction and internal fixation (ORIF) are among the widely used surgical options, but the optimal approach remains unclear. MIPO preserves periosteal blood flow with smaller incisions, while ORIF offers direct visualisation but increases soft-tissue damage. With growing comparative evidence, an updated evaluation of these techniques is necessary.

AIM: To compare the efficacy and safety of MIPO versus ORIF in adult patients with distal tibial fractures.

METHODS: Following PRISMA guidelines and the Cochrane Handbook, we searched in PubMed, Embase, Cochrane, Scopus, and Web of Science till November 2025. Randomised trials and cohort studies comparing MIPO and ORIF were included. Data extraction was performed independently by two reviewers. Risk of bias was assessed using RoB 2 for RCTs and the Newcastle-Ottawa Scale for observational studies. Random-effects models were used to estimate pooled mean differences and odds ratios, and heterogeneity was assessed using I2. Sensitivity and subgroup analyses were conducted by study design.

RESULTS: Nine studies involving 530 patients met our inclusion criteria. No significant differences were found between MIPO and ORIF in AOFAS scores, union time, oper […]

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### Comparative efficacy and safety of aspirin, low-molecular-weight heparin, and rivaroxaban for deep venous thrombosis prophylaxis after hip arthroplasty for femoral neck fracture: a single-center randomized controlled trial.

*Miao S, Yue Q, Zhang X, Lu J, Zhang Y, Lu N, Wei L*

OBJECTIVE: To evaluate the efficacy and safety of aspirin, low-molecular-weight heparin (LMWH), and rivaroxaban for deep vein thrombosis (DVT) prophylaxis after total hip arthroplasty (THA) or bipolar hemiarthroplasty (BHA) for femoral neck fracture, with subgroup analysis stratified by surgical type.

METHODS: This single-center randomized controlled trial (RCT) was conducted between June 2024 and June 2025. A total of 217 consecutive patients with femoral neck fracture undergoing hip arthroplasty were initially screened, and 161 eligible patients were finally enrolled and randomly assigned to the aspirin group, LMWH group, or rivaroxaban group at a 1:1:1 ratio. All patients received standardized intermittent pneumatic compression (IPC) and graduated compression stockings for mechanical DVT prophylaxis, as well as a unified postoperative rehabilitation protocol. The primary outcome was the incidence of lower extremity DVT within 12 weeks postoperatively, confirmed by color Doppler ultrasound. The secondary outcome was the incidence of bleeding-related complications graded by the International Society on Thrombosis and Hemostasis (ISTH) criteria. All patients were followed up at 6, 8, and 12 weeks postoperatively, with routine bilateral lower extremity ultrasound performed at 12 weeks regardless of symptoms. Subgroup analysis was performed on the basis of surgical type (THA ver […]

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### Does the surgical sequence of lumbar spine fusion and total hip arthroplasty affect the dislocation and revision risk? A meta■analysis and systematic review.

*Liu F, Jia C, Wu X, Zhou Y*

BACKGROUND: Patients who underwent primary total hip arthroplasty (THA) after lumbar spine fusion (LSF) had a higher incidence of complications compared with those who had not previously undergone LSF. This study aimed to determine whether the timing of THA before or after LSF impacts the incidence of THA complications.

METHODS: Following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines, we searched PubMed, Web of science, Embase, and Cochrane Library from their inception until 1 October 2025. Data from all eligible published studies meeting the inclusion criteria were extracted and subsequently analyzed using a random-effects model to calculate the rates of all-cause revision, dislocation, periprosthetic joint infections, and aseptic loosening.

RESULTS: Our analysis included 10 studies involving 27,605 patients who underwent THA followed by LSF and 32,002 patients who received LSF prior to THA. There are no significant differences in the rates of dislocation (odds ratio [OR] 1.19, 95% confidence interval (CI) 0.73-1.86, p = 0.45), periprosthetic joint infections (OR 0.91, 95% CI 0.52-1.62, p = 0.76), and aseptic loosening (OR 0.89, 95% CI 0.74-1.08, p = 0.23) between the two surgical sequences, while for all-cause revision, the OR was (OR 1.01, 95% CI 0.70-1.47, p = 0.94), which reached statistical significance.

CONCLUSIONS: Compare […]

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### Effects of anterior cruciate ligament resection and total knee arthroplasty on lower limb alignment and tibiofemoral rotation.

*Fan Y, Chen S, Jiang L, Chen Z, Wu M, Wang C*

BACKGROUND: The anterior cruciate ligament (ACL) is a key structure that restricts excessive anterior translation of the tibia and maintains rotational stability. This study aims to dynamically and quantitatively assess the direct effects of ACL resection and total knee arthroplasty (TKA) on lower limb alignment and tibiofemoral joint rotation under passive, unloaded conditions and in the pathological state of osteoarthritis.

METHODS: This retrospective study included 110 patients with varus knee osteoarthritis who underwent Mako for robotic-assisted TKA. Dynamic intraoperative measurements of the lower limb mechanical axis (flexion angle at maximum extension; varus angle) and tibiofemoral rotation at multiple flexion angles (min to max flexion) were recorded at three intervals: pre-ACL resection, post-ACL resection and post-TKA. The variation in rotation amplitude was calculated for successive flexion intervals.

RESULTS: Post-ACL resection, a significant decrease in the flexion angle at the maximum extension and varus angle (both $p < 0.001$) was observed, which was accompanied by increased tibial internal rotation at flexion angles of 60° and above ($p < 0.05$). Varus alignment was successfully corrected following TKA ($p < 0.001$). However, tibiofemoral rotational kinematics were significantly modified: the amplitude of internal rotation decreased in the initial flexion arc (min […]

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### Comparison of Bankart repair with remplissage and Latarjet procedure for anterior shoulder instability: a systematic review and meta-analysis.

*Nong K, Liu G, Jiang S, Li Q, Kwabena BR, Yuan D, Xiao W, Zhang K et al.*

BACKGROUND: Anterior shoulder instability is the most common type of shoulder instability. For cases with subcritical glenoid bone loss, both Bankart repair with remplissage (BR) and the Latarjet procedure are viable surgical options. The aim of this study was to compare the postoperative outcomes in patients undergoing BR and Latarjet procedure.

METHODS: This review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guideline. The methodological index for nonrandomized studies (MINORS) was used to characterize the methodological quality. A literature search was conducted using PubMed, the Cochrane Library, Embase, Web of Science, and Scopus. The analyzed outcomes included postoperative Rowe score, visual analog scale (VAS) score, the Single Assessment Numeric Evaluation (SANE) score, range of motion (ROM), return-to-sport (RTS) rate, RTS time, recurrence rate, revision rate, and complications. Subgroup analysis was performed on the basis of follow-up duration.

RESULTS: Overall, 12 studies involving 1186 patients were included. The meta-analysis showed that BR was associated with a significantly lower complication rate than the Latarjet procedure. No significant differences were observed in functional outcomes, ROM, recurrence, revision, RTS, or RTS time. Subgroup analysis showed a significant interaction between f […]

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## Risk of periprosthetic joint infection within 1 year following robotic-assisted versus conventional primary total knee arthroplasty: a propensity-score-matched cohort study.

*Tian S, Li Y, Mu W, Ji B, Zhang X, Cao L, Zhao J*

BACKGROUND: Robotic-assisted total knee arthroplasty (RA-TKA) is increasingly being adopted for its ability to enhance bone-resection accuracy and component alignment. However, whether these technical gains influence the risk of periprosthetic joint infection (PJI) remains uncertain, especially in the context of prolonged operative duration. This study aimed to compare the 1-year rate of PJI following conventional total knee arthroplasty (cTKA) and RA-TKA in a propensity-score-matched cohort.

METHODS: We retrospectively reviewed 1284 consecutive patients who underwent primary TKAs at a single centre between 2021 and 2023. The patients were stratified according to surgical technique (cTKA versus RA-TKA) and subsequently matched 1:1 using propensity score analysis (age, sex, body mass index [BMI], American Society of Anesthesiologists [ASA] score, Charlson Comorbidity Index [CCI] score, CCI components and smoking), resulting in 522 pairs (1044 patients) for the final comparative analysis. Operative time and 1-year PJI were assessed using multivariable logistic regression. Infections were stratified according to timing: ≤ 90 days and from 90 days to 1 year after surgery.

RESULTS: The 1-year rate of PJI was 0.77% (4/522) after RA-TKA and 0.96% (5/522) after cTKA (P = 1.000). All PJIs in patients who underwent RA-TKA occurred within 90 days, whereas PJIs in patients who underwent […]

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## Measured resection, gap balancing, and force-sensor-guided total knee arthroplasty result in different femoral and tibial rotational alignment but similar combined knee rotation and clinical outcomes: a randomized controlled trial.

*Hernández-Hermoso JA, Nescolarde L, Yañez-Siller F, Calle-García J, Pérez-Andrés R, García-Perdomo D*

BACKGROUND: Different alignment philosophies and balancing methods may alter femoral and tibial component rotation in distinct ways within the same knee. This study aimed to (1) determine which of three surgical techniques-measured resection (MR), gap balancing with computer-assisted surgery (GB-CAS), or a force-sensor soft-tissue balancing device (FS-STB)-most closely reproduces native femoral, tibial, and combined rotational alignment in mechanically aligned total knee arthroplasty (TKA), and (2) assess whether differences in rotational alignment affect outcomes at 5 years postoperatively.

MATERIAL AND METHODS: A total of 60 patients undergoing primary mechanical alignment TKA were randomly assigned to one of three surgical approaches (n = 20 per group): MR, GB-CAS, or FS-STB. Blinded observers assessed the Knee Society Score (KSS), Western Ontario MacMaster Universities Osteoarthritis Index (WOMAC) score, and hip-knee-ankle angle preoperatively and at the 5-year follow-up visit. Pre- and postoperative two-dimensional (2D)-computed tomography scans were used to measure femoral rotation (BFA), tibial rotation, and combined femur-tibia rotation (TE_PTCA and BC_PTCA). Statistical analyses included paired t-tests, one-way analysis of variance, and effect size calculations.

RESULTS: Femoral rotation remained unchanged in the MR and GB-CAS groups, but decreased slightly (1° exter […]

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## Risser growth plate injury in unstable paediatric pelvic fractures: a multicentre retrospective study.

*Oransky M, Aulisa AG, Roncoroni A, Zoppi AR, Mata M, Rohayem MA, Falciglia F, Toniolo RM*

INTRODUCTION: Unstable pelvic fractures in children are rare but severe injuries and are associated with high-energy trauma. Unlike adults, children have open growth plates, particularly the Risser growth plate (RGP), which is vulnerable to injury. The presence of growth plates, such as the triradiate cartilage and Risser's plate, represents point of vulnerability that determines distinctive fracture patterns, similar to those seen with Salter-Harris growth plate injuries. Lateral compression fracture is the most common mechanism and results in an irreducible crushing of the sacral alae, causing a rotational deformity. Both vertical and lateral compression

displaced pelvic fractures require careful evaluation of the posterior iliac apophysis-associated injury. Failure to treat Risser growth plate injury (RGPI) can lead to severe long-term problems such as limb length discrepancy and scoliosis. This study aimed to determine how commonly RGPI occurs in unstable paediatric pelvic fractures, link it to specific fracture patterns and propose a subgroup type to improve treatment.

MATERIALS AND METHODS: This multicentre retrospective study included 40 children aged up to 12 years with unstable pelvic fractures (AO/OTA 61B and 61C). The patients were treated between 1987 and 2022 at trauma centres in Italy, Argentina and Venezuela. We reviewed patient demographics, injury mechanisms a […]

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### Total femur replacement: function, quality of life, and gait: a multicentric EMSOS study.

*Valentini M, Svehlik M, Leithner A, Bergovec M, EMSOS TFR Study Group, Smolle MA*

BACKGROUND: Total femur replacements (TFRs) involve the replacement of the whole femoral bone, reconstruction of the hip and knee joints, and soft tissue reconstruction for function and joint stability. The aim of this multicentric European Musculo-Skeletal Oncology Society (EMSOS) study is to report on the functional outcome, quality of life, as well as gait analysis in patients with TFR.

MATERIALS AND METHODS: Retrospective data from 12 international participating centers were collected. In total, 65 patients [median (IQR) age at surgery 40 (18-56) years, 60% male] who received a TFR owing to oncologic or nononcologic indications between 1 January 1990 and 31 March 2024 were included. Patient-related outcome measures (PROMs), range of motion (ROM) of the hip and knee, and gait videos were collected.

RESULTS: At a median (IQR) follow-up of 28 (14-126) months, the median (IQR) percentage Musculoskeletal Tumor Society (MSTS) score was 63% (43-73%), 12-Item Short-Form Health Survey (SF-12) was 70% (55-79%), Harris Hip Score (HHS) was 63% (52-77%), and Oxford Knee Score (OKS) was 63% (52-79%); the median (IQR) extension-to-flexion ROM were 80° (70-97.5°, hip) and 70° (40-90°, knee). Gait videos were available for 24/65 patients (37%). The median (IQR) total Edinburgh Visual Gait Score of the operated side was 8.5 (6-13), with deviations from norm in all patients (24/24). Compens […]

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### Vascularized versus nonvascularized free fibular grafts in reconstruction of post-traumatic critical long bone defects, a comparative study.

*Fouaad AA, Massoud AH, Shaban M, Abdel-Naby M, Abulsoud MI*

BACKGROUND: Reconstruction of critical-sized long bone defects is a complex orthopedic challenge. Free fibular grafts, vascularized (FVFG) and nonvascularized (NVFG), are established reconstructive options, but comparative clinical outcomes remain uncertain.

PURPOSE: To compare clinical and radiological outcomes of FVFG and NVFG in patients with post-traumatic critical bone defects more than 10 cm.

METHODS: A randomized controlled trial was conducted with 50 patients assigned equally to FVFG or NVFG groups. The primary outcome was time to union, while secondary outcomes included graft hypertrophy, functional scores (DASH and LEFS), complication rates, and donor site morbidity.

RESULTS: The mean time to union was 5.78 months in the FVFG group and 6.17 months in the NVFG group, showing no statistically significant difference ($p = 0.447$). Rates of graft hypertrophy, functional recovery, and complications were comparable between the groups.

CONCLUSIONS: Both FVFG and NVFG provide effective reconstruction for critical bone defects, with nearly similar healing times and functional outcomes. NVFG may represent a less complex alternative in selected cases.

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### Five-year outcomes of medial open-wedge high tibial osteotomy with lateral supplemental lag screw in patients with obesity.

*Tian YH, Lee KH, Huang CW, Lin SY, Yang JC*

BACKGROUND: Obesity contributes to the accelerated progression of knee osteoarthritis. Medial open-wedge high tibial osteotomy (MOWHTO) is a joint-preserving surgical intervention for unicompartmental knee osteoarthritis; however, its efficacy in patients with obesity remains controversial. This study aimed to evaluate the 5-year clinical and radiographic outcomes of MOWHTO with lateral supplemental lag screw fixation in patients with obesity.

MATERIALS AND METHODS: This retrospective cohort study included patients with obesity who underwent MOWHTO between 2017 and 2020, with a minimum follow-up of 5 years. All procedures were performed by a single surgeon using three-dimensional printed, patient-specific instrumentation, locking plates, and a lateral supplemental lag screw. Clinical outcomes were assessed using the visual analog scale (VAS) and Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC). Radiographic parameters, including the weight-bearing line percentage (WBL%) and medial proximal tibial angle (MPTA), were evaluated.

RESULTS: Significant improvements in VAS and WOMAC scores were observed at all postoperative time points ($p < 0.001$) and were accompanied by improved radiographic alignment, with WBL% shifting toward the Fujisawa point and increased MPTA values. At 5 years, mild regression in alignment was noted; however, the overall correction was […]

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### Outcomes of surgical fixation of osteochondral fracture: a systematic review.

*Pang AS, Oviya R, Tan JJ, Lim KSA, Hui JHP, Tan SHS*

BACKGROUND: Dealing with knee osteochondral defects presents a substantial clinical challenge due to the variable nature of repair outcomes and intricate biomechanical complexities inherent to the knee joint. Numerous surgical alternatives exist for addressing knee osteochondral lesions, yet there is limited evidence available for comparing their respective outcomes.

METHODS: According to the Preferred Reporting Items for Systematic Review and Meta-analyses (PRISMA) guidelines, a systematic literature search was conducted to identify publications from inception to September 2023 on PubMed, Cochrane, and Medline academic search engines with the MeSH terms "osteochondral" AND ("patella" OR "patellar" OR "patellofemoral" OR "femoral condyle lesions" OR "trochlear lesions"). Inclusion criteria were clinical human studies, English language, subjects who underwent surgical management (fixation, reconstruction, or debridement) for knee osteochondral lesions with reported treatment outcomes, and a minimum sample size of ten patients. The methodological index for non-randomized studies (MINORS) was used to evaluate the non-randomized studies' methodological quality. Main outcome measurements Pediatric International Knee Documentation Committee (Pedi IKDC) scale, Tegner-Lysholm Scoring Scale, Tegner Activity Score, Visual Analog Scale, Knee Society Score (KSS) Score, Kujala Score, and F […]

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### Do modular tapered long stems differ in stability and function? A prospective comparison of two stems in complex femoral revision THA.

*Fahmy M, Shawky MA*

BACKGROUND: Revision total hip arthroplasty (rTHA) in the setting of substantial proximal femoral bone loss is technically challenging. Modular tapered fluted stems provide predictable diaphyseal fixation while allowing independent adjustment of version, offset, and limb length. Among these, two commonly used systems-modular tapered fluted stem Type A (Revitan™) and Type B (LIMA Modulus™)-have limited direct comparative evidence. This study aimed to prospectively compare radiographic stem subsidence (primary outcome), as well as functional outcomes, complications, survivorship, and secondary outcomes, between Type A and Type B modular long stems in femoral rTHA.

METHODS: In this single-center randomized prospective study, 110 patients undergoing femoral revision rTHA were randomly assigned to receive either Type A (n = 55) or Type B (n = 55) stems. All procedures were performed by experienced revision surgeons under standardized perioperative and rehabilitation protocols. Radiographs were analyzed for stem subsidence, osseointegration, and limb-length restoration. Functional outcomes were assessed using the Harris Hip Score (HHS), Oxford Hip Score (OHS), and European Quality of Life Visual Analogue Scale (EQ-VAS) at baseline and final follow-up (mean 61.4 months). Complications and stem survivorship were recorded prospectively. Statistical analysis included paired and unpaired […]

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## Clinical comparison of single posterolateral plate with medial-cannulated-screw fixation and double-plate fixation for extra-articular distal humerus fractures.

*Jung HS, Chu MS, Lee JS*

BACKGROUND: Double-plate fixation is the gold standard for extra-articular distal humerus fractures, but it carries a substantial risk of postoperative ulnar neuropathy. Fixation using a single posterolateral plate with a medial cannulated screw may reduce ulnar neuropathy while maintaining fracture stability. This study aimed to compare the clinical outcomes of single-plate-with-medial-screw fixation versus double-plate fixation for extra-articular distal humerus fractures.

MATERIALS AND METHODS: Fifty-six patients who underwent surgery for extra-articular distal humerus fractures (Arbeitsgemeinschaft für Osteosynthesefragen/Orthopaedic Trauma Association [AO/OTA] classification A2 or A3) between January 2018 and August 2024 were divided into a double-plate group and a single-plate-with-medial-screw group. We conducted a retrospective, nonrandomized comparative study. The double-plate fixation was used in 30 patients from January 2018 to October 2021, while the single-plate fixation with a medial screw was used in 26 patients from November 2021 to August 2024. All surgeries were performed using a posterior paratricipital approach. Bony union, radiographic healing, and loss of reduction were evaluated. Postoperative pain scores (visual analog scale at 2 days after the operation), operative time (minutes), elbow range of motion, elbow function (Mayo Elbow Performance Score [MEP […]

PubMed · DOI

## A study on the association between tibial plateau fractures and intra-articular soft-tissue injuries under valgus injury mechanisms.

*Duan S, Wang S, Zhu T, Li Q, Zhang M*

BACKGROUND: While recent investigations have focused on injury mechanism classifications of tibial plateau fractures (TPFs), the association between valgus TPFs and concomitant soft-tissue damage involving menisci and ligaments remains insufficiently elucidated. This study aimed to characterize intra-articular soft-tissue injuries associated with various valgus TPFs and assess the predictive value of lateral plateau depression (LPD) and widening (LPW). Additionally, the analysis extended to other injury mechanisms.

MATERIALS AND METHODS: This study included adult patients with acute tibial plateau fractures who had complete imaging data, excluding patients with open fractures and multiple fractures throughout the body. Imaging examinations were used to assess the fracture mechanism and intra-articular soft-tissue damage.

RESULTS: The study retrospectively analyzed the clinical data of 138 patients with valgus injury TPFs in our hospital. The study compared the incidence of TPFs with intra-articular soft-tissue damage under different injury mechanisms. The result demonstrated that the incidence of valgus hyperextension TPFs combined with medial collateral ligament injuries was relatively high (61.5%) compared with TPFs with other injury mechanism. Multivariable logistic regression and smooth curve fitting revealed significant dose-response relationships of LPD (odds ratio [OR] […]

PubMed · DOI

## The coronoid as the key fragment of trans-ulnar fracture-dislocations of the elbow: Insights from a retrospective cohort comparison using the coronoid-centric Mayo classification system.

*Henssler L, Kerschbaum M, Zanklmaier M, Wieczorek LA, Alt V, Klute L*

BACKGROUND: Trans-ulnar fracture-dislocations of the elbow are rare injuries with complex fracture patterns and variable outcomes. Traditional classification systems offer limited prognostic value. A recently introduced coronoid-centric Mayo classification distinguishes injury subtypes based on coronoid attachment and identifies trans-ulnar basal coronoid (TUBC) fractures as a particularly challenging entity. This study evaluated outcomes across Mayo fracture types and explored factors associated with inferior results in TUBC injuries.

MATERIALS AND METHODS: In this retrospective cohort study, surgically treated trans-ulnar elbow fracture-dislocations managed at a level I trauma center between 2010 and 2022 were identified and classified

according to the Mayo system. Demographic data, injury characteristics, surgical management, radiographic outcomes, and complications were recorded. Functional outcomes were assessed after a minimum follow-up of 12 months using the Mayo Elbow Performance Score (MEPS); Oxford Elbow Score (OES); Quick Disabilities of Arm, Shoulder and Hand Questionnaire (QuickDASH); European Quality of Life Five-Dimension, Five-Level Version (EQ-5D-5L); and range-of-motion measurements. Radiographs were analyzed for union, instability, heterotopic ossification, and post-traumatic osteoarthritis (OA).

RESULTS: A total of 52 patients were included (14 trans-olecr […]

PubMed · DOI

### Feasibility study of a robot-assisted endoscopic technique for plate osteosyntheses of the acetabulum and anterior pelvic ring.

*Hinz N, Althoff G, Thomaschewski M, Bonke LA, Eberenz A, Münch M, Behrends L, Männel G et al.*

BACKGROUND: Minimally invasive stabilization of acetabular and pelvic ring fractures using endoscopic techniques has become increasingly important. The logical advancement of conventional endoscopic techniques is a robot-assisted approach, which benefits from the advantages of robot-assistance systems (e.g., more degrees of freedom for instruments and improved visualization). The aim was therefore to investigate the feasibility of a robot-assisted endoscopic technique for plate osteosyntheses of the acetabulum and anterior pelvic ring.

MATERIALS AND METHODS: In this two-part feasibility study, four different plate osteosyntheses were performed endoscopically using the Hugo™ robot-assisted surgery system, first on ten synthetic pelvic models and then on ten human cadavers. During robot-assisted dissection for the preperitoneal approach, identification of ten relevant anatomical landmarks was also assessed. In both parts, the success, number of drilling errors, and time for each plate were described as learning curves and analyzed using linear regression.

RESULTS: The infrapectineal plate could be successfully performed in 100% of synthetic models, the posterior column plate in 100%, the suprapectineal plate in 90%, and the superior pubic ramus plate in 80%. Learning curves could be observed for the number of drilling errors per plate (e.g., 0.67 to 0, p = 0.009) and the time r […]

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### Favorable subjective clinical outcomes after revision hip arthroscopy for femoroacetabular impingement syndrome despite a high rate of capsular defects and increased capsular thickness.

*Cao Z, Gao G, Lin W, Zhu Y, Zhou X, Wang J, Xu Y*

BACKGROUND: The capsular healing status and capsular thickness changes in patients with femoroacetabular impingement syndrome (FAIS) following revision hip arthroscopy are poorly documented, and their relationship with subjective clinical outcomes remains unclear. The purpose of this work is to evaluate the incidence of capsular defects and changes in capsular thickness using magnetic resonance imaging (MRI) following hip arthroscopy in patients with FAIS, and to compare the subjective clinical outcomes between patients with and without capsular healing after revision hip arthroscopy.

PATIENTS AND METHODS: Consecutive patients with FAIS who underwent revision hip arthroscopy between 2013 and 2023 were included. Patients were categorized into two groups on the basis of capsular healing status after revision hip arthroscopy. Patient-reported outcomes (PROs) were collected preoperatively and at minimum 2-year follow-up, including the modified Harris Hip Score (mHHS), International Hip Outcome Tool-12 (iHOT-12), Hip Outcome Score-Activity of Daily Living Scale (HOS-ADL), Hip Outcome Score-Sport Specific Subscale (HOS-SSS), and visual analog scale (VAS). Achievements in minimal clinically important difference (MCID) and patient acceptable symptom state (PASS), patient satisfaction, and re-revision rates were evaluated and compared between groups. Capsular thickness was assessed by […]

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### From bench to bedside: a novel suture-augmented prosthesis for greater trochanteric fixation in osteoporotic hip arthroplasty.

*Yang M, Yang J, Wang X, Wu Z, Bian S, Sheng R, Liao F, Qin K et al.*

BACKGROUND: Stable fixation of the greater trochanter during hip arthroplasty for unstable osteoporotic intertrochanteric fractures remains a significant challenge. Conventional techniques depend on securing the bone-implant interface, which is particularly compromised in osteoporotic bone. Here, we propose a novel conceptual approach that establishes a direct mechanical bridge from the abductor tendon to the prosthesis, thereby reducing reliance on the fragile bone-implant interface.

METHODS: In this two-stage translational study, we progressed from biomechanical validation to clinical application. First, using a decalcified caprine femoral model simulating osteoporosis, three fixation constructs were compared: locking plate (LP), suture-augmented locking plate (LPSA), and Kirschner-wire tension band (KWTB). Ultimate load and construct stability were evaluated. Biomechanical testing confirmed the principle of suture-mediated load sharing and highlighted the intrinsic weakness of screw fixation in osteoporotic bone. Guided by these results, we designed a novel femoral prosthesis that eliminates screw fixation, employs sutures as the primary load-bearing element, and incorporates integrated suture anchor tunnels. This prosthesis was then assessed in a retrospective series of 15 consecutive elderly patients with osteoporotic intertrochanteric fractures. Clinical outcomes were ev […]

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## Updating radiographic parameters for the healthy growing hip: are current parameters still valid?

*Hütter K, Smolle MA, Butter S, Schroedter R, Tschauner S, Kraus T*

BACKGROUND: This study aimed to evaluate whether the acetabular angle (AA) by Tönnis and lateral center-edge angle (LCEA) by Wiberg-key radiographic parameters in pediatric hip assessment-have changed in a contemporary pediatric population compared with historical reference values, considering trends in earlier skeletal maturation and body composition.

MATERIALS AND METHODS: This retrospective diagnostic accuracy study assessed changes in AA and center-edge angle (CE) angles in a contemporary cohort. A total of 1774 anteroposterior pelvic radiographs (3548 hips) from patients aged 0-18 years without hip pathology were analyzed. Radiographs were obtained between 2006 and 2018 and measured using the Supervisely digital platform. Standardized anatomical landmarks were applied. Data were stratified by age and sex and compared with historical values using two-sample t-tests ($p < 0.05$). To minimize measurement bias, two independent observers conducted all assessments using standardized digital protocols.

RESULTS: A total of 1774 patients aged 0-18 years (mean age, 8.30 ± 5.20 years; 666 females, 1108 males) were evaluated. AA values did not show statistically significant differences in 11 of 16 age groups compared with historical data ($p > 0.05$). In contrast, LCEAs were significantly higher in the contemporary cohort, especially in the 5-8, 9-12, and 12-16 age groups (all $p < 0.01$) […]

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## In-hospital deep vein thrombosis after tibial plateau fractures: incidence, laterality/anatomy, and risk factors in a multicenter retrospective cohort of 3366 patients.

*Yang S, Li G, Li Y, Long Y, Zhang J, Liu L, Zhao Z, Sun C et al.*

BACKGROUND: We aimed to determine the cumulative in-hospital incidence, anatomic distribution, and predictors of duplex ultrasonography-detected lower-extremity deep vein thrombosis (DVT) in patients hospitalized with tibial plateau fractures under routine thromboprophylaxis and protocolized surveillance.

METHODS: We retrospectively analyzed consecutive adults (≥ 18 years) admitted to two level-I trauma centers (November 2014-January 2024). Bilateral duplex ultrasonography was performed on admission/preoperatively (as soon as feasible after arrival and prior to surgery) and repeated serially during hospitalization (approximately every 5-7 days and when clinically suspected). The primary outcome was cumulative in-hospital DVT from admission to discharge; isolated intermuscular calf-muscle vein thrombosis (soleal/gastrocnemius) was recorded descriptively but excluded from the primary endpoint. Multiple injuries (polytrauma) were defined as ≥ 1 additional acute traumatic lesion beyond the index tibial plateau fracture documented on admission. Pulmonary embolism (PE) was assessed only when clinically suspected and confirmed by computed tomography pulmonary angiography (CTPA). Multivariable logistic regression identified independent predictors; receiver operating characteristic (ROC) analysis evaluated discrimination for key continuous predictors.

RESULTS: Among 3366 patients, 675 […]

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### Retained articulating spacers (1.5 stage) in chronic knee periprosthetic joint infections: an analysis of associate factors and outcomes in a single center cohort of patients.

*Balato G, Ascione T, Festa E, De Mauro D, Di Gennaro D, Rosa D, Mariconda M*

BACKGROUND: Two-stage revision remains the gold standard for chronic periprosthetic joint infection (PJI), but it can significantly impact patients' quality of life and functional recovery. The 1.5-stage revision, involving indefinite spacer retention, has emerged as a potential alternative in selected high-risk patients. Our questions were: (1) What are the early functional outcomes, including their infection-free survival? (2) Which pre- and perioperative factors are associated with spacer retention? and (3) What is the mechanical survivorship of retained spacers?

MATERIALS AND METHODS: This was a retrospective observational study including 108 consecutive patients with chronic PJI between 2016 and 2020 who were followed prospectively for clinical outcomes. Functional status and quality of life were assessed using the EuroQol-5-Dimension-5-Level (EQ-5D-5L), Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC), and Knee Society Score (KSS) before spacer placement and prior to scheduled reimplantation. Infection-free status was defined as the absence of clinical, radiographic, and laboratory signs of infection recurrence over a 24-month follow-up. Univariate and multivariate logistic regression were used to identify factors associated with spacer retention.

RESULTS: All patients initially underwent articulating spacer implantation as part of a planned two-st […]

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### Moderate leg length discrepancy: a long-term risk factor for hip osteoarthritis?

*Aprato A, Donis A, Via RG, Scandurra A, Tuè F, Audisio A, Fusini F, Massè A*

BACKGROUND: Leg length discrepancy (LLD) has been implicated as a biomechanical factor contributing to hip osteoarthritis (OA), yet the extent of its influence remains unclear. This study examines the correlation between moderate LLD (≥ 10 mm) and hip OA progression, focusing on asymmetrical OA distribution in patients over 65 years old.

MATERIALS AND METHODS: A retrospective analysis was conducted from a database of 1672 full-length standing X-rays. Patients under 65 years, with deformities, unilateral limb issues, or prosthetics, were excluded; therefore, the study group was composed of 220 patients. Tibial and femoral lengths were measured bilaterally, and hip OA severity was assessed using the Tönnis classification. Statistical analyses included Pearson's Chi-squared test and linear regression to explore correlations between LLD and OA distribution.

RESULTS: Among the sample, 18% showed an LLD ≥ 10 mm. A significant correlation was found between LLD and the asymmetrical distribution of hip OA ($p = 0.002$), with higher OA severity observed in the hypometric limb. Linear regression analysis suggested that each millimeter of LLD corresponded to a 0.74-unit change in OA severity difference between hips.

CONCLUSIONS: This study highlights a significant association between moderate LLD and contralateral hip OA in the elderly, emphasizing the biomechanical impact of asymmetrical […]

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### Authorship, titles and open access as drivers of citation performance in orthopaedics: a scientometric analysis.

*Migliorini F, Vaishya R, Rivera F, Eschweiler J, Kobbe P, Betsch M, Oliva F, Maffulli N*

BACKGROUND: Bibliometric analyses are increasingly used to explore how scientific knowledge is created, disseminated, and perceived. In orthopaedics, research output has expanded rapidly over the past decade, yet the factors determining whether an article achieves wide visibility and scholarly impact remain poorly understood. Beyond the inherent quality of a study, elements such as authorship patterns, title construction, and open access (OA) availability may play an essential role in shaping citation performance. However, evidence in this field is still limited and sometimes contradictory, highlighting the need for large-scale, field-specific analyses.

METHODS: Orthopaedic publications from 2010 to 2020 were identified in Scopus using the keyword 'orthopaedic'. After duplicate removal, 97,806 unique articles were included with complete data on authorship, titles, citation counts, study design, and OA status. Citation rates were normalised per year since publication. Associations between bibliographic features and citation performance were assessed using multiple linear regression, while differences across title styles and study designs were evaluated with comparative statistical testing. Exploratory modelling was performed to identify combinations of authorship and title characteristics linked to the highest predicted citation rates.

RESULTS: Larger author teams were associa […]

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### Interpreting nonsignificant findings: Reassessing the safety of self-locking standalone cages in octogenarian patients undergoing ACDF.

*Abudayeh A, Fishchenko I*

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### A novel, simple, and affordable technique for arthroscopic soft-tissue tenodesis of the long head of the biceps tendon.

*Fossati C, Ioppoli A, Ravaglia R, Vaja F, Menon A, Randelli PS*

BACKGROUND: The purpose of this study was to describe a novel, simple, and implant-free arthroscopic soft-tissue tenodesis technique of the long head of the biceps tendon (LHBT) to the rotator cuff using an absorbable monofilament suture and to evaluate its clinical and functional outcomes at a minimum 1-year follow-up.

METHODS: A retrospective case series of 23 patients (mean age 58.0 ± 7.8 years) who underwent arthroscopic rotator cuff repair with concomitant LHBT soft-tissue tenodesis between June 2021 and June 2023 was analyzed. Functional outcomes were assessed using Constant-Murley Score (CMS), American Shoulder and Elbow Surgeons (ASES) score, Single Assessment Numeric Evaluation (SANE), visual analog scale (VAS) for pain, and the Long Head of Biceps (LHB) score. Supination strength was measured with a handheld dynamometer and compared with the contralateral side. Incidence of Popeye deformity and tenderness over the bicipital groove were recorded.

RESULTS: Objective Popeye deformity was observed in 13% of patients, with subjective concern reported by only 4.3%. Supination strength and LHB scores were similar to the contralateral side (means of 104.65 versus 104.64 N; LHB 93.7 versus 94.6 points). The mean CMS and ASES scores were 90.3 ± 12.4 and 89.6 ± 15.9 points, respectively. The SANE score averaged 87.4 ± 20.9, and the VAS for pain was low (1.65 ± 2.59 cm).

CONCL […]

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### ORIF of three- and four-part proximal humeral fractures: a retrospective study of long-term clinical and complication outcomes of two-plate fixation methods.

*Conteduca J, Gasbarra E, Rausa I, Casto A, Valentino L, Solarino G, Meccariello L, Rollo G*

BACKGROUND: The aim of this study was to compare clinical outcomes and complication rates during long-term follow-up (≥ 6 years) after treatment for three- and four-part proximal humeral fractures using two different types of locking plates.

MATERIALS AND METHODS: A total of 113 patients with three- and four-part proximal humeral fractures who underwent surgery between September 2012 and January 2019 were enrolled retrospectively. Data for 49 patients [ 9 males, 40 females; mean age [standard deviation] (SD ) 68.9 (5.8) years} treated with a PGR (intrauma) plate (group A) and 51 patients [10 males, 41 females; mean age (SD): 66.0 (7.5) years] treated with a PHILOS humeral plate (PHP, group B) were available at the last follow-up. The mean follow-up periods in groups A and B were 10.8 and 10.2 years, respectively, with a minimum follow-up of 6 years. At the final follow-up evaluation, functional outcomes were assessed using the Oxford Shoulder Scale (OSS), Simple Shoulder Test (SST), and Disabilities of the Arm, Shoulder and Hand (DASH) questionnaires. X-ray evaluation was also performed, and complications were recorded.

RESULTS: The mean OSS score in the PGR and PHP groups at follow-up was 42.9 and 39.1, respectively ($p > 0.05$). The SST score (PGR: 7.5 ± 2.0; PHP: 7.3 ± 4.0) and DASH score (PGR: 22.1 ± 5.5; PHP: 20.5 ± 4.0) were similar in both groups ($p > 0.05$). In our serie […]

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### Incidence rates and treatment of the transcervical fracture of the neck of femur in Italy: is total hip arthroplasty an increasingly preferred approach? A population study on trends between 2001 and 2023 based on 1,120,770 hospital discharge records.

*Ciminello E, Romanini E, Venosa M, Cazzato G, Tucci G, Boniforti F, Carpanese L, Cuccu A et al.*

INTRODUCTION: Transcervical femoral neck fractures (TFNFs) are among the most devastating fragility fractures in the elderly. TFNF are associated with excess 1-year mortality rates ranging from 15% to 30%. Treatments include conservative methods, internal fixation, and arthroplasty (partial or total hip arthroplasty). This study aims to analyze the changes in incidences of TFNF in the Italian population between 2001 and 2023 and the evolution of the choices of treatment.

MATERIALS AND METHODS: Using hospital discharge record (HDR) data from 2001 to 2023, records with ICD9-CM codes for femoral neck fractures (820.0 and 820.1) among diagnoses were selected and categorized into four treatment groups: total arthroplasty, partial arthroplasty, fixation, and conservative. Time series were analyzed with stratification by sex and age.

RESULTS: The extracted data included 1,120,724 records of TFNFs, with 871,161 cases treated surgically (total or partial arthroplasty or internal fixation) and 249,563 treated conservatively; the average patient age was 79.1 years, with a higher proportion of women (72.8%). Partial hip arthroplasty was the preferred treatment overall. For younger patients, in the age classes < 45 and 45-54 years, fixation was the most chosen treatment. Over time, the use of the conservative treatment decreased from 27.5% in 2001 to 14.6% of cases in 2023. The use of par […]

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### Does latissimus dorsi tendon transfer provide a normal scapular rhythm and external rotation strength in posterosuperior massive irreparable rotator cuff tears? A kinematic analysis in a retrospective cohort.

*Calvo C, Fiumana G, Brigo A, Ruggiero ED, Bonfatti R, Donà A, Micheloni GM, Giorgini A et al.*

BACKGROUND: Posterosuperior massive irreparable rotator cuff tears (PMIRT) are rare and disabling conditions. When conservative treatment fails, latissimus dorsi tendon transfer (LDTT) is a viable surgical option for symptom relief in carefully selected patients. However, its effectiveness in restoring glenohumeral function and its influence on scapulothoracic rhythm remain subjects of ongoing debate.

PURPOSE: The purpose of this study was to evaluate the clinical outcomes of LDTT and assess its impact on scapulothoracic rhythm using kinematic and electromyographic analysis.

MATERIAL AND METHODS: A total of 18 patients with PMIRT underwent LDTT. Functional scores consisted of Constant-Murley score (CMS), America Shoulder and Elbow Surgeons (ASES) score, and Quick Disabilities of the Arm, Shoulder, and Hand (QuickDASH) score. Electromyography (EMG) activity of the latissimus dorsi muscle was evaluated in conjunction with three-dimensional (3D) kinematic tracking system. Differences in external rotation strength were compared with and without active adduction.

RESULTS: The mean age was 55.9 years (range 40-69 years), with a mean follow-up of 20.4 months (range 6-39 months). At final follow-up, the mean CMS was 68.2 (95% CI 64.9-71.5), ASES was 76.9 (95% CI 66.1-87.6), and QuickDASH was 17.6 (95% CI 9.5-25.6). A significant difference in external rotation strength was observed […]

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### Is there a need for exploration in pulseless supracondylar fractures of the humerus? A systematic review and individual patient data meta-analysis.

*Haoyang C, Chen JG, Lai H, Lim AKS, Tan SHS, Hui JHP*

BACKGROUND: Different approaches have been proposed to treat patients with pulseless supracondylar humeral fractures (SHF). We aim to analyze the current outcomes of patients who have undergone different treatment options for pulseless SHF, namely watchful waiting versus urgent surgical exploration.

METHODS: Electronic databases including PubMed, Embase, and Cochrane Library were explored. Our study included patients from all age groups but only included English-language articles. Single case reports and case studies with adequate description of patient population, injury, and outcomes were included. An individual patient data meta-analysis was done to evaluate key outcomes of surgical exploration versus no surgical exploration in pulseless patients with SHF.

RESULTS: Overall, the data for 1070 individual patients from a total of 48 studies were included. Patients with pulseless SHF with open fractures have a higher probability of requiring vascular intervention ($p < 0.001$) as they are more likely to have disrupted arteries ($p = 0.050$) and more likely to require vascular repairs ($p < 0.001$) than patients with closed fractures. Similarly, patients with pulseless SHF with pucker sign ($p = 0.003$) and ecchymosis ($p = 0.002$) were more likely to undergo surgical exploration. However, the neurological status had no relation to them undergoing surgical exploration ($p = 0.382$), and al […]

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## Increasing complexity and plateauing outcomes in AO type C distal humerus fractures: a 50-year institutional case series.

*Hönck K, Wenzelburger P, Eibinger N, Sadoghi P, Puchwein P*

BACKGROUND: Distal humerus fractures in adults are rare but complex injuries, particularly AO type C3 fractures, often involving comminution, instability, and osteoporotic bone. Demographic trends indicate rising incidence, especially among elderly patients, with increasing functional demands challenging modern osteosynthesis. Open reduction and internal fixation (ORIF) with bicolumnar plating remains the preferred reconstructive approach, although complications remain substantial. This study aimed to evaluate trends in fracture complexity, patient demographics, and functional outcomes over five decades, focusing on the most recent cohort (2009-2018, series E) and comparing with historical cohorts (series A-D, 1969-2008).

METHODS: This retrospective analysis included five consecutive 10-year institutional cohorts (series A: n = 43; B: n = 29; C: n = 47; D: n = 58) of 233 patients with 235 supradiacondylar distal humerus fractures (AO type C1-3) treated with ORIF. Functional outcome parameters (Jupiter and Cassebaum Scores) and demographic data were available for all cohorts; additional parameters (range of motion (ROM), QuickDASH Score) were available for series D and E, Mayo Elbow Performance Score (MEPS), Short Form (SF)-36, and strength were assessed specifically in series E (n = 56; follow-up n = 21). Cross-cohort comparisons of categorical variables used chi-square tests; […]

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## Magnetic resonance imaging-based fat infiltration grading improves reparability prediction in massive rotator cuff tears.

*Hu Q, Zhang G, Wang D, Tung TH, Ding J, Zhou X*

PURPOSE: This study aims to investigate the extent of fat infiltration (FI) in the rotator cuff (RC) muscles in more medial slices from the Y-shaped view (Y-view) on shoulder magnetic resonance imaging (MRI) and assess whether these slices provide a more accurate prediction of the repairability of massive RC tears (massive RCTs).

METHODS: The retrospective study included 57 patients with massive RCTs who successfully underwent arthroscopic surgery between 1 January 2023 and 31 December 2023. Patients were categorized into two groups: the irreparable group and the reparable group. All patients underwent shoulder MRI covering the region from the acromion to the medial border of the scapula in oblique coronal and oblique sagittal orientations. The FI stage of the RC was assessed across three different views to determine which view most effectively predicts the repairability of patients with massive RCTs.

RESULTS: The FI stage of the infraspinatus (IS) muscle across three distinct views demonstrated a significant correlation with the repairability of massive RCTs. The FI stage was more severe in the irreparable group. Receiver operating characteristic (ROC) analysis revealed that the area under the curve for the 1/2 scapula view (0.737) and the most severe view (0.745) exceeded that of the traditional Y-view (0.644). Paired-sample

ROC curve analysis revealed a significant differe […]

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## Mid-term clinical and functional outcomes after reverse shoulder arthroplasty with latissimus dorsi transfer.

*Colombini AG, Rab P, Macken AA, Soares MN, Kimmeyer M, Shirinskiy IJ, Popescu IA, Lafosse L et al.*

BACKGROUND: Although reverse total shoulder arthroplasty (rTSA) with concomitant latissimus dorsi transfer (LDT) has been shown to effectively treat external rotation (ER) deficits, there are limited data regarding its outcomes with modern implants and its impact on activities of daily living (ADLs) requiring ER. The purpose of this study was to assess the mid-term clinical and radiographic outcomes of rTSA with concomitant isolated LDT in patients with an ER lag sign and posterior rotator cuff deficiency.

METHODS: This retrospective cohort study with prospective follow-up included consecutive patients who underwent rTSA with concomitant isolated LDT between 2010 and 2022 with a minimum follow-up of 2 years. Primary outcomes included the resolution of ER lag sign, active ER, and the Activities of Daily Living and External Rotation (ADLER) score. Secondary outcomes included the Auto-Constant Score (CS), Subjective Shoulder Value (SSV), activities of daily living requiring internal rotation (ADLIR) score, visual analog scale (VAS) for pain, and radiographic analysis of standardized radiographs.

RESULTS: In total, 32 procedures in 32 patients were identified. Of these, 22 procedures in 22 patients (68% female, 72.9 ± 8.4 years at surgery) were available for follow-up at 4.8 ± 2.2 years postoperatively (response rate 73%). The ER lag sign resolved in 95.5% of patients, the active […]

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## Partial versus full revision in cementless single-stage exchange for hip periprosthetic joint infection: a multicenter comparative study with 10-year survivorship analysis.

*Mu W, Jarusriwanna A, Xu B, Brown SA, Zhang X, Dawes A, Parvizi J, Cao L*

BACKGROUND: Periprosthetic joint infection (PJI) is one of the most challenging complications following total hip arthroplasty (THA). Traditional management typically involves full revision to ensure comprehensive infection eradication. However, for patients with well-fixed implants, partial revision in a single-stage cementless approach may provide a viable alternative, potentially preserving bone stock and reducing operative time. The relative efficacy of this approach compared with full revision remains to be fully explored. This multicenter study aims to determine whether partial revision in cementless single-stage exchange surgery offers comparable infection control outcomes to full revision for select patients with well-fixed implants.

METHODS: We conducted a retrospective, multicenter cohort study involving 226 patients who underwent cementless single-stage exchange hip arthroplasty for PJI between 1 October 2013 and 31 July 2022. Patients were divided into partial revision (n = 61) and full revision (n = 165) groups. The primary outcome was treatment success, defined as the absence of clinical symptoms and signs of infection at a minimum follow-up of 2 years.

RESULTS: The success rates were 77.0% for the partial revision group and 80.0% for the full revision group, with no significant difference (p = 0.629). Both groups showed comparable 10-year survival rates for ove […]

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# Journal of Orthopaedic Trauma

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## Reproducibility of Two Novel Bone Defect Classification Systems.

*Campbell ST, Agel J, Levack AE, Bernstein M, Wallace SJ, Meinberg E, Quinnan S, Hernandez-Irizarry R et al.*

OBJECTIVES: To determine the reproducibility of 2 novel bone defect classification systems, the Tetsworth classification system and the universal long bone defect (ULBD) classification system.

METHODS: Two independent reproducibility studies were undertaken. A series of cases with bone defects was collected and sent to a panel of orthopaedic surgeons who were members of the OTA or Limb Lengthening and Reconstruction Society who routinely manage bone defects. The surgeons were provided with a measurement calibration scale for defect length assessment. Analysis of reproducibility was performed using a Fleiss multirater kappa model, where a Fleiss kappa of 0 indicated agreement no better than random chance, and 1 indicated perfect agreement. Surgeons also rated their confidence in their rating from 1 to 3, with 3 as "very confident."

RESULTS: For the Tetsworth classification, the Fleiss kappa was 0.264 for the overall classification system (fair agreement) (95% CI, 0.224-0.305). The kappa for the D1, D2, D3 part of the code related to defect completeness (incomplete, subcritical, or critical) was 0.557 (moderate agreement) (95% CI, 0.446-0.648). The kappa for the A, B, C part of the code related to defect shape was 0.402 (moderate agreement) (95% CI, 0.323-0.480). The mean confidence for all cases was 2.46 (range, 1.86-3). For the ULBD classification, the Fleiss kappa for the en […]

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## Technical Tricks to Improve Iliosacral Screw Safety Using Three-dimensional Computed Tomography Imaging.

*Zuelzer DA, Routt MLC*

This technical note describes a method used to facilitate iliosacral screw fixation with three-dimensional (3D) computed tomography (CT) imaging. Preoperative pelvis CT scans were used to generate 3D reconstructions, which were then manipulated into inlet and outlet orientations. The inlet position for S1 was achieved when the anterior cortex of the S1 body became superimposed on itself; the outlet position was when the S1 nerve root tunnel was viewed en face. On the fluoroscopic inlet view of S1, multiple cortical densities were often visible and could be difficult to distinguish as S1 versus other sacral segments. 3D CT helped distinguish the position and shape of the anterior cortical limit of S1 by referencing preoperative 3D CT inlet images that were reproduced intraoperatively on fluoroscopy using the appearance of the anterior pelvic ring and the relative obliquity of the obturator foramen. This technique was especially valuable in the setting of common posterior pelvic anatomic variations-including asymmetry, degenerative osteophytes, severe sacral slope, and dramatic cortical recesses of the upper sacral segment. These variations were often difficult to anticipate with conventional plain film or standard CT imaging alone, underscoring the potential clinical utility of this technique.

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## Preoperative High-Sensitivity Troponin and Risk of In-Hospital Mortality and Cardiac Events After Hip Fracture Surgery.

*Touré A, Okafor K, Medilien B, Mather R, Parry J, Hernandez-Irizarry RC, Bradley CA, Ahn J et al.*

OBJECTIVES: To evaluate whether preoperative high-sensitivity cardiac troponin (hs-cTn) elevation predicted in-hospital cardiac events and mortality after hip fracture surgery, and whether surgical delay modified this relationship.

DESIGN: Retrospective cohort study.

SETTING: Single Level 1 trauma center in the United States.

PATIENT SELECTION CRITERIA: Adult patients who underwent hip fracture surgery between January 1, 2020, and December 31, 2024, and had a preoperative hs-cTn measurement as part of routine perioperative assessment were included.

OUTCOME MEASURES AND COMPARISONS: The primary outcome was a composite of in-hospital cardiac event or mortality. Secondary outcomes were in-hospital mortality and in-hospital cardiac events analyzed separately. Outcomes were compared across preoperative troponin categories defined as normal (<30 ng/L), elevated (30-100 ng/L), and significantly elevated (>100 ng/L), and across surgical delay categories defined as <24, 24-48, and >48 hours.

RESULTS: A total of 510 patients were included. The mean age was 76.3 ± 14.1 years, and 291 patients (57.1%) were female. Preoperative troponin was normal in 424 patients (83.1%), elevated in 58 patients (11.4%), and significantly elevated in 28 patients (5.5%). Thirty-five patients (6.9%) experienced the composite outcome. The composite outcome occurred in 19 of 424 patients (4.5%) with normal […]

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### Ballistic Forearm Fractures are Associated with Higher Neurovascular Injury, Nonunion, and Complications Compared With Blunt Forearm Injuries.

*Barber HF, Pereira DE, Rust BD, Perugini JM, Reaver CN, Guissé NF, Wilson JL, Obey MR et al.*

OBJECTIVES: To characterize injury patterns, treatment differences, and clinical outcomes associated with ballistic and blunt forearm fractures.

DESIGN: Retrospective cohort study.

SETTING: Single Level 1 tertiary trauma center, 2018-2023.

PATIENT SELECTION CRITERIA: The study included adults (≥18) who underwent operative fixation of diaphyseal radius and/or ulna fractures. Exclusion criteria were ipsilateral intraarticular or unstable elbow injuries, pathologic fractures, or fractures involving preexisting implants. Patients were stratified into 2 groups: ballistic and blunt injuries, which were further subclassified as blunt-closed or blunt-open.

OUTCOME MEASURES AND COMPARISONS: Primary outcomes included nerve deficit, complication rate, unplanned reoperation, nonunion, and postoperative infection.

RESULTS: The study included 395 patients with 398 forearm fractures: 81 ballistic fractures in 78 patients [32 years (18-69), 88.9% male], 213 blunt-closed fractures [41 years (18-82), 59.2% male], and 104 blunt-open fractures [46 years (19-92), 71.2% male]. Ballistic fractures had significantly higher rates of nerve deficits (48.2% vs. 6.3%, P < 0.001), complications (38.5% vs. 21.1%, P = 0.020), and unplanned reoperation (30.8% vs. 14.2%, P = 0.012) compared with blunt injuries. Infection rates were similar between ballistic and blunt groups (10.3% vs. 5.8%, P = 0.294). Non […]

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### A Primer on Statistical Certainty and Its Distinction From Clinical Importance.

*Ebramzadeh E, Slobogean GP, Ricci WM, McKellop H*

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### Nail-Plate Constructs Permit Early Weight Bearing and Improved Outcomes Compared With Single Device Constructs in the Treatment of Distal Femur Fractures at One-Year Follow-Up: A Retrospective Analysis.

*Patel UJ, Holloway MR, Guedikian CR, Soin SP, Joseph NM, Ketz JP*

OBJECTIVES: To investigate clinical, radiographic, and patient reported outcomes in patients undergoing surgical fixation for distal femur fractures with single device construct (lateral locked plate or intramedullary nail) or nail-plate combination.

DESIGN: Retrospective cohort study.

SETTING: Single, academic, tertiary Level I trauma center.

PATIENT SELECTION CRITERIA: The inclusion criteria were patients with DFFs (OTA/AO 33) treated operatively with lateral locked plate, IM nail, or nail-plate combination between January 2011 and December 2023 with a minimum follow up of 1 year.

OUTCOME MEASURES AND COMPARISONS: Radiographic fracture healing, complication rates, weight-bearing status, length of stay, gait impairment, and patient-reported outcomes using the Patient-Reported Outcomes Measurement Information System were investigated. Multivariable regression models were used to identify predictors of union, reoperation, discharge disposition, and postoperative complications.

RESULTS: A total of 191 patients were included, with 153 (131 lateral locked plate, 22 intramedullary nail) treated with a single device construct (mean age 72 years [range 22-95], 16% male) and 38 treated with nail-plate combination constructs (mean age 70 years [range 19-94], 16% male). At one-year postoperatively, nail-plate combination versus single device constructs were associated with significa […]

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### Comparative Analysis of Nail Plate Combination Over Alternative Fixation Constructs in Low-Energy Distal Femur Fractures: Recovery of Ambulatory Status and Outcomes.

*Pereira DE, Rust B, Barton D, Eggebrecht Z, Wilson JL, Obey MR, McAndrew CM, Berkes MB*

OBJECTIVES: To compare ambulatory status, union, mortality, and operative complications in patients with distal femur fracture treated with the nail plate combination (NPC) and alternative fixation constructs.

DESIGN: Retrospective Cohort Study.

SETTING: Single Level 1, tertiary referral Academic center.

PATIENT SELECTION CRITERIA: This study included adults with a low-energy distal femur fracture (OTA/AO33A/C and Su 1-3) that received ORIF within 2018-2024. Exclusions were pathologic fractures, antegrade intramedullary devices, and baseline nonambulators (from ambulatory status evaluation only).

OUTCOME MEASURES AND COMPARISONS: The primary outcome was ambulatory status at postoperative intervals using Koval-Walking-Score. Secondary outcomes included radiographic union, mortality, and reoperation.

RESULTS: A total of 404 patients with a low-energy distal femur fracture were included, with 342 included for mobility analysis. Fixation constructs included 189 NPC (mean age 75.0 ± 10.2, 83.1% female), 118 lateral locking plate (LLP) (mean age 73.4 ± 10.2; 76.3% female), 79 retrograde intramedullary nail (rIMN) (mean age 73.7 ± 11.1, 74.7% female), and 18 dual plate combination (mean age 74.6 ± 10.9, 83.3% female) (age P = 0.567, sex P = 0.701). NPC was more frequently selected for Su 2 (56.2%) and Su 3 (65.9%) ( P = 0.001). NPC demonstrated lower average ambulatory impairment […]

PubMed · DOI

### Single Antibiotic Use With Ceftriaxone for Prophylaxis of All Open Fractures.

*Koulopoulos MW, Buseck A, Lynch CP, Lutz K, Brand O, Sommers E, Pompo I, Laychur A et al.*

OBJECTIVES: To compare outcomes for open fractures treated with a prophylactic antibiotic protocol using ceftriaxone (CTX) for all open fractures, referred to as group "CTX," versus a previous protocol using cefazolin for Gustilo I/II fractures and cefazolin plus gentamicin or CTX monotherapy for Gustilo type III fractures only, referred to as group "Classic."

DESIGN: Retrospective cohort study.

SETTING: Single level I trauma center.

PATIENT SELECTION CRITERIA: This study included adult patients undergoing surgery for an open fracture from December 2016 to December 2022 with a minimum follow-up of 3 months.

OUTCOME MEASURES AND COMPARISONS: The primary outcome of deep infection and secondary outcomes of superficial infection rates, nonunion rate, and time to antibiotic administration were compared between overall groups and subgroups based on Gustilo type.

RESULTS: A total of 521 patients were included. Baseline characteristics were similar between groups, with no differences in age (CTX 41.1 ± 19.7 vs. classic 39.5 ± 19.3), sex (68.5% vs. 72.0% male), or comorbidities ( P > 0.05). There were no differences in deep infection (7.8% vs. 12.7%, P = 0.095), superficial infection (7.0% vs. 4.7%, P = 0.427), or nonunion (10.3% vs. 7.9%, P = 0.500) between CTX and Classic groups for all Gustilo types. Similarly for only type I/II fractures, rates of deep infection (3.5% vs. 8.9%, […]

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### Temporal Effects on Intraoperative Bacterial Contamination of Surgical Gloves in Noninfection Orthopaedic Trauma Cases.

*Amirhekmat A, Shimabukuro J, Bittencourt C, Beyer RS, Learned JR, Scolaro JA, Lim PK*

OBJECTIVES: Surgical site infections (SSIs) are among the most serious complications in orthopaedic trauma surgery, contributing to patient morbidity, mortality, and substantial financial costs. With direct inoculation from intraoperative bacterial contamination being a large contributor to these infections, the temporal dynamics of intraoperative bacterial colonization on surgical gloves in orthopaedic trauma cases was investigated.

METHODS: Surgical glove bacterial cultures were taken by firmly pressing the operating room staff member's gloved fingertips onto a sheep's blood agar plate. Cases, without known preexisting infection, were chosen from a single level 1 trauma center at random between 2023 and 2025. Samples were obtained at 2 intraoperative time points: (1) immediately after draping and (2) after implant placement but before final irrigation. These samples were compared with negative controls of freshly donned, but unused sterile gloves. Glove imprints were then incubated for 2 weeks, and bacterial isolates were identified using MALDI-TOF mass spectrometry.

RESULTS: Of 276 total gloves sampled, 70 (25.4% 70/138) demonstrated bacterial growth, compared with 0 controls (0%, 0/138 $P < 0.001$). Contamination was detected in 18.8% of gloves after draping and 31.9% after implant placement. Pelvic procedures had the highest contamination rate by surgical location (58.3%, […]

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### Functional Outcomes After Bipolar Hip Hemiarthroplasty by Using the Direct Anterior Approach Versus Posterior and Direct Lateral Approaches: A Retrospective Cohort Study.

*Tham SYY, Ng JPH, Zhang X, Tan TL*

OBJECTIVES: To compare functional recovery and length of hospital stay in patients undergoing bipolar hip hemiarthroplasty (BHH) for displaced femoral neck fractures using the direct anterior approach (DAA) versus the posterior approach (PA) and direct lateral approach (DLA).

DESIGN: Retrospective cohort study using a prospectively maintained hip fracture registry.

SETTING: Single tertiary care institution in Singapore.

PATIENT SELECTION CRITERIA: Patients aged 60 years and older who underwent BHH for displaced femoral neck fractures (OTA/AO 31-B) between 2020 and 2023 were identified from the institutional hip fracture registry.

OUTCOME MEASURES AND COMPARISONS: Functional outcomes were assessed using the Modified Barthel Index (MBI; higher scores indicate better independence) and Parker Mobility Score (higher scores indicate better mobility) at discharge, 6 and 12 months. Mobility status, length of stay, postoperative complications, reoperation, and 1-year mortality were evaluated. Outcomes were compared between DAA and PA after propensity score matching, with an exploratory comparison with the DLA.

RESULTS: A total of 504 patients were identified; 400 PA, 79 DAA, and 25 DLA. After matching, 67 DAA cases and 134 PA cases were analyzed. Age and sex distribution were similar between DAA and PA groups (80.7 ± 6.3 vs. 81.1 ± 6.5 years, $P = 0.655$; female: 79.1% vs. 75.4%, $P =$ […]

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### Patient-Reported Outcomes after Distal Radius Fractures Differ Across Geographic and Cultural Populations: A Global Systematic Review and Meta-Analysis.

*Bilodeau RE, Bhadouriya RS, Klinger CE, van der List JP, Oquendo YA, Haeberle HS, Victory L, Katzir A et al.*

OBJECTIVES: To assess patient reported outcome (PRO) heterogeneity across different cultural and geographical regions through a systematic review and meta-analysis of PROs after volar plate fixation of distal radius fractures.

DATA SOURCES: PubMed, EMBASE, and Cochrane Central Register of Controlled Trials were queried for randomized controlled trials from 2010 to 2023 reporting Disabilities of the Arm, Shoulder, and Hand (DASH), QuickDASH, Patient-Rated Wrist Evaluation (PRWE), or Visual Analog Scale (VAS) PROs.

STUDY SELECTION: Randomized controlled trials with treatment arms consisting of open reduction and internal fixation using a volar locking plate for adult distal radius fracture (OTA/AO 23) and reported DASH, QuickDASH, PRWE, or VAS PROs were included. Studies not published in English or that provided insufficient extraction data were excluded.

DATA EXTRACTION: Studies were characterized by country of origin, continent of origin, and cultural region of origin. Extracted data included PRO outcomes, patient demographics, and follow-up ranges.

DATA SYNTHESIS: Random-effects meta-analysis was performed with restricted maximum likelihood method, and subgroup analyses were performed to determine differences in pooled PROs across geographical and cultural regions.

RESULTS: A total of 52 treatment arms met inclusion criteria for DASH analysis, 33 for PRWE, and 23 for VAS. […]

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AOTS

## Archives of Orthopaedic & Trauma Surgery

Volume 146, Issue 1 · August 2026 · 120 new articles

### Avascular necrosis of the femur following COVID-19: an updated systematic review.

*Azadvari M, Akbari N, Hatam S, Rastkar M, Razavi SZE*

BACKGROUND: One of the complications of COVID-19 infection is avascular necrosis of the femoral head. To date, various studies have reported a wide range of osteonecrosis of the femoral head in patients with COVID-19. The previous systematic review included only 14 studies. So, this systematic review aims to update the findings of these studies.

METHODS: Databases such as PubMed, Scopus, EMBASE, PsycINFO, Web of Science, and Google Scholar were systematically searched by two independent researchers on April 1st, 2024.

RESULTS: A literature search revealed 1229 records, 161 full-texts were evaluated, and finally, 55 studies remained for systematic review. Fifty-five studies were included in this systematic review. Four studies were cross-sectional studies, six were retrospective studies, and the remaining (45studies) were case reports or case-series studies. Totally, 864 cases were included. Except for 12 studies, 43 studies reported the administration of steroid therapy in included cases. The most common underlying conditions were hypertension, followed by diabetes. In 274 (31.7%) of included cases, AVN was bilateral, and both hips were involved. In some studies, non-surgical treatments-including supplement therapy, anticoagulant therapy, physiotherapy, and hyperbaric oxygen therapy-were administered. Several studies reported symptomatic improvement in patients treated with t […]

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### From fluoroscopy to artificial intelligence: three decades of percutaneous pelvic ring fixation-a bibliometric and science-mapping analysis (1996-2026).

*Vicenti G, Ferorelli D, Loiodice M, Sirago G, Solarino G, Dell'Erba A*

INTRODUCTION: Percutaneous pelvic-ring fixation requires implant placement within narrow osseous corridors. Navigation, robotics and artificial intelligence (AI) have successively been proposed to improve accuracy and reduce radiation exposure. This study mapped three decades of research on these technologies, characterising publication trends and the field's intellectual, conceptual and social structure.

MATERIALS AND METHODS: The Web of Science Core Collection (primary dataset, with parsed cited references) and Scopus (robustness comparison) were searched on 8 June 2026. Searches combined pelvic-fixation and technology terms and targeted articles, reviews and conference papers; degenerative-spine, deformity, arthroplasty and isolated acetabular records were excluded, while adjacent WoS categories retained by platform indexing were evaluated in sensitivity analysis. Bibliometrix analyses comprised performance indicators, collaboration patterns, co-citation, keyword co-occurrence, thematic mapping and trend analysis on a harmonised keyword field.

RESULTS: The primary corpus comprised 271 documents from 97 sources, with 1311 authors and 16.3 mean citations per document. Output rose from low single digits before 2010 to 25-29 documents per year in 2022-2023; an exploratory logistic model indicated continued rapid growth. The Journal of Orthopaedic Trauma and Injury dominated th […]

PubMed · DOI

### Skin color representation in orthopedic literature.

*Goyal RK, Adio AA, Nadiminti H, Black EM, Khan AZ, Horneff JG, Hill BW, Abboud JA*

INTRODUCTION: Underrepresentation of skin color in educational materials can undermine diagnostic accuracy and contribute to disparities in orthopedic care for minority populations. While underrepresentation of darker skin types has already been identified in orthopedic literature, this study assessed whether underrepresentation is apparent in open access journals, is uniform across all orthopedic subspecialties, differs between figure types, and has shifted over time as awareness of skin of color disparities has grown.

MATERIALS AND METHODS: 40,674 figures were accessed from 14 open-access orthopedic journals and assigned a subspecialty and a country of origin based on author affiliation. Automatic filters narrowed the dataset to 4,809 figures. Each figure was processed by a GPT-4o artificial intelligence (AI) pipeline to obtain skin presence (yes/no), figure type (clinical, intraoperative, illustration), Fitzpatrick skin type (I-VI), and a confidence score (1-5). Agreement was high with human review for skin presence and figure type and moderate with a board-certified dermatologist for skin tone, with disagreement biased toward darker ratings. 2,025 skin-containing figures were analyzed.

RESULTS: Among 2,025 figures, most depicted light skin (65.6%, Fitzpatrick I-II), 29.6% medium (III-IV), and only 4.8% dark (V-VI). This demonstrated severe overrepresentation of light skin […]

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### Open and endoscopic decompression for cubital tunnel syndrome: long-term clinical outcomes, prognostic factors, and comparison of anterior transposition techniques.

*Ertem H, ■ireci C, Al F, Alyousef A, Çiçek ■B, Ertem K*

BACKGROUND/OBJECTIVE: The optimal surgical treatment for cubital tunnel syndrome (CuTS) remains controversial, and comparative evidence regarding endoscopic-assisted anterior transposition is lacking. This study compared the clinical and functional outcomes of open and endoscopic decompression, evaluated open versus endoscopic-assisted anterior transposition, and identified patient-related prognostic factors associated with postoperative outcomes.

METHODS: Fifty-six patients who underwent surgery for CuTS between 2010 and 2023 were retrospectively reviewed. Nineteen patients underwent open in situ decompression and 37 underwent endoscopic decompression. Outcomes were assessed using Quick DASH, VAS, Wilson-Krout grading system, patient satisfaction, and job change. An additional comparative analysis was performed in patients requiring anterior transposition because of intraoperative ulnar nerve instability (open, n = 16; endoscopic-assisted, n = 11). Smoking, body mass index (BMI), occupation, and preoperative McGowan grade were evaluated as prognostic factors.

RESULTS: Median follow-up was 138 months (range, 56-180) in the open group and 76 months (range, 27-156) in the endoscopic group. No statistically significant differences were detected between open and endoscopic decompression in postoperative Quick DASH, VAS, Wilson-Krout grades, patient satisfaction, job change, or co […]

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### A feasible and cost-effective method to induce distal radius fractures in full-length forearm human cadaveric specimens.

*Krüger J, Terndrup M, Thomsen MG, von Keudell A, Andersen JA, Maleitzke T*

INTRODUCTION: Reported rates of loss of reduction in nonoperatively treated distal radius fractures (DRF) range from 17 to 71%. Thus, realistic biomechanical models are warranted to test novel immobilization devices that prevent loss of reduction in nonoperative fracture management. Currently available human cadaveric specimen models have several limitations: (i) disrupted soft tissue integrity, (ii) difficulty in replicating complex fracture

patterns, and (iii) high technical demands and costs. Therefore, the aim of this study was to establish a feasible and accessible method for inducing DRFs, that accurately reflects in-vivo biomechanical fracture conditions in tissue-intact cadaveric specimens, suitable for testing of novel immobilization devices.

MATERIALS AND METHODS: Fifteen upper extremity cadaveric specimens were included. Specimens were prepared as full■length forearm models. Using a repurposed strength training tower as a vertical drop tower, fractures were induced by proximally administering 120-135 J impacts. CT scans were performed pre- and post-fracture induction. All fractures were classified according to the AO/OTA classification.

RESULTS: DRFs were successfully induced in all 15 specimens. A single 120 J impact produced a DRF in 11 out of 15 cases (73%). A second impact was administered in 4 out of 15 specimens (27%). Eleven of the 15 induced DRFs (73%) were […]

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### Tibiofemoral and patellofemoral kinematics after kinematic versus mechanical alignment in varus TKA-a dynamic knee rig study.

*Simon JM, Siedow TM, Thorwächter C, Petri TE, Bauer L, Holzapfel BM, Müller PE, Niethammer TR*

INTRODUCTION: Kinematic alignment (KA) in total knee arthroplasty (TKA) of the varus knee shows promising clinical outcomes by preserving individual joint anatomy. However, it remains unknown whether this results in knee joint kinematics that more closely resemble the native baseline condition than mechanical alignment (MA). Therefore, this study evaluated the direct influence of KA versus MA on tibiofemoral and patellofemoral kinematics and contact biomechanics in the varus knee during a standardized dynamic muscle-loaded squat.

MATERIALS AND METHODS: Seven fresh-frozen human cadaveric knee specimens with constitutional varus (-1.4° to -8.3°) were tested in a dynamic knee rig (30° to 130° flexion) under muscle-controlled loading. Kinematics, retropatellar contact area, peak pressure, and quadriceps force were continuously recorded across three successive states within the same specimen: native, KA TKA, and MA TKA using a medial-pivot implant design. Statistical analysis was performed using one-dimensional statistical parametric mapping, including non-parametric SnPM1D ANOVA and pairwise SnPM{F} tests for kinematics, and paired-samples SPM{t} tests for contact mechanics and quadriceps force.

RESULTS: Between alignment techniques (KA vs. MA), no statistically significant differences emerged regarding kinematics and patellofemoral biomechanics following Holm-Bonferroni correcti […]

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### Research culture methods detect more anaerobes compared to conventional clinical workflows in periprosthetic joint infection: a pilot study of 14 cases.

*Visperas A, Lu Q, Lehman B, Foo SS, Chen W, Vachharajani V, Stappenbeck T, Piuzzi N*

INTRODUCTION: Periprosthetic joint infection (PJI) remains one of the most challenging complications of total joint arthroplasty, often complicated by limitations in diagnostic sensitivity. Standard clinical culture methods fail to identify pathogens in 20%-50% of cases, leading to potential under- or overtreatment. This pilot study aimed to compare the microbial detection capabilities of standard clinical culture methods with enhanced research laboratory techniques, and to explore whether additional microbial findings might have the potential to influence antibiotic treatment planning.

MATERIALS AND METHODS: Thirteen patients undergoing surgery for PJI, as defined by the 2018 International Consensus Meeting (ICM) on Infection criteria, were prospectively enrolled at a single academic medical center between August 2023 and September 2024. Intraoperative synovial fluid, periprosthetic tissue, and implant swabs were collected and processed using both standard clinical workflows and optimized research protocols. Culture results and treatment decisions were reviewed in collaboration with Infectious Disease specialists.

RESULTS: Both methods identified primary organisms. However, research techniques detected additional organisms not captured by clinical cultures ($p = 0.013$). Despite these differences, overall detection rates were comparable (57.1% vs. 59.5%; $p = 0.963$), with the r […]

PubMed · DOI

## A composite inflammation-nutrition-coagulation score and 1-year mortality in geriatric proximal femur fractures: a retrospective cohort study.

*Bozgeyik B, Kaya O, Önder C, Atmaca E, Çakar A*

INTRODUCTION: This study aimed to evaluate the prognostic value of a composite score integrating the neutrophil-to-lymphocyte ratio (NLR), serum albumin, and D-dimer levels for predicting 1-year mortality in geriatric patients with proximal femur fractures.

MATERIALS AND METHODS: This retrospective cohort study included 300 patients aged ≥ 65 years who were admitted with proximal femur fractures and underwent surgical treatment between November 2023 and March 2026. Demographic, clinical, and laboratory data were collected from hospital records. A composite score based on NLR, albumin, and D-dimer was calculated. An exploratory unweighted 0-3 composite score was constructed from cohort-derived ROC thresholds for NLR, albumin, and D-dimer. Logistic regression, receiver operating characteristic (ROC) curve analysis, net reclassification improvement (NRI), integrated discrimination improvement (IDI), Kaplan-Meier survival analysis, Cox regression, calibration analysis, and decision curve analysis (DCA) were performed to evaluate predictive performance.

RESULTS: A total of 300 patients were analyzed, of whom 75 (25.0%) died within 1 year. Non-survivors were older and had significantly higher NLR and D-dimer levels and lower albumin levels (all $p < 0.001$). In mutually exclusive adjusted regression models, higher NLR and D-dimer levels, lower albumin levels, and a higher composite s […]

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## Closed reduction in distal radius fractures: manual manipulation vs. finger trap traction-a prospective randomized controlled clinical trial.

*Benedikt S, Horling L, Bode S, Stock K, Schmidle G, Arora R*

BACKGROUND: Initial treatment of distal radius fractures (DRF) can be either performed without reduction or with a reduction method as finger-trap traction (FTT) only or by additional manual reduction (MR) under hematoma block. There is no clear evidence on which reduction method is superior, however, MR is more invasive and seems to lead to higher pain levels and more post reduction complications. The objective of this study was to evaluate the clinical relevance of MR compared to FTT in acute DRFs.

MATERIALS AND METHODS: Study design was a prospective, single-centre, balanced randomized [1:1], and parallel grouped, controlled trial. Inclusion criteria were patients with a DRF type A2, A3 or C according to the AO classification and a dorsal tilt of ≥ 10°. The patients were randomized either in the FTT or in the MR group. Depending on the patient's functional demand and the fracture configuration, conservative or operative treatment was initiated. Over one year, clinical and radiological data were assessed at six follow-ups.

RESULTS: A total of 53 patients were included. Regarding radiological results neither FTT nor MR achieved sufficient reduction results (median dorsal tilt was - 16 to -18 after one year) and none of the techniques was superior. The clinical results between the conservatively treated groups were comparable and showed no significant difference. Complication […]

PubMed · DOI

## Coronal alignment-related mediolateral load distribution and polyethylene wear in a medial-pivot total knee arthroplasty design.

*Lesnicar C, Kistler M, Thorwächter C, Tural I, Simon JM, Niethammer TR, Holzapfel BM, Woiczinski M*

INTRODUCTION: While mechanical alignment in total knee arthroplasty aims for a neutral hip-knee-ankle angle, kinematic alignment preserves the native joint line, often allowing residual varus or valgus. Clinically, constitutional variations in alignment affect the coronal joint moments and the mediolateral load distribution, but the impact on polyethylene wear remains unclear.

MATERIALS AND METHODS: Wear of a medial pivot total knee arthroplasty insert (MS 5C®, implantcast GmbH, Germany) was evaluated in a knee wear simulator replicating the walking cycle. Inserts were tested under three conditions representing overall lower-limb coronal alignment and the associated mediolateral load distribution: neutral (60% medial / 40% lateral), varus (70% / 30%), and valgus (50% / 50%), with three (n = 3) dynamically loaded inserts and one load-soak control per condition. Testing followed ISO 14243-1:2009 over 5 million cycles.

Gravimetric wear and wear patterns were assessed.

RESULTS: A numerical trend was observed among the groups: The varus-related load distribution demonstrated the lowest overall wear (4.83 ± 1.06 mg/million cycles), whereas the valgus group produced the highest wear (9.43 ± 1.47 mg/million cycles); the neutral group was intermediate (6.81 ± 3.07 mg/million cycles). However, these differences did not reach statistical significance. Visual inspection revealed medial-d […]

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## Novel bicolumnar fixation with primary cup implantation for pelvic discontinuity in revision hip arthroplasty: a comparative clinical study.

*Abul MS, A■■r M, Y■lmaz H, Hekim Ö, Sevim ÖF, Tuncay ■*

BACKGROUND: Pelvic discontinuity is a rare but challenging condition in revision hip arthroplasty. Traditional cage-based reconstruction techniques often rely solely on mechanical fixation and may be associated with implant migration, loosening, and mechanical failure. Alternative strategies aimed at restoring pelvic continuity and achieving biological fixation may improve construct stability and clinical outcomes.

METHODS: This retrospective study included 19 patients with intraoperatively confirmed pelvic discontinuity who underwent revision total hip arthroplasty between 2014 and 2024. Patients were divided into three groups according to the reconstruction technique used: Group 1 (n = 8) underwent anti-protrusio cage reconstruction alone, Group 2 (n = 6) received additional anterior quadrilateral surface plating prior to cage implantation, and Group 3 (n = 5) underwent restoration of pelvic continuity through anterior and posterior column fixation followed by implantation of a porous-coated primary acetabular cup. Clinical and radiological outcomes were assessed with a minimum follow-up of 2 years, including complications, loosening, reoperation, and reoperation-free survival.

RESULTS: Nineteen patients (mean age 71.2 years) with intraoperatively confirmed pelvic discontinuity were included. Aseptic loosening rates differed significantly among the three groups (P = 0.039), […]

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## Arthroplasty centers logistically separated from general orthopedic departments were associated with a more successful implementation of a standardized day-case setup : A Danish cohort study of 6718 procedures from a fast-track multi-center collaboration.

*Jensen CB, Danielsen O, Varnum C, Jakobsen T, Andersen MR, Bieder MJ, Overgaard S, Jørgensen CC et al.*

BACKGROUND: Day-case hip and knee arthroplasty can reduce bed occupancy and improve cost-effectiveness, which is crucial for addressing the increasing demand among hip and knee osteoarthritis patients. This study investigated how different arthroplasty center setups were associated with the implementation and day-of-surgery (DOS) discharge rates of a standardized day-case setup.

MATERIALS AND METHODS: We included unilateral primary hip and knee arthroplasty patients from seven fast-track arthroplasty centers, all implementing a standardized day-case setup in September 2022 with identical eligibility and discharge criteria. Centers were categorized as: Satellite Arthroplasty Centers (SACs): Only primary arthroplasties, separate logistics. Elective Arthroplasty Centers (EACs): Complex primaries, revisions, separate logistics. Arthroplasty Unit Orthopedic Departments (AUOs): Primary and revision procedures, integrated into a general orthopedic department. We compared DOS discharge rates from September 2022 to September 2023 with a baseline period (July 2019-December 2019) using crude percentages. We compared the DOS discharge rate for day-case-eligible patients in study period between the center types using mixed effects models.

RESULTS: We included 6718 patients (1357 SAC, 4309 EAC, 1052 AUO) during the study period and 3974 in the baseline period. DOS discharge rates increased […]

PubMed · DOI

## Preoperative coronal tibiofemoral subluxation does not affect 2-year outcomes after mobile-bearing medial unicompartmental knee arthroplasty for anteromedial osteoarthritis.

*Kinoshita T, Bunyoz KI, Mortensen KRL, Bredgaard Jensen C, Price A, Gromov K, Troelsen A*

INTRODUCTION: Coronal tibiofemoral subluxation (CTFS) is frequently observed in patients with medial knee osteoarthritis; however, its clinical relevance and the acceptable degree of CTFS when considering

unicompartmental knee arthroplasty (UKA) remain unclear. This study aimed to determine whether preoperative CTFS influences short-term postoperative clinical outcomes in patients with anteromedial osteoarthritis (AMOA) undergoing medial UKA.

METHODS: In this retrospective cohort study, 457 consecutive knees with AMOA who underwent mobile-bearing UKA performed at a single center between August 2016 and May 2023 were analyzed. CTFS, defined as medial translation of the femur relative to the tibia, was measured on preoperative weight-bearing anteroposterior radiographs and expressed as the percentage of tibial plateau width. Patients were categorized into mild (0-5%), moderate (≥ 5% to < 10%), and severe (≥ 10%) CTFS groups. Clinical outcomes were evaluated using the Oxford Knee Score (OKS) preoperatively and at 3 months, 1 and 2 years postoperatively. Pairwise comparisons of changes in OKS were performed using a general linear model with Bonferroni-adjusted post hoc tests, and adjusted mean differences were calculated controlling for age, sex, body mass index, and preoperative OKS.

RESULTS: At 2 years, mean OKS improvements were 16.6 ± 8.9, 14.8 ± 8.3, and 16.0 ± 9.3 in the mi […]

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### Radial head arthroplasty restores axial construct stiffness in a Monteggia-like injury model: a biomechanical study.

*Weber MM, Hockmann JP, Rosenkranz TA, Leschinger T, Müller LP, Rausch V*

INTRODUCTION: Monteggia-like lesions are rare but complex elbow injuries involving fractures of both the proximal ulna and the radial head. While the influence of ulnar alignment and soft tissue integrity on radial head stability has been extensively studied, the effect of radial head integrity on the stability of ulnar fixation remains poorly understood. Given the high rates of nonunion and revision surgery after proximal ulna fractures, improved understanding of forearm biomechanics in these combined injuries is highly clinically relevant as it influences the surgical strategy in primary care of these injuries. Accordingly, this study aimed to evaluate the biomechanical impact of radial head resection and radial head arthroplasty on forearm stiffness in a simulated Monteggia-like injury model. We hypothesized that radial head resection would reduce construct stiffness compared with both the native state and radial head arthroplasty, and that arthroplasty would partially restore stiffness toward native levels MATERIALS AND METHODS: Eight fresh-frozen human cadaveric elbow specimens were biomechanically tested under axial loading. After baseline testing in the native state, a standardized segmental comminution model was created and stabilized using locking plate osteosynthesis. Sequential testing was then performed after radial head resection and after implantation of a modular […]

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### Correction: Single double-loaded anchor repair of isolated Lafosse III subscapularis tears in patients not involved in physically demanding activities provides satisfactory functional and radiologic outcomes at minimum 5-year follow-up.

*Aral F, Oklaz EB, Ahmadov A, Asiret E, Delice C, Oklaz EBG, Akdulum I, Kanatli U*

PubMed · DOI

### Comparison of surgically treated floating elbow injuries and supracondylar humeral fractures in children.

*Liu B, Ren P, Su Y*

OBJECTIVES: Floating elbow is a rare injury type in pediatric fractures. This study compared the differences in injury mechanisms, treatment modalities, complications and prognosis between floating elbow injury and supracondylar humerus fractures (SHF) in children in a surgically treated cohort.

METHODS: The retrospective analysis included patients with floating elbow and SHF treated at our hospital between May 2016 and May 2023. Sex, age, body mass index (BMI), mechanism of injury, affected limb, fracture displacement, presence of neurovascularinjury, type of injury, Gartland classification, and acute compartment syndrome (ACS) were analyzed. Elbow function was evaluated using the Mayo Elbow Performance Score (MEPS). ACS was defined by clinical criteria including pain, paresthesia, and/or compartment pressure > 30 mmHg.

RESULTS: A total of 113 children were enrolled, including 27 in the floating elbow group (22 males, 5 females, median age 8 years [IQR 6-11 years]), and 86 in the SHF group (58 males, 28 females, median age 8 years [IQR 5-10 years]). Mechanism of injury (p = 0.001), fracture type (p = 0.003), and preoperative nerve injury (p = 0.001) differed significantly between the two groups. ACS incidence differed significantly between open and closed fractures in the floating elbow group (p = 0.028); however, this finding is based on a small number of events.

CONCLUSIO […]

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### Platelet-rich plasma for knee osteoarthritis: a systematic review and meta-analysis of structural outcomes.

*Pasqualotto E, Pasqualotto T, Lunelli L, de Sousa FB, Funchal LFZ, Dos Reis FB*

PURPOSE: To assess the impact of platelet-rich plasma (PRP) on imaging-based structural outcomes in patients with knee osteoarthritis (OA).

METHODS: PubMed, Embase, and the Cochrane Library were systematically searched for randomized controlled trials (RCTs) published up to May 2026 evaluating imaging-based structural effects of PRP in knee OA. Risk ratio (RR) was used to evaluate binary outcomes and standardized mean differences (SMDs) or mean differences (MDs) for continuous endpoints, with 95% confidence intervals (CIs).

RESULTS: A total of 12 RCTs involving 1,200 patients were included. PRP was associated with greater improvement in patellofemoral cartilage thickness/volume compared with control treatments (SMD 0.94; 95% CI, 0.53 to 1.34), whereas no significant differences were observed for medial tibial cartilage thickness/volume or cartilage defect progression. Magnetic resonance imaging (MRI)-based structural scores showed no between-group differences in Whole-Organ Magnetic Resonance Imaging Score (WORMS) osteophytes or subchondral cysts, although PRP demonstrated a significant reduction in bone marrow edema (SMD - 0.40; 95% CI, - 0.71 to - 0.09). Synovitis did not differ overall between groups (SMD - 0.22; 95% CI, - 0.43 to 0.00).

CONCLUSION: In this systematic review and meta-analysis, PRP demonstrated limited and inconsistent effects on imaging-based structural o […]

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### First-in-human prospective pilot study of a bipolar dual-center total wrist arthroplasty: early mechanical stability and clinical outcomes.

*Israeli R, Gorelick L, Gannot G, Oron A*

PURPOSE: Total wrist arthroplasty (TWA) is limited by implant loosening, instability, and non-physiological load transfer, partly because most contemporary implants rely on a single center of rotation and do not preserve native wrist kinematics. This first-in-human pilot study evaluated the feasibility, early mechanical stability, and clinical outcomes of a novel bipolar dual-center TWA system designed to preserve radiocarpal and midcarpal motion.

METHODS: This prospective single-center pilot case series included five consecutive patients with end-stage wrist arthritis treated with a bipolar dual-center TWA implant using cement fixation, as per the device's intended use. The outcome measures included pain assessed using the pain subscale of the Patient-Rated Wrist Evaluation (PRWE) questionnaire (normalized to a 0-10 scale), total PRWE score, DASH score, range of motion, grip strength, radiographic implant stability, complications, and reoperation. Clinical outcomes were assessed at the most recent follow-up visit.

RESULTS: The mean follow-up was 19.2 months (range, 12-24 months), with 24-month data available for three patients. Implant survivorship was 100%, with no loosening, migration, dislocation, or implant revision observed on radiographs. One patient experienced a traumatic postoperative fall without loss of implant fixation. The mean PRWE-derived pain score improved f […]

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### The reliability and validity of the 30-Second Fast-Paced Walk Test in patients with total knee arthroplasty.

*Unver B, Ozcan D, Sevik Kacmaz K, Karatosun V*

BACKGROUND: Walking speed is a significant measure of functional ability and recovery after total knee arthroplasty (TKA). The 30-Second Fast-Paced Walk Test (30SFW) is a simple, low-cost, and easy-to-administer assessment method that evaluates short-distance walking ability.

OBJECTIVE: To investigate the reliability and validity of the 30SFW test in patients with TKA.

METHODS: This study was designed as a repeated-measures study conducted by one assessor. 38 patients with TKA who had undergone primary TKA at least 6 months earlier (mean age 70.11 ± 10.19 years) were included in the study. The 30SFW, the Hospital for Special Surgery (HSS) Knee Score, and Short Form (SF)-12 Quality of Life Questionnaire were administered to the patients. The 30SFW was administered twice at a single center on the same day, with an hour-long rest interval between sessions.

RESULTS: The test-retest reliability of the 30SFW was excellent. The intraclass correlation coefficient (ICC [2, 1]), minimal detectable change (MDC95), and standard error of the measurement (SEM) values were 0.98, 2.91, and 1.05, respectively, at the 95% confidence level. The Pearson correlation coefficient of the 30SFW with the physical component of the SF-12 was 0.45 ($p = 0.005$).

CONCLUSION: The 30SFW has excellent test-retest reliability and moderate concurrent validity in assessing walking ability in patients with TKA. […]

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## Net present value analysis of total salary compensation in hand surgery: academic versus non-academic careers.

*Moore JW, Minerva AJ, Daley DN, Daly CA, Silvestre J*

PURPOSE: Studies comparing salary compensation of hand surgeons in different practice settings have been limited. This study compared projected lifetime salary compensation between academic and non-academic hand surgeons.

METHODS: Compensation data were sourced from the 2025 Medical Group Management Association (MGMA) and American Medical Group Association (AMGA) salary benchmarks. Net present values of theoretical career paths in hand surgery were calculated assuming a 32-year career and a 3% discount rate. Sensitivity analyses were performed with variable discount rates and career lengths. Monte Carlo simulations were performed.

RESULTS: Academic practice demonstrated lower lifetime salary compensation than non-academic practice across career trajectories, sensitivity analyses, and Monte Carlo simulations using MGMA data with minimal differences using AMGA data. With MGMA, academic practice ($12.6 million) resulted in 19.7% lower lifetime salary compensation than non-academic practice ($15.7 million). With AMGA, academic practice ($15.0 million) resulted in 2.7% higher lifetime salary compensation than non-academic practice ($14.6 million). Monte Carlo simulations demonstrated lower lifetime salary compensation for academic practice with MGMA ($12.9 million versus $16.0 million, 19.4% difference, $P < 0.001$) but comparable compensation with AMGA data ($15.1 million versus $1 […]

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## Complicated labral injuries in female patients with developmental dysplasia of the hip are associated with localized stress concentration in the superior acetabulum.

*Ogawa Y, Shimizu T, Ohashi Y, Sugawara Y, Takahashi D, Iwasaki N*

INTRODUCTION: Developmental dysplasia of the hip (DDH) alters joint mechanics and increases the risk of labral injury, compromising hip stability and leading to abnormal stress distribution and cartilage degeneration. However, the relationship between labral injury severity and joint loading stress in DDH remains unclear. This study investigated the effect of labral injury severity on stress distribution in DDH patients using computed tomography osteoabsorptometry (CT-OAM) and finite element analysis.

MATERIALS AND METHODS: We analyzed 54 hips from 45 female patients undergoing joint-preserving surgery for DDH. Labral injuries were classified as simple or complicated on radial MRI. Subchondral bone density distribution was evaluated with CT-OAM, while finite element analysis of pelvic and femoral bones with cartilage was performed under a single-leg stance to assess contact pressure and area. Statistical analyses examined associations between labral injury severity and stress distribution.

RESULTS: Superior injuries showed significantly increased high-density area (HDA) in the superior acetabulum, indicating localized stress concentration (simple injury vs. complicated injury, 64.4% vs. 80.3%, $p < 0.001$). However, finite element analysis revealed no significant differences in contact pressure or area (simple vs. complicated, 5.3 N/mm² vs. 6.2 N/mm², $p = 0.538$), which we inter […]

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## Postoperative alignment and patients reported outcome measures improvement after high tibial osteotomy: a significant interaction with osteoarthritis severity.

*Itou J, Itoh M, Kuwashima U, Oda T, Okazaki K*

PURPOSE: Optimal postoperative alignment in high tibial osteotomy (HTO) remains controversial due to inconsistent target definitions and limited evidence supporting osteoarthritis (OA) severity-based strategies. This study evaluated clinical outcomes across postoperative weight-bearing line (WBL) categories stratified by OA severity, hypothesizing that overcorrection would be detrimental in early OA but not in advanced OA.

METHODS: This retrospective single-center study included 181 knees that underwent open-wedge or closed-wedge HTO between 2017 and 2023 with a minimum 2-year follow-up. Knees were classified as early OA (Kellgren-Lawrence [KL] ≤ 2) or advanced OA (KL ≥ 3). The corrective angle was defined as the planned angular change required to shift the mechanical axis to the target postoperative alignment, determined by aiming for a postoperative %WBL of 62.5% on double-limb standing full-length radiographs. Overcorrection was defined as %WBL ≥ 70%. Clinical and radiographic outcomes were assessed at baseline and final follow-up. Multivariable regression models identified factors associated with changes (Δ) in Forgotten Joint Score-12 (FJS-12) and Knee Injury and Osteoarthritis Outcome Score (KOOS), and logistic regression evaluated achievement of minimum clinically important difference (MCID) and patient acceptable symptom state (PASS). Each knee was treated as an indepe […]

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## Analysis of serological markers and synovial fluid cytology in aseptic revision total knee arthroplasty.

*Khury F, Zhong J, Ruff G, Sarfraz A, Bahlouli L, Aggarwal VK, Schwarzkopf R*

BACKGROUND AND AIM: Recent studies examining synovial fluid analysis to differentiate causes for aseptic revision total knee arthroplasty (rTKA) were limited to aseptic loosening and instability. This study aims to establish specific serological and synovial profiles for various rTKA etiologies.

METHODS: We retrospectively reviewed patients who underwent aseptic rTKA at a single academic institution between 2012 and 2023. Inclusion required complete serum inflammatory markers (erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP)) and synovial fluid differentials within three months pre-operatively. Periprosthetic joint infections were strictly excluded utilizing the 2018 International Consensus Meeting criteria. In total, 156 patients were included and stratified into five aseptic etiologies: aseptic loosening (50.6%), instability (28.2%), arthrofibrosis (16.7%), liner wear (2.6%), and component malposition (1.9%). Serological markers were evaluated utilizing a multivariate general linear model adjusting for patient demographics, while synovial markers were assessed via Welch's Analysis of Variance (ANOVA) and Games-Howell post-hoc testing.

RESULTS: Multivariate analysis demonstrated that the underlying rTKA indication was not a significant independent predictor of elevated ESR ($P = 0.661$) or CRP ($P = 0.947$). Welch's ANOVA revealed significant differences in the […]

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## Clinical outcomes and feasibility of a fast-track protocol after total ankle replacement: a prospective study.

*Vena LM, Moggio L, Cofano E, Longo UG, van Dijk P, van Dijk CN, Gasparini G, Mercurio M*

PURPOSE: Postoperative recovery after total ankle replacement remains challenging, and evidence supporting the use of fast-track protocols in this setting is limited. The purpose of this study was to evaluate the short-term clinical outcomes and safety of a fast-track protocol following total ankle replacement.

METHODS: A prospective observational study was conducted on consecutive adult patients undergoing primary total ankle replacement through an anterior approach between January 2023 and June 2024. All procedures were performed by a single experienced surgeon using the same implant. The fast-track protocol included early postoperative mobilization, multimodal pain management, bleeding control, and structured physiotherapy with standardized discharge criteria. Ankle function was assessed using the American Orthopaedic Foot and Ankle Society's (AOFAS) Ankle Hindfoot Scale preoperatively and at hospital discharge. Secondary outcomes included time to active ankle mobilization, time to assisted ambulation with crutches, length of hospital stay, and wound healing evaluated using the ASEPSIS wound score.

RESULTS: Twenty-two patients were included. Median AOFAS Ankle Hindfoot Scale scores improved significantly from 29.0 preoperatively to 68.0 at discharge. Active ankle mobilization was achieved after a median of two days, assisted ambulation after three days, and the median hosp […]

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## Patient-specific management in high-risk revision total hip arthroplasty: mid-term outcomes of a next-generation constrained acetabular liner.

*Khury F, Luebbering B, Ruff G, Bahlouli L, Sarfraz A, Schwarzkopf R, Aggarwal VK*

BACKGROUND: Managing recurrent instability in revision total hip arthroplasty (rTHA) requires a personalized approach, as traditional constrained liners often yield suboptimal results due to locking-mechanism failure and limited impingement-free range of motion. This study aimed to evaluate the mid-term clinical outcomes of a novel constrained acetabular liner (CAL) when utilized as a patient-specific solution for a high-risk cohort undergoing complex rTHA. By tailoring the constraint to individual biomechanical deficits, this next-generation device seeks to mitigate historic failure modes.

METHODS: We retrospectively reviewed 32 patients who underwent 37 rTHA procedures with a next-generation CAL at a single urban academic center between March 2017 and October 2023. Average follow-up was 4.7 years (range, 0.3 to 9.7 years). The most common indication for rTHA was instability (70.3%), followed by infection (16.2%), aseptic loosening (10.8%), and polyethylene wear (2.7%). Just over one third of patients (37.5%) had undergone a prior revision. Preoperative constructs included fixed-bearing (37.8%), dual-mobility (32.4%), CAL (24.3%), static spacer (2.7%), and hemiarthroplasty (2.7%).

RESULTS: In most procedures (70.3%), the CAL was implanted into a pre-existing acetabular cup, whereas 29.7% involved both cup and liner revision. Following rTHA, no cases of implant wear, breakage […]

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## Diagnostic accuracy of standard practice wrist MRI for scapholunate ligament injuries: a retrospective arthroscopic correlation study.

*Spodzieja S, Boever J, Rau J, Bruckner T, Müller H, Wulf J, Unglaub F, Cavalcanti Kussmaul A*

INTRODUCTION: The scapholunate ligament is the most injured intercarpal ligament. Untreated tears can lead to carpal instability and subsequent radiocarpal and midcarpal osteoarthritis. While wrist arthroscopy remains the gold standard, the diagnostic performance of standard practice preoperative MRI in clinical practice warrants further investigation.

METHODS: We retrospectively analysed patients who underwent preoperative wrist MRI followed by wrist arthroscopy between 2013 and 2021. MRI findings regarding scapholunate ligament integrity were compared with arthroscopic findings serving as reference standard. Sensitivity and specificity were calculated for detection of any scapholunate ligament lesion (partial or complete tear).

RESULTS: A total of 889 patients were included. Complete paired data were available for 864 patients with a mean age of 42.4 ± 15.4 years (range 10-83 years). 51.3% were male. MRI classified the ligament as intact in 83.9%, partially ruptured in 9.1%, and completely ruptured in 6.9%. Arthroscopy identified intact ligaments in 78.2%, partial tears in 12.9%, and complete tears in 8.9%. In this cohort, MRI demonstrated high specificity (91.1%, 95% CI 89.0-93.3) but limited sensitivity (42.0%, 95% CI 35.0-49.1) for scapholunate ligament lesions.

CONCLUSION: In standard clinical practice, preoperative wrist MRI shows high specificity but insufficient sen […]

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## Correction: Robotic-arm assistance in revision total hip arthroplasty improves the accuracy of acetabular component positioning: a cadaver-based study.

*Fitzgerald K, Borukhov I, Chandra V, Smitterberg C, Mont MA, LiArno S*

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## Expert evaluation of ChatGPT-generated responses to patient questions in conventional and fast-track total knee arthroplasty.

*Maciag B, Jegierski D, Frankiewicz A, Tarkowski P, ■arnovsky K, Mietlicki F, Stolarczyk A*

BACKGROUND: Fast-track total knee arthroplasty (TKA) asks patients to understand early mobilization, early discharge, opioid-sparing analgesia, and selective use of drains and catheters. We evaluated expert-rated quality of ChatGPT responses to patient questions about these pathway-specific concepts alongside conventional TKA questions.

MATERIALS AND METHODS: In this cross-sectional evaluation, three arthroplasty surgeons independently scored responses to 50 conventional TKA and 20 investigator-developed fast-track questions in four domains on 5-point scales. The conventional questions were selected by a multidisciplinary panel; the fast-track questions did not undergo the same selection process. Response-level means were compared descriptively and with exploratory Mann-Whitney U tests.

RESULTS: Seventy responses received 210 expert assessments. Recalculated mean ± SD scores across individual ratings were 4.99 ± 0.15 for accuracy, 4.98 ± 0.14 for completeness, 4.98 ± 0.17 for safety, and 5.00 ± 0.07 for readability. Each of the 20 fast-track responses received a score of 5 from each of the three raters in all four domains (Supplementary Table S1). Nevertheless, question-set construction differed and scores clustered at the ceiling, so the exploratory comparison does not establish equivalence.

CONCLUSIONS: Experts gave high ratings to the evaluated responses, including those […]

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## Epidemiology, determinants, and microbiological profile of implant-associated infections following intramedullary nailing in a tertiary hospital in Ethiopia.

*Zegeye AM, Belay SA, Kebede EB, Alemayehu BT, Zeleke SS, Molla YD*

BACKGROUND: Implant-associated infections after intramedullary nailing (IMN) are a major cause of morbidity in patients with long bone fractures. This study aimed to determine the magnitude, determinants, and microbiological profile of such infections in a tertiary hospital in Ethiopia.

METHODS: A hospital-based cross-sectional study was conducted among 260 patients who underwent IMN for lower extremity long bone fractures between January 2021 and December 2023. Clinical data, operative details, and microbiological results were collected, and logistic regression was applied to identify determinants of infection.

RESULTS: The overall infection rate was 8.46% (95% CI: 5.38-12.53; 22/260), comprising 4.62% deep and 3.85% superficial infections. We found that emergency surgery and hospital stay longer than 14 days were independently associated with an increased risk of infection. Among the limited number of culture-positive infections, Gram-negative bacteria predominated, with Escherichia coli as the most frequently identified pathogen.

CONCLUSION: The infection rate was higher than in many other settings, underscoring the need for improved perioperative infection prevention and reduced hospital stays. Findings highlight the importance of perioperative optimization and hospital infection control in LMICs.

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## Notch-referenced mediolateral verification of tibial tunnel position in anterior cruciate ligament reconstruction: a 3D imaging validation study.

*Nam VT, Thang VD, Huy LD, Phuc LH, Anh LN, Thinh LD, Long CH, Huyen NN et al.*

BACKGROUND: Non-anatomic tibial tunnel placement remains a leading contributor to anterior cruciate ligament reconstruction (ACLR) failure. Mediolateral (ML) tunnel positioning is particularly difficult intraoperatively because no consistent bony landmark defines the lateral footprint border.

PURPOSE: To evaluate the accuracy of a novel intraoperative Notch-Referenced Mediolateral (NRML) verification technique by comparing the ML position of the tibial tunnel center (TTC) on postoperative 3D CT with the native tibial footprint center (TFC) on preoperative 3D MRI.

METHODS: In this retrospective cross-sectional study, 108 consecutive patients underwent primary all-inside single-bundle ACLR with intraoperative NRML verification. This arthroscopic technique uses the intercondylar notch as a visual reference to verify mediolateral tibial tunnel position before definitive drilling. TFC and TTC were independently measured on preoperative 3D MRI and postoperative 3D CT using a standardized quadrant-grid protocol by two blinded observers. The primary outcome was ML deviation (tunnel - footprint). MRI-CT agreement and measurement reliability were assessed with intraclass correlation coefficients (ICC) and Bland-Altman analysis.

RESULTS: The anatomical TFC was located at 46.1 ± 2.3% of tibial plateau width in the ML direction, versus 44.8 ± 2.0% for the TTC. Mean ML deviation was -1.01 […]

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## Preoperative pain and achievement of the minimal clinically important difference for physical function one year after total hip arthroplasty: a prospective cohort study.

*Gajardo-Burgos R, Arteaga San Martín R, Fuentes J, Monrroy Uarac M, Saavedra-Santiesteban C, Bascour-Sandoval C, de Villarreal ES*

PURPOSE: To examine the association between preoperative social, clinical, and biological variables and the achievement of the minimal clinically important difference (MCID) on the WOMAC-function subscale one year post total hip arthroplasty (THA).

METHODS: In this prospective cohort study, 85 patients (aged ≥ 65) completed baseline preoperative assessment, and 53 completed the 12-month follow-up (32 lost to follow-up; 66.0% achieved MCID). The outcome was achieving WOMAC-function MCID (≥ 26.54 points). Multivariable logistic regression adjusted for baseline pain, age, sex, and BMI were performed. ROC curves evaluated discriminative capacity.

RESULTS: An exploratory ROC analysis identified a baseline pain threshold of 6.5 on the Numerical Rating Scale yielded an AUC of 0.740 (70% sensitivity, 77% specificity).

CONCLUSIONS: Preoperative pain intensity independently associated with clinically meaningful functional improvement post-THA. An exploratory pain threshold of 6.5 on the NRS may provide an exploratory benchmark for future risk stratification.

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## The boundary between microbial detection and implant-related infection: a structured narrative review of analytical credibility, biological persistence, host response, and clinical causality.

*Chang-Sheng L, Feng-Jun L, Huan-Sheng L, Qiang H, Zong-Ke Z*

BACKGROUND: Unexpected microbial signals are frequently encountered in patients with orthopaedic implants who do not satisfy validated criteria for confirmed infection. The central interpretive problem is not limited to periprosthetic joint infection (PJI); it also arises in fracture-fixation and spinal instrumentation, and it involves conventional culture, sonication, histopathology, broad-range polymerase chain reaction (PCR), 16 S rRNA sequencing, and metagenomic next-generation sequencing (mNGS).

METHODS: We performed a structured narrative review of literature identified in PubMed, Embase, and Web of Science, updated through 27 June 2026. Eligible publications addressed microbial or biofilm-associated signals in orthopaedic implants without unequivocal infection at the time of sampling, or reported outcomes after unexpected positive cultures. Evidence was organized by implant type, indication for implant removal, detection method, host-response evidence, and longitudinal outcome. Particular attention was paid to the distinction between routinely removed fracture-fixation devices and failure-related arthroplasty or spinal revisions.

RESULTS: Routine removal of fracture-fixation devices provides the least confounded clinical window and demonstrates that microbial signals may be recovered from implants in the absence of recognized failure or overt infection. By contrast, ar […]

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## Fewer poor joint awareness results with cementless TKA compared with the same-design cemented implant.

*De Berardinis L, Leggieri F, Salari P, Polizzotti G, Baldini A*

BACKGROUND: Comparisons of cemented and cementless TKA have focused on mean outcomes and generally reported equivalence. We hypothesized that a 3D-printed full-titanium cementless TKA and the same-design cemented implant would not differ significantly in average joint awareness but would differ in the proportion of patients with the poorest results, and we compared the two both at the mean and across the outcome distribution.

METHODS: Prospective two-cohort study of 109 primary unilateral TKAs (54 cementless, 55 cemented) for end-stage knee osteoarthritis, with complete 24-month follow-up. Exclusion criteria were valgus deformity and bilateral surgery. The primary outcome was the FJS-12, analysed both at the mean (multivariable linear regression, with propensity score overlap weighting as sensitivity analysis) and across its distribution (quantile regression). Secondary outcomes were PASS achievement (FJS-12 ≥ 33.3), OKS, and time to functional satisfaction.

RESULTS: The cementless group was younger ($P < 0.001$) with more men ($P = 0.008$). Mean FJS-12 was not different between groups after adjustment ($\beta = 7.4$, 95% CI - 1.8 to 16.6, $P = 0.115$), although the confidence interval did not exclude a difference exceeding the minimal clinically important difference; this finding was reproduced by overlap weighting. On quantile regression, cementless patients scored significantly higher […]

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## Elective primary total hip arthroplasty in rheumatoid arthritis versus osteoarthritis: in-hospital outcomes from the national inpatient sample, 2016-2021.

*Finkel B, Gan-Or H, Ankol S, Tawatha M, Behrbalk E*

BACKGROUND: Whether coded rheumatoid arthritis (RA) among patients undergoing elective total hip arthroplasty (THA) is associated with excess short-term inpatient morbidity in contemporary U.S. practice remains uncertain.

METHODS: We performed a retrospective cohort study using the National Inpatient Sample (2016-2021). Adults undergoing elective primary THA were identified using ICD-10-CM/PCS codes. The exposure cohort included admissions with coded RA in any diagnosis position. The comparator cohort included admissions with narrow primary hip osteoarthritis (OA) as the principal diagnosis and no RA. Propensity scores were estimated from demographic, socioeconomic, hospital, and comorbidity variables. Inverse probability of treatment weighting (IPTW) with survey weighting was used for the primary analysis, followed by doubly adjusted regression models that included age and all variables with persistent post-weighting imbalance greater than 0.10 (female sex, osteoporosis, chronic obstructive pulmonary disease, chronic anemia, depression, and race). Additional sensitivity analyses included stricter weight truncation, a restriction analysis, conventional multivariable modeling, and sex-stratified analyses addressing residual sex imbalance.

RESULTS: A total of 274,582 admissions met eligibility criteria; 274,539 were included in the final analytic cohort after exclusion of 43 re […]

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## Research trends in total knee arthroplasty in India: a five-year bibliometric analysis.

*Shetty S, Vijayan S, Mohandas Rao KG, Rao SK, Nayak Y, George BM*

BACKGROUND: Total Knee Arthroplasty (TKA) is one of the fastest-growing orthopaedic procedures in India, driven by demographic transitions, increased life expectancy, and the rising burden of knee osteoarthritis. With expanding procedural volumes, periodic evaluation of scientific output is essential to identify evolving research priorities, institutional leadership, and collaborative patterns. This study aimed to perform a bibliometric analysis of Indian TKA research published over the past five years.

METHODS: Publications were retrieved from the Scopus database using a structured search strategy for TKA-related terms. The search was restricted to studies published between 1 January 2021 and 30 December 2025 and conducted in an Indian setting. Bibliometric indicators assessed publication trends, document types, journals, authorship, institutional productivity, and state-wise contributions. Network-based bibliometric

visualizations were generated using VOSviewer.

RESULTS: A total of 860 publications were identified, showing a steady increase in research output with peak productivity in 2025 (n = 221). Maharashtra and Delhi were the leading contributors, and the All-India Institute of Medical Sciences was the most productive institution. Original research articles accounted for 69.5% of publications. Emerging themes included robotic-assisted surgery, AI-based modeling, and ad […]

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### Round versus tape sutures with interference fit anchors perform similarly in cadaveric glenoid and greater tuberosity biomechanical testing.

*Dimitroff JD, Smith C, Hanish S, Oladeji LO, Bezold W, Nuelle CW, Smith MJ*

PURPOSE: To determine whether suture shape (flat-tape vs. round-wire) influences biomechanical performance at the suture-anchor-bone interface in interference-fit anchors used for rotator cuff and glenoid labrum repair.

METHODS: Four matched pairs of human cadaveric shoulders were used for the glenoid model, with six anchor sites per glenoid. Five matched pairs of cadaveric humeri were used for the rotator cuff model, with three anchor sites per humerus. Each model tested multiple suture-anchor constructs under cyclic loading followed by load-to-failure. Biomechanical outcomes included cyclic displacement, yield characteristics, ultimate failure strength, stiffness, and force at fixed displacements. A linear mixed-effects model was used to compare suture types, with suture type and anchor position as fixed effects and specimen as a random effect.

RESULTS: In the glenoid/labrum model, there were no significant differences among suture types in cyclic displacement (cycles 1, 50, or 100; p = 0.376, 0.101, 0.109), displacement at yield (p = 0.406), yield force (p = 0.880), force at 1 mm (p = 0.264) or 5 mm (p = 0.085), or ultimate failure strength (p = 0.204). Stiffness (p = 0.050) and force at 2 mm displacement (p = 0.033) differed at the overall model level, however, no significant pairwise differences were observed. In the humerus/rotator cuff model, displacement at cycle 50 ( […]

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### Association of systemic bone mineral density and tibial cortical thickness with postoperative subsidence after surgical treatment of tibial plateau fractures.

*Hesmerg MK, Benner JL, Joosse P, Keijser LCM*

INTRODUCTION: Tibial plateau fractures are common fractures above the age of 50 years old and are frequently considered osteoporosis-associated fractures. However, the association between bone mineral density (BMD) and postoperative articular subsidence remains unclear. The primary objective was to assess whether there was an association between systemic BMD or tibial cortical thickness and articular subsidence after surgical treatment of tibial plateau fractures. Secondary objectives were to evaluate preoperative fracture displacement and the prevalence of reduced BMD in this cohort according to WHO criteria.

MATERIALS AND METHODS: In this retrospective cohort study, patients treated surgically for low-energy tibial plateau fractures between 2017 and 2024 who underwent Dual-Energy X-ray Absorptiometry (DEXA) within 12 months before or after injury were included. DEXA-derived T-scores were used to assess systemic bone density and Tibial Cortical Thickness (TCT) was used as a measure of tibial cortical bone status. Fracture subsidence and preoperative displacement were measured in the coronal plane using articular depression. Multivariable linear regression analyses were performed adjusting for age, BMI and fracture comminution. The primary outcome was also adjusted for preoperative articular depression.

RESULTS: Fifty-nine patients were included in the analysis. There were 57 […]

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### Comorbidity burden and fracture pattern as predictors of early postoperative medical complications following bipolar hemiarthroplasty for displaced femoral neck fractures: a retrospective cohort study.

*Miyamoto S, Yoshii Y, Ishii T*

INTRODUCTION: Postoperative medical complications substantially affect outcomes after bipolar hemiarthroplasty (BHA) for displaced femoral neck fractures, with an impact comparable to surgical factors. This study aimed to

identify risk factors for postoperative medical complications, focusing on comorbidities, polypharmacy, and fracture pattern.

MATERIALS AND METHODS: This retrospective cohort study included 310 hips in 297 patients undergoing BHA. Baseline characteristics, comorbidities, and medication profiles were evaluated using the Charlson Comorbidity Index (CCI), Elixhauser Comorbidity Index (ECI), and comorbidity-polypharmacy score (CPS). The primary outcome was medical complications within 30 days. Multivariable logistic regression was used to identify independent predictors, and 1-year survival was assessed using Kaplan-Meier analysis.

RESULTS: Medical complications occurred in 53.7% of hips, most commonly delirium (17.0%). In the CCI-based model, age at surgery (odds ratio [OR] 1.05; 95% confidence interval [CI], 1.01-1.08; p = 0.006), displaced transcervical fractures (AO/OTA 31B2.1-31B2.2) (OR 2.07; 95% CI, 1.18-3.61; p = 0.011), displaced basicervical fractures (AO/OTA 31B2.3-31B3) (OR 4.67; 95% CI, 1.45-15.00; p = 0.010), a medium CCI category (OR 3.58; 95% CI, 1.10-11.60; p = 0.034), and a very high CCI category (OR 12.80; 95% CI 2.27-72.10; p = 0.004) were in […]

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### Hemi-hamate arthroplasties for proximal interphalangeal joint fracture dislocations: long-term follow-up in acute, subacute and chronic cases.

*Cansü E, Ilgaroglu S, Msangi WJ, Yurga E*

INTRODUCTION: Hemihamate arthroplasty (HHA) method is one of the recommended treatment methods for unstable proximal interphalangeal joint (PIPJ) fracture dislocations The primary goal of this method is to achieve a stable and well-aligned joint by reconstructing the palmar lip. The aim of this study is to present mid to long term results of patients treated with HHA method in the acute, subacute and chronically.

MATERIALS AND METHODS: 22 patients was operated with this technique by single surgeon between 2013 and 2025. Patients were classified as acute (< 3 weeks), subacute (3-6 weeks), and chronic (> 6 weeks) based on the time elapsed between the date of injury and the date of surgery. Patients were assessed clinically, radiologically and subjectively.

RESULTS: The mean ROM of the PIPJ was 80.6° (20-100°), the flexion was 91.6° (65-100), and the extension lag was 11° (0-50). Grip strength on the operated side was 91.7% of the average on the unaffected side (74-110). The mean rest pain score according to the VAS (0 least-10 most) was 0.6 (0-3), and the activity pain score was 2 (0-5). The mean Q-DASH scores assessing functional level were 9.55 and the mean Mayo scores were 81.4. When patients were asked about their satisfaction after surgery, 12 patients responded very good, two patients good, and four patients moderate.

CONCLUSIONS: With HHA, reasonable PIPJ ROM and good f […]

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### Partial triceps tendon transfer for dynamic stabilization in posterolateral rotatory instability of the elbow: a biomechanical study.

*Pucher CJ, Lanzerath F, Kahmann SL, Hackl M, Müller LP, Leschinger T*

PURPOSE: Posterolateral rotatory instability (PLRI) of the elbow results from lateral collateral ligament complex (LCLC) insufficiency. In the setting of failed conventional ligament reconstruction, no established salvage procedure is available. This study biomechanically evaluated partial triceps tendon transfer to the supinator crest as a dynamic stabilization concept for PLRI.

METHODS: Nine fresh-frozen human elbow specimens underwent cyclic external rotational loading of 0.5 Nm. Specimens were tested sequentially in the native state, an LCLC/common extensor origin-deficient unstable state, after transosseous LCLC repair, and after partial triceps tendon transfer to the supinator crest with the LCLC detached. Testing was performed under combined biceps/triceps loading conditions with external forearm rotation relative to the humerus as the primary outcome. A two-factor repeated-measures analysis of variance (RM-ANOVA) compared unstable, repaired, and tendon-transfer conditions across loading conditions. The native state was assessed unloaded and reported separately in a one-factor RM-ANOVA.

RESULTS: Intervention significantly affected external rotation ($F(2,16) = 33.31$, $p < 0.001$, $\eta 2p = 0.81$). Across loading conditions, partial triceps transfer significantly reduced external rotation compared to the unstable state

(mean difference [MD]=-3.38°, p = 0.005), while LCLC repair […]

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### Development and validation of a model to predict full thickness rotator cuff tear.

*Wang T, Tian Q, Lu B, Guo J, Yin B, Zhu Y*

INTRODUCTION: Full-thickness rotator cuff tear (FTRCT) often requires surgical treatment. Magnetic resonance imaging (MRI) is the diagnostic gold standard; however, it is costly and resource-intensive. We aimed to develop and validate a clinical prediction model for FTRCT in patients with RCI, with the goal of identifying those at high risk at the initial visit.

METHODS: This prospective study consecutively enrolled patients clinically diagnosed with RCI from May 2025 to September 2025. All patients underwent shoulder MRI to confirm the presence or absence of FTRCT. Potential predictors involving demographic characteristics, clinical history, and physical examination findings, were collected. Predictor selection was performed using least absolute shrinkage and selection operator (LASSO) regression, followed by multivariable logistic regression to construct a risk prediction model, and then a nomogram was developed.

RESULTS: 357 patients were included, MRI confirmed FTRCT in 166 patients (46.5%). Patients were randomly divided into a training set (n = 249) and a test set (n = 108). LASSO regression identified 10 candidate predictors, and multivariable logistic regression identified physical labor, history of trauma, age, daily pain score, pain duration, and physical examination shoulder strength as independent predictors of FTRCT. The nomogram achieved an AUC of 0.910 (95% CI, […]

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### Minimally invasive circumferential fixation using TIFI and INFIX for unstable pelvic ring injuries: a prospective evaluation of clinical and radiological outcomes.

*Fahmy M, Sadek FZ, Khaled SA, Said MS, Kassem E*

PURPOSE: To evaluate the clinical and radiological outcomes of minimally invasive circumferential fixation using combined trans-iliac internal fixation (TIFI) and anterior internal fixation (INFIX) in unstable pelvic ring injuries.

METHODS: A prospective study was conducted between 2022 and 2023 including 30 patients with unilateral unstable pelvic fractures (Tile B2 and C1). Patients were followed for a minimum of 12 months (mean 22 ± 4.7 months). Functional outcomes were assessed using the Majeed Score and Pelvic Outcome Score. Radiological assessment included displacement measurements, reduction quality, maintenance of reduction, and time to union.

RESULTS: Thirty patients (mean age 32.2 ± 14.9 years) were included. Mean operative time was 103.5 ± 10.5 min, blood loss 335 ± 65.8 mL, and fluoroscopy time 52 ± 14.7 s. Pelvic displacement improved from 15.8 ± 4.6 mm preoperatively to 3.6 ± 2.4 mm postoperatively, with a mean correction of 12.2 ± 4.5 mm and minimal loss of reduction (0.3 ± 0.8 mm). At one year, mean Majeed and Pelvic Outcome scores were 89.1 ± 5.6 and 34.6 ± 2.6, respectively. Mean time to union was 13.4 ± 2.1 weeks, with 90% excellent-good reductions (Matta criteria) and 100% union. Complications included superficial infection (13.3%), lateral femoral cutaneous nerve irritation (13.3%), and asymptomatic heterotopic ossification (3.3%).

CONCLUSION: Circumfere […]

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### Radiographic accuracy of implant positioning in unicompartmental knee arthroplasty, does an imageless robotic system make a difference?

*Montagna A, Andriollo L, Cinelli V, Marescalchi M, Matascioli L, Benazzo F, Rossi SMP*

BACKGROUND: Robotic-assisted technologies and evolving implant designs have been introduced in unicompartmental knee arthroplasty (UKA) to improve the accuracy and reproducibility of component positioning. However, the relative contribution of robotic assistance compared with conventional instrumentation to radiographic outcomes in experienced, high-volume centers remains unclear.

METHODS: This retrospective comparative study included 96 consecutive medial UKAs performed at a single high-volume institution. Patients were stratified into three groups: Journey I UKA implanted with NAVIO imageless

robotic assistance (n = 38), Journey II Uni UKA implanted using conventional instrumentation (n = 27), and Journey II UK UKA implanted with CORI imageless robotic assistance (n = 31). Standardized standing long-leg weight-bearing radiographs were obtained preoperatively and postoperatively. Radiographic evaluation included hip-knee-ankle (HKA) angle, femoral component coronal angle (FCCA), tibial component coronal angle (TCCA), and sagittal alignment parameters. Tibial coronal alignment was interpreted according to the Cartier metaphyseal alignment philosophy. Intergroup comparisons were performed using appropriate statistical tests.

RESULTS: Baseline demographic characteristics and preoperative radiographic parameters were comparable among the three groups (all p > 0.05). Postoperativ […]

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### The perforator-based axial adipofascial turnover flap combined with juncturae tendinum grafting for complex dorsal phalangeal defects.

*Ozer K*

INTRODUCTION: Complex dorsal finger soft tissue defects remain a challenging reconstructive problem, particularly when associated with exposure of bone and extensor apparataus. Advances in the anatomical understanding of the dorsal cutaneous vascular system of the digits have allowed adipofascial flaps in this region to evolve from random-pattern to axial-pattern designs, improving vascular reliability. However, a structured and comprehensive approach for managing these combined defects has not been clearly defined in the literature. This study aims to present a holistic loco-regional strategy that optimizes the reconstructive ladder while minimizing donor-site morbidity and technical complexity.

MATERIALS AND METHODS: This clinical series includes seven fingers with complex soft tissue defects involving the dorsal aspect of the middle and distal phalanges. All cases were managed using an ipsilateral loco-regional protocol consisting of three components: (1) an extended axial adipofascial turnover flap for soft tissue coverage, (2) tendon reconstruction using juncturae tendinum harvested from the fourth intermetacarpal space, and (3) full-thickness skin grafting from the wrist. Patients were followed for a minimum of six months. Functional outcomes were evaluated by range of motion and complication rates.

RESULTS: Complete flap survival was achieved in all cases. All fingers […]

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### Operative treatment of high-grade medial collateral ligament injury and chronic medial knee instability: a systematic review.

*Xia Y, Zhao Y, Dou X, Wang X, You H, Chen L, He Y, Zhang J et al.*

High-grade medial collateral ligament (MCL) injury and chronic medial knee instability can compromise both valgus and rotational control, especially in cruciate or multiligament injury. This systematic review evaluated operative treatment according to clinical phenotype rather than MCL grade alone. PubMed/MEDLINE and Web of Science Core Collection were searched from January 1, 2000, to August 10, 2026; a Cochrane CENTRAL update was completed on September 7, 2026. Eligible clinical reports included operative treatment of acute high-grade MCL/medial-complex injury or chronic symptomatic medial instability. Risk of bias was assessed with RoB 2, ROBINS-I, or the Joanna Briggs Institute case-series checklist. Thirty-five reports were included: 5 randomized or quasi-randomized trials, 9 nonrandomized comparative studies, and 21 case series. Routine acute medial surgery during ACL reconstruction was not consistently superior across heterogeneous grade III populations. In contrast, selected displaced distal or Stener-like lesions were plausible repair targets because interposition can prevent healing in apposition. Chronic symptomatic valgus and/or anteromedial rotatory insufficiency was most often treated with reconstruction, with directional improvement in objective stability and patient-reported outcomes in case series. Evidence for augmentation, the optimal reconstruction construct […]

PubMed · DOI

### Correction: Recovery of gait after ankle fracture surgery following early mobilization and weight-bearing.

*Bauer L, Schlegel AS, Scheuer A, Böcker W, Polzer H, Baumbach SF*

PubMed · DOI

## Factors associated with long-term patient-reported outcomes after periacetabular osteotomy.

*Shoji T, Ueki S, Kaneta H, Morita H, Kozuma Y, Adachi N*

INTRODUCTION: Long-term survivorship and patient-reported outcomes after periacetabular osteotomy (PAO) have been reported; however, the relationships of long-term PROMs with preoperative cartilage degeneration and postoperative three-dimensional hip morphology remain incompletely understood. This study evaluated long-term PROMs and examined their associations with preoperative cartilage T2 values and postoperative hip morphology after PAO.

MATERIALS AND METHODS: This retrospective study included 42 hips in 42 patients with early-stage osteoarthritis who underwent PAO for developmental dysplasia of the hip and were followed for more than 10 years. Clinical outcomes were assessed preoperatively and at final follow-up, whereas the Japanese Orthopaedic Association Hip Disease Evaluation Questionnaire (JHEQ) was assessed at final follow-up. Hip morphology was evaluated using computed tomography (CT), and cartilage degeneration was assessed using preoperative T2 mapping magnetic resonance imaging. Associations with final JHEQ scores were analyzed using correlation analyses and multivariable linear regression with JHEQ scores treated as continuous outcomes.

RESULTS: The mean final total JHEQ score was 56.4 ± 19.6 points. In the multivariable linear regression model for total JHEQ (adjusted [Formula: see text]), higher preoperative acetabular T2 values ([Formula: see text], 95% CI [ […]

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## Clinical, etiological, and microbiological differences between foot and non-foot osteomyelitis.

*Kundakci B, Dalgic Y, Dalkir KA, Olke HC, Mirioglu A, Tekin M*

INTRODUCTION: Osteomyelitis presents with heterogeneous clinical, etiological, and microbiological characteristics depending on anatomical location. While foot osteomyelitis is commonly associated with diabetic foot infections and contiguous spread, non-foot osteomyelitis often arises from post-traumatic, hematogenous, or postoperative causes. However, direct comparative analyses between these two entities remain limited.

MATERIALS AND METHODS: A retrospective analysis was conducted on 202 adult patients who underwent surgical treatment for osteomyelitis between 2011 and 2026. Patients were divided into foot (n = 89) and non-foot (n = 113) osteomyelitis groups. Clinical, etiological, microbiological, and treatment-related variables were compared. Only intraoperatively obtained deep tissue and/or bone cultures were included.

RESULTS: Patients with foot osteomyelitis were older and had a significantly higher prevalence of diabetes mellitus (49.4% vs. 5.3%, p < 0.001), which was identified as an independent predictor of foot osteomyelitis (OR 14.15, 95% CI 4.32-46.42; p < 0.001). Infections in foot osteomyelitis predominantly developed via contiguous spread (73.0%), whereas non-foot osteomyelitis showed more heterogeneous etiologies, including post-traumatic and hematogenous causes (p < 0.001). Microbiologically, Staphylococcus aureus was the most common pathogen in non-foot ost […]

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## Identification of patient subgroups by hierarchical cluster analysis in femoral neck fractures and surgical outcome analysis.

*Ipek E, Altunta■ Y, Balkanl■ B, Bozca MA, Unutmaz H, Demirkale ■, Eren OT*

INTRODUCTION: Femoral neck fractures are a heterogeneous entity with wide variability in patient profiles and outcomes. Single-variable classification may not capture the multidimensional interactions that influence prognosis. This study aimed to identify clinically meaningful patient subgroups using hierarchical cluster analysis and to compare mortality, postoperative mobility, and complication rates between these subgroups.

MATERIALS AND METHODS: This retrospective cohort study included 672 patients treated for intracapsular femoral neck fractures classified as AO Foundation/Orthopaedic Trauma Association (AO/OTA) 31-B between 2009 and 2023, including 586 who underwent hemiarthroplasty (HA) and 86 who underwent total hip arthroplasty (THA). Unsupervised hierarchical clustering was performed using the Gower distance matrix and Ward's D2

linkage method based on 20 preoperative and intraoperative clinical and surgical variables, excluding outcome measures. The optimal number of clusters was determined by silhouette analysis. Mortality predictors were assessed using Cox proportional hazards regression, and predictors of good postoperative mobility were evaluated by binary logistic regression.

RESULTS: Two distinct clinical phenotypes were identified: a high-risk cluster (n = 607; mean age 78.3 years, American Society of Anesthesiologists [ASA] physical status III 86.2%, HA 96.4 […]

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### High variability in resistance to valgus deformation among knee braces indicated for medial collateral ligament injuries: a standardized bench-model biomechanical comparison.

*Pechlivanidou E, Stacheter M, Wahl P, Jacobi M*

INTRODUCTION: To quantify valgus deformation and per-load mechanical stiffness of commercially available knee braces indicated for medial collateral ligament (MCL) injuries, and to evaluate whether mechanical performance is associated with material classification and qualitative stability assessment.

MATERIALS AND METHODS: Nine knee braces identified through a national market survey were tested on a custom lower limb mechanical surrogate under progressive valgus loading at 5.7, 11.4, and 19.4 Nm. Angular deformation was measured using video-based motion tracking, and per-load brace stiffness (Nm/°) was computed at each load level.

RESULTS: At maximum applied torque (19.4 Nm), per-load stiffness ranged from 1.6 to 6.5 Nm/° across the nine devices - a fourfold difference between the stiffest and most compliant brace - with median valgus deformation of 8° (range 3-12°). In an exploratory, underpowered between-category comparison, rigid braces showed numerically higher stiffness than semi-rigid and soft devices ($\varepsilon^2$ = 0.34-0.45); this comparison is reported for hypothesis generation only. Qualitative ratings showed only moderate, inconsistent correlations with measured performance.

CONCLUSIONS: Commercially available knee braces indicated for MCL injuries demonstrate highly variable mechanical resistance to valgus deformation under standardized bench conditions. The present sampl […]

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### The role of the plantar fascia, flexor hallucis longus, and fibro-osseous tunnel in functional hallux rigidus.

*Wang M, Hunter Aitchison A, Orriols AM, Ridenour RM, Easley ME, Adams SB*

BACKGROUND: Functional hallux rigidus is characterized by limited first metatarsophalangeal (MTP) dorsiflexion during weight bearing conditions, but normal motion otherwise. This phenomenon alters the biomechanics of the foot, potentially leading to first MTP pain and trauma to the MTP cartilage. The purpose of this study is to assess the role of the flexor hallucis longus (FHL), plantar fascia (PF), and fibro-osseous tunnel in functional hallux rigidus.

METHOD: Fourteen cadaveric lower limbs were analyzed. The Dananberg technique was used to dorsiflex the great toe and radiographs were taken after each of the following manipulations, measuring the dorsiflexion angle of the first MTP. The plantar fascia was transected with a Z-cut, shortened approximately 10 mm, then repaired. The fibro-osseous tunnel was released. The FHL was transected with a Z-cut then shortened approximately 7 mm.

RESULTS: Average passive dorsiflexion of the first MTP is 23.8° (95% CI 19.7-27.9) prior to manipulation. Shortening the plantar fascia and FHL did not decrease the MTP dorsiflexion significantly beyond baseline ($p>0.05$). Lengthening the plantar fascia increased the dorsiflexion by 6.7° (28.1% increase) and lengthening the FHL increased dorsiflexion by 7.0° (29.4% increase). The difference between the two methods was not significant ($p>0.05$). Releasing the fibro-osseous tunnel alone increased th […]

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### Mix and match strategies in revision total hip arthroplasty: a review of techniques and outcomes.

*Ashkenazi I, Khury F, Konan S*

PURPOSE: This review evaluates the clinical indications, surgical rationale, and outcomes of six targeted component-retention strategies in revision total hip arthroplasty (THA) designed to reduce operative morbidity and

preserve host bone stock.

METHODS: This is a narrative review. A structured but non-systematic search of PubMed/MEDLINE and Scopus was used to identify contemporary clinical cohorts, joint registry data, and biomechanical analyses, with the evidence organised sequentially across acetabular and femoral subsites.

RESULTS: Acetabular retention of stable metal-on-metal shells with dual-mobility bearings minimizes surgical trauma and yields exceptionally low dislocation rates (0% to 4%). Cementing highly cross-linked polyethylene (XLPE) or dual-mobility liners into secure or revision cups successfully decouples biological fixation from alignment, ensuring high mid-term survivorship. On the femoral side, trunnion adapter sleeves protect the head-neck junction during isolated bearing exchanges, demonstrating reliable 10-year survivorship (85%). For unavoidable stem extractions with sound cement mantles, cement-in-cement revision offers a rapid pathway; utilizing high-fatigue-resistant cobalt-chromium alloys instead of stainless steel structurally mitigates the rare risk of cantilever stem fracture. Lastly, for uncontained bone loss or profound osteopenia, extended l […]

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## Complications following operative versus nonoperatively managed isolated distal third ulna shaft fractures: a retrospective single-institution cohort study.

*Schultz K, Wang JE, Manley B, Peek K*

PURPOSE: This study aimed to compare radiographic and clinical outcomes of operatively versus nonoperatively managed isolated distal third ulna fractures.

METHODS: This is a retrospective cohort study. Operatively and nonoperatively managed isolated distal third ulna shaft fractures on forearm radiographs in adult patients with a minimum of six-week follow-up were identified at an academic level 1 trauma center over seven years from 2017 to 2024. Outcomes included rates of radiographic evidence of union, pain scores, nonunion, tendinopathies, removal of hardware, or heterotopic ossification at final follow-up. Outcomes were compared between cases managed operatively and those managed nonoperatively.

RESULTS: A total of 56 isolated distal third ulna shaft fractures were identified with a mean follow-up duration of 30.5 weeks, including 12 (21.4%) managed operatively and 44 (78.6%) managed nonoperatively. The mean age was 49.0 years in the nonoperative group and 38.6 years in the operative group (p = 0.115), and the proportions of males were higher in the nonoperative group (61.4% vs. 16.7%, p = 0.006). At final follow-up, radiographic evidence of union was present in 77.3% of nonoperative cases and 91.7% of operative cases (p = 0.266). The complication rate was lower in the nonoperative group (15.9% vs. 41.7%), although the difference was not statistically significant (p = 0.0 […]

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## Radiation exposure for planning deformity corrections with hexapods: comparison of conventional X-ray, standard CT, and ultra-low-dose CT.

*Schüll D, Baumgartner H, Krüger B, Thelen J, Keller G*

INTRODUCTION: Hexapod systems are among the most accurate devices for correcting complex bone deformities. Postoperative planning is carried out using web-based software solutions based on at least two X-ray images in orthogonal planes. Recent studies have shown that inaccuracies in determining mounting parameters and the spatial relationship between the reference ring and the bone may lead to measurement errors. Computed tomography images potentially offer greater accuracy in this regard, but are associated with increased radiation exposure. The aim of this study was to compare radiation exposure from conventional radiography (X-ray), standard computed tomography (CT), and ultra-low-dose CT (ULD-CT) in the context of planning deformity corrections using a hexapod system and to evaluate suitability and comparability of these imaging modalities for treatment planning.

MATERIALS AND METHODS: Between March 1, 2022, and August 31, 2024, 28 patients were treated with a hexapod system for post-traumatic deformities at a Level I trauma center. All patients for whom standardized radiographs (anteroposterior and mediolateral views), conventional CT scans, and ULD-CT scans were available were included in this retrospective analysis. The study population thus comprised n = 18 patients. The effective radiation dose (µSv) was calculated separately for each examination. The values were comp […]

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## Operative versus conservative management of distal humerus fractures in patients aged 65 years and older: a target-trial-informed retrospective cohort study of five-year mortality.

*Murad H, Shehadeh M, Igbaria I, Mahamid MK, Yassin M, Mahamid A*

INTRODUCTION: Older patients with distal humerus fractures die more often after conservative than after operative treatment in unadjusted comparisons; whether this reflects treatment or the selection of healthier patients for surgery is unresolved. We estimated the association between operative versus conservative management and five-year all-cause mortality in patients aged ≥ 65 years.

MATERIALS AND METHODS: Retrospective comparative cohort study at two tertiary trauma centres (2012-2025). Cases were ascertained by automated screening of codes and free text, then adjudication of all uncertain records against source notes and injury radiographs, masked to treatment and vital status. Patients were classified by management initiated within 14 days as operative (n = 75) or conservative (n = 78); vital status was registry-linked. The primary estimand was the five-year restricted mean survival time (RMST) difference; principal adjusted estimates were inverse-probability-weighted (IPW) RMST and survival differences, requiring no proportional-hazards assumption. Cox, window and landmark analyses are supportive.

RESULTS: Among 153 patients (52 deaths), conservatively managed patients were older (median 83 vs. 73 years, $p < 0.001$). Unadjusted five-year mortality was 41.9% versus 18.4%: RMST difference + 11.2 months (95% CI + 5.6 to + 16.8), log-rank $p = 0.006$. After weighting for age, […]

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## Identification of biomechanical alterations associated with nonunion after tibial shaft fractures using musculoskeletal simulation based on motion capture and instrumented insole data.

*Wahl T, Ganse B, Warmerdam E, Hildebrand F, Hofmann UK, Praster M*

INTRODUCTION: Musculoskeletal simulations based on gait analysis data provide access to parameters such as joint reaction forces, muscle forces, and push-off moments, which cannot directly be measured in clinical practice. Applying these methods may help evaluate weight-bearing patterns and identify mechanical risk factors associated with the development of nonunion. This study aimed to investigate differences in the longitudinal development of forces in the musculoskeletal system between cases with and without union in tibial shaft fractures.

MATERIALS AND METHODS: Longitudinal 3D marker-based motion capture and instrumented insole data of thirteen patients (8 union; 5 nonunion) with tibial shaft fractures were used to generate individualized musculoskeletal models using the AnyBody™ software.

RESULTS: In a linear mixed-effects model, the fractured limb showed impaired biomechanical parameters compared with the contralateral limb, including knee joint reaction forces (KJRFs), muscle forces, and push-off moments (all $p < 0.001$). During follow-up, parameters of the fractured leg increased over time, including KJRFs ($p = 0.011$), muscle forces ($p \leq 0.030$), and ground reaction force ($p = 0.008$). The union group exhibited greater muscle forces, particularly in the soleus ($p = 0.014$) and gastrocnemius muscles ($p = 0.049$). The nonunion group exhibited significantly greater asymmetry […]

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## Biomechanical comparison of innovative suture techniques for repair of Achilles tendon rupture in a porcine flexor tendon model.

*Weigand L, Liebrand B, Scheible S, Meffert R, Hoelscher-Doht S*

INTRODUCTION: In recent years, evidence-based progressive rehabilitation protocols following the surgical treatment of Achilles tendon rupture have been shown to be safe and appear superior to more restrictive treatment regimens. These protocols mainly rely on early mobilization and weight bearing. To enable this during postoperative rehabilitation, a biomechanically stable surgical treatment must be ensured. Different solutions for suturing ruptured Achilles tendons are currently favored by orthopedic surgeons and have been investigated in the past, including promising further developments of traditional approaches such as the combination of pretension and the Bunnell technique, as well as minimally invasive or percutaneous techniques using systems like the Arthrex PARS. However, the best solution in this regard remains unclear.

METHODS: We analyzed the biomechanical stability of four different suture techniques using porcine deep flexor tendons as a biomechanical model for human Achilles tendons. The tendons were harvested, prepared and sutured according to their respective group (A: Bunnell, B: Bunnell + pretension, C: PARS-to-PARS, D: PARS + anchor). The four groups (n = 12) underwent progressive, cyclic, uniaxial tensile loading in the biomechanical testing machine (ZwickRoell Z020).

RESULTS: Group D (PARS + anchor) showed the highest maximum force at ultimate failure, r […]

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### Comparative outcomes of delta-on-delta and delta-on-highly cross-linked polyethylene bearings in total hip arthroplasty: a nationwide cohort study.

*Lee YK, Kang D, Kim H, Kang E, Cho J, Park JW*

INTRODUCTION: The purpose of this study was (1) to evaluate the national trends of Delta ceramic and highly crosslinked polyethylene (HXLPE) liner in total hip arthroplasty (THA), and (2) to compare survivorship of Delta ceramic versus HXLPE liner when used with a Delta ceramic head.

MATERIALS AND METHODS: Using the K-NHIS database, we identified all patients who underwent primary THA using Delta ceramic head in South Korea between 2003 and 2013 (n = 16,250). Cumulative incidences of dislocation, periprosthetic joint infection (PJI), and periprosthetic fracture were evaluated according to bearing type: Delta-on-Delta ceramic (DoD) and Delta-on-HXLPE (DoHXLPE). Survivorship for revisions was calculated to determine differences. Multivariable Cox proportional hazard regression models were used for comparisons, and propensity score matching was additionally performed as a sensitivity analysis.

RESULTS: A total of 15,299 patients underwent THA with Delta ceramic head between 2003 and 2013. Delta liner was used in 14,255 patients (93.2%) and HXLPE in 1,044 patients (6.8%). Utilization of Delta liner has noticeably increased since 2007. The DoHXLPE group had a higher risk of PJI (aHR = 2.10, 95% CI = 1.29-3.41) compared to the DoD group within 1 year. There were no significant differences in risk of dislocation (aHR = 1.41, 95% CI = 0.97-2.05) and periprosthetic fracture (aHR = 0.9 […]

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### A comparative study on the clinical efficacy and complications of proximal femoral nail antirotation and femoral neck system in the treatment of basicervical femoral neck fracture (AO/OTA 31-B3).

*Jiang X, Lin D, Huang Y, Liu J, Lin F*

OBJECTIVES: Basicervical femoral neck fractures (FNFs) are prone to internal fixation failure owing to their inherent biomechanical instability. This study compares the efficacy of femoral neck system (FNS) and proximal femoral nail anti-rotation (PFNA) in treating basicervical FNF.

MATERIALS AND METHODS: A retrospective analysis was conducted on 99 patients with basicervical FNF admitted to fuzhou second general hospital from January 2020 to June 2024. Patients were divided into two groups based on the internal fixation method used: the FNS group (53 cases, average age 63.83 ± 18.80 years) and the PFNA group (46 cases, average age 69.78 ± 16.65 years). Main observation indicators included reduction quality and internal fixation-related complications, while secondary indicators included tip-apex distance, surgical time, occult blood loss, wound length, postoperative hospital stay, mortality, and Harris hip score.

RESULTS: The average follow-up time was 24.03 ± 9.4 months. Preoperative data were comparable between groups (P > 0.05). The PFNA group had a significantly lower overall incidence of implant-related complications than the FNS group (2.17% vs. 16.98%, P = 0.035), and the distribution of implant-related complication subtypes also differed significantly between groups (P = 0.042). No significant differences were observed in fracture reduction quality, tip-apex distance, […]

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### Management and outcome of displaced acetabular fractures in the oldest-old: a retrospective analysis of orif and acute total hip arthroplasty.

*Mamoutakis I, Horst K, Alabdulrahman H, Groven RVM, Bläsius FM, Hildebrand F, Berk T, Bolierakis E*

PURPOSE: The optimal surgical management of geriatric acetabular fractures remains controversial. This study compared short-term outcomes following open reduction and internal fixation (ORIF) or acute total hip arthroplasty (THA) with or without additional osteosynthesis (THA ± ORIF) and evaluated the impact of a new treatment algorithm (standard operative procedure, SOP) that favors the use of THA in elderly and medically comorbid patients.

METHODS: In this retrospective cohort study, patients aged ≥ 70 years with isolated trauma-associated unilateral displaced acetabular fractures who underwent surgical treatment at a single Level I trauma center between January 2016-March 2019 (old SOP) and January 2022-December 2025 (new SOP) were included. Patients underwent either ORIF or THA ± ORIF. The transitional implementation phase of the new SOP was excluded. The primary outcome was postoperative time to gait mobilization. Secondary outcomes included pain at discharge, functional mobility, complications, mortality, quality of fracture reduction, and conversion to THA following initial ORIF. Because treatment allocation followed different institutional algorithms across the study periods, the study primarily evaluates an evolving treatment strategy rather than directly comparing ORIF and THA.

RESULTS: Sixty patients (median age 82 years, IQR 78-87; median Charlson Comorbidity Inde […]

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## Race, ethnicity, and language are underreported in hip fracture CPGs.

*Mulakaluri A, Snyder EM, Young B, Kamal RN, Shapiro LM*

INTRODUCTION: Clinical Practice Guidelines (CPGs) are systematically developed recommendations based on peer-reviewed literature that guide patient care. Considering the impact of sociodemographic factors on patient outcomes, this study aims (1) to review the racial, ethnic, and linguistic representation of patients in literature utilized to inform the CPGs for hip fractures in older adults and (2) to assess the use of linguistically- and culturally-adapted Patient Reported Outcome Measures (PROM) in these studies.

MATERIALS AND METHODS: Publications listed under the American Academy of Orthopaedic Surgeons (AAOS) CPGs for the Management of Hip Fractures in Older Adults were identified and extracted. Variables extracted included: title, author, publication year, inclusion/exclusion criteria, country/ies of study, language of study, patient language, race and ethnicity, patient language proficiency, outcome measures utilized, translation or cultural-adaptation of outcome measures, and clinical trial registration page. The extracted variables were reported descriptively.

RESULTS: Of the 196 unique articles included, 12/196 (6.12%) reported race/ethnicity data and 9/196 (4.59%) reported patient language, of which 6/9 (66.7%) reported patient language proficiency. Of the studies that reported patient language or language proficiency, 8/9 (88.9%) specified this data in their inclu […]

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## Level-dependent changes in cervical multifidus composition following anterior cervical discectomy and fusion: bridging quantitative MRI and clinical perspectives.

*Folkerts T, Tripathi R, Caffard T, Verna B, Zhu J, Sama AA, Cammisa FP, Hughes AP*

INTRODUCTION: Fatty infiltration (FI) of the cervical paraspinal muscles is associated with greater disease severity and less favorable outcomes in degenerative cervical spine disease. However, longitudinal data on whether muscle composition changes following anterior cervical discectomy and fusion (ACDF) are scarce. This study aimed to evaluate changes in FI and functional cross-sectional area (fCSA) of the cervical multifidus and rotatores after ACDF and to assess whether these changes differ across surgical levels.

MATERIALS AND METHODS: This single-center retrospective study analyzed patients who underwent ACDF for degenerative cervical stenosis between 2015 and 2018. Patients were included if both preoperative and postoperative cervical spine MRI (obtained 1-5 years after surgery) were available. The multifidus and rotatores colli were segmented bilaterally on T2-weighted axial images. Measurements were categorized by position within the fusion construct as upper (most cranial fused vertebra), middle (intervening vertebrae), and lower (most caudal fused vertebra) levels. FI and fCSA were quantified using Otsu thresholding. P-values were adjusted for multiple testing using the Benjamini-Hochberg procedure.

RESULTS: Thirty-five patients met inclusion criteria (median age 58 years, 40% female, 1-4 fused levels). Bilateral FI decreased significantly at the lower level (52.1 […]

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## Superior reduction quality without improved outcomes: open versus closed reduction for displaced femoral neck fractures in adults under 60 years.

*Chi JE, Liu CH, Lo YH, Lai CY, Chen IJ, Yu YH*

INTRODUCTION: The optimal reduction strategy for displaced femoral neck fractures (DFNFs) in non-elderly adults remains controversial. Although open reduction may improve radiographic alignment, its impact on outcomes remains unclear.

MATERIALS AND METHODS: Adults aged < 60 years with DFNFs who were treated with internal fixation between 2017 and 2023 were retrospectively reviewed. Patients were categorized into closed and open reduction groups. Reduction quality was assessed using the Garden alignment index and the Haidukewych reduction criteria. Multivariate analyses were performed for the overall cohort and patients with Pauwels type III fractures. Outcomes included non-union, osteonecrosis, femoral neck shortening, and major reoperations.

RESULTS: In total, 98 patients were included, among whom 83 and 15 underwent closed and open reduction, respectively. Although open reduction achieved a higher rate of excellent reduction (53.3% vs. 16.9%, p = 0.012), it was not associated with lower rates of non-union or osteonecrosis. In the overall cohort, age was associated with non-union (odds ratio [OR], 1.09; p = 0.024), whereas poorer reduction quality was associated with major reoperation (fair: OR 7.03, p = 0.012; poor: OR 47.6, p = 0.007). In patients with Pauwels type III fractures, poorer reduction quality predicted conversion to arthroplasty (fair: OR 9.84, p = 0.014), and […]

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## Longer hospital stay but similar one-year mortality and readmission after combined proximal humerus and proximal femur fragility fractures: a Propensity-Score-Matched Cohort Study.

*Murphy S, Kellett C, Roberts D, McGurk C, Ahern E, O'Loughlin P*

BACKGROUND: Fragility Proximal Humerus (PH) fractures are common and often associated with significant morbidity and mortality. Unlike Proximal Femur (PF) fractures, PH fractures are often omitted from established orthogeriatric multidisciplinary team (MDT) pathways. Combined Proximal Humerus and Proximal Femur fractures (PH&PF) are rare. Combined injuries would be expected to result in worse outcomes compared to isolated PH fractures due to greater injury burden. This study assessed outcomes between isolated PH and combined PH&PF fractures, exploring whether differences in care pathways may contribute to outcome.

METHODS: A retrospective propensity score-matched cohort study was performed over a 2-year period identifying patients sustaining isolated fragility PH fractures and over a 5-year period identifying patients sustaining combined PH&PF fractures secondary to low-energy trauma. Propensity score matching was performed using age, sex, comorbidity burden (as defined by Charlson Comorbidity Index (CCI)), mobility status, cognitive impairment and fracture pattern. Primary outcome was 1-year mortality. Secondary outcomes included 1-year readmission and length of stay (LoS). Mortality was analysed using Firth logistic regression, readmission using clustered logistic regression, and LoS using Gamma regression with cluster-robust standard errors.

RESULTS: Sixty-five patients we […]

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## Are kneeling stress radiography and the posterior sag sign associated with subjective flexion instability after posterior-stabilized total knee arthroplasty?

*A■k■n M, Ça■lar Ö, Tokgözo■lu AM, Atilla B*

INTRODUCTION: Flexion instability is a common but underdiagnosed cause of patient dissatisfaction after posterior-stabilised total knee arthroplasty (TKA). Standard clinical tests have limited reliability for this condition. We investigated whether kneeling stress radiography and selected physical examination findings can help identify it.

MATERIALS AND METHODS: We retrospectively reviewed 145 posterior-stabilised TKA knees in 119 patients, divided into two groups by the presence of subjective knee instability. The posterior sag sign, 90° anterior drawer test, and kneeling stress radiographic parameters-particularly the change in tibiofemoral translation under load

(Δ-TPFA)-were compared between groups. Post-hoc analyses included ROC curve analysis, diagnostic accuracy with Wilson 95% confidence intervals, and multivariable logistic regression adjusted for age, sex, BMI, follow-up duration, implant type, and lumbar pathology.

RESULTS: Thirty knees (20.7%) reported subjective instability. Mean Δ-TPFA was 1.40 ± 3.75 mm in the unstable group versus - 1.17 ± 3.28 mm in the stable group (p = 0.001; AUC 0.68, 95% CI 0.56-0.80) and remained an independent predictor on multivariable analysis (OR 1.30 per mm, 95% CI 1.11-1.52, p < 0.001). The posterior sag sign had a specificity of 82.6% (95% CI 74.6-88.4%) and was independently associated with instability (OR 3.26, 95% CI 1.54-6.91, […]

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### Pelvic injuries in elderly trauma patients: epidemiology, injury characteristics, in-hospital care and factors influencing mortality.

*Enocson A, Lundin A, Granqvist M*

BACKGROUND: Pelvic and acetabular fractures are increasing in incidence in elderly patients. Whether the fracture contributes independently to mortality, or merely accompanies the factors that drive it, remains contested, and large national data on elderly trauma patients are limited. The primary aim of this study was to determine factors influencing 30-day mortality in elderly trauma patients, with particular focus on pelvic injuries. Secondary aims included to analyse the influence of these injuries on in-hospital care and discharge destination.

METHODS: All patients aged ≥ 60 years admitted as trauma alerts in Sweden between 1 January 2022 and 31 December 2024 were identified through the Swedish Trauma Registry (SweTrau). The primary outcome was 30-day mortality. The association between pelvic injury (pelvic or acetabular fracture) and 30-day mortality was assessed using multivariable logistic regression, adjusting for gender, ASA class, NISS, and head injury.

RESULTS: Of 11,624 patients (median age 76 years, 39.7% males), 776 (6.7%) sustained a pelvic injury. Patients with pelvic injury more frequently sustained high-energy trauma, had a higher median NISS (17 vs. 11), longer hospital length of stay (6 vs. 3 days), higher ICU admission rate (30.9% vs. 20.0%), and was less frequent discharged direct home (34.0% vs. 55.8%; all p < 0.001). Crude 30-day mortality was 16.1% vs […]

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### Pain, quality of life, and functional outcomes of total hip arthroplasty in female patients with crowe type III-IV developmental dysplasia of hip: retrospective cohort study.

*Güzel N, Genç AS, Gümü■ NE, Ta■k■n O, Güzel A*

INTRODUCTION: The aim of this study was to investigate physical and psychosocial recovery including pain and quality of life measures, as well as lower limb functional out-comes in patients with Crowe type III-IV developmental dysplasia of the hip (DDH) who underwent total hip arthroplasty (THA) with subtrochanteric shortening osteotomy.

MATERIALS AND METHODS: This study was conducted with a total of 45 female patients aged 32-65 with Crowe type III-IV DDH. Pre-op 30-second chair-stand Test (30-s CST), 6-minute walk test (6MWT) and Visual Analogue Scale (VAS) were assessed clinically, whereas Harris Hip Score (HHS), EuroQol 5-Dimension 5-Level (EQ-5D-5 L) scale, Forgotten Joint Score-12 (FJS-12), Hip disability and Osteoarthritis Outcome Score (HOOS), and SF-36 Quality of Life Scale (SF-36) scores were measured and recorded based on patient self reports. At least two years after the surgery, the same measurements were repeated in the patients.

RESULTS: At the mid-term follow-up, a significant increase was observed in the 30-s CST and 6MWT distance compared with the pre-op period (p < 0.01). Furthermore, statistically significant differences were observed in the VAS, HHS, EQ-5D-5 L, FJS-12, HOOS and SF-36 scores (p < 0.01).

CONCLUSIONS: An evaluation after THA with subtrochanteric shortening osteotomy in female patients revealed a significant improvement in functional perform […]

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### Bioabsorbable vs. titanium screws in first metatarsophalangeal joint arthrodesis: a pilot randomized controlled trial with 2-year follow-up.

*Partio N, Mattila V, Waris V, Mäenpää H*

INTRODUCTION: To assess the feasibility and preliminary clinical outcomes of bioabsorbable screw fixation in first metatarsophalangeal joint arthrodesis compared to conventional titanium screw fixation.

METHODS: In this prospective pilot randomized controlled trial, 30 patients were allocated 1:1 to bioabsorbable or titanium cannulated lag screw fixation. Arthrodesis was performed using a cup-and-cone technique at two Finnish hospitals. Full weight-bearing in an orthopaedic shoe was allowed after six weeks. Outcomes included AOFAS and VA-FAS FA-VAS scores, union rates, and complications assessed at 3, 6, 12, and 24 months.

RESULTS: Both groups showed significant functional improvement with no between-group differences. At 24 months, mean AOFAS scores were 89.1 (SD 2.7) in the titanium group and 90.0 (SD 0.0) in the bioabsorbable group (p = 0.23). Mean FA-VAS total scores (sum of pain, function, and complaints subscales) were 1948 (SE 59.4) in the titanium group and 1914 (SE 61.4) in the bioabsorbable group (p = 0.655). Union rate was 90% overall; nonunion occurred in one titanium and two bioabsorbable patients, all requiring revision. No implant-related complications or hardware removal surgeries were recorded. Degradation of bioabsorbable screws caused no adverse clinical or radiographic reactions.

CONCLUSIONS: Bioabsorbable screws appear safe and feasible in first metatars […]

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### Intraoperative patellofemoral load varies between total knee replacement and patellofemoral joint resurfacing, and across TKR implant systems.

*Phillipson L, Naybour J, Watson A, London N, Barrett D*

INTRODUCTION: Patellofemoral joint loading is influenced by implant design and surgical execution during knee replacement. Altered patellofemoral mechanics have been implicated in postoperative symptoms, yet the magnitude and variability of intraoperative load changes across different replacement procedures and implant systems remain incompletely characterised. The aim of this registry study was to describe and quantify intraoperative patellofemoral joint load changes following total knee replacement (TKR) and patellofemoral joint resurfacing (PFJR), focusing on cumulative and compartment-specific load redistribution.

MATERIALS AND METHODS: This observational registry study analysed prospectively collected intraoperative patellofemoral joint load measurements from 291 knee replacement procedures. Load was recorded in medial, lateral, superior, and inferior compartments before and after trial component implantation. Absolute and percentage changes in compartmental and cumulative load were calculated. Descriptive analyses examined the distribution and variability of load change across procedure type and implant system, with implant comparisons restricted to TKR procedures. No formal hypothesis testing was pre-specified, and no adjustment for potential confounding variables was undertaken.

RESULTS: Cumulative patellofemoral load change demonstrated substantial inter-individual v […]

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### Cortical hypertrophies in cementless short stem total hip arthroplasty in young patients under 60 years and elderly patients over 75 years-analysis of outcome and risk factors for cortical hypertrophies.

*Schwarz PM, Stützle S, Stadler C, Holzbauer M, Pisecky L, Gotterbarm T, Luger M*

BACKGROUND: Cortical hypertrophy (CH) is a common radiological finding associated with short-stem total hip arthroplasty (THA), but its clinical significance remains unclear. This study evaluates differences in outcomes between young patients under 60 years of age and elderly patients over 75 years of age who underwent cementless short stem THA. The aim was to evaluate if there are differences in these patients depending on the presence of CH.

METHODS: A retrospective analysis of 208 short-stem THAs performed between 2014 and 2017 was conducted. Patients were divided into younger (< 60 years, n = 119) and older (> 75 years, n = 89) groups. Clinical outcomes, including Harris Hip Scores (HHS) and Oxford Hip Scores (OHS), were compared between patients with and without CH. Multivariate regression analysis was performed to identify risk factors for CH, focusing on delta hip offset, canal fill index, and stem alignment.

RESULTS: Younger patients with CH demonstrated significantly better HHS (96.1 ± 7 vs. 90.8 ± 14.3, p = 0.010) and OHS (45.8 ± 5.2 vs. 42.9 ± 8.8, p = 0.028) compared to those without CH. In older patients, the presence of CH did not significantly impact outcomes. Regression analysis identified younger age (p = 0.012), higher Canal Fill Index III (p = 0.047), and stem alignment (p = 0.049) as significant predictors of CH.

CONCLUSION: Cortical hypertrophy is associ […]

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### Comparative effectiveness of single-stage versus two-stage revision for hip prosthetic joint infection: emulation of the INFORM target trial using electronic health record data.

*Zhang C, Li Y, Su C, Shen Y, Lu S*

OBJECTIVE: To estimate the comparative effectiveness of single-stage versus two-stage revision for hip prosthetic joint infection (PJI) in Chinese tertiary hospitals by emulating the INFORM randomized trial using the target trial emulation framework.

DESIGN: Retrospective cohort study emulating a target trial, reported per the TARGET guideline. The protocol and analysis plan were written before analysis but were not prospectively registered.

SETTING: Three tertiary hospitals in China. Index revision surgeries were performed between 1 January 2018 and 30 June 2022, with follow-up data locked on 31 December 2023, ensuring a minimum potential follow-up of 18 months for every patient.

PARTICIPANTS: 754 adults with confirmed hip PJI undergoing revision surgery (280 single-stage; 474 two-stage), identified from electronic health records and confirmed by manual chart review. Treatment groups reflected the intended surgical strategy documented before or at the index operation.

MAIN OUTCOME MEASURES: Primary outcome: Harris Hip Score (HHS) at 18 months.

SECONDARY OUTCOMES: surgically treated reinfection, reoperation, mortality, and complications. The primary analysis used inverse probability of treatment weighting (IPTW) on the full cohort with multiple imputation of missing covariate and outcome data; 1:1 propensity score (PS) matching (280 pairs) was confirmatory. Patients who di […]

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### Cemented dual mobility component in newly implanted porous tantalum shells for severe acetabular defects: medium-term outcomes from a high-volume single-surgeon series.

*Loppini M, Di Matteo V, Bulgarelli A, La Camera F, Chiappetta K, Morenghi E, Grappiolo G*

BACKGROUND: The management of severe acetabular bone defects remains challengin due to extensive bone loss and poor soft tissue quality during revision total hip arthroplasty (rTHA). The aim of this study was to assess clinical and radiological outcomes of the use of dual mobility components (DMC) cemented into a newly implanted porous tantalum (PT) acetabular shell during acetabular rTHA.

METHODS: Patients who underwent cementation of a DMC into a newly implanted PT cup between 2014 and 2020 were included. Bone defects were classified according to the Paprosky classification. The primary outcome was to determine the reintervention-free survival rate. The secondary outcome was clinical and radiographic assessment at last follow-up.

RESULTS: Thirty-six patients (36 hips), with an average age of 67.9 years (range, 46-90) were included. Paprosky type III defects (61.1%) were more common than type II defects (38.9%), with type IIIa (36.1%) and IIIb (25.0%) predominating. Median follow-up was 67.3 months (range; 17-109). The dislocation rate was 8.3% (n = 3). Four (11.1%) of 36 hips underwent acetabular component revision: 2 (5.5%) for infections, 1 (2.7%) for aseptic loosening, and 1 (2.7%) for dislocation. Cup survival rate was 89.08% (CI 69.54% to 96.38%) at 60 months. The Harris hip score (HHS) improved from 31.8 ± 12.0 to 87.0 ± 14.2 points (p < 0.001). The vertical and horiz […]

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### From load-bearing to load-sharing: contact-driven transitions in locked plate constructs.

*Hollensteiner M, Schindler M, Greinwald M, Mühling M, Baumeister D, Augat P*

Locked plate osteosynthesis functions as an internal fixator, yet load-induced plate-bone and fragment-fragment contacts - key transitions from load-bearing to load-sharing - remain poorly characterized. This study quantified their effects on construct stiffness, plate strain, and interfragmentary motion under axial loading. Twenty-seven configurations were tested using epoxy-glass surrogate bone fixed with titanium LCP plates, varying fracture gap (3-9 mm), working length (49-121 mm), and plate-bone distance (0-3 mm). Digital image correlation measured strain and motion. Contact events were identified manually from characteristic transitions in the force-displacement response, supported by corresponding changes in plate-surface strain progression. Geometric parameters dominated initial stiffness (125-442 N/mm), decreasing with plate elevation and working length in elevated setups. Within the investigated load range up to 600 N, 16/27 configurations showed non-linear transitions associated with contact events: fragment-fragment contact caused pronounced stiffness increases and strain stagnation at the fracture gap, whereas plate-bone contact was associated with local plate-surface strain stagnation at the location overlying the contact while strain at the fracture gap continued to increase. Interfragmentary motion stagnated after contact, confirming contact-dependent shifts in […]

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## Factors associated with periprosthetic joint infection following total knee arthroplasty: an updated systematic review and meta-analysis.

*Gao H, Deng Y, Ma Y, Li X, Zhang F, Mei Y*

BACKGROUND: This study aimed to systematically evaluate factors associated with periprosthetic joint infection (PJI) following total knee arthroplasty (TKA), and thereby to provide evidence-based references for clinical prevention and perioperative risk stratification.

METHODS: Computerized searches were conducted in the following databases from their inception until May 26, 2025: PubMed, Web of Science, Embase, the Cochrane Library, CINAHL, China National Knowledge Infrastructure, Wanfang Database, Chinese Scientific Journal Database, and Chinese Biomedical Literature Database. Two researchers independently screened the literature, extracted data, and assessed study quality. The methodological quality was assessed using the Newcastle-Ottawa Scale. Quantitative synthesis was performed when at least two studies reported comparable exposure definitions and sufficient comparator information; otherwise, narrative synthesis was used. Review Manager 5.4 software was used for the primary analysis. This study is registered on PROSPERO (CRD420251079339).

RESULTS: A total of 25 observational studies were included in the qualitative synthesis, of which 24 contributed to the primary quantitative synthesis. Quantitatively pooled factors associated with PJI included male sex (OR = 1.39, 95% CI 1.27-1.51), BMI ≥30 kg/m2 (OR = 5.72, 95% CI 2.65-12.36), prolonged operative time, type 2 diabet […]

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## Early surgery of trochanteric femoral fractures in patients on direct oral anticoagulants (DOACs) reduces the length of stay (LOS) and the 1-year mortality rates without increasing the risk for revision surgery: a matched-pair analysis.

*Weihs V, Laggner R, Humenberger M, Duma A, Frossard M, Hajdu S*

BACKGROUND: It remains unclear whether early surgical fixation of trochanteric femoral fractures in patients with direct oral anticoagulation (DOAC) therapy has an impact on the long-term outcome.

METHODS: A retrospective cohort study on patients with trochanteric femoral fractures and DOAC therapy between 2016 and 2024 was conducted. Patients were divided into two groups according to their time to surgery and matched with age and ASA score as covariates. The primary outcome of interest was the impact of early surgery on the long-term outcome.

RESULTS: Of 233 included patients, 102 patients were assigned in each group after propensity score matching. Patients with early surgery had lower in-hospital complication rates (16.0% vs. 26.3%; $p = 0.077$) and a significantly shorter median length of stay (LOS) (11.5 days (IQR 6) vs. 15 days (IQR 6); $p < 0.001$). No difference in the rates of revision surgery (9.8% vs. 5.9%; $p = 0.298$) was observed. Kaplan-Meier curves revealed a significant difference in the mortality rates during the follow-up period ($p = 0.0057$) with lower in-hospital (2.0% vs. 6.9%; $p = 0.088$), 30-day (3.9% vs. 9.8%; $p = 0.097$) and 1-year mortality rates (20.6% vs. 31.4%; $p = 0.079$).

CONCLUSION: Early surgical fixation of trochanteric femoral fractures in patients on DOAC therapy leads to a significant reduction in the LOS without increasing the risk for peri- and […]

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### Platform-specific learning curve and early tibial radiographic agreement of ROSA®-assisted medial unicompartmental knee arthroplasty in an experienced high-volume surgeon: a consecutive case series.

*Petrillo S, Ardiri D, Marullo M, Romagnoli S*

PURPOSE: To evaluate the platform-specific learning curve and early tibial radiographic agreement of imageless ROSA®-assisted medial unicompartmental knee arthroplasty (UKA), with short-term clinical findings assessed exploratorily.

METHODS: This retrospective consecutive case series included 30 patients who underwent ROSA®-assisted medial UKA with the Persona® Partial Knee implant between January and April 2026 at a high-volume arthroplasty referral centre. A continuous learning-curve cumulative summation (LC-CUSUM) analysis of total surgical duration was performed against an independently defined surgeon-specific historical benchmark derived from institutional operative-register skin-to-skin times for manual UKA performed by the same surgeon (20 ± 4 min). Adequate performance was defined as a mean duration of 25 min, inadequate performance as 30 min, and the reference standard deviation as 4 min. Type I and type II error targets were 0.05 and 0.10, respectively; the decision boundary was calibrated by Monte Carlo simulation over a 30-case monitoring horizon. Robotic time and a supportive comparison of cases 1-5 versus 6-30 were secondary efficiency analyses. Postoperative radiographic measurements were performed by an observer blinded to the planned and robot-validated ROSA® values. Agreement was quantified using Bland-Altman bias and 95% limits of agreement.

RESULTS: Mean […]

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### Cartilage disruption patterns associated with percutaneous intramedullary screw fixation for metatarsal fractures: a cadaveric study.

*Keith B, Burke J, Hynes K, Baker J, Huh J*

PURPOSE: Intramedullary screw fixation is an emerging technique for operative metatarsal fractures, yet the extent and location of articular cartilage disruption from retrograde screw insertion remain undefined. This study quantifies cartilage surface disruption and maps injury locations in the metatarsal head following screw placement.

METHODS: Fourteen fresh-frozen cadaveric feet (56 lesser metatarsals) underwent standardized retrograde intramedullary screw fixation using 3.6 mm screws. Following dissection, the total articular surface area and cartilage defect were measured to calculate the percentage of disruption. The cartilage surface was divided into quadrants (dorsal-to-plantar) and thirds (medial-to-lateral) to determine screw trajectory.

RESULTS: Mean cartilage disruption increased progressively from the 2nd to 5th metatarsals: 10.17%, 11.2%, 14.93%, and 17.43%, respectively. All screws were located within the middle third in the medial-to-lateral plane. In the dorsal-to-plantar plane, the majority of screws were positioned in the 2nd and 3rd quadrants. No screws were placed in the medial or lateral thirds.

CONCLUSION: Retrograde intramedullary screw fixation results in measurable cartilage disruption, with a tendency for screw placement in the middle third in the medial-to-lateral plane and 2nd and 3rd dorsal-to-plantar quadrants. While the relative cartilage surf […]

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### Locking plate fixation for proximal humerus fractures: are four proximal screws adequate? A clinical and radiological analysis of 105 consecutive cases.

*Michelitsch C, Stillhard PF, Zindel C, Sommer C, Acklin YP*

INTRODUCTION: Proximal humerus fractures are common injuries, and achieving stable fixation is essential to prevent complications such as varus collapse and screw perforation. While optimal screw positioning has been emphasized biomechanically, clinical evidence regarding the required number of screws in the humeral head,

particularly when minimally invasive aiming devices limit placement to four screws, remains limited. This study evaluated whether fixation with four cranial screws provides adequate clinical and radiological stability.

MATERIALS AND METHODS: A retrospective cohort study was conducted including 105 patients with displaced proximal humerus fractures treated with plate osteosynthesis. Patients were grouped according to the number of proximal screws used: four screws (n = 58) or more than four screws (n = 47). The primary outcome was the Constant-Murley Score. Secondary outcomes included secondary screw perforation, head-shaft angle maintenance, restoration of medial cortical support, and avascular necrosis. Mean follow-up was 18 ± 6 months.

RESULTS: Functional outcomes were comparable between groups, with Constant-Murley Scores of 78 ± 7 in the four-screw group and 73 ± 13 in the multi-screw group (p = 0.23). Rates of screw perforation (8.6% vs. 10.6%; p = 0.72) and postoperative head-shaft angles were likewise similar, and no significant alignment change occur […]

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### Nighttime intramedullary nailing of femoral fractures is not associated with worse outcomes: a 13-year Level I trauma center cohort.

*Abialevich A, Benkovich V, Osinsky V, Adiniaev Y, Tzaytlin I, Heinemann Y, Acker A*

BACKGROUND: Whether nighttime orthopedic trauma surgery adversely affects outcomes remains controversial. Evidence specifically addressing intramedullary fixation across femoral fracture locations remains limited. This study evaluated whether nighttime surgery was associated with a higher rate of recorded postoperative complications than daytime surgery.

METHODS: This retrospective observational cohort included adult patients with nonperiprosthetic femoral fractures treated with intramedullary fixation between January 2010 and January 2023 at a single Level I trauma center. Of 311 screened records, 23 were excluded because the patient was younger than 18 years (n = 17) or the fracture was periprosthetic (n = 6), leaving 288 procedures (242 daytime and 46 nighttime). The primary outcome was any recorded postoperative complication during available follow-up. Secondary outcomes included radiographic abnormality, delayed union (> 24 weeks), nonunion, revision surgery, and mortality. Surgeon experience was categorized as senior surgeon or resident. Exploratory fracture-location subgroup analyses and multivariable logistic regression were performed.

RESULTS: Nighttime patients were younger (35.1 ± 19.1 vs. 46.6 ± 24.6 years, p < 0.001), more frequently male (82.6% vs. 64.0%, p = 0.014), more often injured by a high-energy mechanism (80.4% vs. 56.6%, p = 0.002), and underwent surger […]

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### Patient-reported outcomes after lateral unicompartmental and total knee arthroplasty: a propensity score-matched cohort study.

*Hariri M, Sorbi R, Moradi B, Weishorn J, Trefzer R, Reiner T, Walker T, Nees TA*

INTRODUCTION: Comparative evidence on patient-reported outcomes (PROs) after lateral unicompartmental knee arthroplasty (UKA) and total knee arthroplasty (TKA) remains limited. We compared PROs between cohorts matched for demographic characteristics and baseline knee function.

MATERIALS AND METHODS: Prospectively collected single-center data were retrospectively analyzed. Lateral UKA was performed for isolated lateral-compartment osteoarthritis and TKA predominantly for multicompartmental disease. One-to-one propensity score matching incorporated age and body mass index, exact matching on sex, and restrictions on age and preoperative Oxford Knee Score (OKS), yielding 31 pairs. Postoperative OKS was the primary outcome. Secondary outcomes included OKS improvement, Forgotten Joint Score (FJS), University of California, Los Angeles (UCLA) activity score, OKS minimal clinically important difference (MCID) achievement, and FJS patient acceptable symptom state (PASS) attainment.

RESULTS: All matching covariates achieved acceptable balance (absolute standardized mean differences < 0.20). Postoperative OKS did not differ significantly (UKA, 38.9 ± 7.0; TKA, 39.8 ± 7.5; mean paired difference [TKA minus UKA], 0.94; 95% confidence interval [CI], - 2.57 to 4.44; p = 0.590). No significant differences were detected for OKS improvement (difference, 0.77; 95% CI, - 2.62 to 4.17; p = 0.644) […]

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## Pre- and postoperative 4D CT analysis of thumb CMC joint kinematics after first metacarpal extension osteotomy.

*Guebeli A, Honigmann P, Dobbe JGG, Keller M*

INTRODUCTION: First metacarpal extension (Wilson) osteotomy combined with modified Brunelli ligamentoplasty is performed for early thumb carpometacarpal (CMC1) osteoarthritis to improve pain and function. The purpose of this study was to quantify in vivo changes in CMC1 joint contact distribution and kinematics after this combined procedure and to determine whether postoperative patterns resemble those of healthy volunteers. We hypothesized that the procedure would reduce cumulative palmar contact during opposition-retroposition.

MATERIALS AND METHODS: In a prospective single-center observational cohort study, six adults with early CMC1 osteoarthritis (Eaton-Littler I-II) underwent standardized preoperative and 6-month postoperative four-dimensional computed tomography (4D CT) of the CMC1 joint during active opposition-retroposition. Five healthy volunteers served as reference with a single 4D CT scan. First metacarpal-trapezium motion was registered across frames. Joint contact was estimated using a 2.0 mm proximity surrogate on the trapezial surface. The primary outcome was change in cumulative palmar contact area. Secondary outcomes included change in dorsoradial contact, temporal variability of contact curves, and contact-centroid path length and mean position. Within-subject pre- versus postoperative comparisons were performed; healthy scans provided reference patterns. […]

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## Minimal kyphotic progression following pediatric vertebral fractures: implications for reduced radiographic surveillance.

*Wegmann S, Lenz M, Babasiz T, Eysel P, Mueller LP, Weber M*

INTRODUCTION: Pediatric spinal fractures are rare injuries (14.24 per 100,000 per year), representing only 0.6-3% of all spinal injuries. While most stable compression fractures are managed conservatively, concerns exist regarding progressive vertebral collapse and kyphotic deformity. Unlike extensive adult literature, pediatric-specific data on radiographic progression following conservative management remain limited, creating uncertainty about the clinical utility of routine follow-up imaging in this radiosensitive population.

MATERIALS AND METHODS: Retrospective single-centre cohort study of 54 pediatric patients (mean age 12.3 years) with traumatic vertebral body fractures diagnosed between 2012 and 2024. Kyphosis angles were measured using Cobb methodology at baseline and follow-up intervals (1, 3, and 6 weeks). The primary outcome was the change in kyphotic angle; secondary outcomes included the identification of predictors of progression. Paired t-tests and multivariable linear regression analysis were performed.

RESULTS: Kyphosis angles remained essentially unchanged across all follow-up intervals (ΔFU1 + 0.19°, ΔFU2 + 1.27°, ΔFU3 - 0.14°), with no statistically significant differences detected ($p > 0.05$ at all timepoints). No clinical or radiographic factors predicted progression of kyphosis. Only three patients (5.6%) required treatment escalation. Subgroup analyses […]

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## Correction: Different sagittal cutting plane orientations do not affect posterior tibial slope in open wedge high tibial osteotomy: a Sawbone model study.

*Gurbanov E, Cochard B, Bauer DE, Miozzari HH, Tscholl PM*

PubMed · DOI

## Resident autonomy in orthopedic trauma: independent femoral IMN by senior residents yields comparable outcomes to fellowship-trained surgeons.

*Balziano S, Nesher G, Ben Nun N, Segev T, Freidlander A, Prat D*

INTRODUCTION: Intramedullary nailing (IMN) of femoral shaft fractures is a fundamental cornerstone procedure in orthopedic trauma and an important benchmark of resident surgical competency. While independent resident operating is essential for skill development, its safety in complex trauma cases remains debated. This study evaluated whether senior residents can safely perform femoral IMN without direct supervision compared with fellowship-trained trauma surgeons.

MATERIALS AND METHODS: A retrospective cohort analysis was performed of adult patients undergoing IMN for isolated femoral shaft fractures between 2015 and 2023 at a level I trauma center. Cases were grouped by primary surgeon: senior residents (postgraduate year ≥ 4) versus attending trauma surgeons. Demographics, fracture characteristics, operative parameters, complications, and revision rates were compared. To minimize selection bias, 1:1 propensity score matching (PSM) was performed based on preoperative covariates including age, comorbidity, fracture type, injury mechanism, and severity.

RESULTS: A total of 207 patients met inclusion criteria; 150 (75 per group) were matched. Operative duration was longer in resident-led cases (median 120 vs. 91 min, p = 0.003). Postoperative hospital stay (5.0 vs. 6.0 days, p = 0.674) and weight-bearing protocols (p = 0.218) were similar. Overall complication rates did not dif […]

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### Racial and ethnic disparities in perceived health and barriers to care among adults with hip osteoarthritis: a national analysis of 16,000 + patients from the all of us program.

*Khan ST, Devadiga SA, Emara AK, Brahmbhatt S, Elmenawi KA, Delaney PG, Piuzzi NS*

OBJECTIVE: To explore racial and ethnic disparities in perceived health, healthcare access, and perceived barriers among adults with hip osteoarthritis (OA) in a large, nationally representative U.S.

METHODS: Cross-sectional analysis of 16,743 adults with hip OA from the All of Us Research Program. Participants self-identified as Black, Hispanic, or White. Outcomes included self-reported general, mental, physical, and social health, provider communication, and structural barriers to care. Between-group comparisons used Chi-square tests.

RESULTS: Black and Hispanic participants were younger, had lower income and education, and relied more on Medicaid than White participants (all p < 0.001). They reported higher rates of unaffordable deductibles, difficulty accessing follow-up care, insurance coverage issues, transportation challenges, and medication cost-related nonadherence (all p < 0.001) Table 3. Minority groups also more often experienced poor provider respect, trust, and communication. Across all health domains, Black and Hispanic participants reported poorer outcomes and greater emotional distress (all p < 0.001).

CONCLUSION: Significant racial and ethnic disparities exist in structural and perceptual barriers to care among adults with hip OA. These findings identify potentially modifiable barriers that warrant further study and may inform future efforts to improve equi […]

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### Postoperative periprosthetic fractures associated with compressive osseointegration: a systematic review of the Zimmer Biomet Compress® device.

*Hashimoto K, Nishimura S, Goto K*

The Zimmer Biomet Compress® Compliant Pre-Stress (CPS) device uses a compressive osseointegration technique for endoprosthetic fixation and provides an alternative to stemed implants. Despite having benefits, such as bone stock retention, periprosthetic fractures continue to remain a concern. Therefore, this systematic review presents the prevalence, risk factors, and outcomes of the CPS device and periprosthetic fractures after treatment. Medical databases (PubMed, Embase, and Google Scholar) were searched to identify studies reporting complications related to the CPS device. Peer-reviewed studies with at least 2 years of follow-up for fracture rates, revision results, and implant survivorship were included. The primary data collected were fracture incidence and timing of complications, and the CPS device was compared with the traditional cemented stem. This review highlights important studies, including a cohort of 221 patients and pediatric data from 36 patients. Adult populations showed a periprosthetic fracture rate of 2.7-3.9%. A distinct temporal pattern emerged: the frequency of fractures was most pronounced within the first 2 years after surgery. In the pediatric context, spindle survivorship was 86.3% at 5-10 years, despite higher rates of mechanical complications. Notably, for most fracture cases, the bone-implant interface was radiographically stable, providing easi […]

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### Hip resurfacing arthroplasty at a minimum of 14 years: revision is concentrated among female patients with dysplasia and small components.

*Lin YC, Chang CH, Hsieh PH, Huang YY, Chen DW, Lau NC*

INTRODUCTION: Hip resurfacing arthroplasty (HRA) is a bone-preserving option for young, active patients. The influence of the underlying diagnosis on long-term outcome remains unclear, particularly the distinction between primary osteoarthritis (OA), developmental dysplasia of the hip (DDH) and non-OA conditions.

MATERIALS AND METHODS: We retrospectively reviewed 104 consecutive metal-on-metal HRAs in 101 patients (three bilateral) performed by one surgeon between 2006 and 2011, stratified into Primary OA (n = 50), DDH (n = 29) and Non-OA (n = 25: avascular necrosis [AVN] n = 15, inflammatory arthritis n = 5, post-traumatic arthritis n = 3, Legg-Calvé-Perthes disease n = 2). AVN was analysed additionally as an independent group, and exploratory analyses restricted to female patients and to cups ≤ 50 mm addressed the collinearity of dysplasia, sex and size. Survivorship used Kaplan-Meier estimation with revision as the endpoint.

RESULTS: Implant status was known for all 104 hips; the 91 unrevised assessed hips had a mean follow-up of 16.2 ± 1.5 years. Overall 15-year survivorship was 95.2%. Primary OA, Non-OA and AVN each showed 100%; DDH 82.8% (versus Primary OA p = 0.002, versus Non-OA p = 0.031). All five revisions occurred in female DDH patients with a cup ≤ 50 mm; DDH survivorship remained lower within female patients (p = 0.025) and within small cups (p = 0.006), with no […]

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## Elective arthroplasty on "high-risk" calendar dates: no evidence for worse outcomes in 43,000 procedures.

*Walter N, Lowenberg DW, Heiss C, Lau EC, Rupp M*

BACKGROUND: Concerns regarding the timing of elective surgery remain common among patients and may influence surgical scheduling. Superstition-associated calendar dates such as Friday the 13th are frequently perceived as "high-risk" time points, yet evidence regarding their impact on arthroplasty outcomes is lacking. This study investigated whether primary total hip arthroplasty (THA) and total knee arthroplasty (TKA) performed on these dates are associated with differences in revision risk, mortality, or procedure frequency.

METHODS: A nationwide retrospective cohort study was conducted using Medicare physician service records (2009-2019). Primary THA and TKA procedures were identified in a 5% sample of Medicare beneficiaries aged ≥ 65 years. Outcomes included aseptic revision, septic revision, and mortality. Competing-risk survival analyses (Fine-Gray) and multivariable Cox regression models adjusted for demographic, clinical, and socioeconomic covariates were applied. Procedure volumes were compared between superstition-associated and control dates.

RESULTS: A total of 945 arthroplasties were performed on Fridays falling on the 13th. No significant differences were observed in aseptic revision, septic revision, or mortality compared with control dates. For Friday the 13th, hazard ratios were 0.80 (95% CI 0.44-1.47) for aseptic revision, 0.83 (95% CI 0.31-2.25) for septic r […]

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## Evaluation of a novel strap fixation method for biomechanical testing of sheep Achilles tendons.

*Basci O, Aydemir S, Husemoglu RB, Çeltik M*

BACKGROUND: Reliable fixation of tendon specimens is a major challenge in biomechanical testing, as slippage and tissue damage can reduce accuracy and reproducibility. Although various clamp- and suture-based fixation methods have been reported, there is still no widely accepted, practical, low-cost technique that preserves tendon integrity during testing. This study aimed to compare commonly used fixation techniques and to evaluate whether a novel strap-based fixation method provides a more stable and tissue-preserving alternative for biomechanical testing of sheep Achilles tendons.

METHODS: Forty fresh-frozen cadaveric sheep Achilles tendons obtained from a local butcher were used, and the specimens were stored at - 20 °C until testing. After controlled thawing at room temperature, all tendons were repaired using the Bunnell suture technique with No. 0 Vicryl. Four different proximal fixation methods were evaluated: (1) sutured fixation with strap, (2) direct clamp fixation, (3) Kocher clamp fixation, and (4) Krackow suture fixation. All specimens were mounted on a universal testing machine and subjected to a load-to-failure protocol at a constant loading rate of 10 mm/s. Maximum force (N), displacement at failure (mm), and mechanical failure mechanisms were recorded for each sample. Data were analyzed using the Kruskal-Wallis test with

Bonferroni-corrected Mann-Whitney U te […]

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### Implementation and early results of same-day hip and knee replacement in a public hospital without prior same-day surgical experience, a feasibility study.

*Jørgensen CC, Sveen TM, Johansen B, Sander M, Steen-Hansen C, Sundstrup MS, Rothe C, Lange KH*

BACKGROUND: Hip and knee replacements are increasingly performed as same-day surgery (SDS). However, the feasibility of SDS pathways in departments without prior experience is unknown. We report on the implementation process, feasibility and early results of implementing a SDS protocol in a department without prior experience with same-day hip and knee replacement.

METHODS: The study design was based on Bowens feasibility study framework on implementation processes. Data were collected prospectively through patient reported questionnaires and medical records during, and from the local Department of Digitalisation and Analytics after implementation. Outcome data included discharge on day of surgery, 30-days primary-care contacts, 90-days readmissions and patient satisfaction.

RESULTS: Implementation was initiated as a dedicated fellowship from September 2023 to October 2024. Patients were admitted and discharged through a dedicated day-surgical department with transfer to the regular ward in case of admission. The first SDS patient underwent surgery the 4th of September 2023. By October 2024, 150 of 713 operated patients (21.0% CI:18.2-24.2) had SDS. Of these, same-day discharge was achieved in 125 (83.3% CI:76.6-88.5) patients of which 6 (4.0% CI:0.9-7.1) were readmitted within 90-days. From October 2024 to October 2025, 178 of 800 patients were planned for SDS (22.3% CI:19.5 […]

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### Relationship of foot anthropometry with Modified Pirani score during Ponseti correction of idiopathic congenital talipes equinovarus: a prospective observational study.

*Suresh M, Pinto DJV, Rajasekharan P*

BACKGROUND: The Ponseti method is the gold standard treatment for idiopathic congenital talipes equinovarus (CTEV). Clinical monitoring during correction primarily relies on the Modified Pirani score, which evaluates deformity severity based on clinical findings. However, objective quantitative assessment of morphological foot changes during treatment remains limited. This study aimed to evaluate the relationship between serial foot anthropometric measurements and Modified Pirani scores during Ponseti correction of idiopathic clubfoot.

METHODS: A prospective observational cohort study was conducted in 30 children with idiopathic unilateral CTEV undergoing Ponseti correction at a tertiary care center between June 2024 and February 2026. Five anthropometric parameters were measured weekly for eight weeks: great toe to midheel distance (GT-MH), little toe to midheel distance (LT-MH), second toe to midheel distance (ST-MH), medial malleolus to midheel distance (MM-MH), and lateral malleolus to midheel distance (LM-MH). Modified Pirani scores were assessed simultaneously by two independent observers. Pearson correlation analysis and inter-observer reliability testing were performed.

RESULTS: Significant improvement was observed in all anthropometric parameters and Pirani scores over the study period ($p < 0.001$). Mean GT-MH increased from 80.73 ± 3.36 mm to 87.56 ± 3.41 mm, LT-MH f […]

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### Union outcomes and associated factors in adolescent scaphoid fractures.

*Torun Ö, Girgin AB, Ka■■kç■ M, Acar A, Çevik HB*

PURPOSE: Scaphoid fractures (SF) in adolescents differ from adult fractures in fracture pattern, healing capacity, and skeletal maturity. Limited data exist on factors affecting union and healing time in this population. This study evaluated clinical, radiological, and treatment-related factors associated with union and healing time in adolescent SF.

METHODS: This study included 77 adolescents (aged 10-18 years) treated for SF. Chronological age, bone age, fracture characteristics, treatment modality, union status, and healing time were recorded. Fractures were classified by anatomical location, displacement, and time to diagnosis (< 6 weeks or ≥ 6 weeks). CT-based

morphometric measurements, including scaphoid height, lateral intrascaphoid angle, and dorsal cortical angle, were obtained from initial computed tomography images. Functional outcomes were assessed using the QuickDASH and patient-rated wrist evaluation (PRWE) scores.

RESULTS: Union was achieved in 63 patients (81.8%), with a median union time of 8 weeks (range, 6-10 weeks). Union rates were highest in distal pole fractures (100%) and lowest in proximal pole fractures (33.3%). Non-displaced fractures had higher union rates than displaced fractures (92.5% vs. 56.5%), and fractures diagnosed within 6 weeks demonstrated better union outcomes. Surgical treatment, performed in 16.9% of patients, was associated with lowe […]

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### The effect of surgeon volume on outcomes after total hip arthroplasty: a systematic review and meta-analysis.

*Meissner N, Ortwig K, Ramos-Pascual S, Agu C, Stoeve J, Schrednitzki D, Rolvien T, Halder AM*

PURPOSE: To synthesize, critically appraise and systematically review studies comparing outcomes after primary total hip arthroplasty (THA) performed by low- vs. high-volume surgeons, and to propose evidence-based threshold definitions for these categories.

METHODS: This review followed the PRISMA guidelines and was registered in PROSPERO. Medline and Embase were searched on March 24, 2025. Prospective and retrospective clinical studies were included if they reported THA thresholds for low- and high-volume surgeons. Articles not written in English or German were excluded. Outcomes were pooled and presented in forest plots. Methodological quality was assessed.

RESULTS: Of 872 identified articles, 18 were included, reporting on 796,061 patients undergoing THA. Methodological quality was high in all studies (>6 of 7 points). Thresholds used to define high- vs. low-volume were determined using arbitrary values (n = 11), tertiles (n = 1), quartiles (n = 3), percentiles (n = 2), or receiver operating characteristic curves (n = 1). High-volume surgeons had significantly lower 1-month readmission rates [odds ratio (OR) = 1.6; p = 0.033] and 3-month mortality rates (OR = 1.8; p = 0.008). In contrast, there were no significant differences between high- and low-volume surgeons for 3-month revision rates (OR = 2.0; p = 0.228), 12-month revision rates (OR = 1.8; p = 0.228), and 3-month re […]

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### Hexapod-assisted limb salvage after talectomy: outcomes of tibiocalcaneal arthrodesis using the TL-hex system.

*Guerrero Serrano JA, Sánchez Morata E, García López JM, Mencía González S, Vilá Y Rico J*

BACKGROUND: Severe talar destruction due to infection, trauma, avascular necrosis, or neuroarthropathy represents a major reconstructive challenge. Talectomy followed by tibiocalcaneal arthrodesis (TCA) is a limb-salvage option in selected high-risk patients, although optimal fixation remains controversial. Modern hexapod external fixators offer computer-assisted multiplanar control that may improve alignment and fusion after talectomy. The aim of this study was to evaluate the clinical and radiographic outcomes of talectomy followed by TCA using the TL-HEX system in a series of high-risk patients.

METHODS: A retrospective observational study was conducted between January 2020 and January 2024 including all patients who underwent total talectomy followed by TCA using a TL-HEX circular external fixator. Indications included talar destruction secondary to infection, avascular necrosis, post-traumatic collapse, Charcot neuroarthropathy, and aseptic loosening of total ankle arthroplasty. Primary outcomes were radiographic fusion and hindfoot alignment. Secondary outcomes included time to union, functional scores (AOFAS, EFAS), pain (VAS), complications, and patient satisfaction. Preoperative and final follow-up outcomes were compared using paired statistical tests, with significance set at $p < 0.05$.

RESULTS: Thirteen patients (mean age 57 years) were included with a mean follow-u […]

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### Radiographic changes in midfoot alignment after hallux valgus correction with and without concomitant second tarsometatarsal arthrodesis: a comparative study.

*Kim J, Kyung MG, An Y, Kim NY, Lee KM, Lee DY*

INTRODUCTION: Second tarsometatarsal (TMT) joint osteoarthritis occurs in a subset of patients with hallux valgus (HV). While second TMT arthrodesis is performed to address symptomatic arthritis, its association with midfoot alignment changes during HV correction remains unclear. This study aimed to compare changes in radiographic arch parameters between patients undergoing HV correction with and without concomitant second TMT arthrodesis.

MATERIALS AND METHODS: This retrospective comparative study included 73 feet with HV and medial column malalignment (lateral Meary's angle > 10°). The TMT group (n = 37) underwent HV correction combined with second TMT arthrodesis for symptomatic osteoarthritis, while the control group (n = 36) underwent isolated HV correction. These patients were matched to achieve comparable baseline characteristics. Radiographic parameters of the transverse arch (M2-M5 angle, C1-M2 distance) and medial column alignment (lateral Meary's angle) were evaluated preoperatively and at a minimum 1-year follow-up. Functional outcomes were assessed as a secondary measure.

RESULTS: The TMT group demonstrated greater changes in transverse arch parameters compared with the control group. The M2-M5 angle decreased by 2.1° (95% CI, 1.4° to 2.7°) in the TMT group, whereas no significant change was observed in controls (-0.2°; 95% CI, -0.7° to 0.4°). The C1-M2 distance […]

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## Coronal shear fractures of the distal humerus in children and adolescents: proposal for a modified classification.

*Lieber J, Esser M, Fernandez FF, Schmittenbecher PP, Schneidmueller D, Strohm P, Kaiser MM, Kraus R*

PURPOSE: Coronal shear fractures (CSF) of the capitulum and trochlea humeri are rare injuries during growth age. No systematic, prospective analysis of the classification, procedure, and outcome has yet been found in the literature.

MATERIALS AND METHODS: Based on the results of a multicentre study of CSF in growing patients, we propose a redesign of the AO Paediatric Comprehensive Classification of Long-Bone Fractures (PCCF) for children and adolescents. We also assessed inter-rater agreement consistency with the modified classification and calculated the intra-class correlation coefficient (ICC) and its 95% confident interval (CI).

RESULTS: A detailed classification has been proposed that refers to the location of the fragment, the extent to which the trochlea is involved, and the distinction between single and multiple fracture injuries. This modified AO classification distinguishes between five types of injury (13-E/3.1I - 13-E/3.1IV and 13-E/3.2), replacing the current code 13r-E/8.1. Inter-rater reliability using this modified classification was moderate (ICC 0.452 (95% CI 0,284; 0,661). This is more accurate than all previous classifications tested for CSF.

CONCLUSION: A revised classification of coronal shear fractures of the distal humerus in children and adolescents is required. Our proposed classification considers the various types of this injury and their implic […]

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## Correction: Prevalence and risk factors for stem subsidence following reverse shoulder arthroplasty using on-lay press-fit short stems.

*Nishiura R, Nakazawa K, Manaka T, Hirakawa Y, Ito Y, Terai H*

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## Reverse shoulder arthroplasty vs. hemiarthroplasty for proximal humerus fracture: in-hospital outcomes in a national trauma registry.

*Raizah A*

BACKGROUND: Reverse shoulder arthroplasty (RSA) has rapidly supplanted hemiarthroplasty for acute proximal humerus fracture, driven by superior functional outcomes and lower revision rates. Whether this shift has altered perioperative morbidity at the population level remains poorly characterized. This study compared in-hospital outcomes between RSA and hemiarthroplasty in a national trauma cohort.

METHODS: This retrospective cohort study used the National Trauma Data Bank (2017-2022). Adults ( > = 18 years) admitted with proximal humerus fracture who underwent RSA (n = 2657) or hemiarthroplasty (n = 428)

during the index hospitalization were included. The primary outcome was non-home discharge; secondary outcomes were any in-hospital complication, extended length of stay exceeding 7 days, and intensive care unit (ICU) admission. Multivariable logistic regression and inverse probability of treatment weighting (IPTW) were used to address confounding by indication.

RESULTS: Among 3085 patients (median age, 73 years; 2304 [74.7%] female), RSA patients were older (median, 74 vs. 64 years; P<.001) and more frequently female (77.3% vs. 58.6%; P<.001). After multivariable adjustment, RSA was not significantly associated with non-home discharge (adjusted odds ratio [aOR], 1.20; 95% confidence interval [CI], 0.93-1.55), any complication (aOR, 1.01; 95% CI 0.58-1.77), extended length o […]

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### Impact of cognitive impairment on postoperative delirium and discharge disposition following knee arthroplasty: a Japanese nationwide propensity score-matched study.

*Mori Y, Tarasawa K, Tanaka H, Kanabuchi R, Mori N, Fushimi K, Aizawa T, Fujimori K*

PURPOSE: As populations age, the number of older patients undergoing knee arthroplasty continues to increase. Cognitive impairment is prevalent among older adults and has been associated with adverse perioperative outcomes; however, its impact on postoperative delirium and discharge disposition following knee arthroplasty remains insufficiently defined, particularly in large Asian populations.

METHODS: We conducted a nationwide retrospective cohort study using Japan's Diagnosis Procedure Combination database from April 2016 to March 2023. Older patients who underwent primary total knee arthroplasty (TKA) or unicompartmental knee arthroplasty (UKA) were identified. Cognitive impairment was defined using ICD-10 diagnostic codes at admission. The primary outcome was postoperative delirium, and secondary outcomes included discharge to home, length of hospital stay, and perioperative blood transfusion. Propensity score matching (1:1) was performed to adjust for demographic factors, comorbidities, and surgical characteristics. Multivariable logistic regression and sensitivity analyses were conducted.

RESULTS: Among 259,319 eligible patients, 3,934 matched pairs were identified after propensity score matching. Patients with cognitive impairment had a significantly higher risk of postoperative delirium compared with those without cognitive impairment (absolute risk difference, 3.2%; […]

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### Association of primary tumor surgery with survival outcomes in patients with bone metastases across multiple cancer types.

*Xu PH, Shi B, Li T, Tian M, Fang M, Cheng M, Huang W, Ren F et al.*

BACKGROUND: Synchronous bone metastases (BM) have a notable impact on the survival rates of patients across various cancer types. While primary tumor surgery (PTS) has shown potential benefits, its impact across different cancers and patient subgroups needs further clarification. This study aimed to investigate the survival benefits of PTS across various cancer types.

METHODS: This retrospective cohort study analyzed data from the Surveillance, Epidemiology, and End Results (SEER) database, covering 53,015 patients diagnosed with BM from 2010 to 2019. Propensity score matching (PSM) was performed to reduce baseline imbalance. Overall survival (OS) and cancer-specific survival (CSS) were evaluated using Kaplan-Meier analysis and Cox proportional hazards regression.

RESULTS: PTS substantially improved overall and cancer-specific survival in patients with lung, kidney, breast and bladder cancers. All surgical modalities benefited breast and renal cancer survival. In prostate cancer, PTS produced no overall survival gain; survival varied by surgery type, being better after prostatectomy and worse with local resection. For liver cancer, PTS improved cancer-specific but not overall survival, and the small matched cohort necessitates circumspect interpretation.

CONCLUSION: The findings indicate that PTS can significantly enhance survival in selected patients with BM, particularly i […]

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### Liquid nitrogen-treated autograft combined with megaprosthesis versus total femoral replacement for large-volume malignant femoral bone tumors: a comparative study.

*Sang NTQ, Trung ND, Huy LD, Phuc LH, Thang VD, Anh LN, Hung LT, Hien TTT et al.*

BACKGROUND: Limb-salvage surgery for extensive femoral osteosarcoma presents major reconstructive challenges. Conventional options such as total femoral replacement (TFR) and long-segment biological reconstructions are associated with significant complications. We evaluated the clinical outcomes of a novel hybrid approach combining a liquid nitrogen-treated autograft with a megaprosthesis (LN-MP) and compared them with those of TFR.

MATERIALS AND METHODS: This retrospective study included 28 patients (range 8-24 years old) with malignant femoral bone tumors treated between 2020 and 2024. Patients underwent either LN-MP (n = 10) or TFR (n = 18). Perioperative outcomes, functional results (Musculoskeletal Tumor Society score-MSTS), bone union, oncological control, and survival were analyzed.

RESULTS: LN-MP demonstrated better functional outcomes (MSTS 27.0 ± 2.1 vs. 24.4 ± 2.2, p = 0.014), shorter operative time, and lower blood loss compared with TFR. Length of stay and follow-up duration were comparable. Continuous disease-free survival was higher in the LN-MP group (90% vs. 44%, p = 0.058), with lower recurrence (10% vs. 28%). Bone union was achieved in 70% of LN-MP cases at a mean of 9 months, with longer graft length associated with delayed union.

CONCLUSIONS: LN-MP reconstruction may serve as a joint-preserving alternative to total femoral replacement for patients with e […]

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## Tibial bone transport using external fixation in adults: a systematic review.

*Giuli C, Di Pietro G, Farine F, Bocchi MB, Vitiello R, Palmacci O*

INTRODUCTION: Tibial bone transport using external fixation is widely employed for the reconstruction of large and complex tibial bone defects; however, indications and surgical strategies remain highly heterogeneous. Tibial bone transport using external fixation is a demanding but reliable reconstructive technique, characterized by low failure rates and generally favorable outcomes despite frequent complications. Adequate soft-tissue management, selective docking-site grafting, and early functional rehabilitation were commonly reported. Further prospective studies are needed to standardize treatment strategies and improve clinical decision-making. MATERIALS AND METHODS: A systematic review was conducted in accordance with PRISMA guidelines in April 2025, searching PubMed/MEDLINE and the Cochrane Library. Extracted data included patient demographics, comorbidities, indications, defect length, surgical technique, frame type, external fixation time, healing index, complications, failures, and ASAMI bone and functional outcomes. Methodological quality was assessed using the MINORS score and results were synthesized descriptively.

RESULTS: A total of 3543 patients from 89 studies were included. Post-traumatic infection or osteomyelitis accounted for 58.5% of indications. The mean defect length was 7.04 cm. Bifocal bone transport was performed in 82.4% of cases, and circular exter […]

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## Accuracy and risk factors for poor cup orientation in total hip arthroplasty: a comparison between supine and lateral registration using a portable navigation system.

*Matsumoto S, Tetsunaga T, Yamada K, Okuda R, Masada Y, Yamamoto T, Okazaki Y, Ozaki T*

INTRODUCTION: Accurate acetabular cup orientation is essential in total hip arthroplasty (THA). Portable inertial navigation may improve reproducibility, but accuracy in lateral decubitus THA may depend on the pelvic registration workflow. We compared cup placement accuracy between supine and lateral registration workflows using NAVBIT (Smith & Nephew PLC) and explored factors associated with large navigation errors.

MATERIALS AND METHODS: We retrospectively reviewed 110 consecutive primary cementless THAs performed in the lateral decubitus position using NAVBIT. The supine-registration workflow included 52 hips (supine pelvic registration followed by repositioning), and the lateral-registration workflow included 58 hips (registration and cup implantation in the lateral decubitus position). NAVBIT recorded radiographic inclination (RI) and radiographic anteversion (RA) intraoperatively; postoperative RI and RA were measured via computed tomography. Accuracy was defined as the absolute difference between navigation records and postoperative measurements. We compared navigation accuracy outcomes between propensity score-matched groups. Outliers were defined as an

absolute error > 5° in RI or RA; factors associated with outliers were evaluated in the overall cohort.

RESULTS: After propensity score matching, 46 matched pairs (n = 92) were analyzed. Postoperative cup orientation w […]

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### Differences in the coronal plane alignment of the knee (CPAK) measurements using long-leg radiographs and computed tomography.

*Kawaguchi K, Tay ML, Young S*

INTRODUCTION: Coronal plane alignment of the knee (CPAK) parameters are increasingly used to characterize native coronal alignment and guide alignment strategies in total knee arthroplasty (TKA) based on long-leg radiographs (LLR). Computed tomography (CT) based robotic TKA software enables calculation of CPAK parameters based on anatomical reference points. However, potential differences in CPAK measurements between LLR and CT remain incompletely defined. This study compared CPAK coronal alignment parameters and phenotype distributions between LLR and CT, and evaluated factors associated with measurement discrepancies.

METHODS: This was a radiographic subanalysis of a prospective randomised controlled trial including 241 patients undergoing robotic-assisted TKA. Medial proximal tibial angle (MPTA) and lateral distal femoral angle (LDFA) were measured and arithmetic hip-knee-ankle-angle (aHKA), joint line obliquity (JLO) and CPAK phenotype distribution were calculated and classified from preoperative LLR and CT. Discrepancy groups were defined as LLR = CT (≤ 2°), LLR < CT (> 2°) and LLR > CT (> 2°). Associations between discrepancy groups and preoperative deformity and flexion contracture were assessed.

RESULTS: Compared with LLR, CT demonstrated a more varus MPTA (mean 86.7° vs. 87.1°, p = 0.04), a more valgus LDFA (87.1° vs. 87.8°, p < 0.01) and a lower JLO (173.9° vs. 174. […]

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### Conservative and surgical treatment of wrist osteoarthritis: a systematic review.

*Boever J, Unglaub F, Wulf J, Spies CK, Langer MF, Kußmaul AC*

INTRODUCTION: Wrist osteoarthritis leads to pain and functional impairment. Despite conservative treatment options, surgery remains a central therapeutic component. In Germany, the number of patients rose from over 472,000 cases in 2018 to more than 580,000 in 2024. This review examines the most common surgical treatment approaches, focusing on Proximal Row Carpectomy (PRC) and Four-Corner Fusion (4CF), discussing indications, contraindications and potential complications.

METHODS: A systematic and narrative literature review was conducted on treatment options for wrist osteoarthritis. The systematic review focusing on PRC and 4CF followed PRISMA guidelines and included searches in PubMed, Cochrane and Embase. Included were RCTs, systematic reviews and meta-analyses from January 2010 to May 2025 that compared both procedures in terms of pain, range of motion, grip strength, function, complication rates and reoperation rates.

RESULTS: Eight studies were included. Both procedures significantly reduced pain and improved function. PRC showed advantages in range of motion and lower complication rates, while 4CF was associated with higher grip strength. Partial arthrodesis (e.g., scaphotrapeziotrapezoid (STT) arthrodesis or radioscapholunate (RSL)) arthrodesis may be selectively used but are linked to higher nonunion rates. Total wrist arthrodesis or wrist arthroplasty may be consi […]

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### Factors associated with selection of total hip arthroplasty versus hemiarthroplasty for femoral neck fracture in older adults: a nationwide analysis of 99,084 cases.

*Maman D, Steinfield Y, Berkovich Y*

BACKGROUND: Surgical management of displaced femoral neck fractures in older adults typically involves hemiarthroplasty or total hip arthroplasty (THA). Although clinical guidelines suggest that THA may be considered in selected healthier and cognitively intact patients, real-world procedure selection varies widely. This study evaluated patient factors associated with selection of THA versus hemiarthroplasty in a contemporary U.S.

METHODS: A retrospective cohort study was conducted using the 2022 Nationwide Readmissions Database (NRD). Patients ≥ 65 years hospitalized with femoral neck fracture were identified using ICD-10-CM codes. Those treated with internal fixation or non-arthroplasty procedures were excluded. Weighted analyses characterized demographics, comorbidities, and hospital utilization between THA and hemiarthroplasty groups. A multivariable logistic regression model identified factors independently associated with receiving THA. All analyses accounted for NRD survey design.

RESULTS: Among 142,013 operative cases, 99,084 met inclusion criteria (81.8% hemiarthroplasty; 18.2% THA). Patients receiving THA were younger (76.6 vs. 81.9 years), had shorter length of stay (6.06 vs. 7.24 days), and were more frequently discharged home (17.5% vs. 6.2%). Metabolic conditions were associated with increased odds of THA, including obesity (OR 1.17) and sleep apnea (OR 1.12). F […]

PubMed · DOI

## Effect of distal screw number on axial deformation and fragment-specific stability in AO/OTA 2R3 C2.1 distal radius fractures.

*Dietz L, Spiegel C, Kohler FC, Biedermann U, Zderic I, Feist A, Gueorguiev-Rüegg B, Lenz M et al.*

PURPOSE: To investigate the biomechanical influence of distal screw number in palmar plate fixation of unstable intra-articular distal radius fractures with radiodorsal metaphyseal comminution.

METHODS: Nine matched pairs of cryopreserved human radii were prepared using a standardized AO/OTA 2R3 C2.1 fracture model with a defined radiodorsal metaphyseal defect zone. For each pair, specimens were randomized to palmar fixation using a 2.4-mm variable-angle locking plate with either four or six distal screws. After embedding, constructs were subjected to cyclic axial loading (150 N, 5,000 cycles) simulating early postoperative rehabilitation. Global axial deformation and fragment-specific three-dimensional interfragmentary motion were recorded using an optical motion-tracking system. Statistical analysis was performed using non-parametric tests ($p < 0.05$).

RESULTS: Two specimens from one donor pair were excluded, leaving 16 radii (8 pairs) for analysis. Bone mineral density and initial construct stiffness did not differ between groups. The six-screw configuration showed significantly lower axial deformation after cyclic loading ($p = 0.047$) and lower radial fragment-shaft translation at both time points ($p = 0.028$; $p = 0.045$). Radial fragment-shaft rotation was lower in the six-screw group at baseline ($p = 0.018$), but no rotational difference remained after cyclic loading. In the […]

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## Crowe severity and femoral stem geometry are associated with lower-limb alignment after total hip arthroplasty for unilateral developmental dysplasia of the hip-related secondary osteoarthritis: valgus alignment trajectories.

*Afacan MY, Gurcinar MG, Davulcu CD, Unlu MC, Kaynak G*

BACKGROUND: Total hip arthroplasty (THA) for secondary osteoarthritis related to developmental dysplasia of the hip (DDH) restores hip mechanics. We aimed to quantify pre-to-postoperative changes in ipsilateral knee and global lower-limb alignment after unilateral THA to determine whether alignment changes differ by Crowe classification and femoral stem geometry.

METHODS: This retrospective cohort study included 74 patients who underwent unilateral primary THA for DDH-related secondary osteoarthritis, with a mean follow-up of 45.4 months. The predefined primary radiographic outcome set comprised preoperative-to-postoperative changes and final follow-up values in coronal lower-limb alignment parameters, including mechanical and anatomic lateral distal femoral angles (mLDFA and aLDFA), mechanical medial proximal tibial angle (mMPTA), joint line convergence angle (JLCA), global offset (GO), hip-knee-ankle angle (HKA), knee mechanical axis deviation ratio (KMADR), and femoral mechanical-anatomic axis angle (FMMA). The contralateral reference limb was used for side-to-side radiographic comparison. Patients were stratified by Crowe classification (I-II vs. III-IV), and femoral stem geometry analyses were performed according to conical, wedge, and rectangular stem designs. Clinical outcomes included hip/knee visual analog scale, Harris Hip Score, Knee Society Score, Oxford Knee Score […]

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## Can ChatGPT pass the polish national medical specialization examination in orthopedics and traumatology?

*Maci■g B, Bujak K, Jegierski D, Stolarczyk A*

INTRODUCTION: Artificial intelligence (AI) has evolved rapidly in recent years and is becoming increasingly integrated into many areas of medicine. In the medical field, these systems have attracted considerable attention because of their potential applications in clinical decision support, medical education, scientific communication, and postgraduate training. The aim of the present study was to evaluate whether ChatGPT could achieve a passing score on the Polish National Medical Specialization Examination (Panstwowy Egzamin Specjalizacyjny, PES) in orthopedics and traumatology and to determine how different prompting strategies influenced its performance.

MATERIALS AND METHODS: Authors systematically assessed the performance of ChatGPT-4 across five consecutive official PES exams (from Autumn 2023 to Spring 2025) in orthopedics and traumatology. Each exam was administered using three distinct prompting strategies: Professor prompt, Specialist prompt, Resident prompt. For each exam session (e.g., Spring 2024, Autumn 2023), the prompts were submitted to ChatGPT sequentially. The model's responses were evaluated against the official answer key published by the Polish Center of Medical Exams (Centrum Egzaminów Medycznych, CEM).

RESULTS: The results were averaged across all prompts. The overall accuracy was 81% (range: 68.33%-85.83%). The highest score (85.83%) was recorded mult […]

PubMed · DOI

## Two-year outcomes following concomitant ACL and extra-articular ligament reconstruction: a nationwide analysis of over 5,000 patients.

*Abid R, J Burkhart R, J Moyal A, M Adelstein J, Kodsy M, G Calcei J, E Voos J, M Apostolakos J*

PURPOSE: The anterior cruciate ligament (ACL) is the most commonly injured and reconstruction ligament in the United States. Associated injuries are common, oftentimes including components of the extra-articular knee. However, complications following concomitant ACL and extra-articular ligaments (EAL) has not been defined in a large-scale, population-based sample. We aimed to analyze outcomes following concomitant ACL and EAL reconstruction using the TriNetX nationwide database.ep METHODS: Deidentified data from the TriNetX Collaborative Network was used to identify all patients who received ACL and EAL reconstruction on the same day. Demographic information including age, sex, race, and geographic data were collected. Rates of medical and orthopaedic complications were analyzed at 30, 90, and 180 days and at 2 years. A propensity matched sub-analysis comparing outcomes patients with concomitant and staged repair was performed. Patients were identified using Current Procedural Terminology (CPT) coding.

RESULTS: We identified 5944 patients who received concomitant ACL and EAL reconstruction with postoperative outcomes available at 30, 90, and 180 days. 5040 patients had data available at 2 years. Serious medical complications like surgical site infection, deep vein thrombosis, acute kidney failure, and sepsis were rare at each point of follow-up. Rates of revision surgery were […]

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## Recovery of gait after ankle fracture surgery following early mobilization and weight-bearing.

*Bauer L, Schlegel AS, Scheuer A, Böcker W, Polzer H, Baumbach SF*

BACKGROUND: Ankle fractures are among the most common fractures in adults. Despite significant advancements in surgical techniques, the postoperative treatment regimens remain conservative. This study aims to objectively evaluate gait recovery following early mobilization and weight-bearing after surgical ankle fracture treatment using instrumented gait analysis.

METHODS: This prospective, single-armed, longitudinal study enrolled adult patients following surgical treatment for isolated ankle fractures with early pain dependent weightbearing. Gait analysis, utilizing a treadmill integrated with a pressure plate and IMU-based motion capture systems, was conducted at 6 weeks, 3-, 6-, and 12-months post-surgery, assessing spatio-temporal and kinematic parameters. Results were compared longitudinally and against a healthy control group.

RESULTS: Longitudinal data were available for 44 patients. Significant (p<.001) improvements of self-paced walking velocity (SPV) were observed from 6 weeks (2.2 ± 1.1 km/h), to 3 (2.9 ± 0.9 km/h) and 6 months (3.4 ± 0.9 km/h). The spatio-temporal parameters showed significant changes between 6 weeks and all further time points only at SPV. Kinematic parameters showed significant longitudinal improvements at SPV for hip, knee, and ankle motion (including all three ankle planes). Patient gait parameters at 6 and 12 months were comparable to healthy […]

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## A comparative analysis of three knee arthrodesis techniques to manage failed total knee arthroplasty due to infection: intramedullary nailing, LRS external fixator, and compression plating.

*Rajasekaran RB, Ponniah HS, Palanisami DR, Sundaram VP, Palanivelayutham SS, Jayaramaraju D, Dhanasekaran S, Rajasekaran S*

INTRODUCTION: This study compared external fixation (EF), intramedullary nailing (IMN), and compression plating (CP) for knee arthrodesis (KA) following failed total knee arthroplasty (TKA) due to periprosthetic joint infection, focusing on complications, radiological, and functional outcomes.

METHODS: Patients aged ≥ 18 who underwent knee arthrodesis following failed TKA due to infection from 2008 to 2023 were included. Four surgeons used three techniques: EF, IMN, and CP. Retrospective data on demographics, surgical techniques, complications, and outcomes were collected. Postoperative patient-reported outcome measures (PROMs), pain, and functional status were assessed. Radiographic data was reviewed for limb shortening and union status.

RESULTS: 46 patients (mean age 61.9 years, BMI 28.2), predominantly male (84.8%) were involved in this study. Twenty patients underwent EF, 16 underwent CP, and 10 underwent IMN, all for infected primary TKA. Pseudomonas was the most common infecting organism (15.2%). Average follow-up was 29.1 months. Bony fusion was achieved in 41 knees (89.1%), with five requiring additional surgery; four eventually required amputation due to non-fusion. EF group had the shortest time to union and no major complications or amputations. Time to union was significantly shorter in the CP group compared to the IMN group ($p < 0.001$), and EF group also had a si […]

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## Accuracy of iliac crest reference pin placement in total hip arthroplasty: a CT-based analysis of risk factors for cortical perforation.

*Masada Y, Tetsunaga T, Yamada K, Inoue T, Okuda R, Tetsunaga T, Okazaki Y, Ozaki T*

INTRODUCTION: Accurate iliac crest reference pin placement is essential for safe navigation-assisted total hip arthroplasty (THA). Although complications are uncommon, inaccurate placement may cause neurovascular injury. This study aimed to evaluate pin insertion accuracy using postoperative computed tomography (CT), identify risk factors for cortical perforation, and determine whether CT-detected cortical perforation was associated with postoperative acetabular cup positioning accuracy.

MATERIALS AND METHODS: We retrospectively analyzed 200 primary THAs performed using portable or CT-based navigation between December 2022 and July 2025. Pin position was assessed on CT 1 week postoperatively and classified as complete intraosseous insertion or cortical perforation. Risk factors were examined using univariable and multivariable logistic regression analyses. Perforation direction was evaluated using Firth's penalized logistic regression. Postoperative cup positioning accuracy was assessed using CT-based radiographic inclination (RI), radiographic anteversion (RA), and absolute errors from target angles of 40° for RI and 15° for RA.

RESULTS: Cortical perforation occurred in 70 of 200 hips (35.0%): 32 medial and 38 lateral. On a pin-by-pin basis, 113 of 400 pins (28.3%) demonstrated perforation. In multivariable analysis, smaller ASIS width, lateral decubitus positioning, and 4-m […]

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## Long-term outcomes of arthroscopic single-anchor fixation versus microfracture for osteochondral fractures of the knee weight-bearing area in adolescents and young adults.

*Yang W, Pan Q, Wang H, Peng Y, Meng C, Huang W*

PURPOSE: Osteochondral fractures involving the weight-bearing area of the knee in young patients require stable restoration of the native osteochondral fragment whenever possible. This study aimed to evaluate the long-term clinical and radiological outcomes of arthroscopic single-anchor fixation and to compare them with those of microfracture.

METHODS: This retrospective comparative study included 24 adolescents and young adults with osteochondral fractures of the femoral condyle or trochlear region treated between 2016 and 2017. Twelve patients underwent arthroscopic single-anchor in situ fixation, and 12 underwent microfracture. The fixation technique used one double-loaded suture anchor to obtain multi-point compression of the osteochondral fragment while preserving the subchondral plate. Clinical outcomes were assessed using the Lysholm, International Knee Documentation Committee, and Tegner scores. Magnetic resonance imaging was evaluated using the MOCART 2.0 score. The mean follow-up was 93.7 months.

RESULTS: The single-anchor group showed a significantly higher intraoperative contact area ratio than the microfracture group (96.5% ± 3.2% vs. 85.8% ± 5.6%, P < 0.001). At final follow-up, the single-anchor group had superior Lysholm scores (91 ± 5 vs. 83 ± 8, P = 0.009), IKDC scores (86 ± 6 vs. 78 ± 7, P = 0.010), and Tegner activity levels (5.0 ± 1.1 vs. 3.9 ± 0.9, P = 0 […]

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## Isometric points in cerclage for patellar tendon repair: a 3-dimensional, weightbearing computed tomography kinematic simulation.

*Zindel C, Hodel S, Luo Y, Meisterhans M, Fucentese SF, Vlachopoulos L*

PURPOSE: The gold standard for the management of ruptured patellar tendon is the surgical repair with nonabsorbable sutures and additional reinforcement e.g. cerclage wires as described by McLaughlin in 1947. Despite the early description, the exact placement of such cerclage wires remains unclear. The aim of the study was (1) to analyze how the cerclage elongates during knee flexion and (2) to define the insertion points with the most isometric behavior at the patella and tibial tuberosity (TT).

METHODS: 3D simulations of 10 healthy knees in the range from 0° to 120° of flexion using weightbearing computed tomography scans were generated. 12 insertion points were defined at the patella in equal distances along the lateral and medial border from the proximal to the distal pole. 42 insertion points were defined at the tibial tuberosity (TT) in a 3.0 × 3.0 cm large grid around its center. An automatic string simulation algorithm was used to calculate the length for all possible combinations of the cerclage from 0° to 120° flexion. The most isometric combination of insertion points was found using the isometric score.

RESULTS: The most isometric insertion at the TT is distally ($p < 0.01$) and towards the center of the TT ($p < 0.01$). The most isometric points at the TT is located 61 mm distal to the joint line and 5 mm from the anterior border of the TT in a true lateral view fluo […]

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## Patient mortality exceeds implant revision risk with equivalent survival following unicompartmental versus total knee arthroplasty in octogenarians.

*Innocenti M, Leggieri F, Massenzi M, Stimolo D, Akkaya M, Matassi F, Civinini R*

BACKGROUND: While traditional surgical decision-making emphasizes long-term implant survivorship for octogenarians knee reconstruction, this paradigm may be inappropriate when patient mortality risk substantially exceeds revision risk. This study compared mid-term mortality between unicompartmental knee arthroplasty (UKA) and total knee arthroplasty (TKA) in octogenarians, hypothesizing that implant type would not significantly influence patient survival.

METHODS: A retrospective cohort study was conducted on 238 consecutive octogenarian patients (age ≥ 80 years) who underwent primary knee arthroplasty at a single tertiary orthopaedic center between January 2018 and December 2020. Patients with revision arthroplasty, fracture, tumor, inflammatory arthropathy, or incomplete data

were excluded. The primary outcome was all-cause mortality at minimum 5-year follow-up. Secondary outcomes included age-stratified mortality (80-84, 85-89, ≥ 90 years) and identification of independent mortality predictors. Statistical analyses included independent t-tests, chi-square/Fisher's exact tests, standardized mean differences, and multivariable logistic regression adjusting for age, follow-up duration, and registry year.

RESULTS: Of 238 patients, 185 (77.7%) underwent TKA and 53 (22.3%) underwent UKA. Overall mortality was 35.3% (84/238): 33.5% in TKA versus 41.5% in UKA (OR = 1.41, 95% CI: 0 […]

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### Short- to mid-term clinical outcomes and return to sport and work after arthroscopic debridement of the extensor carpi radialis brevis in patients with chronic lateral epicondylopathy.

*Siebenlist S, Doucas A, Hatt FL, Lappen S, Vieider RP, Lacheta L, Kadantsev P*

BACKGROUND: This study aimed to evaluate the short- to mid-term clinical outcomes following arthroscopic extensor carpi radialis brevis (ECRB) debridement in patients with chronic epicondylopathy humeri radialis (cEHR). We hypothesized that arthroscopic ECRB debridement would result in favorable short- to mid-term patient-reported outcomes, improving pain levels, strength, range of motion (ROM) and patient satisfaction.

METHODS: A retrospective case series was conducted on patients who underwent arthroscopic ECRB debridement for cEHR with symptoms > 6 months. Patients were assessed at a minimum of 12 months postoperatively using the Mayo Elbow Performance Score (MEPS), Patient-Rated Tennis Elbow Evaluation (PRTEE), Disabilities of the Arm, Shoulder, and Hand (DASH) score, and Visual Analog Scale (VAS) for pain. Pre- and postoperative scores were compared using DASH and VAS. Maximum isometric wrist extension strength and postoperative ROM were measured bilaterally. Return to sport and work, patient satisfaction, and complications were documented.

RESULTS: Of the 35 eligible patients, 30 completed follow-up (86% follow-up rate; mean age 48.4 ± 6.6 years; range: 36-65 years; 47% male). The mean duration of symptoms before surgery was 12 months (IQR 8-20, range 6-62). The mean follow-up duration was 28.13 ± 10.9 months (range 12-45). At final follow-up, median outcome scores were […]

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### Complications and outcomes after operative treatment of low-energy pelvic ring and acetabular fractures.

*Salonpää KB, Nurkkala J, Lantto I, Kakko S, Kaakinen T, Aitavaara-Anttila M, Liisanantti J*

BACKGROUND: Fragility fracture of the pelvis (FFP) with a low-energy mechanism of injury is relatively common among the growing geriatric population. The incidence of postoperative complications after surgically treated FFP is not well documented. The aim of this study was to report complications after operatively treated FFP and to determine whether these complications would have a detrimental effect on outcomes.

METHODS: Retrospective study of 200 consecutive patients who underwent operative treatment at Oulu University Hospital, Oulu, Finland, between 1.1.2010 and 15.4.2021 for FFP with a low-energy injury mechanism. Postoperative complications were recorded, and associations with outcomes were depicted.

RESULTS: Postoperative complications were recorded in 31 (15.5%) patients. Patients with complications had higher intraoperative bleeding (OR 2.8 (1.0-7.9), p = 0.048) and higher ASA classification 3.4 (1.3-8.4), P = 0.009). Patients with complications had longer hospital stays, but no clear association between complications and mortality or discharge location was observed.

CONCLUSIONS: The incidence of postoperative complications was comparable with that reported in previous literature. Patients who experienced complications had longer hospital stays; however, no definitive association between complications and outcomes was identified.

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### Minced cartilage implantation for cartilage regeneration: a survey of current clinical practices.

*Yilmaz T, Leutheuser S, Niemeyer P, Behrendt P, Roessler P, Gille J, Mumme M, Vogt S*

INTRODUCTION: The objective of this study is to evaluate how minced cartilage implantation (MCI) is currently applied in clinical settings, focusing on the selection criteria, procedural execution, handling of concomitant joint pathologies, and postoperative management.

MATERIAL AND METHOD: An online survey was conducted among all 4,915 members of the German Society for Arthroscopy and Joint Surgery (AGA). Invitations with a survey link were emailed, and the questionnaire was accessible anonymously from January 10 to February 20, 2024. A total of 927 responses were received and analyzed, resulting in a 19% response rate. Demographic data were described descriptively; survey answers were reported as percentages. The 25-item questionnaire, developed by AGA and the Quality Circle for Cartilage Repair & Joint Preservation (QKG e.V.), included single- and multiple-choice questions on experience with cartilage therapy, minced cartilage procedure application and postoperative care, and the overall assessment of this technique within cartilage repair.

RESULTS: MCI has emerged as one of the three most commonly used cartilage repair techniques in the knee joint. Defect sizes up to 4 $cm^2$ represent the largest treated group. The application of the procedure is quite heterogeneous, with some practitioners using a shaver and others manually mincing with a scalpel. Fixation methods also var […]

PubMed · DOI

## Variability in United States online rehabilitation protocols after open and endoscopic hip abductor repair.

*Anwunah C, Rosenberg S, Hartwell M*

PURPOSE: The purpose of this study was to evaluate readily accessible U.S. based online physical therapy protocols for open and endoscopic hip abductor repairs and to assess the variability in post-operative rehabilitation protocols.

METHODS: Online physical therapy protocols from U.S. based orthopaedic institutions for gluteus medius/minimus repair were collected using a web-based query utilizing search terms "hip abductor repair rehabilitation protocol" and "gluteus repair rehabilitation protocol." Published protocols from both Electronic Residency Application Service (ERAS) academic orthopaedic surgery programs and private groups were collected. All publicly available protocols were included. Protocols were analyzed using a standardized spreadsheet to assess rehabilitation components and timelines. Descriptive statistics (median values, percentages, ranges) were calculated.

RESULTS: 54 rehabilitation protocols were assessed, including 22 (41%) from ERAS-affiliated academic programs or associated physicians. Postoperative weight-bearing was addressed in 53/54 (98%) protocols, with 41 (77%) recommending toe-touch/touchdown/flat-foot/20-lb weightbearing. Median duration of restricted weight-bearing was 6 weeks (range, 2-8 weeks). Regarding range of motion restrictions, 37 (69%) protocols limited hip flexion to 90°, 42 (78%) advised against passive adduction past neutral, and […]

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# Injury

Volume 57, Issue 12 · October 2026 · 37 new articles

## Is surgeon technique a modifiable risk factor for heterotopic ossification after posterior acetabular fracture fixation? A multi-surgeon study.

*Kumar DS, Luther L, Schaffer NE, Litten RM, Traven SA, Moore RS, Moreno-Diaz AF, Velasco-Castro J et al.*

INTRODUCTION: Heterotopic ossification (HO) is a well-recognized complication following operative fixation of acetabular fractures through a posterior approach. Established risk factors largely reflect nonmodifiable patient and injury characteristics. Whether surgeon-controlled aspects of operative technique influence HO remains poorly defined. The purpose of this study was to determine if the individual surgeon and specific technique variables are associated with HO after accounting for established patient and injury factors.

METHODS: We retrospectively reviewed 1292 patients who underwent acetabular fracture fixation through a Kocher-Langenbeck approach at two Level 1 trauma centers between 2010 and 2025 by 14 fellowship-trained orthopaedic traumatologists. Patients receiving any form of HO prophylaxis were excluded. Four surgeon-controlled variables were examined: aggressiveness of gluteus minimus debridement, trochanteric osteotomy, thrombin gel use, and closed suction drainage. HO was graded on final radiographs by two independent reviewers using the Brooker classification, with third-reader adjudication for disagreements, and categorized as little-to-no (grade 0-I), moderate (grade II), or severe (grade III-IV). Mixed-effects logistic regression models with surgeon as a random effect were used to assess the associations between patient, injury, and surgeon-controlled tec […]

PubMed · DOI

## Impact of sarcopenic obesity phenotype on long-term survival in geriatric trauma: A nationwide cohort study using weight-adjusted waist index.

*Kim HG, Ju UC, Kim JY, Jung E*

BACKGROUND: As the global population ages, identifying effective prognostic markers for elderly trauma patients has become crucial. While Body Mass Index (BMI) is commonly used, its utility is limited by the "obesity paradox," failing to differentiate between muscle and fat. This study aimed to evaluate the Weight-Adjusted Waist Index (WWI), a surrogate marker for sarcopenic obesity, as a predictor of 20-year mortality following heavy trauma in the elderly.

METHODS: We analyzed a nationwide cohort of 18,366 elderly patients (age■65) from the National Health Insurance Service-National Sample Cohort (NHIS-NSC) who experienced heavy trauma (hospitalization, surgery, or ICU admission) between 2002 and 2022. Patients were categorized into WWI tertiles. Multivariable Cox proportional hazards models were used to estimate hazard ratios (HRs) for 20-year mortality, adjusting for demographics, lifestyle factors, comorbidities, and laboratory findings.

RESULTS: During the 20-year follow-up, the highest WWI tertile (T3) was associated with significantly higher mortality compared to the lowest tertile (T1). In the fully adjusted model, the HR for 20-year mortality in the T3 group was 1.72 (95% CI 1.55-1.91). Notably, the risk was more pronounced in obese patients (BMI■25 kg/m2, HR 2.15) and those with low physical activity. Furthermore, high WWI was associated with a higher prevalence of […]

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## Articular compliance and tolerance to residual incongruity after intra-articular fracture: A hypothesis-driven narrative review.

*Dhar SA, Dar TA, Dar RA, Farooq S, Maajid A, Zargar S, Mughal I*

BACKGROUND: Residual articular incongruity after intra-articular fracture is widely held to cause post-traumatic osteoarthritis (PTOA), yet tolerance of a given step-off differs between joints and between regions of the same joint. Reduction targets are nonetheless expressed as uniform millimetre thresholds, and no tissue-level mechanism has been offered for the variation.

HYPOTHESIS: The compliance of the osteochondral unit - cartilage thickness, cartilage material properties and subchondral bone stiffness together - modulates tolerance of residual incongruity, and incongruity is more informatively expressed relative to local cartilage thickness than in absolute millimetres.

METHODS: Narrative review with a documented search of PubMed/MEDLINE (inception to 14 October 2026) covering cartilage and subchondral properties of opposing surfaces, experimental and computational studies of articular incongruity, and clinical studies relating residual displacement to outcome. Four searches returned 3928 records, 3579 after deduplication; 54 sources informed the synthesis. This is not a systematic review; reporting follows SANRA.

FINDINGS: Small, localised contact-stress elevations appear tolerable; step-offs remodel less completely as their height increases; and cumulative contact-stress exposure above a threshold predicts PTOA after tibial plafond fracture. Opposing surfaces differ […]

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### Lumbopelvic fixation for high-energy sacral fractures and spinopelvic dissociation: A systematic review and meta-analysis of complications, unplanned revision and implant removal.

*Patel HA, Patel S, Suresh RI, Yang E, Lindley AK, Kidd G, Schneider MB, Zareef U et al.*

INTRODUCTION: Secondary procedures after lumbopelvic fixation (LPF) include planned implant removal, removal for symptoms and revision for complications. Their differing indications, together with limited comparative evidence, complicate counselling and surgical planning.

METHODS: PubMed, Embase, Scopus and Cochrane CENTRAL were searched to May 2026 (PROSPERO CRD420251036375) for comparative and single-arm studies (≥10 LPF patients). High-energy injuries and fragility fractures were considered separately. Random-effects models summarised complications, unplanned revision, implant removal and within-study comparisons of surgical approaches and fixation strategies.

RESULTS: Twenty-five studies were included (22 high-energy, 3 fragility). Across 19 high-energy LPF cohorts (564 patients), pooled surgical-site infection was 9.3% (95% CI 6.1-14.1), implant loosening or failure was 5.3%, nonunion occurred in 4 of 389 patients (1.0%) and loss of reduction in 2 of 355. Pooled unplanned revision was 11.9% (95% CI 8.7-16.0; prediction interval 5.3-24.3), most often for infection or wound complications. Implant removal was 79.0% in routine-removal series and 17.3% in selective-removal series. Four studies compared minimally invasive and open LPF. Pooling three studies showed less blood loss with minimally invasive LPF (mean difference -271 mL, 95% CI -367 to -175) and shorter operative t […]

PubMed · DOI

### Fracture characteristics in coronal shear fractures of the distal humerus: Clinical significance of medial trochlear involvement.

*Ishihara N, Saeki M, Fujihara Y, Saeki S, Tanaka H, Ota H, Kumagai H, Yazaki N et al.*

BACKGROUND: Coronal shear fractures of the distal humerus exhibit diverse fracture-extension patterns that are not fully characterized by conventional classification systems. The clinical significance of fracture extension into the medial or lateral column remains incompletely understood. This study investigated fracture-extension patterns beyond the conventional Dubberley classification and evaluated their associations with clinical outcomes and bone-related complications.

METHODS: This multicenter retrospective study included surgically treated coronal shear fractures of the distal humerus with a minimum follow-up of 12 months at four affiliated institutions between 2012 and 2024. Fractures were classified using the AO/OTA and Dubberley systems. Computed tomography (CT) images were reviewed to assess medial trochlear involvement, lateral column involvement without posterior condylar comminution, and isolated trochlear fractures. Clinical outcomes included elbow range of motion (flexion, extension, and arc of motion) and postoperative complications. Univariate logistic regression analysis was performed to identify factors associated with bone-related complications.

RESULTS: Among 374 surgically treated distal humeral fractures, 35 coronal shear fractures were included in the final analysis. Medial trochlear involvement was identified in 8 fractures (23%) and was most frequen […]

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## Global research on cricket injuries (1982-2025): A bibliometric analysis.

*Vaishya R, Singh A, Gupta BM, Vaish A, Dhillon MS, Parmar Y, Orchard J*

BACKGROUND: Cricket is one of the most widely played sports worldwide and carries a significant burden of musculoskeletal injuries, particularly among fast bowlers. Over the past four decades, research in cricket injury epidemiology, biomechanics, workload monitoring, and surveillance has expanded substantially.

OBJECTIVE: To conduct a comprehensive bibliometric analysis of global cricket injury research, identifying growth trends, influential publications, leading contributors, collaboration networks, and major thematic domains.

METHODS: Publications from 1982 to 2025 were systematically analyzed using bibliometric indicators including annual growth, authorship productivity, institutional and country contributions, citation impact, total link strength (TLS), and co-authorship networks. The top 100 cited papers and the top 10 highly cited papers (HCPs) were examined in detail to identify landmark contributions and research priorities.

RESULTS: A sharp rise in publications occurred after 2003, with nearly 90% of output produced in the last two decades. Australia, the United Kingdom, and South Africa led in productivity, citation impact, and collaborative link strength. The top 10 HCPs accounted for 10% of papers but 37.25% of total citations, underscoring concentrated scholarly influence. Fast bowlers consistently demonstrated the highest injury incidence, particularly lumbar […]

PubMed · DOI

## Machine learning models with SHAP explanations for predicting neurosurgical intervention and six month functional outcome after traumatic brain injury.

*Zhou Z, Cheng H, Liu H, Zhang D, Wang Q, Shao X*

BACKGROUND: Admission information can help describe surgical treatment and subsequent functional outcomes following traumatic brain injury (TBI). We developed interpretable machine learning models to address these two tasks.

METHODS: This retrospective study included 673 adults evaluated at a single center between December 2018 and December 2023. Outcomes recorded were neurosurgical intervention and six-month Glasgow Outcome Scale status, dichotomized as favorable (scores 4-5) or unfavorable (scores 1-3). Patients were randomly allocated to training and internal validation sets in a 7:3 ratio. Least absolute shrinkage and selection operator (LASSO) regression was used to select predictors for the XGBoost and random forest models. Receiver operating characteristic (ROC) analysis, along with classification and prediction error metrics, was employed for model evaluation. Shapley additive explanations (SHAP) were used to interpret model predictions.

RESULTS: There were 75 unfavorable outcomes and 66 surgical interventions. The reported validation accuracy for XGBoost was 87.56% for predicting functional outcomes and 95.02% for surgical interventions. Validation Brier scores were 0.083 and 0.035, respectively. The Glasgow Coma Scale (GCS), pupillary light reflex, and blood glucose levels contributed significantly to functional outcome predictions, while GCS, fibrinogen, and white […]

PubMed · DOI

## Geography matters: Variations in neck trauma between urban and rural level-II trauma centers.

*Goldstein AL, Nino E, Biton D, Kruyer VN, Bar-Haim R, Laks S, Soroksky A, Levit B et al.*

BACKGROUND: Geodemographic factors create divergent clinical patterns between urban and rural settings. Level-II trauma centers are a vital trauma-care infrastructure for both urban and rural communities. This study compares neck trauma patterns, resource utilization, and outcomes at these distinct geodemographic nodes.

METHODS: Retrospective analysis (2014-2024) of neck trauma at an urban center (UC, n = 65) and a rural center (RC, n = 60). Demographics, mechanism, treatments, interventions, and outcomes were analyzed.

RESULTS: RC patients sustained more blunt, transport-related multisystem trauma with higher ISS. UC patients predominantly presented with localized penetrating injuries. Rural patients had higher ICU admission rates, longer ICU stays, more transfers to level-I centers yet underwent fewer operative interventions, invasive airway procedures, and CT angiography.

CONCLUSIONS: There are significant variations in neck trauma between urban and rural level-II trauma centers. An "intervention paradox" exists were rural centers stabilize high-severity blunt trauma for transfer, while urban centers provide definitive care for penetrating wounds. These findings necessitate geodemographically tailored protocols. Understanding geodemographic variations is a critical aspect of optimizing trauma systems.

LEVEL OF EVIDENCE: II Retrospective cohort.

PubMed · DOI

### Limb preservation and optimization: A whole-patient approach to comprehensive care after catastrophic extremity injuries.

*Cook JL, Rucinski K, Nuelle JAV, Royse LA, London DA, Lalonde SP, Schweser KM, Crist BD*

INTRODUCTION: The objective of this study was to describe the Limb Preservation and Optimization Center's (LPOC) components and mechanisms in treating catastrophic extremity injuries and report the initial patient demographics, indications, and outcomes using data from an established registry.

METHODS: A retrospective observational study was performed including all patients entered in the LPOC registry since January 1, 2014. Patient demographics, injury types and treatments, and outcomes, including functional outcomes. Descriptive statistics were calculated to determine proportions, means, standard deviations, and ranges.

RESULTS: 992 patients were included with a mean final follow-up of 20.5 months (SD:22.1, range 0-114); 178 (17.9%) underwent an amputation. Of those who reported RTW, 164 patients (71.6%) reporting RTW at full capacity. Lower extremity injuries were most common (74.9%). Demographic, health insurance, living setting, and presentation variables were similar among LPOC patients. Fractures and soft tissue injuries predominated across all injury categories with fracture fixation and wound coverage treatments most common. Upper extremity joint injuries and all types of combined injuries were associated with the highest reoperation rates. Treatment failure rates ranged from 11% to 100% with fracture and bone-related injuries being associated with the most favorable […]

PubMed · DOI

### Epidemiological characteristics and comparison of clinical outcomes between conservative and surgical treatment for rib fractures in South China: A 10■year retrospective cohort study of 4085 cases with propensity score■matching analysis.

*Zhang T, Cai Q, Liang B, Song X*

PURPOSE: To systematically describe the epidemiological characteristics of rib fractures in South China and compare the clinical characteristics, hospital stay, and total costs between patients receiving conservative versus surgical treatment, thereby providing evidence■based support for clinical decision■making and public health prevention.

METHODS: A retrospective analysis was performed on 4085 hospitalized patients with rib fractures admitted to the Fifth Affiliated Hospital of Southern Medical University between January 2014 and December 2023. Data on gender, age, injury mechanism, fracture type, complications, blood group, and occupational distribution were collected. A subgroup analysis comparing conservative (n = 586) versus surgical treatment (n = 137) was conducted among 723 patients admitted between January 2021 and December 2023. Propensity score matching was used to balance baseline characteristics between the two groups, and differences in hospital stay, total costs, and injury mechanisms were compared.

RESULTS: (1) Epidemiological characteristics: Of the 4085 patients, 68.13% (2783/4085) were male, and the 41-60 year age group accounted for the highest proportion (50.35%). Autumn and winter were the peak seasons for injury (54.71%). Traffic accidents were the leading cause of injury (51.18%), followed by falls (21.82%) and height■related falls (19.13%). Multiple […]

PubMed · DOI

### Segmental phase angle as predictor of activities of daily living and muscle health in patients with osteoporotic fracture A retrospective cohort study.

*Ito Y, Yoshimura Y, Nagano F, Matsumoto A*

PURPOSE: This study aimed to determine whether the segmental phase angle is associated with clinical outcomes such as activities of daily living (ADL), muscle strength, and skeletal muscle mass in patients with osteoporotic fractures.

METHODS: This was a retrospective cohort study of patients with osteoporotic fractures admitted to convalescent rehabilitation wards. The primary outcome was a Functional Independence Measure motor (FIM-motor) score at discharge, and secondary outcomes were handgrip strength and skeletal muscle mass index (SMI) at discharge. Multivariate analysis was used to adjust for confounders and to examine the association between segmental phase angle and outcome.

RESULTS: A total of 115 subjects (97 women, age 81.0 ± 10.0 years) were included in the study. The median baseline phase angle was 3.6°. Multiple regression analysis adjusted for confounders showed that arm phase angle was independently positively associated with handgrip strength (β=0.250, P < 0.001) and SMI (β=0.215, P < 0.001) at discharge. Leg phase angle was independently positively associated with FIM-motor score (β=0.077, P = 0.045), handgrip strength (β=0.174, P < 0.001), and SMI (β=0.154, P < 0.001) at discharge.

CONCLUSIONS: The leg phase angle was associated with ADL, muscle strength, and skeletal muscle mass in patients with osteoporotic fractures at discharge. Segmental phase angle […]

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## Union rate and time to union following platelet-rich plasma therapy in fracture nonunion: A systematic review and meta-analysis.

*Regmi A, Saini R, Baral S, Jain VK, Kulshrestha S, Iyengar KP*

BACKGROUND: Fracture nonunion remains a challenging clinical problem associated with prolonged disability and repeated surgical interventions. Platelet-rich plasma (PRP) has been proposed as a biological adjunct to enhance bone healing; however, clinical evidence remains inconsistent. This study aimed to evaluate the effectiveness of PRP in the management of fracture nonunion.

METHODS: A systematic review and meta-analysis were conducted in accordance with PRISMA guidelines (PROSPERO Registration: CRD420261366849). PubMed, Embase, and Scopus were searched from inception to March 2026. Studies evaluating PRP in fracture nonunion and reporting union rate or time to union were included. Random-effects meta-analyses were performed to compare PRP with control interventions. Risk ratios (RR) were calculated for union rates and standardized mean differences (SMD) for time to union. Additional single-arm analyses were performed to estimate pooled outcomes in PRP-treated cohorts.

RESULTS: Twenty-one studies comprising 877 patients were included. Two comparative studies (n = 119) demonstrated a higher likelihood of union with PRP (RR 1.17, 95% confidence interval [CI] 1.01-1.36, p < 0.05) without significant heterogeneity. Six studies (n = 250) showed no significant difference in time to union between PRP and control groups (SMD -0.72, 95% CI -1.48-0.05), with substantial heterogeneity […]

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## Province-wide assessment of emergency care capacity and skills in a rural provincial health system: A facility-based cross-sectional assessment in Battambang, Cambodia.

*Takaiso H, Nomura K, Watanabe T, Nitta Y, Heak A, Im S, Okubo Y, Miyachi T*

BACKGROUND: Emergency care capacity remains critically limited in low- and middle-income countries (LMICs), particularly in rural areas where specialists and infrastructure are scarce. In Cambodia, stroke and ischemic heart disease are leading causes of mortality, and trauma burdens are increasing alongside rapid infrastructure development. Evidence on emergency equipment availability and frontline provider skills at the facility level in rural provinces remains limited.

METHODS: A facility-based cross-sectional assessment was conducted across all seven public referral hospitals in Battambang Province, Cambodia, in January 2023. Emergency equipment availability was evaluated through structured interviews and direct visual confirmation against the World Health Organization (WHO) Guidelines for Essential Trauma Care. Clinical skill examinations were administered to physicians (two scenario-based cases: acute stroke with airway compromise; traumatic injury with tension pneumothorax and intra-abdominal bleeding) and nurses (a Basic Life Support [BLS] scenario assessed in pairs). Hospitals were stratified by Complementary Package of Activities (CPA) level: CPA1 (basic care only), CPA2 (including surgery and intensive care), and CPA3

(highest provincial level).

RESULTS: At Complementary Package of Activities (CPA)1 (basic care only) hospitals, intubation equipment and 12-lead elect […]

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### A scoping review evaluating artificial intelligence applications across various injury prevention domains.

*Othman Y, Kasabwala D, Jayagopi K, Lemoine J, Park D, Hamza D, Elkbuli A*

INTRODUCTION: Artificial intelligence (AI) has become increasingly utilized for injury prevention across diverse clinical and community settings including machine learning (ML). This scoping review aimed to synthesize the current literature on the applications and predictive performance of AI across multiple injury prevention domains.

METHODS: This scoping review searched five major databases, including PubMed, Embase, ScienceDirect, ProQuest, and the Cochrane Library, to identify studies evaluating AI applications for injury prevention published through May 2026. Eligible studies included original U.S.-based, English-language research across self-harm and suicide, abuse and violence, fall prevention, and sports injury prediction. Studies were excluded if they lacked measurable prevention outcomes or sufficient methodological detail, or focused on injury treatment rather than prevention, to ensure relevant and interpretable evidence for synthesis.

RESULTS: Nineteen studies met the inclusion criteria. ML models generally outperformed or matched traditional screening methods for self-harm and suicide prediction (AUC 0.81-0.82), while multimodal approaches achieved the highest predictive accuracy (AUC 0.90). In abuse and violence prevention, AI predicted intimate partner violence with an AUC of 0.88 and identified cases before formal disclosure. AI also improved fall risk assess […]

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### Early fecal diversion versus no initial diversion in blunt open pelvic fractures: A systematic review and meta-analysis of pelvic infection and mortality.

*Ferrari F, Ancieta MDH, Laura LYS, Botti M*

BACKGROUND: The benefit of routine early fecal diversion in blunt open pelvic fractures remains uncertain. We examined its association with pelvic infection and short-term mortality.

METHODS: MEDLINE/PubMed, Embase, Scopus, Web of Science Core Collection, and CENTRAL were searched from inception through 12 August 2026, with citation searching. Comparative observational studies were synthesized using crude risk ratios (RRs), REML random-effects models, modified Hartung-Knapp confidence intervals, and Mantel-Haenszel sensitivity analyses without continuity correction. Two reviewers independently selected studies, extracted data, and assessed ROBINS-I risk of bias and GRADE certainty.

RESULTS: Eighteen observational comparative reports were retained. Potential cohort overlap prevented a valid unique overall participant total. Eight cohorts contributed to broad pelvic infection (171 participants; seven estimable): RR 0.78 (95% CI 0.35-1.70; $I^2$=19.6%). The stricter outcome-definition analysis included five cohorts (88 participants; four estimable): RR 0.72 (95% CI 0.07-6.97). Excluding two author-reported infected hematomas through an existing replacement sensitivity gave RR 0.84 (95% CI 0.34-2.05). Eight mortality cohorts (236 participants) yielded RR 0.82 (95% CI 0.33-2.07; $I^2$=19.0%). Post-2006 evidence remained inconclusive for infection (three cohorts; RR 0.74, 95% CI 0.009-58 […]

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### Enhanced wound healing potential of a carboxymethyl cellulose/gelatin-based hydrogel containing decellularized amniotic membrane and Matricaria chamomilla extract: A multifunctional composite dressing.

*Zamani S, Ghorbani M, Molzemi S, Farahani MK, Raei M*

BACKGROUND: Chronic wound healing remains a complex biological process influenced by excessive inflammation, oxidative stress, microbial invasion, and insufficient tissue regeneration. Therefore, the design of multifunctional biomaterials that integrate structural support with biological regulation has emerged as a promising strategy in advanced wound care. This study sought to develop a composite hydrogel incorporating decellularized

amniotic membrane (dAM)-derived extracellular matrix components alongside Matricaria chamomilla extract, with the goal of achieving enhanced regenerative and therapeutic outcomes.

METHODS: A CMC/GEL-Chamomilla-dAM hydrogel was prepared using decellularized dAM and subsequently evaluated in terms of its physicochemical and biological characteristics. The effectiveness of the decellularization process and preservation of ECM components were first examined, followed by characterization of hydrogel morphology, rheological properties, swelling and degradation behavior, and pH stability. Its biological performance was further assessed through hemocompatibility, cytocompatibility, antioxidant, anti-inflammatory, and antibacterial assays, as well as a full-thickness wound model.

RESULTS: Incorporation of dAM and chamomile extract resulted in a stable hydrogel with favorable physicochemical and biological characteristics. The decellularization procedure […]

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### Can posterolateral tibial plateau fractures be treated with an anterolateral approach? A biomechanical study on novel plating concepts in artificial tibia models.

*Zderic I, Llano L, Menze J, Altmann M, Breceda A, Herbst E, Krause M, Luo CF et al.*

BACKGROUND: The aim of this study was to evaluate biomechanically whether fixation of posterolateral tibial plateau fractures (PLTPFs) through the more accessible modified anterolateral approach and by using a newly designed lateral plate with or without posterior plate extension or supplemental anteroposterior screw fixation, can provide comparable stability to direct posterolateral buttress plating.

METHODS: PLTPFs were simulated on 40 artificial tibiae and subsequently fixed with 5 different plating techniques (n = 8), namely using Variable Angle Locking Compression Plate (VA-LCP) Lateral Proximal Tibia Plate 3.5 (VA Lateral), Variable Angle Optimized Locking Technology (VOLT) Lateral Proximal Tibia Plate 3.5 (VOLT Lateral), VOLT Lateral Proximal Tibia Plate 3.5 with Posterolateral Extension (Hug), VOLT Posterolateral Proximal Tibia Plate 3.5 (PL), and VOLT Lateral Proximal Tibia Plate 3.5 with two anteroposterior parallel 3.5 mm locking screws placed superiorly to the plate's lateral-to-medial rafting locking screws (Jail) using a dedicated guide block. Biomechanical testing was performed under progressively increasing cyclic loading to failure. Interfragmentary movements were monitored via motion tracking.

RESULTS: Fragment displacement and its posterior tilt over the first 40,000 cycles did not differ significantly between Hug, PL, and Jail (all $p > 0.999$). Hug, PL, and […]

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### Where is trauma care delivered in New York state? A statewide analysis of distribution and outcomes by trauma center level.

*Patel P, Mon NO, Angus LDG, Szydziak E, Cardozo-Stolberg S, Swezey E, Alamprese S, Angus T et al.*

BACKGROUND: Understanding where and how trauma care is delivered is critical to preventing and managing its disease burden. This study identified patterns in trauma care delivery by characterizing the relationship between trauma presentations and outcomes as a function of trauma center level, including trauma presentations at non-trauma centers.

METHODS: A retrospective review of trauma inpatients aged ≥ 15 years across New York State from 2016 through 2020 was performed, utilizing a comprehensive New York State all-payer data reporting system, the Statewide Planning and Research Cooperative System (SPARCS). The primary endpoint for this study was mortality. Secondary endpoints included overall complications, discharge dispositions, and hospital length of stay.

RESULTS: A total of 361,141 patients were included with 42.1% presenting to Level 1 and 37.3% to Non-Trauma Centers. Older adults were more frequently treated at Level 3 and Non-Trauma Centers, while younger patients and high-energy mechanisms (motor vehicle crashes and penetrating injuries) were more commonly managed at Level 1 and Level 2 Centers. The association between trauma-center group and mortality varied by severity of illness (SOI) (interaction $p < 0.0001$). Compared with Level 3 and Non-Trauma Centers, Level 1 and 2 Centers had higher adjusted mortality odds among hospitalizations with major SOI (aOR 1.43, 95 […]

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### Variations in hip fracture management at trauma vs. non-trauma centers.

*Harris B, Benson E*

BACKGROUND: Hip fractures represent a growing public health burden. Optimal management relies on timely surgical intervention coordinated by multidisciplinary care and appropriate resource availability, all of which may vary between hospital settings. Differences between trauma hospitals and non-trauma hospitals may influence outcomes, yet these variations have not been well characterized.

METHODS: We conducted a retrospective cohort study of hip fracture cases over a 2-year period (October 2022 to August 2024), comparing patient and procedural characteristics between a Level II trauma hospital (TH) (n = 104) and a non-trauma hospital (NTH) (n = 31). Polytrauma patients were excluded. Data were extracted from the electronic medical record (EMR). Primary outcomes were hospital length of stay (LOS) and time from admission to surgery. Group comparisons utilized chi-square or Fisher's exact tests for categorical variables and Mann-Whitney U or Welch t-tests for continuous variables, as appropriate. A multivariable random forest model was used to evaluate predictors of operative duration.

RESULTS: ASA physical status distribution differed significantly between facilities (p = 0.03), with more ASA > 2 patients at the NTH (76.6% vs. 65.7%). Osteosynthesis was performed in 73.3% of NTH cases and 82.2% of TH cases; arthroplasty comprised 26.7% and 17.8%, respectively. LOS differed sig […]

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## Comprehensive treatment approaches for traumatic tattoos: An integrative review.

*Haik J, Levi Y, Kornhaber R, Cleary M, Beylin D, Tasigiorgos S, Harats M*

BACKGROUND: Traumatic tattoos result from an injury that drives external pigment into the dermis. They often affect exposed sites such as the face and extremities. Pigment may persist for years within the dermis, complicating management and limiting cosmetic outcomes. Consequently, no optimal treatment technique has been identified, and long-term data remain limited.

AIM: To synthesise evidence on treatments for traumatic tattoos and to evaluate effectiveness and outcomes over the past 25 years.

DESIGN: Integrative review with narrative synthesis following Whittemore and Knafl's five stages.

DATA SOURCES: An electronic search of PubMed and Scopus was conducted from 2000 to November 2025 to identify original peer-reviewed research focusing on treatments for traumatic tattoos and to evaluate their effectiveness and outcomes.

RESULTS: Twenty-eight articles met the inclusion criteria, representing 27 studies, comprising largely case reports and small series, with one retrospective cohort study. Treatment approaches are grouped into laser-based therapies, acute or delayed mechanical debridement (including hydrosurgery), and other surgical or mechanical methods. Q-switched Nd:YAG, ruby, alexandrite, fractional CO2, and picosecond lasers commonly achieve good to excellent clearance with mostly transient adverse effects. Early debridement within 24 h often prevented or markedly red […]

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## Patterns of depression screening, referral, and behavioral health follow-up after orthopaedic trauma.

*Reddy S, Cahill M, Schimizzi G, Fram B, Erdman MK, Christiano A*

OBJECTIVE: To evaluate baseline patterns of depression, behavioral health referral, and behavioral health treatment among orthopaedic trauma patients at a Level I trauma center.

METHODS: Design: Retrospective Cohort, Secondary Analysis Setting: Single Level 1 Trauma Center Patient selection criteria: Adult orthopaedic trauma patients who completed an ambulatory postoperative visit between September 2024 and November 2025. Outcome measures and comparisons: Demographic variables, PHQ-9 screening results, injury characteristics, behavioral health referral patterns, and referral completion were extracted from institutional electronic health records. Primary outcomes were the proportion of patients screening positive for depression (PHQ-9 >10), with moderately severe to severe depression (PHQ-9 ≥15) triggering behavioral health referral, and subsequent referral completion rates. Secondary outcomes included referral patterns by injury mechanism and six-month orthopaedic complications or unplanned healthcare utilization.

RESULTS: Among 528 patients, 44.7% (236/528) screened positive for depressive symptoms, and 21.2% (112/528) demonstrated moderately severe to severe depression triggering a behavioral health referral. 33.9% (38/112) of referred patients completed a behavioral health visit compared with 67.6% (115/170) of referred non-trauma orthopaedic patients. The severe depressio […]

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## Predictors of 30-day mortality and complications following surgical treatment of pathologic hip fractures.

*Thomas G, Njoku-Austin CO, Almeida ND, Sinha S, Schrand TV, Mirvish A, Singh AK, Hoogervorst P*

Pathologic hip fractures present complex challenges and are associated with higher postoperative complication rates and mortality compared with non-pathologic fractures. This study aimed to compare 30-day outcomes between pathologic and non-pathologic hip fracture patients and identify factors associated with 30-day mortality and postoperative complications among patients with pathologic hip fractures using the National Surgical Quality Improvement Program database. A retrospective analysis was conducted from 2006 to 2023. Patients aged ≥ 18 years undergoing hip fracture surgery were included. Patients with non-pathologic hip fractures were compared with those with pathologic hip fractures. Univariate analyses and multivariable logistic regression models were used to identify factors associated with pathologic hip fracture 30-day mortality and complications. Among 204,675 hip fracture patients, 6159 (2.9%) had pathologic hip fractures. Patients with pathologic hip fractures had higher rates of 30-day mortality (7.8% vs. 5.1%, $p < 0.001$), any postoperative complication (35.3% vs. 30.2%, $p < 0.001$), cardiac/transfusion complications (27.5% vs. 22.2%, $p < 0.001$), renal complications (0.5% vs. 0.3%, $p = 0.017$), and venous thromboembolic complications (3.4% vs. 1.7%, $p < 0.001$). Infectious complications and neurological complications were not significantly different between groups. […]

PubMed · DOI

## Video-assisted resident education is associated with improved ankle joint reduction: A prospective controlled study.

*Bilodeau RE, Hall AJ, Como M, Cautela F, Rowe J, Joseph D, Reddy RP, Hogan MV et al.*

BACKGROUND: Proper fracture reduction and splinting technique is critical for soft tissue protection and fracture stability in the acute management of ankle fractures. However, residency education for acute ankle fracture reduction and splinting techniques remains poorly standardized and streamlined. This study evaluated the effect of implementing a video-assisted ankle fracture management education program administered to first-year orthopaedic residents on ankle joint reduction and splint quality.

METHODS: This was a prospective controlled study conducted at a single level I trauma center. All consecutive ankle fractures (OTA/AO: 4F, 44) requiring the placement of a bivalved short leg cast or AO splint from January 1, 2019, to June 30th, 2022 were included. Exclusion criteria included skeletal immaturity, missing imaging, delayed fracture presentation more than two weeks from injury, concurrent ipsilateral injury/polytrauma interfering with ankle reduction, open fracture, or ankle fractures treated without an AO splint or cast. The primary outcome was joint reduction quality defined as radiographic post-reduction tibiofibular overlap, tibiofibular clear space, and medial clear space. Secondary outcomes included splint padding thickness and ankle dorsiflexion angle to assess splint quality.

RESULTS: A total of 280 consecutive ankle fracture cases were included for evaluation […]

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## Opportunistic assessment of bone quality using CT-derived hounsfield units in postmenopausal women with femoral trochanteric fractures.

*Terada C, Yokoo S*

BACKGROUND: CT-derived Hounsfield unit (HU) measurements may provide an opportunistic assessment of bone density when dual-energy X-ray absorptiometry (DXA) is difficult to obtain during acute treatment of femoral trochanteric fractures. We compared three femoral-head regions of interest (ROIs) and explored their ability to identify markedly low total hip bone mineral density (BMD).

METHODS: This single-center retrospective study included 87 women aged ≥ 65 years with AO/OTA type 31A1 (excluding A1.1) or 31A2 femoral trochanteric fractures treated with intramedullary nailing. Admission CT HU values were measured in three femoral-head ROIs on coronal and axial images and compared with contralateral hip DXA T-scores measured approximately 1 week after surgery. Pearson correlations and receiver operating characteristic (ROC) analyses were performed. The prespecified total hip T-score thresholds were ≤ -2.5 and ≤ -3.3; the latter was used as an operational definition of markedly low BMD in this study.

RESULTS: Coronal whole-femoral-head HU values (ROI 1) showed a moderate correlation with total hip T-scores (r = 0.569). For the ≤ -2.5 threshold, 74 women met the criterion and only 13 did not; the ROI 1 coronal AUC was 0.841. For the ≤ -3.3 threshold, 51 women met the criterion and 36 did not. A coronal ROI 1 threshold of < 158.7 HU had an AUC of 0.765 (95% CI, 0.660-0.870), sensi […]

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## Thoracic irrigation in managing traumatic hemothorax: A concise review of techniques and practice variations.

*Indorewala Y, Prashar S, Nishida C, Nasef Y, Othman Y, Haddadi M, Elkbuli A*

BACKGROUND: Retained hemothorax following tube thoracostomy occurs in up to one-third of thoracic trauma patients, often necessitating additional interventions. Thoracic irrigation has emerged as a potential adjunct to improve pleural drainage. The objective of this concise review is to evaluate the role of thoracic irrigation in the management of traumatic hemothorax by highlighting current standards of care and variations in techniques.

METHODS: This review searched five search engines, screening studies published before September 26, 2025. Studies focusing on thoracic irrigation in the setting of trauma and hemothorax were included.

RESULTS: A total of six studies were included. Thoracic irrigation in the management of traumatic hemothorax showed significantly lower odds of secondary intervention (OR: 0.16, p < 0.001) and shorter hospital length of stay (7 days vs. 9, p = 0.04) in patients who received thoracic irrigation compared with standard chest tube drainage. Most patients underwent irrigation using medium to large-bore tubes, although one study showed no association between tube size and outcomes. Patients receiving ≥ 1000 milliliters (mL) of irrigation had more hospital-free days compared to those receiving < 1000 mL (p = 0.007), while another study found no significant differences.

CONCLUSION: Thoracic irrigation shows promise as an adjunct to tube thoracostomy i […]

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## Do radiographic surrogates of pre-bend appropriateness influence outcomes after titanium elastic nailing in pediatric femoral mid-shaft fractures?

*Sar■ta■ TB, Yücel MH, Parlak S, Ar■c■ V, Tan A, Yalç■nkaya A, Terzi E, ■ahin A et al.*

PURPOSE: To investigate whether radiographic technical parameters reflecting pre-bend appropriateness are associated with radiographic and clinical outcomes after titanium elastic nailing (TEN) in pediatric femoral mid-shaft fractures.

METHODS: This retrospective cohort included 85 children aged 5-11 years treated with TEN between 2015 and 2025. Constructs were classified as criteria-met (n = 38) or criteria-unmet (n = 47) using nail symmetry, a bend apex-fracture apex distance ratio ≤ 10%, and a metaphyseal spread ratio ≥ 50%. Separate exploratory multivariable models evaluated the composite classification and its components, adjusting for clinical and technical covariates.

RESULTS: Median coronal angulation was lower in the criteria-met group (2.0° vs 5.0°; p = 0.005). The adjusted mean difference (criteria-unmet minus criteria-met) was 3.42° (95% CI 0.98-5.87; p = 0.007). Other group comparisons were inconclusive. Exploratory component analysis associated greater spread with greater limb length discrepancy (3.61 mm per increase of 10 %age points; 95% CI 1.81-5.40; q = 0.002), although technical parameters were highly correlated. The complication risk ratio for criteria-unmet versus criteria-met constructs was 1.41 (95% CI 0.45-4.48).

CONCLUSION: Meeting the study-specific radiographic criteria was associated with lower coronal angulation, but its clinical importance remai […]

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## Scapular fracture and risk of great vessel injury in geriatric blunt trauma: A TQIP analysis.

*Peisepar M, Rafieezadeh A, Prabhakaran K, Nair H, Kannichamy V, Harikrishnan S, Shnaydman I, Kirsch J et al.*

INTRODUCTION: Scapular fractures are markers of high-energy blunt thoracic trauma and may indicate severe intrathoracic injury. Despite the substantial mortality linked with great vessel injuries (GVI), data specifically examining scapular fracture and GVI in the geriatric population remain limited. We hypothesized that scapular fracture, particularly when accompanied by concomitant thoracic fractures, is independently associated with GVI in geriatrics with blunt trauma.

METHODS: We conducted a retrospective cohort study using the American College of Surgeons Trauma Quality Improvement Program (ACS-TQIP) database (2017-2023). Patients aged ≥ 65 years with a blunt mechanism of injury and AIS severity 2-5 in the neck, thorax, or upper extremity were included. GVI was defined using specific AIS codes for major thoracic and cervical vessels. Multivariable logistic regression evaluated association between scapular fracture patterns and GVI, including isolated scapular fractures, those with concomitant sternal, rib or flail chest fractures. Models adjusted for age, sex, race, mechanism, ISS, shock index, and modified frailty index. Vessel-specific analyses were also performed. Primary outcome was GVI. Secondary outcomes were mortality and LOS.

RESULTS: A total of 677,134 patients were included. Median age was 76 [70-82] years; 54.1% were female. Scapular fractures were identified i […]

PubMed · DOI

## Orthopedic trauma in patients with opioid use disorder: Current concepts in perioperative management.

*Stout JW, Keenan J, Rehman S*

BACKGROUND: Opioid use disorder (OUD) is an important challenge in orthopedic trauma, where acute injury intersects with preexisting opioid exposure, tolerance, complex pain, and barriers to continuity of addiction treatment. OUD has been associated with higher rates of complications, readmission, and treatment disruption, yet OUD-specific perioperative management remains inconsistently incorporated into orthopedic trauma practice. This review synthesizes the epidemiology, outcomes, and perioperative management evidence for this population.

METHODS: A clinically oriented current concepts review synthesized peer-reviewed literature, prioritizing studies published from 2013 onward. PubMed and Embase were searched, supplemented by Google Scholar, Semantic Scholar, and manual review of reference lists from relevant articles. Studies examining perioperative outcomes, multimodal analgesia, and management of medications for OUD (MOUD) relevant to orthopedic trauma were included; studies limited to pediatric, elderly, or prenatal populations were excluded.

RESULTS: Preinjury opioid exposure is common among orthopedic trauma patients, whereas formally diagnosed OUD is less prevalent, affecting approximately 1% of operative trauma patients and up to 2.3% of patients with open lower-extremity fractures. OUD and opioid-related disorders are associated with higher rates of infection, nonu […]

PubMed · DOI

## The downside of the pull-up: A 10-year national analysis of emergency department visits, 2015-2024.

*Shabbir R, Layton B, Kang Z, Kim B, Rehman S, Galos DK*

BACKGROUND: To characterize the epidemiology and clinical patterns of injuries related to pull-up and chin-up bars in a nationally representative emergency department (ED) dataset.

METHODS: Retrospective cross-sectional analysis of the U.S. Consumer Product Safety Commission National Electronic Injury Surveillance System from January 1, 2015, to December 31, 2024. Cases were identified using product code 3277 with narrative filters for "pull-up," "chin-up," and "doorway bar," excluding barbells and playground equipment. NEISS sampling weights produced national estimates. Annual totals with 95 % confidence intervals were calculated and reported. Injury characteristics were summarized by age group, body region, diagnosis, location, disposition, and a narrative-derived mechanism category (fall, struck by bar, slip or detachment, other).

RESULTS: An estimated 40,987 ED visits were attributed to pull-up bar injuries, averaging about 4000 per year, with a peak in 2024 (5421; 95 % CI, 5315-5527). Patients were predominantly male (66.1 %) with a bimodal age distribution centered on children 5-14 years and adults 18-34 years. Craniofacial trauma predominated, with head and face together comprising nearly one-half of presentations, followed by upper-extremity involvement, including shoulder, wrist, and forearm. Laceration, internal head injury, and fracture were the leading diagnoses, […]

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### How the cause of surgical delay affects mortality in fragility hip fractures: A cohort study.

*Senan R, Karthik P, Dhilon P, Howard T, Rayan F, Kirmani S, Shyamsundar S*

BACKGROUND: Prompt surgery for fragility hip fractures is a cornerstone of care and a key metric within the Best Practice Tariff (BPT) in the United Kingdom. Yet, since 2011, time to theatre has steadily increased, BPT adherence has fallen, and paradoxically, mortality has reduced. It is still unclear if surgical delay is an independent predictor of mortality. Emerging evidence suggests that the causes of surgical delays may be more important determinants of mortality than the delay itself.

METHODS: A cohort study of 1769 hip fracture patients above the age of 60 was conducted in a single centre in England. This study collected demographic data, timing to theatre (<36 h vs ≥36 h), reasons for any delays, 30-day mortality and 1-year mortality between March 2020 and December 2024.

RESULTS: The most common cause of delay was insufficient theatre capacity (N = 365, 20.6%), followed by delays awaiting medical stabilisation (N = 102, 5.8%). Age ($X^2$ 71.638 $p < 0.001$), ASA grade ($X^2$ 37.045 $p < 0.001$), sex ($X^2$ 27.721 $p < 0.001$), pre-admission residence ($X^2$ 36.772 $p < 0.001$), and reason for delay ($X^2$ 27.495 $p = 0.007$) were significant predictors of mortality. Delays due solely because of theatre unavailability did not increase 30-day (OR 0.784 $p = 0.398$) or 1-year mortality (OR 1.123 $p = 0.474$). In contrast, delays due to medical optimisation were associated with markedly higher 30-day […]

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### The blunt truth: The role of marijuana in PTSD after intentional traumatic injury.

*Kern JA, Wilson C, Joshi SA, Ghio M, O'Neill K, Reese J, Babb R, Emili P et al.*

BACKGROUND: Survivors of interpersonal violence experience high rates of post-traumatic stress disorder (PTSD), and rates of marijuana use are rising among young adults. There is however, limited understanding of marijuana use patterns among survivors of interpersonal violence with PTSD. We evaluated relationships between PTSD and marijuana use following injury and hypothesized that PTSD-positive patients would report elevated follow-up marijuana use.

METHODS: Between 10/2019 and 7/1/2024, we prospectively collected data on patient demographics, PTSD, substance use (including marijuana), and PROMIS-29 quality of life domains for adult trauma patients admitted to an urban, academic, level-1 trauma center. Participants were screened at baseline and at 6- or 12-month follow-up. Injury characteristics were collected from the institutional trauma registry. Among the intentionally injured, we compared patients by follow-up PTSD screening results. We excluded patients who screened positive for PTSD at baseline. Using multivariable logistic regression we assessed relationships between follow-up PTSD and marijuana use, including demographic and injury characteristics as covariates. Our primary outcome was dichotomized follow-up marijuana use.

RESULTS: Of 105 participants injured by intentional trauma with complete follow-up data, 69 screened negative for PTSD at baseline and therefore […]

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### Pediatric flexor tendon injuries: A 13-year retrospective review of injury characteristics, treatment, and outcomes.

*■brahimov J, Bag■r M, Bolat ■, Kocaer Ö, Bozkurt O*

BACKGROUND: Pediatric flexor tendon injuries are rare but potentially devastating. Due to low incidence, optimal perioperative management remains poorly defined, and evidence is scarce. This study evaluates the injury characteristics, surgical treatment, age-specific functional outcomes, and challenges in this population.

METHODS: A retrospective review was conducted on 79 pediatric patients (mean age 6.84 ± 4.27 years; 67.1% male) who underwent surgical repair for flexor tendon lacerations between 2010 and 2023. Outcomes were assessed using Total Active Motion (TAM), Strickland score, grip/pinch strength, Semmes-Weinstein monofilament, two-point discrimination, and QuickDASH questionnaire, with a mean follow-up of 59.9 ± 32.8 months.

RESULTS: Zone 2 was the most common injury site (45.6%), with glass being the predominant mechanism (67.1%). Nerve injury was present in 35.4% of cases. Overall, good/excellent results were achieved in 81.9% (TAM) and 89.0% (Strickland) of patients. The 0-5 years age group had significantly lower Strickland scores than the 6-12 years group (74.28% vs. 82.9%, $p < 0.05$). Zone 5 injuries demonstrated significantly better outcomes than Zone 2 and 3 ($p < 0.001$). The overall complication rate was low (5.1%), with a rerupture rate of 2.5%.

CONCLUSION: Pediatric flexor tendon repairs yield favorable functional outcomes with low complication rates. Age- […]

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### Quantitative decision-support strategies to enhance emergency medical service accessibility in high-risk freeway tunnels.

*Jung S, Qin X*

Crashes inside road tunnels often result in more severe outcomes due to spatial constraints that delay prehospital emergency medical services (EMS) access. Emergency evacuation facilities, such as cross passages and emergency parking areas, are critical for ensuring timely EMS response during traffic congestion caused by in-tunnel crashes. This study presents a quantitative approach to develop targeted strategies for enhancing such facilities to improve EMS accessibility in tunnels. Nationwide in-tunnel crash data from Korean freeways over a four-year period were analyzed, and the limited number of severe crash cases was addressed using SMOTE-based data augmentation to balance severity levels in the prediction model. A SHAP-based XGBoost classifier was applied to identify the influence of key features related to EMS accessibility and emergency evacuation facilities on crash severity. The results showed that crashes occurring more than 1 km from the tunnel entrance and EMS response coverage within a 10-15 km range have a significant impact on increasing crash severity. The presence of emergency cross passages, three or more lanes, emergency parking areas, and properly spaced CCTV installations were found to help reduce crash severity. Based on these findings, tunnels with particularly limited EMS accessibility were prioritized, and tailored facility improvement measures were pro […]

PubMed · DOI

### Acellular fish-skin grafting complex soft tissue defects in orthopedic trauma: Technique and outcomes of a consecutive case series.

*Kallevang JK, Faherty MS, Scheschuk J*

This article describes a surgical technique for applying an acellular fish-skin graft (Kerecis Omega3 Wound; Kerecis, Ísafjörður, Iceland) in orthopedic trauma and reports outcomes from a consecutive series of patients treated at an urban Level I trauma center. After institutional review board approval, we reviewed 27 adult patients with complex extremity wounds who were treated with acellular fish-skin grafting between January 1st, 2023, and December 30th, 2025. Wound etiologies included traumatic soft tissue injuries, postoperative wound dehiscence, necrotizing soft tissue infections, and chronic osteomyelitis with associated soft tissue compromise. Acellular fish-skin grafting was indicated in this series of patients for full-thickness skin loss (or deeper), including those with exposed tendons, bone, or ligaments, in whom flap reconstruction was contraindicated. Patients underwent serial irrigation and debridement followed by application of an acellular fish-skin graft. The primary outcome was complete wound healing without flap-based soft-tissue reconstruction. Secondary outcomes included the number of graft applications, duration of negative-pressure wound therapy (NPWT), time to split-thickness skin grafting (STSG), and all complications. Acellular fish-skin grafting facilitated wound healing in most patients, including those with infection, exposed bone, tendon, and/or […]

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### Peripheral nerve blocks reduce opioid requirements in the first 24 h following surgical fixation of acute tibial shaft fractures.

*Khan UA, Quigley C, Schulman M, Holland J, Ghidei S, Kim A, Schulman J, Gaski G et al.*

OBJECTIVES: To evaluate the effect of peripheral nerve blocks (PNB) on postoperative opioid use, length of stay (LOS), and incidence of acute compartment syndrome (ACS) in patients presenting with acute tibial shaft fractures treated with intramedullary nail (IMN) fixation.

METHODS: Design: Retrospective cohort study Setting: Level 1 Academic Trauma Center from 2018 to 2022. Patient Selection Criteria: Patients aged 18-79 years with acute AO/OTA 42A-C tibial shaft fractures treated with IMN within 7 days of injury Outcome Measures and Comparisons: The primary outcome was postoperative opioid use, measured in morphine milligram equivalents (MME), during hospital stay, up to a maximum of 72 h after surgery. Patients who received a preoperative PNB were compared to those who did not receive a block. Secondary outcomes included postoperative LOS and incidence of ACS.

RESULTS: 246 patients were eligible for analysis of which 78 (32%) received a PNB (PNB=yes) and 168 (68%) did not (PNB=no). Mean MME were significantly lower in the PNB=yes group compared to the PNB=no group at 24 h (73 vs. 138, $p < 0.001$). Controlling for age, Injury Severity Score (ISS), and time to surgery, a multivariable regression model demonstrated a 36% reduction in MME per hour for patients that received PNBs. There were 3 cases of ACS in PNB=No and 1 in PNB=Yes. All 3 patients without a PNB were diagnosed w […]

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## From static fracture classification to dynamic clinical phenotypes: Feature-based longitudinal annotation and retrieval of comparable cases.

*Regazzoni P, Jupiter J, Giannoudis PV, Oca FD*

Traditional surgical classifications provide standardized languages for describing clinical conditions at diagnosis, but they necessarily simplify complex and varying biological and mechanical phenomena into predefined categories. In fracture surgery, the AO/OTA classification has substantially improved communication and comparability by providing a standardized morphological framework. However, patients with apparently similar morphology at diagnosis may follow substantially different clinical trajectories after treatment. We propose a complementary approach based on feature-based annotations and longitudinal documentation of the complete treatment course. Rather than representing a patient solely by a static classification at time zero, this approach describes a dynamic clinical phenotype, comprising patient characteristics, fracture features, intraoperative events, postoperative course and outcome. Such structured longitudinal data enable the identification of comparable real-world cases beyond conventional classification subgroups. Artificial intelligence may facilitate the integration and analysis of these multidimensional data and, importantly, help identify when apparently similar clinical trajectories begin to diverge. This framework can support surgical education and shared decision-making by providing easy access to comprehensively documented comparable cases.

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## Gabapentinoid-opioid co-prescription after long bone fracture fixation is associated with increased 1-year nonunion.

*Mahatme RJ, Moore SA, Gangavaram A, Moon E, Khadilkar S, Woolnough T, Ellis JD, Beltran MJ*

PURPOSE: Gabapentinoids are increasingly incorporated into multimodal analgesia after orthopaedic surgery, yet their impact on outcomes following fracture fixation remains unclear. This study evaluated the association between early postoperative gabapentinoid-opioid co-prescription and clinical outcomes after operative fixation of closed long bone fractures.

METHODS: Using TriNetX, we identified opioid-naïve and gabapentinoid-naïve adults undergoing intramedullary or plate-and-screw fixation of isolated closed diaphyseal tibia, humerus, or femur fractures. Patients prescribed gabapentin or pregabalin and opioids within three days after surgery were compared with patients receiving opioids alone. Propensity score matching (1:1) incorporated demographics, medical comorbidities, fracture location, fixation method, mental health disorders, and substance use disorders. Primary outcomes were prolonged (30-90 days) and chronic (90-180 days) postoperative opioid use. Secondary outcomes included 30-day emergency department (ED) visits and readmissions, and 1-year nonunion and reoperation.

RESULTS: After matching, each cohort consisted of approximately 2000 patients (range, 1946-2012). Gabapentinoid-opioid therapy was associated with higher rates of prolonged (21.0% vs 16.1%; RR 1.31; $p < 0.001$) and chronic (9.8% vs 7.4%; RR 1.33; $p = 0.007$) postoperative opioid use. Gabapentinoid-opio […]

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# European Journal of Trauma & Emergency Surgery

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## Fixation of patellar fractures using bioabsorbable hydroxyapatite/poly-L-lactide cannulated screws and subperiosteal nonabsorbable sutures.

*Kobayashi M, Usami T, Takada N, Iwata H, Murakami H, Katsuragawa R, Kuroyanagi G*

INTRODUCTION: Bioabsorbable implants offer a promising alternative to metallic fixation for patellar fractures, potentially avoiding hardware-related complications. In 2023, a new generation of bioabsorbable hydroxyapatite/poly-L-lactide (HA/PLLA) cannulated screws was introduced, featuring a low-profile (2 mm) trident-shaped head and increased root diameter for enhanced strength. This study evaluates the clinical outcomes of a surgical technique combining these redesigned screws with subperiosteal nonabsorbable sutures.

MATERIALS AND METHODS: Between April 2023 and May 2025, 19 consecutive patients with transverse patellar fractures (AO/OTA 34-C1/C2) underwent open reduction and internal fixation. Two bioabsorbable cannulated screws were used. High-strength nonabsorbable sutures were passed through the screw channels and a central bone tunnel using a double-loop sliding knot for compression. The sutures were systematically routed subperiosteally to minimize tissue irritation. Full weight-bearing with a brace began on postoperative day one. Clinical and radiographic outcomes were assessed. This study was approved by the institutional review board of Kainan Hospital.

RESULTS: The mean maximum articular surface gap was 8.7 ± 5.2 mm. The mean follow-up was 8.6 ± 3.0 months. Bone union was achieved in all cases. At the final follow-up, the mean knee flexion was 138.7° ± 13.0° and […]

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## Impact of pre-existing bleeding disorders on outcomes in patients with traumatic brain injury.

*Yamaji F, Coimbra BC, Zakhary B, Coimbra R*

PURPOSE: Traumatic brain injury (TBI) remains a major global public health problem. This study aimed to evaluate the association between pre-existing bleeding disorders and clinical outcomes among patients with TBI. We hypothesized that pre-existing bleeding disorders would be associated with worse outcomes following TBI.

METHODS: We conducted a retrospective cohort study using the 2021-2023 National Inpatient Sample (NIS) to evaluate the association between pre-existing bleeding disorders and clinical outcomes among patients aged 15 years or older hospitalized with TBI.

RESULTS: Mortality was significantly higher in the bleeding disorder group (13.9% vs. 4.8%; $p < 0.001$), the median length of stay was longer (5 days vs. 3 days; $p < 0.001$), median hospitalization costs were substantially higher (US $20,582 vs. US $13,515; $p < 0.001$), and discharge home was less frequent (44.8% vs. 61.6%; $p < 0.001$). After adjustment for age, comorbidity burden, injury severity, and other clinically relevant covariates, pre-existing bleeding disorders remained independently associated with higher odds of in-hospital mortality (adjusted OR, 2.31; 95% CI, 1.99-2.69).

CONCLUSION: Pre-existing bleeding disorders were independently associated with higher odds of in-hospital mortality after TBI. Patients with these disorders also experienced worse clinical outcomes. Early recognition and appropriat […]

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## 24-hour risk updating for major ICU complications after severe trauma: a prospective cohort study.

*Sanchez C, Cachafeiro L, Millan P, Polo E, Garcia L, Sarmiento C, Vejo J, Lopez A et al.*

PURPOSE: Conventional trauma prognostic models rely primarily on admission data and cannot account for subsequent physiological response to treatment. We evaluated whether routinely collected physiological information available at 24 h provides incremental prognostic information for major intensive care unit (ICU) complications in severely injured patients.

METHODS: We conducted a prospective cohort of adults with severe trauma (Injury Severity Score [ISS] ≥ 15) admitted to the ICU between January 2024 and April 2025. The primary outcome was a composite of major respiratory, hemodynamic, renal, or septic complications during the ICU course. TRISS was compared with an admission ridge model incorporating ISS, Glasgow Coma Scale, hemoglobin, lactate, and international normalized ratio, and a 24-h updated ridge model additionally incorporating 24-h lactate and change in creatinine. Performance was internally validated with 1,000 bootstrap resamples.

RESULTS: Among 109 patients, median ISS was 29 (IQR 20-38), and 73 (67.0%) developed major ICU complications. Optimism-corrected AUCs were 0.788 for TRISS, 0.873 for the admission model, and 0.887 for the 24-h updated model; Brier scores were 0.178, 0.140, and 0.131. Compared with the admission model, the 24-h updated model improved performance (ΔAUC 0.016, 95% bootstrap interval 0.004-0.030; ΔBrier - 0.010, - 0.017 to - 0.005). Twent […]

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## Addition of a McLaughlin cerclage to locking plate fixation of distal patellar pole fractures reduces interfragmentary displacement - a biomechanical human cadaveric study.

*Deichsel A, Große-Allermann A, Horst K, Kittl C, Lodde M, Gueorguiev B, Drenchev L, Skulev HK et al.*

PURPOSE: To investigate whether the addition of a McLaughlin cerclage to locking plate fixation of distal patellar pole fractures provides improved biomechanical construct stability and reduced interfragmentary displacement.

METHODS: A distal patellar pole fracture was simulated in sixteen fresh-frozen human cadaveric knees by a horizontal osteotomy in the distal third of the patella. Subsequently, locking plate fixation was performed. In a randomized half of the specimens, an additional McLaughlin wire cerclage was applied. The specimens were then subjected to 3000 cycles of extension starting at 90° of flexion under a maximum tensile load of 320 N. Using a Digital Image Correlation (DIC) system, the interfragmentary displacement between the proximal and distal fragments was measured at 1, 10, 20, 30, 50, 100, 500, 1000, 2000, and 3000 cycles. Statistical analysis was performed using repeated-measures ANOVA and Welch's unpaired t-t test at 3000 cycles.

RESULTS: An additional McLaughlin cerclage significantly reduced interfragmentary displacement under cyclic loading in comparison to locking plate fixation alone. After 3000 cycles, displacement was 2.27 ± 1.84 mm with plate fixation alone and 0.36 ± 0.22 mm with additional McLaughlin cerclage (mean difference 1.91 mm; 95% CI 0.38-3.45 mm; p = 0.0215).

CONCLUSION: An additional McLaughlin cerclage reduced interfragmentary dis […]

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## Trauma care performance and patient outcomes - insights from a major trauma centre in Sweden.

*Eriksson J, Larsson E, Arnason B, Ohlén D, Hedberg M, Jonsson Fagerlund M, von Oelreich E*

PURPOSE: Trauma is a leading cause of death among individuals under 45 years of age. Evaluating trauma system performance over time is important but inherently difficult due to concurrent changes in patient demographics, case-mix and care processes. The aim of this study was to describe temporal changes in patient demographics, injury severity and trauma care at a major trauma centre in Sweden from 2017 to 2023. We hypothesised that risk-adjusted 30-day mortality decreased and that time to computed tomography was reduced over the study period.

METHODS: A retrospective cohort study was conducted using data from the Karolinska Trauma Registry, including all trauma patients aged > 15 years fulfilling criteria for full trauma team activation, admitted between January 2017 and August 2023. Multivariable logistic regression was used to assess changes in 30-day mortality over time, adjusted for age, sex, injury severity category, ASA score and injury type. Time from arrival to computed tomography was analysed by linear regression and, because the distribution was skewed, by quantile (median) regression.

RESULTS: A total of 5,615 trauma patients were included. Over the study period, the patient population became older, and ASA scores increased, while injury severity and the proportion of penetrating injury remained unchanged. In multivariable logistic regression with 2017 as referenc […]

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### Chest tube fixation strength depends on suture caliber and fixation technique rather than tube diameter: a biomechanical cadaver study.

*Fricke JJ, Schöbel T, Meißner R, Schleifenbaum S, Kleber C, Metelmann I, Krämer S*

BACKGROUND: Chest tube dislodgement remains a relevant complication in emergency, trauma, and intensive care medicine. In a previous biomechanical study, the Modified Johannesburg technique using Mersilene USP 1 demonstrated superior fixation strength compared with other chest tube fixation techniques. The present study aimed to determine whether suture size, suture material, and chest tube diameter influence fixation strength.

METHODS: A prospective interventional biomechanical cadaveric study was performed using fresh porcine thoracic wall specimens. A total of 320 load-to-failure tests were conducted under standardized laboratory conditions. Two fixation techniques, Modified Johannesburg and Purse String, were compared using two chest tube diameters (20/28 Ch) and four non-absorbable braided suture configurations: Mersilene USP 1, Mersilene USP 2 - 0, Silk USP 1, and Silk USP 2 - 0. Each experimental group consisted of 20 specimens. Specimens were tested in a uniaxial testing machine until construct failure. Force [N] and elongation [mm] at failure were measured. The experiments were videotaped, and the failure mechanism was analyzed.

RESULTS: The Modified Johannesburg technique demonstrated significantly higher pull-out forces compared to the Purse String technique (125.0 vs. 66.4 N; $p < 0.001$), as did larger suture sizes (USP 1 vs. 2 - 0; $p < 0.001$) and Mersilene compare […]

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### Changes in patellar height following suprapatellar and infrapatellar tibial nailing: a retrospective comparative cohort study.

*Olivieri R, Hernández J, Soto JM, Bustos F, Díaz R, Rojas C, Laso JI*

PURPOSE: Patella baja has been described after infrapatellar tibial nailing and may contribute to postoperative knee morbidity; however, whether the nail insertion approach influences postoperative patellar height remains unclear. This study compared longitudinal changes in patellar height (PH) after suprapatellar and infrapatellar intramedullary nailing for tibial shaft fractures. We hypothesized that the suprapatellar approach would better preserve patellar height.

METHODS: A retrospective comparative cohort study included 114 tibial shaft fractures treated with intramedullary nailing between 2016 and 2023, using a suprapatellar (n = 86) or infrapatellar (n = 28) approach. Patellar height was assessed using the Caton-Deschamps Index (CDI) on preoperative and final follow-up radiographs. The primary outcome was the longitudinal change in CDI (ΔCDI). Fracture union, complications, and reoperations were secondary outcomes.

RESULTS: The mean reduction in CDI was greater following infrapatellar than suprapatellar nailing (ΔCDI - 0.116 ± 0.173 vs. -0.011 ± 0.107; $p = 0.001$). After adjustment for baseline CDI and follow-up duration, this association remained significant (adjusted $\beta$=-0.055; 95% CI - 0.107 to - 0.004; $p = 0.035$), although sensitivity analyses showed attenuation after additional adjustment for sex. Fracture union rates were similar between approaches (93.0% vs. 92.9% […]

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### Intraarticular calcaneal fractures: biomechanical analysis of two fixation constructs.

*Ziegenhain F, Zderic I, Zondervan R, Richards RG, Gueorguiev B, Rammelt S, Schepers T, Swords M*

PURPOSE: The aim of this biomechanical study was to investigate the fixation stability of a small low-profile 2.7 mm variable-angle locking anterolateral calcaneal plate supplemented with two additional independent cortex screws versus a traditional small 3.5 mm locking calcaneal plate in a standardized Sanders IIc fracture model.

METHODS: Sixteen fracture models cocomposed of three polymer blocks (fragments) each and simulating Sanders IIc intra■articular calcaneal fractures were assigned to two groups. Group 1 (control) received a small 3.5 mm locking calcaneal plate, while Group 2 received a small 2.7 mm variable■angle locking anterolateral calcaneal plate with two additional independent cortex screws. All constructs were instrumented using prefabricated pilot holes and standardized screw configurations. Biomechanical testing was conducted implementing a 3■point-bending setup and progressively increasing cyclic loading to failure. Angular interfragmentary movements were quantified via motion tracking along with cycles to clinically relevant failure.

RESULTS: Initial stiffness (N/mm) in Group 1 (65.2 ± 34.7) was significantly lower versus Group 2 (232.5 ± 48.6), p = 0.020. Although angular interfragmentary movements around the anteroposterior axis were significantly greater in Group 1 versus Group 2 (p < 0.001), Group 1 demonstrated significantly smaller angular interfragme […]

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### Correction: Time to surgery and its impact on length of stay and mortality for neck of femur fractures at a major trauma centre.

*Tang LY, Ho SYS, Zhou AK, Geetala R, Krkovic M*

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### Clinical outcomes of combined MPFL-reconstruction with MPTL-augmenting patellar distalization using hamstring autograft versus isolated MPFL in adolescents with patella alta: a retrospective comparative study.

*Salem H, Ali EMS, Metwally O, Salem M, El-Aidy S, Sebaei M, Hatem A*

BACKGROUND: Patella alta is a major risk factor for recurrent patellar dislocation in adolescents. This study aimed to compare the clinical outcomes of isolated medial patellofemoral ligament (MPFL) reconstruction with those of MPFL reconstruction augmented by medial patellotibial ligament (MPTL)-distalization, both using a semitendinosus hamstring autograft, in adolescent patients with patella alta.

METHODS: A retrospective cohort study was conducted on 28 adolescent patients (mean age 11.2 ± 1.6 years; 9 males, 19 females) treated for recurrent patellar dislocation between May 2020 and July 2025. All patients had patella alta (Caton-Deschamps index > 1.2) and a history of at least one dislocation, with varying degrees of trochlear dysplasia (Dejour types A-B). All patients underwent MPFL reconstruction using a semitendinosus hamstring autograft harvested with preservation of its native distal tibial insertion, fixed via a single oblique transpatellar tunnel and femoral interference-screw fixation proximal to the distal femoral physis. Group A (n=14) underwent isolated MPFL reconstruction. Group B (n=14) underwent the same MPFL reconstruction with an additional MPTL-augmenting patellar distalization step using the same graft, secured by trans-osseous periosteal suture at the lateral patella. Surgical techniques emphasized physeal spare during reconstruction. Preoperative and […]

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### Health economic evaluation of polytrauma treatment in the current aG-DRG system - Results of a Level I Trauma Center.

*Hierl K, Thiedemann C, Kerschbaum M, Popp D, Alt V*

PURPOSE: Resource-intensive polytrauma care is associated with high treatment and readiness costs and poses a financial challenge for hospitals. The aim of this retrospective single-center study of the years 2020 to 2022 was to analyze polytrauma reimbursements and costs.

METHODS: Based on billing and clinical data of 201 polytrauma cases from a German University Hospital and Level I Trauma Center, revenues and costs of the eleven MDC 21 A polytrauma DRGs were analyzed. Ratios of the revenue in relation to LOS, ventilation hours, OR time and implant costs as well as personnel costs were calculated. Contribution margins were determined as an upper-bound estimate.

RESULTS: DRG reimbursement was highest for the most severely injured patients (50,852 €, DRG W36Z). It was lowest for patients who died within five days (2,214 €, DRG W60Z) or without surgeries (4,056 €, DRG W61B). The contribution margin was also highest for the most severely injured patients (52,775 €, DRG W36Z). Revenues in relation to ICU LOS (3,455 €/day, DRG W36Z) and ventilation time (238 €/hour, DRG W01B) were lowest for the

most severe polytraumas. Regarding OR time and implant costs, reimbursement was lowest for injuries of the spine, pelvis and extremities with 62 €/min (DRG W02B) and 5 €/€ (DRG W02A).

CONCLUSION: Treatment of polytrauma patients with long ICU stay, extended mechanical ventilation and mult […]

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### Patterns, and outcomes of airbag -related penetrating injuries: a retrospective analysis from a level 1 trauma center.

*Khoschnau S, AlMadani A, Mekkodathil A, Al-Hassani A, El-Menyar A, Peralta R, Consunji R, Kloub A et al.*

BACKGROUND: The deployment of defective airbag inflators can generate high-velocity metallic fragments that cause face, neck or chest penetrating injuries with potentially lethal consequences. This study aimed to describe the injury patterns, management, and outcomes of suspected airbag-related penetrating injuries (ARPI) among patients in a Middle Eastern country.

METHODS: A retrospective observational study of patients presenting with penetrating injuries suspected of being related to defective airbag deployment. Data were obtained from the Qatar National Trauma Registry (QNTR), electronic medical records, emergency medical services (EMS), emergency department (ED), and Mortuary. Data were collected and analyzed descriptively.

RESULTS: A total of 16 patients had suspected ARPIs between April 2019 and February 2026. The mean age was 33.1 ± 8.3 years, and 93.8% were male. Metallic foreign bodies (FBs) from airbag inflator fragmentation were documented in 11/16 (68.8%) patients by imaging, and surgical removal while in 4/16 patients there were only entry-exit wounds after deployment of airbags following car crash. In one case, airbag deployed while mechanic was repairing a car steering wheel. Penetrating neck (43.8%), thoracic (31.3%), and maxillofacial injuries (31.3%), predominantly mandibular fractures, were most common. Surgical intervention was required in 11 (68.8%) pati […]

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### Guideline adherence in helicopter versus ground emergency medical services: an indication-based analysis of 129,783 patients from the TraumaRegister DGU®.

*Faul P, Hagebusch P, Lefering R, Schweigkofler U, Aras I, Egert A, Koch DA*

BACKGROUND: Helicopter emergency medical services (HEMS) are associated with higher intervention rates and improved access to specialized trauma centers compared with ground emergency medical services (GEMS). However, it remains unclear whether these differences reflect increased intervention intensity alone or more consistent implementation of guideline-recommended care. This study aimed to compare rates of selected guideline-recommended prehospital trauma interventions between HEMS and GEMS using a nationwide trauma registry.

METHODS: This retrospective multicenter cohort study was based on data from the TraumaRegister DGU® (TR-DGU) from 2016 to 2024. Adult patients (≥ 16 years) with severe trauma, defined as maximum Abbreviated Injury Scale (MAIS) ≥ 3, receiving physician-staffed prehospital care and primary admission to German trauma centers were included. After exclusion of secondary transfers, missing key data, and treatment in local trauma centers (level 3), 129,783 patients remained, of whom 25.0% were transported by HEMS. Rates of selected interventions recommended by the German S3 guideline on the treatment of polytrauma/severe injuries were assessed using predefined indication-based subgroups derived from available registry variables as surrogates for potential clinical indications to improve comparability between transport modalities. Analyses were performed descri […]

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### Preoperative prevalence and risk factors for thromboembolic events following tibial plateau fractures: a retrospective analysis of an underestimated risk.

*Herold J, Lorenz ED, Eymann H, Lohse F, Kamin K, Walther EJ, Mäder M, Panchyrz I et al.*

BACKGROUND: Tibial plateau fractures (TPF) are complex intra-articular injuries carrying a high risk of deep vein thrombosis (DVT), muscular vein thrombosis (MVT) and pulmonary embolism (PE). Preoperative thrombosis is frequently clinically silent yet may cause serious complications, whereas its detection delays definitive surgery.

This study assessed the prevalence and risk factors of preoperative venous thromboembolic events (VTE) in surgically treated TPF before and after implementation of routine duplex ultrasound screening.

METHODS: In this retrospective single-centre study at a level 1 trauma centre, all adult patients surgically treated for traumatic TPF (AO 41B/C) between January 2015 and August 2025 were reviewed (547 patients, 554 fractures). Until August 2020 duplex ultrasound (DUS) was performed only on clinical suspicion; from August 2020 onward routine preoperative DUS was progressively implemented. VTE were confirmed by DUS or CT angiography. Prevalences are reported per fracture and patient characteristics per patient; regression analyses were performed at fracture level, except for the patient-related model.

RESULTS: With DUS restricted to clinical suspicion (cohort A; 246 fractures), the detected prevalence of preoperative VTE was 3.7% (95% CI 1.9-6.8) and that of DVT 2.4% (1.1-5.2). Under routine screening (cohort B; 205 screened fractures) it was 20.0% (15 […]

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## Emergency surgery for blunt abdominal trauma in polytrauma - the impact of time to laparotomy on outcomes.

*Rausch G, Kalbas Y, Halvachizadeh S, Zumsteg J, Shehu A, Mikova E, Lehmann K, Mica L et al.*

PURPOSE: Blunt abdominal trauma (BAT) is common in polytrauma patients. Emergency surgery is required for haemodynamically significant solid organ injury and intra-abdominal hollow organ injury. The effect of time to surgery on outcomes in patients requiring emergency laparotomy remains uncertain.

MATERIALS & METHODS: Adult polytrauma patients (ISS > 16) who underwent emergency laparotomy for blunt abdominal trauma between 1996 and 2020 from a trauma registry of a single level one trauma centre were included. Patients were divided into two groups based on the time elapsed from admission to emergency laparotomy: "Super Early" (≤ 60 min) and "Early" (60-180 min). Length of stay (LOS), intensive care unit (ICU)-LOS, ventilation time, re-look operations and mortality were assessed. Adjusted logistic regression was used to determine the independent predictive parameters, including timing of surgery, for mortality.

RESULTS: The study included 190 polytrauma patients with concomitant blunt abdominal trauma (mean age 42 years (± 17), median ISS of 36 (IQR: 19)). There were 95 patients in each group. Patients in the "Super Early" group had a compromised haemodynamic status and a higher ISS. Liver injury and serous membrane injury were more common in the super early group. The mortality rate was significantly higher in the "Super Early" group (51.6%) compared to the "Early" group (26.3 […]

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## High-velocity trauma by social media: a new mechanism of injury in pediatric traumatology?

*Sennert M, Schoof B, Körner M, Le M, Kolman J, Madadi-Sanjani O, Wintges K*

PURPOSE: Over the past two decades, social media challenges have emerged as a significant cause of injuries among adolescents. This study evaluates a sudden cluster of severe upper limb injuries resulting from the viral "Superman challenge" to raise clinical awareness of digital behaviors in pediatric trauma.

METHODS: A retrospective study at a tertiary pediatric trauma center evaluated patients aged < 18 years admitted in December 2024 following the "Superman challenge."

RESULTS: Eight patients were admitted within a 10-day window following high-energy injuries from the challenge. The cohort showed a female predominance (62.5%) with a mean age of 13.25 (±1.03) years. The high-velocity mechanism led to atypical, severe injury patterns distinct from routine accidental falls. We observed an unusual cluster of severe distal forearm fractures in five patients, which uniformly required surgical treatment. Three patients even suffered bilateral distal radius fractures. Additional associated injuries included fractures of the ulnar styloid process (n = 5) and concomitant TFCC tears (n = 2), reflecting the high-energy nature of the challenge.

CONCLUSIONS: The rapid, viral spread of physical social media challenges such as the 'Superman Challenge' can lead to dangerous injury mechanisms involving high-velocity trauma among children and adolescents. These trends could lead to sudden, […]

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## Surgical stabilization of rib fractures in a high-altitude trauma cohort: in-hospital clinical outcomes and resource use in a propensity score-weighted retrospective cohort study.

*Wan Q, Sun Q, Li W, Yundan C, Hu L, Shen Y*

PURPOSE: Surgical stabilization of rib fractures (SSRF) is increasingly used in the management of traumatic rib fractures, but its in-hospital outcomes in high-altitude trauma settings remain uncertain. We evaluated the association of SSRF with in-hospital pulmonary complications and length of stay in a high-altitude (≈ 3,500 m) trauma system.

METHODS: We conducted a single-center retrospective cohort study of adults with CT-confirmed traumatic rib fractures admitted from 2015 to 2023 and managed with SSRF or non-operative management. Confounding was addressed using stabilized inverse probability of treatment weighting (IPTW) with augmented outcome models; overlap weighting, propensity score matching, and time-dependent exposure analysis were used as sensitivity analyses. The primary outcome was a composite of in-hospital pneumonia, atelectasis, or respiratory failure. Pulmonary risk ratios (RRs) were estimated using modified Poisson regression, while ICU and hospital length of stay were analyzed using log-transformed weighted models.

RESULTS: Among 324 patients, 121 underwent SSRF and 203 received non-operative management. Before adjustment, SSRF patients had greater injury severity. In IPTW-augmented analyses, the pulmonary estimate was imprecise (RR 0.86, 95% CI 0.23-3.21; P = 0.822), and no significant difference was detected in ICU length of stay (exp[β] 1.04, 95% CI 0.9 […]

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## Pelvic hemorrhage in the context of demographic change: frequency and CT-based characterization of pelvic bleeding.

*Gerhardt F, Höch A, Notov D, Henkelmann J, Zeidler S, Einhorn S, Kleber C*

PURPOSE: Traumatic pelvic hemorrhage is a major driver of morbidity and mortality in pelvic fractures. Classic literature described pelvic hemorrhage as predominantly venous, whereas arterial injury is considered less common. We evaluated the frequency, anatomical distribution, and presumed sources of pelvic hemorrhage in a contemporary trauma cohort using contrast-enhanced CT (CECT).

METHODS: We conducted a single-center, retrospective cohort study of consecutive patients with pelvic fractures treated at a level-I trauma center between 2020 and 2022. All patients underwent CECT at initial presentation. Two experienced pelvic trauma surgeons independently reviewed all scans. Radiologically relevant hemorrhage was defined by contrast extravasation, soft-tissue asymmetry/displacement, or a contained intrapelvic hematoma. Patients with and without hemorrhage were compared regarding age, sex, trauma mechanism, anticoagulation, and fracture type. The presumed bleeding sources were categorized as arterial, venous, cancellous, or mixed.

RESULTS: Radiologically relevant hemorrhage occurred in 95/512 (18.6%) patients and was more frequent after high-energy trauma and in men. Iliac sites predominated, whereas presacral bleeding was uncommon. The presumed bleeding source was cancellous in 44.2%, mixed in 29.5%, arterial in 18.9%, and isolated venous in 7.4%. Overall mortality was 7.0% a […]

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## The new trochanteric nail TFN-A has a higher complication rate in the treatment of proximal femoral fractures than the predecessor model PFN-A.

*Kuttner H, Schläppi M, Meier C, Benninger E, Wahl P*

AIMS: Trochanteric fractures (TF) are among the most frequent fractures and have major impact on the affected patients and the healthcare system. Cephalomedullary nails (CMN) are widely used for these fractures. Complications remain frequent, either due to comorbidities or concerning the implant. The goal of this retrospective, single-centre study was to compare the complication rates of two CMN from one manufacturer.

MATERIALS AND METHODS: All internal fixations TF performed between 01/2017 and 07/2021 in our institution were analysed retrospectively. Follow-up was at least two years. The CMN were the PFN-A or the TFN-A. Demographic data, all reoperations and revisions were recorded. Reoperations and revisions were estimated using the Kaplan-Meier method, with subgrouping as minor reoperation, minor revision or major revision, including

conversion to total hip arthroplasty.

RESULTS: A total of 492 CMN, used in 475 patients, could be included. Among these, 248 were PFN-A and 244 were TFN-A. The overall reoperation rate was 15%, 12.9% for the PFN-A and 17.2% for the TFN-A, respectively. The TFN-A group had significantly more major revisions (hazard ratio 3.57 (95% confidence interval 1.42-9.94; p = 0.004). No relevant differences were found regarding demographic data or the quality of reduction and internal fixation.

DISCUSSION: The TFN-A had a significantly higher rate of co […]

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### Comparison of two immobilization techniques for radiological outcomes in fifth metacarpal neck fractures: a prospective controlled study.

*Süner U, Tunçez M, Sa■■ro■lu D, Önder Y, Özdemir M, Bulut T*

PURPOSE: Although fifth metacarpal neck fractures are commonly managed conservatively, there is no consensus regarding the optimal immobilization strategy. The aim of this study was to compare the radiographic outcomes of the Jahss Position Splint (JPS) and the conventional Ulnar Gutter Splint (UGS) in the treatment of fifth metacarpal neck fractures.

METHODS: A total of 78 adult male patients treated for isolated closed fifth metacarpal neck fractures between January 2023 and June 2024 were prospectively evaluated. Patients were allocated to either the JPS group (n = 42) or the UGS group (n = 36). Patients were assigned to JPS or UGS according to the predefined odd/even-calendar-day duty schedule of two surgeons, each of whom routinely applied one of the techniques. All patients underwent closed reduction using the Jahss maneuver followed by three weeks of immobilization. Radiographic outcomes included initial fracture angulation (IFA), post-reduction angulation (PRA), final follow-up angulation at 12 weeks (FFA), metacarpal shortening, and malunion rates.

RESULTS: No significant differences were observed between the groups regarding age or IFA (p > 0.05). Correction during reduction and splint application was greater with JPS than with UGS (15.2 ± 9.6° vs. 8.9 ± 11.0°; p = 0.009), whereas subsequent loss of reduction during immobilisation did not differ significantly betwee […]

PubMed · DOI

### Surgical management of orbital wall fractures in juveniles: a European multicenter study.

*Boffano P, Lahtinen A, Brucoli M, Ruslin M, Melle A, Stathopoulos P, Maniati D, Dediol E et al.*

INTRODUCTION: Isolated orbital floor fractures in juveniles are a challenge, as children and adolescents may experience diplopia and limited ocular motility, even with little to no periorbital edema or visible signs of injury to the eye. The aim of the present study is to assess the demographic and clinical variables, and the outcomes of pediatric patients affected by isolated orbital wall fractures at several European oral and maxillofacial surgery centers.

METHODS: Pediatric patients (less than 18 years) with hospitalized and surgically treated for unilateral isolated orbital blow out fractures of the orbital floor and/or the medial wall were included. The following data were recorded for each patient: gender, age, etiology of trauma, site of orbital wall fracture(s), timing of intervention, surgical approach, orbital reconstruction material/implant, post-operative complications RESULTS: A total of 51 pediatric patients (45 boys, 6 girls) with a mean age of 12.1 years were included. Most patients (43 children, 84.3%) presented with an isolated orbital floor fracture. Clinically, the most frequently observed sign was diplopia / double vision (43 patients, 84.3%). More than two thirds (68.6%) of cases were surgically treated within the first two days from injury. In 6 patients (11.8%) one or more persistent postoperative disturbances were found six months postoperatively, incl […]

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### Management and outcomes of moderate-to-severe pelvic ring fractures: a 10-year experience from an Australian Level 1 Trauma Centre.

*Tay TH, Rosley M, Patel B, Wullschleger M*

PURPOSE: Moderate-to-severe pelvic ring fractures are associated with high morbidity and mortality despite advances in resuscitation, development in trauma systems and multispecialty haemorrhage control. This study aimed to describe the management strategies and clinical outcomes of moderate-to-severe pelvic ring fractures over a 10-year period at an Australian Level I Trauma Centre.

METHODS: A retrospective observational cohort study was performed of patients aged ≥ 16 years with moderate-to-severe pelvic ring fractures (Abbreviated Injury Scale (AIS) ≥ 3) treated between January 2015 and December 2024. Data were obtained from the institutional prospectively collected trauma registry.

RESULTS: A total of 313 patients met the inclusion criteria. The mean age was 47.0 years and 69.3% were male. Motor vehicle (35.1%) and motorcycle collisions (24.9%) accounted for 60.1% of injuries. Median ISS was 26 (IQR 19-34), with 70.6% sustaining pelvic AIS 4-5 injuries. 20.1% of patients underwent at least one haemorrhage-control intervention, including angioembolisation (14.4%), preperitoneal pelvic packing (5.8%) and external fixation (5.8%). Angioembolisation utilisation increased from 6.7% in 2015 to 40.0% in 2024. In-hospital mortality was 7.7%, with traumatic brain injury accounting for 62.5% of deaths and haemorrhagic shock for only one death.

CONCLUSION: Moderate-to-severe pelvic […]

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## Clinical validation of the DRoPH score: a preoperative tool for estimating reduction difficulty in proximal humerus fractures.

*Barthel C, Rudolph F, Pape HC, Allemann F*

BACKGROUND: Proximal humerus fractures are common injuries that are frequently associated with considerable technical challenges during surgical reduction and fixation. Although anatomical reduction is crucial for fracture healing and functional recovery, the anticipated difficulty of achieving satisfactory reduction is difficult to estimate preoperatively. The Difficulty of Reduction of Proximal Humerus Fractures (DRoPH) score was recently developed through a structured expert consensus process to estimate anticipated reduction difficulty based on predefined fracture- and patient-related characteristics. The aim of this study was to perform an initial clinical validation of the score.

METHODS: We retrospectively analyzed patients who underwent plate fixation of proximal humerus fractures at a tertiary trauma center between January 2014 and December 2024. DRoPH scores (0-10 points) were stratified into low (0-3), moderate (4-6), and high (7-10) difficulty categories. Validation was based on operative duration, postoperative reduction quality according to the criteria described by Schnetzke et al., and postoperative complications. Multivariable regression analyses were performed to assess the independent association of the DRoPH score with the primary outcomes after adjustment for age, sex, and surgical approach. Inter- and intra-rater reliability were assessed using intraclass […]

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## Renal trauma in a pediatric Level 1 trauma center: morbidity, mortality, and the role of renal scintigraphy.

*Jansen SAM, Haouli B, Backes M, Tromp TN, Vriens D, Edwards MJR, Uitert AV, Nelen SD*

PURPOSE: Pediatric renal trauma is uncommon but potentially serious, and is often managed conservatively. Complications may occur, and the clinical value of renal scintigraphy during follow-up remains unclear. This study aimed to assess morbidity and mortality after pediatric renal trauma and to assess the clinical value of renal scintigraphy during follow-up.

METHODS: This retrospective single-center cohort study included pediatric trauma patients (< 18 years) with renal injuries admitted to a Level 1 trauma center in the Netherlands between January 1, 2001, and December 31, 2024. Mortality and in-hospital complications were assessed, and associations between renal scintigraphy abnormalities and complications during follow-up were evaluated.

RESULTS: A total of 72 patients were included, of whom all injuries resulted from blunt trauma. Conservative management was applied in 88% of cases; embolization and nephrectomy were performed in 8% and 3% of patients, respectively. In-hospital mortality was 11% and was attributed exclusively to concomitant traumatic brain injury. Complications occurred in 25% of patients, primarily infections (19%), followed by hypertension during

follow up (4%). Renal scintigraphy was performed in 53% of patients, predominantly in those with high-grade injuries; abnormalities were frequently observed (77%) but did not correlate with the occurrence of h […]

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### Measurement properties of pulmonary function measurement and patient-reported outcome measures in patients with multiple rib fracture without flail segment; validation study alongside the multicenter FixCon trial.

*Luttjeboer ML, Van Diepen MR, Prins JTH, Van Wijck SFM, Verhofstad MHJ, Wijffels MME, Van Lieshout EMM, FixCon study group*

BACKGROUND: The reliability and validity of pulmonary function, pain and health related quality of life (HR-QoL) as outcome measures for patients with rib fractures, remain uncertain. Our objective was to evaluate measurement properties in patients with multiple rib fractures by comparing them with generic HR-QoL instruments (EuroQoL-5D-5L [EQ-5D-5L], Short Form-12 [SF-12]).

METHODS: A multicenter RCT in adults (n = 180) was performed. Spirometry was performed at 30 days, 3, 6, and 12 months after trauma. Patients completed five questions on pain, EQ-5D-5L, SF-12, and two anchor questions on improvement in pulmonary function and thoracic complaints since previous measurement. Reliability, construct validity, responsiveness and floor and ceiling effects were evaluated for all follow-up time-points.

RESULTS: EQ-5D and SF-12 (sub)scales showed adequate internal consistency. Construct and longitudinal validity were adequate for all (sub)scales, except for construct validity of the SF-12 Physical Component Summary. Ceiling effects were present for all pain questions from 30 days onwards, for EQ-5D-5L utility score (US) from 3 months onwards. Responsiveness analysis indicated that spirometry values showed trivial (tidal volume), small (inspiratory capacity, FEV1), or moderate (FVC) degree of change from 30 days to 12 months. All five pain levels, EQ-5D-5L and SF-12 (sub)scales show […]

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### Prehospital severity and in-hospital mortality in trauma patients managed by the Friuli-Venezia Giulia helicopter emergency medical service: a 6-year retrospective cohort study.

*Tomasino S, Pegani C, Orso D, Ursella C, Tullio A, Bacchetti G, Molinari C, Marangone A et al.*

BACKGROUND: Data on in-hospital mortality among trauma patients managed by Helicopter Emergency Medical Services (HEMS) in Italy remain limited. We aimed to describe the epidemiology and in-hospital mortality of trauma patients transported after management by the Friuli-Venezia Giulia HEMS and to assess prehospital factors associated with death.

METHODS: We conducted a retrospective observational cohort study including trauma patients managed by HEMS FVG and transported to hospital, either directly by HEMS or after continuation of transport by ground emergency medical services, between 2018 and 2023. The primary outcome was in-hospital mortality. Jeffreys-prior penalized logistic regression was used to reduce sparse-data bias and instability from separation. The RTS-based model was designated as the primary model and the NACA-based model as a secondary sensitivity analysis. Model performance was assessed internally, and the functional forms of both age and RTS were examined using restricted cubic splines.

RESULTS: Of 3,357 trauma records in the source cohort, 205 prehospital deaths were excluded from the in-hospital mortality analysis, leaving 3,152 transported patients. Outcome data were available for 3,143 patients, of whom 107 died in hospital (3.4%). In the primary RTS-based model, older age, lower RTS, tracheal intubation, and prehospital CPR were associated with mortali […]

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### Surgeon-perceived intraoperative complexity is associated with early adverse outcomes after major emergency abdominal surgery: a prospective cohort study.

*Jensen LR, Degett TH, Solstad TU, Nielsen LS, Kokotovic D, Jensen TK, Burcharth J*

BACKGROUND: Intraoperative complexity in major emergency abdominal surgery lacks a standardized definition and is rarely quantified prospectively. Whether surgeon-perceived intraoperative complexity is associated with

short-term postoperative outcomes remains unknown.

METHODS: In this single-center prospective cohort study, 411 patients undergoing major emergency abdominal surgery were included. Immediately after surgery, the operating surgeon rated perceived intraoperative complexity on a 0-10 scale. The association between surgeon-perceived intraoperative complexity and the primary outcome, 30-day unplanned reoperation, was investigated using Fine-Gray regression. The model was adjusted for age, sex, body mass index, performance status, and intraoperative pathology. Secondary outcomes were 30-day mortality, length of stay, and Comprehensive Complication Index.

RESULTS: Unplanned reoperation occurred in 15% and 30-day mortality in 11%. Higher surgeon-perceived intraoperative complexity was associated with reoperation (sHR 1.20 per point; 95% CI 1.08-1.34). Complexity was also associated with prolonged LOS (β 1.47 days per unit increase; 95% CI 0.93-2.01), higher CCI category (OR 1.24 per unit increase; 95% CI 1.14-1.35), and increased 30-day mortality (OR 1.25; 95% CI 1.07-1.45).

CONCLUSION: Surgeon-perceived intraoperative complexity was associated with reoperation, morbid […]

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### The challenges and opportunities of major trauma care in Germany - results of multidisciplinary expert interviews of the LeAf trauma study.

*Neubert A, Hempe S, Bieler D, Jaekel C, Gaeth C, Höfer C, Bartha S, Schwojer E et al.*

BACKGROUND: The recovery process of major trauma survivors involves various sectors within a complex system of different medical service providers and stakeholders who accompany and support patients throughout the recovery pathway. The aim was to explore experiences and opinions of experts, who are actively involved in the care of major trauma patients in the German healthcare system.

METHODS: Semistructured in-depth group interviews were used to investigate experiences and opinions of experts with the focus on exploration of expert perspectives on opportunities and challenges of the cross sectoral care. An exploratory analysis was performed with an artificial intelligence-based coding software. In a second step, a system map was developed, and all inter- and intra-sectoral challenges and all opportunities for the optimization of processes were mapped.

RESULTS: 39 experts from 13 medical specialties and occupational groups who were interviewed highlighted that intersectional issues especially across and within sectors pose a significant challenge to patient care. Four challenges were highlighted that significantly influence patient care: (1) Communication between sectors; (2) Complexity of transitions between sectors; (3) Inadequate aftercare and (4) Lack of psychosocial care.

DISCUSSION: In conclusion five key learnings can be derived which could help to transform the healt […]

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### Costs and health-related quality of life of non-operative and surgical treatment modalities for displaced distal radius fractures: a comprehensive overview based on the CAST study.

*Barvelink B, Heemskerk SCM, Verhaar JAN, Reijman M, Colaris JW, CAST study group*

PURPOSE: The aim of this study was to provide a comprehensive overview of costs and health-related quality of life (HRQoL) associated with three treatment modalities for common displaced distal radius fractures (DRFs): (1) non-operative treatment, (2) direct plate fixation and (3) delayed plate fixation in case of fracture redisplacement during follow-up.

METHODS: A cost and HRQoL analysis was performed based on data collected during the CAST study, a multicenter cluster-randomized controlled trial including adult patients with a displaced DRF, randomized between immobilization with a plaster splint or circumferential cast. Included patients were prospectively followed for one year. Operatively treated patients also continued questionnaire follow-up. From a societal perspective, healthcare costs, patient and family costs, and indirect (i.e. productivity) costs were collected. Patients completed questionnaires on HRQoL (EQ-5D-5 L), healthcare consumption and work productivity.

RESULTS: In total, 693 patients were included in this study (non-operative: 402, direct plate fixation: 165, delayed plate fixation: 126). The estimated mean total costs per patient were €2,212 for non-operative treatment, €4,339 for direct plate fixation and €4,393 for delayed plate fixation HRQoL improved between 3 and 12 months but did not reach pre-injury HRQoL in all groups.

CONCLUSIONS: This study […]

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### Peri-periprosthetic femoral fractures: a single-centre case series of a rare and challenging clinical scenario.

*Matteotti R, Dalla Vecchia G, Masotti A, Monticelli L, Nardi M, Fiore L*

PURPOSE: Periprosthetic femoral fractures (PFFs) are an increasingly common complication of hip arthroplasty. A poorly defined scenario is a new femoral fracture after a previously treated PFF has united, which we term a peri-periprosthetic femoral fracture.

METHODS: We retrospectively reviewed 154 surgically treated PFF episodes in patients aged over 65 years at our institution between 2011 and 2021. A peri-periprosthetic fracture was defined as a new acute femoral fracture after documented radiographic union of a previously treated PFF. Clinical, surgical, and radiographic data were reviewed.

RESULTS: Seven patients (4.5% of the institutional PFF cohort) met the criteria. Mean age was 83.1 years and 6 (85.7%) were female. The mean interval to the peri-periprosthetic fracture was 37.3 months (median 27; range 7-105). By the Vancouver classification, 3 fractures were type B1, 1 type B3, and 3 type C. Treatment was individualised to fracture configuration, prosthetic stability, hardware, and bone stock. Radiographic union was documented in all 4 patients with adequate follow-up, at a mean of 4.3 months. Five patients regained ambulation; one with pre-existing limitations did not, and ambulatory status was unavailable for one. Medical complications occurred in 2 patients, and no major orthopaedic complication required reoperation. Mortality was 14.3% at one year and 42.9% at tw […]

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### In-hospital mortality in hip fracture patients undergoing surgery later than 24 hours post admission might rather reflect the impact of comorbidities and treatment limitations than additional risks imposed by the mere timely delay itself.

*Babayi M, Müller CW*

PURPOSE: Early surgery for proximal femoral fractures is widely recommended, but the association between surgical delay and short-term mortality remains uncertain because observational studies are vulnerable to confounding by patient characteristics and treatment limitation.

METHODS: We retrospectively studied adults with operatively treated proximal femoral fractures at a DGU-certified regional trauma center in Germany (2022-2025). Time to surgery was measured from emergency department registration to operative start. Delays > 24 h were classified as medical or non-medical where sufficiently documented. The primary outcome was in-hospital mortality. Adjusted analyses included age, sex, ASA category, fracture type, concomitant orthopaedic injury, and treatment limitation (DNR/DNI; not time-anchored to surgery) and used Firth penalized logistic regression.

RESULTS: Among 927 patients, 871 (94.0%) underwent surgery within 24 h; 53 (5.7%) died in hospital. Crude mortality was 5.6% after surgery within 24 h and 7.1% after surgery > 24 h. After adjustment, no clear independent association between surgery > 24 h and mortality was detected (aOR 1.03, 95% CI 0.29-2.92). Higher age, male sex, advanced ASA class, and treatment limitation were associated with mortality. In exploratory analyses, mortality was 19.0% among medically delayed patients and 0% among patients with non-medical d […]

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### Blast injury simulation in pigs using an air-driven shock wave generator: effects of shock waves of lateral shock wave exposure.

*Kiriu N, Sekine Y, Sasa R, Yamamura K, Aoki M, Nagamura T, Nishii K, Saitoh D et al.*

PURPOSE: We have hypothesized that respiratory arrest caused by temporary brainstem dysfunction is the primary cause of death during the hyperacute phase of blast injury and have demonstrated that this is highly likely. To date, we have examined the effects of shock waves on pigs by exposing them to shock waves from both the dorsal and ventral sides. Therefore, in this study, we investigated the effects of lateral shock wave exposure in pigs and identified priority regions for protection against blast injuries.

METHODS: Eight male hybrid pigs (age, 10-12 weeks; mean weight, 36.6 kg) were anesthetized and fitted with a femoral arterial line and subcutaneous electrocardiography leads for continuous monitoring of vital signs and respiration. Pigs were fixed prone with the right lateral thorax aligned to the aperture. A compressed air-driven shock tube generated a shock wave (driving pressure: 3.0 MPa). Arterial blood gases were obtained before and 0, 3, 5, and 10 min post-exposure. Whole-body computed tomography was performed pre- and post-exposure. Survivors were observed for 3 h, then underwent necropsy. Regarding lung injury, since "primary blast lung injury" is defined as "radiological and clinical evidence of acute lung injury occurring within 12 h of exposure and not due to secondary or tertiary injury", this study focuses on organ injury and mortality in the hyperacute and […]

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### Cumulative operative burden is associated with poorer function after segmental bone transport for lower-extremity long-bone defects.

*Y■ld■z M, Köro■lu M, Karaba■ ME, Çiçek ■B, Stevenson T, Harma A*

OBJECTIVE: Segmental bone transport offers biological reconstruction and limb preservation for extensive lower-extremity long-bone defects, but treatment often involves prolonged external fixation, planned stages, and unplanned reoperations. We evaluated reconstructive and functional outcomes and examined the association between cumulative operative burden and final patient-reported function.

METHODS: This retrospective single-centre study screened 44 patients treated between January 2012 and March 2023. Thirty-six patients, representing 37 reconstructed long bones, had complete final outcome ascertainment. ASAMI bone outcome was reported per reconstructed bone (n = 37); limb preservation and all functional and multivariable analyses were patient-level (n = 36). Outcomes included limb preservation, ASAMI bone and functional results, the Lower Extremity Functional Scale (LEFS), the Karlström-Olerud score, healing indices, treatment-related events, and the cumulative number of formal operations. Multivariable linear regression, bootstrap resampling, sensitivity analyses, and exploratory receiver operating characteristic analysis were performed.

RESULTS: Mean defect length was 10.32 ± 4.67 cm and the mean defect-to-bone length ratio was 26.3% ± 12.4%. Patients underwent a mean of 8.08 ± 4.21 operations (median, 7; range, 3-18). Mean follow-up after external-fixator removal was 5 […]

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### The utilisation of international consensus definitions of the disease polytrauma: a systematic review.

*Kim SY, Xu Z, Balogh ZJ*

PURPOSE: The disease of polytrauma is associated with significant healthcare burden and poor prognosis. The validated Newcastle and the Berlin consensus definitions of polytrauma have been proposed more than a decade ago for standardising clinical, scientific and advocacy related communications. This systematic review aims to evaluate the uptake of the Newcastle and the Berlin definitions of polytrauma since their introduction.

METHODS: This systematic review (PROSPERO: CRD420251119950) adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. A literature search of original research studies reported in English on polytrauma patients from 1st January 2015 to 31st of December 2025 was conducted, using Medline, Embase, Scopus, Pubmed, CINAHL and CENTRAL. Specific focus was on the utilisation of the Berlin and the Newcastle definitions, but all explicitly stated definitions of polytrauma were analysed.

RESULTS: A total of 2354 studies were identified through initial literature search. 1491 studies were excluded after screening by title and abstract. The full text review of 863 studies was performed and resulted in 88 eligible studies after the exclusion of 732 studies with no explicit definitions of polytrauma and 43 studies on non-human subjects. Ten studies (11.4%) used the Berlin definition of polytrauma, and six studies (6.8%) rep […]

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### Adhesive small bowel obstruction in the virgin abdomen: comparable phenotype, divergent management.

*Cioffi SPB, Martinez-Casas I, Marrano E, Cimbanassi S, Cimino MM, Kurihara H, Mohseni S, Kaplan LJ et al.*

BACKGROUND: In small bowel obstruction (SBO), the absence of prior abdominal surgery is commonly interpreted as a marker of higher-risk pathology and is often associated with a lower threshold for early operative intervention. Whether this assumption reflects differences in disease phenotype or decision-making remains uncertain.

METHODS: This post hoc analysis of a prospective, multicenter cohort study included consecutive adults with CT-confirmed mechanical SBO. Analyses were restricted to patients with adhesion-related SBO to enable comparison between virgin abdomen adhesive SBO (vaASBO) and prior-surgery adhesive SBO (psASBO). Clinical presentation, imaging findings, management strategies, and 60-day outcomes, including patient-reported outcome measures (PROMs), were evaluated. Analyses were associative and unadjusted.

RESULTS: Among 1,737 patients, 417 (24.0%) had no prior abdominal surgery; adhesions accounted for 30% of cases in this subgroup. The adhesion-restricted cohort included 982 patients, of whom 127 (12.9%) had vaASBO. Clinical and radiological features were similar between groups; however, management differed substantially. Patients with vaASBO had higher rates of direct-to-surgery (+ 15.0% points), lower use of water-soluble contrast (-19.1% points), and higher rates of nonoperative management (NOM) failure (+ 25.1% points). Among patients undergoing surgery […]

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## What to do with K-wires in displaced upper extremity fractures in children - where to bury or not.

*Frodl A, Borbely M, Karvouniaris N, Mühlenfeld N, Siegel M, Kuminack K, Pietsch C, Wagner FC et al.*

Kirschner wires (K-wires) are the standard fixation method for displaced pediatric upper extremity fractures, yet no consensus exists on whether wire ends should be buried subcutaneously or left exposed epicutaneously. While epicutaneous K-wires offer cost-effectiveness and avoid second anesthesia, concerns remain regarding infection risk and psychological distress. To systematically review and compare complication rates between buried and exposed K-wire fixation in pediatric upper extremity fractures, including infection, refracture, skin erosion, and secondary implant displacement. We conducted a systematic review following PRISMA guidelines, searching Cochrane, PubMed, EmBase, and Ovid databases. Inclusion criteria were: modified Coleman Methodology Score (mCMS) > 60, upper extremity fractures treated with K-wire fixation, and age < 16 years. Studies including adult patients and biomechanical studies were excluded. Meta-analysis compared outcomes across 12 studies encompassing 1,977 patients. Buried K-wires demonstrated significantly lower overall infection rates (OR = 1.74; 95% CI: 1.19-2.56; p = 0.004). However, no significant difference was observed in deep surgical siteinfections (OR = 1.56; 95% CI: 0.60-4.0; p = 0.36). Exposed wires showed significantly lower skin irritation rates (OR = 0.09; 95% CI: 0.04-0.21; p < 0.00001). No significant differences were found in seco […]

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## Exploring risk groups for concomitant injuries in facial fracture patients: a data-driven analysis.

*Mustasilta R, Kultanen E, Puolakkainen T, Haapanen A, Thorén H, Snäll J*

PURPOSE: Patients with facial fractures are a diverse group with varying injury mechanisms and extent of injury. These patients often suffer from concomitant injuries (CIs) of different severity outside the facial region, some of which are life-threatening and may require urgent care. Therefore, early identification of patients at risk for CI is important. This study aimed to investigate exploratory risk groups for concomitant injuries in patients with facial fractures, focusing on the impact of injury energy, sex and age on the prevalence and type of concomitant injury.

METHODS: In this retrospective cohort study, a data-driven analysis was used to perform statistical analysis and identify risk groups for different CIs. The study included 4170 patients with facial fractures evaluated at tertiary trauma centres from January 2013 to October 2020.

RESULTS: 70.9% of patients with facial fractures were male. Patients' mean age was 45.7 years. Of the injury mechanisms, 36.0% were classified as high-energy trauma. In all, 31.7% of facial fracture patients had some type of CI. Female patients had more CIs than male patients (34.0% vs. 30.8%). The occurrence of CIs increased progressively with age, peaking in the age group 90-94 years, where over 55% of patients experienced CIs. However, considerable variation was observed in CI types across age groups.

CONCLUSION: Due to the high C […]

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### Hydroxyapatite/poly-L-lactide mesh tube versus β-tricalcium phosphate granules in locking plate fixation of proximal humeral fractures: a retrospective comparative study.

*Iwata H, Takada N, Usami T, Hayashi Y, Murakami H, Kuroyanagi G*

PURPOSE: Proximal humeral fractures in older adults are often accompanied by metaphyseal bone defects, leading to reduction loss after plate fixation. Fibular allograft struts provide effective augmentation. However, their availability is limited. This study aimed to compare the outcomes of using hydroxyapatite/poly-L-lactide (HA/PLLA) mesh tubes as endosteal struts and β-tricalcium phosphate (β-TCP) granules for augmentation in locking plate osteosynthesis for proximal humeral fractures.

METHODS: This retrospective study included 42 adult patients who underwent locking plate fixation for acute, closed proximal humeral fractures at a single institution between 2013 and 2022. Among them, 20 patients received endosteal strut-HA/PLLA mesh tube (group E), and 22 received β-TCP granules (group G). Demographics, radiographic outcomes (humeral head height and neck-shaft angle), range of motion, complications, and surgical parameters were compared between the two groups.

RESULTS: Group E showed a significantly smaller mean change in humeral head height than group G (1.5 ± 3.1 vs. 4.0 ± 4.0 mm, p = 0.01). No significant difference in the mean change in neck-shaft angle was observed between the two groups (p = 0.31). Furthermore, no significant differences were observed in shoulder range of motion, complication rates, operative time, or blood loss between the two groups.

CONCLUSION: T […]

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### The burden of blast injuries on a level i trauma center: a comparative analysis of blast versus blunt trauma.

*Quint E, Barsky G, Osyntsov A, Replyanski I, Shaked G*

PURPOSE: We hypothesized that blast trauma results in a greater institutional resource burden than blunt trauma.

METHODS: This retrospective study compared blast trauma patients (N = 321) to blunt trauma patients (N = 9,172). A Propensity Score Matching (PSM) on Age, Sex and Injury Severity Score (ISS) was executed, yielding 1,481 matched blunt controls. The primary composite endpoint was Institutional Resource Burden, defined as requiring one or more of the following high-intensity resource metrics: prolonged hospital length of stay (LOS >7 days), ICU admission, or need for urgent surgical intervention (<6 hours from admission). Secondary endpoints included in-hospital mortality, shock-trauma room triage, any surgical intervention, multiple operations (≥ 3), massive transfusion protocol (MTP) activation, requirement for mechanical ventilation, and ICU/hospital LOS.

RESULTS: Blast patients demonstrated a higher rate of the primary composite institutional burden (78.2 vs. 51.2, OR = 3.42, 95% CI: 2.59 - 4.51, p <0.0001). Blast mechanism independently predicted shock-trauma room triage (OR = 10.60, 95% CI: 8.18 - 13.75, p <0.0001), urgent surgery (OR = 7.49, 95% CI: 5.73 - 9.80, p <0.0001), ≥3 operations (OR = 12.40, 95% CI: 8.59 - 17.90, p <0.0001), MTP activation (OR = 11.68, 95% CI: 6.48 - 21.05, p <0.0001), and ICU admission (OR = 5.54, 95% CI: 4.37 - 7.04, p <0.0001). Thes […]

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### Comparison of the shetty test and ottawa ankle rules for foot and ankle fractures in the emergency department: a prospective validation study.

*Gumustas I, Altinbilek E, Topcu H, Avlac A, Oztekin A, Karaketir M, Ozturker N*

PURPOSE: The Shetty test has been proposed as a simpler alternative to the Ottawa Ankle Rules (OAR) for excluding fracture in foot and ankle trauma and was reported to have perfect sensitivity in its derivation study. We compared its diagnostic accuracy with that of the OAR against imaging in an emergency department population.

METHODS: One hundred consecutive patients with acute foot and/or ankle trauma were enrolled after at least one of two attending physicians judged radiography indicated. Both index tests were performed independently by two blinded emergency physicians; where they differed, the patient was classified positive if either was positive. The

reference standard was fracture on radiography, or on CT obtained on clinical suspicion. McNemar and DeLong tests were used, and inter-test agreement Cohen kappa.

RESULTS: Fractures occurred in 26 patients (26%); 8 (30.8%) were occult, detected only on CT. The Shetty test had lower sensitivity than the OAR (76.9% vs. 92.3%) but higher specificity (67.6% vs. 18.9%). It missed 6 of 26 fractures, including two talar and one base-of-fifth-metatarsal fracture, all of which were managed with plaster-splint immobilisation. Used alone it would have avoided 56 radiographs while missing 6 fractures, versus 16 avoided and 2 missed for the OAR. Agreement was poor (kappa 0.19).

CONCLUSION: The Shetty test did not reproduce the sensit […]

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### Maxillofacial fractures associated with electric scooter trauma: a cohort study from a national level I trauma center.

*Tousidonis M, Gonzalez-Moure B, Fernandez-Acosta D, Rey-Valcarcel C, Perez-Diaz MD, Troulis MJ, Hintze JM, Ochandiano S et al.*

PURPOSE: Electric scooters (e-scooters) have become a widely used mode of urban transportation, but their increasing popularity has been accompanied by a rise in trauma-related emergency department visits. Craniofacial injuries represent a particularly relevant subset due to the exposed riding position and the limited use of protective equipment. This study analyzed the epidemiological characteristics, fracture patterns, associated injuries, and treatment requirements of patients presenting with facial fractures after e-scooter accidents.

METHODS: A retrospective observational study was conducted at a tertiary university hospital between January 2020 and December 2024. Adult patients presenting to the emergency department with facial fractures directly related to e-scooter accidents were included. Demographic variables, helmet use, alcohol and drug involvement, fracture distribution, associated systemic injuries, and treatment modalities were assessed.

RESULTS: A total of 42 patients with facial fractures were identified. Mean age was 29.4 years, and 76.2% were male. Helmet status was documented in 10/42 patients (23.8%); among documented cases, 3/10 (30.0%) reported helmet use. Alcohol status was documented in 14/42 patients (33.3%); among documented cases, 10/14 (71.4%) were positive. Drug status was documented in 15/42 patients (35.7%); among documented cases, 8/15 (53.3%) […]

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### The radial nerve bridging plate technique for humeral shaft fractures: an approach facilitating plate removal without impeding fracture union - a case series.

*Putzeys G, Nugteren LHT, Reynders-Frederix P, Hartog DD*

BACKGROUND: Removing a plate of the humeral shaft with the radial nerve on top of the plate is challenging and carries a high risk for iatrogenic radial nerve palsy (iRNP). This study aimed to assess the incidence of iRNPs following plate removal of a radial nerve bridging plate (RNBP) and to evaluate outcomes of this RNBP in treatment of a humeral shaft fracture.

METHODS: This was a single-center retrospective case series. Patients treated between May 2013 and December 2025 with a RNBP for a humeral shaft fracture or nonunion were included. The primary outcome measure was the incidence of iRNP after removal of the RNBP. Secondary outcome measures were the incidence of iRNP, malunion and nonunion after primary surgery.

RESULTS: Of 48 included patients, the mean follow-up duration was 24 months. In 11 patients the plate was removed, no iRNP occurred in these patients. Of 41 patients who had no primary RNP after the fracture, seven patients experienced a temporary iRNP after primary surgery. All of the RNPs fully recovered with a mean recovery time of 12.3 weeks (range: 1-32). There were three nonunions, one of which was a pathological fracture, and one was a nonunion of a surgical neck fracture in a multilevel fracture.

CONCLUSION: The RNBP technique allows safe and uneventful removal of the RNBP without iRNP, while maintaining the efficacy of plate fixation to treat humeral […]

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## The limited value of routine radiographs in the follow-up treatment of proximal humerus fractures.

*Elwakel AA, Willinge GJA, Goslings JC, Twigt BA*

PURPOSE: Proximal humerus fractures (PHFs) are common injuries in adults and occur particularly often in older patients. In our hospital, patients are routinely scheduled for follow-up radiographs at approximately two and six to eight weeks after injury, although the management yield of this practice is uncertain.

METHODS: We performed a retrospective cohort study of consecutive adults (≥ 18 years) presenting with PHFs to the OLVG emergency department (ED) between January and April 2022. Patients with multiple fractures, pathological fractures, fractures initially treated or followed up at another hospital, and fractures other than proximal humerus fractures were excluded. The primary outcome was a documented treatment-plan change during the observed outpatient treatment period, including conversion from conservative to operative treatment, cancellation of planned surgery, or additional surgery after initial operative treatment. The primary outcome proportion was reported with a two-sided exact 95% confidence interval. Descriptive variables included fracture classification, comminution, imaging use, follow-up timing, secondary displacement, and treatment duration.

RESULTS: One hundred patients were included (mean age 60.4 ± 18.0 years; 60% female). Most fractures were one-part (63%) or two-part (33%), 45% showed comminution, and 87% of patients were treated conservatively. Th […]

PubMed · DOI

## Waiting time and mortality after hip fracture: findings from a large single-center cohort study.

*Sundet M, Martinsen M, Sexton J, Sundin U, Lillegraven S*

PURPOSE: Time to surgery after hip fracture is a potentially modifiable factor that may influence mortality, but the evidence for an association is conflicting for waiting time < 48 h. The aim of this study was to investigate the association between waiting time to surgery and mortality, with waiting time categorized into 12-hour intervals.

METHODS: We analysed all hip fractures operated at a large Norwegian hospital between 2015 and 2023. Waiting time from admission to surgery was categorized into 12-hour intervals. Mortality at one week, one month, three months (pre-specified primary endpoint) and one year was assessed using multivariable logistic regression, adjusting for age, sex, nursing home residency, Charlson Comorbidity Index (CCI), fracture type, calendar year of surgery, and direct-acting oral anticoagulant (DOAC) use. Model fit and discrimination were assessed for all models.

RESULTS: The study included 4136 hip fractures. At the pre-specified primary endpoint of three months, mortality was numerically higher in patients with waiting times > 12 h, but this did not reach statistical significance for any waiting-time category (adjusted odds ratio [AOR] 1.16-1.25, all p > 0.2). Among the secondary, exploratory endpoints, the one-year timepoint showed increased mortality with increased waiting: patients operated between 12 and 24 h had significantly higher one-year mo […]

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## Diagnostic accuracy and predictive value of hard signs for operative or endovascular management and major injury in penetrating neck trauma: a single-centre retrospective cohort study.

*Barrientos ML, Zapata CAL, Toro DAM, López CAD*

PURPOSE: Hard signs guide operative decision-making in penetrating neck trauma (PNT), yet their accuracy and predictive value remain incompletely defined. We quantified the diagnostic accuracy of hard signs for major injury and for operative or endovascular management, and explored multivariable risk estimation.

METHODS: Single-centre retrospective cohort of 310 consecutive patients with PNT (June 2013-July 2025). Hard and soft signs were assigned by the attending surgeon at initial evaluation and abstracted retrospectively from the clinical record. The primary outcome was major injury (confirmed vascular or aerodigestive injury). Operative or endovascular management was analysed as a secondary outcome describing institutional practice, not as an independent diagnostic property, because the same signs informed both the index test and the treatment decision.

We estimated sensitivity, specificity, predictive values and likelihood ratios with exact binomial confidence intervals, and fitted exploratory multivariable models (STARD/TRIPOD).

RESULTS: Major injury occurred in 92/310 (29.7%) and operative or endovascular management in 96/310 (31.0%). Hard signs were specific but insensitive for major injury (sensitivity 55.4%, 95% CI 44.7-65.8; specificity 88.1%, 83.0-92.1) and for management (sensitivity 59.4%, 48.9-69.3; specificity 90.7%, 85.9-94.2). Outcome frequency rose across n […]

PubMed · DOI

### Epidemiology of spinal fractures in polytrauma patients.

*Malan SJ, Meakes S, Fischer A, Balogh ZJ*

PURPOSE: The epidemiology of spinal fractures in polytrauma, population characteristics and their hospital outcomes are not well defined. This study aimed to describe the epidemiology and outcomes of spinal fractures in a consensus defined polytrauma cohort.

METHODS: A 13-year (2009-2021) retrospective study was conducted on Newcastle Definition based polytrauma patients at a Level-1 centre. Polytrauma patients were categorised based on the presence of vertebral fractures and compared in demographics, injury mechanism, injury severity and pattern, physiological parameters, treatment aspects, and outcomes.

RESULTS: From the 1425 polytrauma patients, 632 (44.4%) sustained spinal fractures. These patients were older (48 vs. 38 years, $p < 0.001$) with identical injury severity (ISS = 29), were more frequently admitted (62.8% vs. 57.4%, $p = 0.037$) and also stayed longer in ICU (6 vs. 3 days, $p < 0.001$) and in hospital (14.5 vs. 12 days, $p = 0.002$). Mortality did not differ significantly (10.4% vs. 13.5%, $p = 0.08$). The lumbar spine was the most frequently fractured segment (7.3%), followed by thoracic (3.3%), cervical (3.3%) and sacral (3.1%) levels. Spinal cord injury occurred in 41 (6.5%) spinal fracture patients and was associated with a high ISS and prolonged hospitalisation [ISS:34(IQR 20.5); ICU LOS:10(IQR21.75); Hospital LOS:22(IQR41)]. Spinal fracture patients that classifi […]

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### Age-dependent injury mechanisms and injury patterns in pediatric window falls.

*Badina A, Deygas A, Saliba I, Corbu A, Da Costa EP, Georges S, Wicart P*

PURPOSE: Pediatric defenestration is a high-energy trauma mechanism in which age influences injury patterns and clinical severity. While previous studies have focused primarily on epidemiology and mortality, detailed analyses of age-related injury distribution and surgical burden remain limited. This study aimed to characterize age-dependent injury patterns, neurological severity, and hospital resource utilization following pediatric window falls.

METHODS: We conducted a retrospective single-center cohort study including all children aged 0-18 years admitted for defenestration between 2008 and 2023. Demographic data, fall height, intentionality, injury distribution, Glasgow Coma Scale (GCS), fracture patterns, and number of surgical procedures were collected. Multivariable logistic regression identified predictors of severe neurological impairment ($GCS \leq 8$), and Poisson regression assessed determinants of surgical burden.

RESULTS: A total of 333 children were included. Age distribution was bimodal, with predominantly accidental falls in young children (median 4.6 years) and intentional falls in adolescents (median 13.7 years). Younger children more frequently sustained craniofacial injuries and presented with lower admission GCS values, whereas adolescents more often sustained combined lower limb and spinal injuries. Fall height correlated negatively with GCS ($\rho = -0.32$) and […]

PubMed · DOI

### Clinical outcomes and risk factors in the surgical treatment of distal femur fractures.

*Höller LF, Höller S, Wittur P, Sehmisch S, Lehmann W, Weiser L*

PURPOSE: Distal femoral fractures are rare but severe injuries with high complication rates, particularly in elderly or multimorbid patients. This study aimed to analyze complication profiles and identify risk factors for surgery-related and non-surgery-related complications following operative treatment.

METHODS: A retrospective cohort study was conducted including all adult patients undergoing surgical treatment of distal femur fractures at a Level I trauma center between 2012 and 2018. Demographic, pre-operative, operative, and postoperative characteristics were analyzed. Risk factors were assessed using univariate statistics.

RESULTS: A total of 56 patients with 60 distal femur fractures were included. Surgery-related complications occurred in 11.7% of cases, while non-surgery-related complications were observed in 58.3%. In univariate analysis, gender ($p = 0.004$), age ($p = 0.014$), BMI ($p = 0.006$), osteoporosis ($p = 0.047$), open fracture ($p < 0.001$), high-energetic trauma (0.031) and time to surgery ($p = 0.024$) were significant risk factors for surgery-specific complications. For non-surgery-related complications, need of an external fixator for primary surgical care and type of operation were significant risk factors in univariate analysis. In multivariable logistic regression analysis, need of an external fixator was the only significant risk factor for developing non-s […]

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## Hip fracture surgery within 24 hours: no association between time to surgery and mortality in a real-world cohort.

*Pankratz C, Wolters M, Cintean R, Gebhard F, Schuetze K, Eickhoff A*

PURPOSE: Early surgery is recommended for hip fractures, and in Germany, operative treatment within 24 h has been mandatory since 2021. While delays beyond 24-48 h are associated with worse outcomes, it remains unclear whether further acceleration within a 24-hour timeframe improves survival. This study evaluated mortality and perioperative risk factors within a strictly implemented early-treatment pathway.

METHODS: This retrospective single-centre cohort study included 1,051 consecutive patients with femoral neck and trochanteric fractures treated surgically within 24 h of admission (2016-2020). Time to surgery was grouped into < 6 h, 6-12 h, and > 12-24 h. The primary outcome was one-year mortality; secondary outcome was two-year mortality. Mortality data were obtained from the national citizen registry. Multivariable logistic and Cox proportional hazards regression analyses were performed for one-year mortality.

RESULTS: Mean time to surgery was 10.2 ± 6.2 h. One-year mortality was 29.5% and increased to 40.1% at two years. Mortality did not differ significantly between time-to-surgery groups (< 6 h: 30.9%, 6-12 h: 26.4%, > 12-24 h: 31.5%; $p = 0.258$). This finding was consistent across all adjusted analyses. In contrast, non-surgical complications were the strongest independent predictors of mortality, particularly pneumonia (OR 2.1), renal failure (OR 5.2), and cardiac ev […]

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## Small bowel obstruction in older adults: similar nonoperative success but worse outcomes across management strategies - insights from the ESTES SnapSBO study.

*Bunino FM, Cimino MM, Martinez Casas I, Bellio G, Porta M, Cayre L, Mohseni S, Kaplan LJ et al.*

BACKGROUND: Small bowel obstruction (SBO) is a common surgical emergency. It remains unclear whether adverse outcomes in older patients reflect impaired obstruction resolution, differences in disease phenotype, or reduced physiologic tolerance.

METHODS: We performed a prespecified age-stratified analysis of the multicenter SnapSBO cohort, which consecutively enrolled adults with radiologically-confirmed mechanical SBO, stratified as younger (< 65 years) or older (≥ 65 years). Presentation, etiology, management, and outcomes were compared using risk differences and odds ratios. Exploratory analyses evaluated patients aged 65-79 years and ≥ 80 years.

RESULTS: Among 1,731 patients, 729 (42.1%) were aged ≥ 65 years. Older adults had more non-adhesive etiologies, including neoplastic and inflammatory obstruction, and more imaging features of severe disease. They were less likely to undergo a trial of nonoperative management (67.9% vs. 72.7%; absolute risk difference, - 4.8%; $P = 0.037$). However, among those managed nonoperatively, resolution rates were similar (68.8% vs. 69.5%). Despite this, older adults experienced higher rates of complications (16.7% vs. 6.7%), aspiration pneumonia (3.8% vs. 1.0%), and in-hospital mortality (6.9% vs. 3.6%). Those aged ≥ 80 years showed similar management but substantially higher mortality and complications.

CONCLUSIONS: Older adults achieved s […]

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## Pancreatic stent and transpapillary pseudocyst drainage via endoscopic retrograde cholangiopancreatography for children with blunt pancreatic duct injury.

*Kouchi K, Fumita T, Nakata C, Kagaya K*

BACKGROUND: Nonoperative management (NOM) in blunt pancreatic injury (BPI) is frequently complicated by pseudocyst formation and pancreatic atrophy. To address these complications, we have utilized a pancreatic stent via endoscopic retrograde cholangiopancreatography (ERCP) for children with PDI since 2019.

AIM: We present our experience with ERCP-based NOM for PDI and provide a literature review of stent therapy in pediatric PDI.

MATERIALS AND METHODS: Ten children with BPI treated between January 2007 and December 2025 were classified according to the 2024 American Association for the Surgery of Trauma pancreatic injury grades: IB (n = 3), IIA (n = 3), IIIA (n = 2), and IIIB (n = 2). Pancreatic stents were placed in two IIIA cases and one IIIB case. The IIIB case subsequently underwent transpapillary pseudocyst drainage (TPPD) due to a pseudocyst infection.

RESULTS: NOM was successful in all low-grade injuries (< II) without complications. One IIIB patient treated without stenting developed a pseudocyst requiring cyst-gastrostomy; 1-year follow-up demonstrated severe pancreatic atrophy. In the two IIIA stent cases, the duct injury was successfully bridged: one patient had no pseudocyst formation, and the other developed a pseudocyst that resolved spontaneously; both showed preserved pancreatic parenchyma. The IIIB patient treated with TPPD experienced rapid cyst resolution […]

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## Impact of pre-injury anticoagulation on outcomes in middle-aged trauma patients: a matched-pair analysis of the TraumaRegister DGU®.

*Koch DA, Faul P, Jakobi T, Schnetz M, Egert A, Münzberg M, Lefering R, Schweigkofler U et al.*

PURPOSE: Pre-injury anticoagulation is a well-known risk factor associated with adverse outcomes in geriatric trauma patients, but its impact in younger populations remains unclear. The study aimed to evaluate the association between pre-injury antithrombotic therapy and clinical outcomes in trauma patients aged 40-60 years.

METHODS: Data were obtained from the TraumaRegister DGU® (2015-2023). Patients aged 40-60 years from European trauma centers with ISS ≥ 9 were included. Patients were categorized according to the presence or absence of pre-injury antithrombotic therapy. A matched-pair analysis was performed based on age group, sex, ASA classification, and injury severity by body region. The primary endpoint was in-hospital mortality; secondary outcomes included coagulopathy, transfusion requirements, emergency surgery, and organ failure. Analyses were primarily descriptive.

RESULTS: A total of 64,215 patients were included, of whom 6.7% (n = 4,326) received antithrombotic therapy. Antiplatelet agents were most common (68%), followed by direct oral anticoagulants (DOAC) (18%) and Vitamin-K-Antagonists (VKA) (11%). Coagulopathy was more frequent in anticoagulated patients, particularly in those receiving VKA and DOAC. In unmatched analyses, mortality was slightly higher in anticoagulated patients (6.7% vs. 5.0%) and showed a non-significant trend towards higher mortality in […]

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## Correction: Impact of routine use of the GFAP and UCH-L1 TBI blood test for management of mild TBI on patient care in a European emergency department.

*Brüning-Wolter F, Wolff N, Schrader M, Musaelyan K, Chandran R, Hoffmann M, Marino JA, McQuiston B et al.*

PubMed · DOI

## From rib fixation to dynamic chest wall reconstruction: historical lessons for modern surgical stabilization of rib fractures.

*Sutori D, White TW, Bauman ZM, Whitbeck SS, Spering C*

PURPOSE: Surgical stabilization of rib fractures (SSRF) has re-emerged as an important component of chest wall trauma care, yet its history is often presented as a simple timeline rather than as a source of operative and biomechanical lessons. This literature, problem-oriented historical review re-examines earlier rib fixation strategies in order to identify the mechanical and operative problems they were intended to solve, why many disappeared,

and which principles remain relevant to contemporary SSRF practice.

METHODS: A literature review was conducted focusing on historically important fixation techniques and conceptual milestones in the management of serious rib fractures and flail chest. PubMed/MEDLINE and Google Scholar were searched from database inception to 19 March 2026 using terms related to rib fracture fixation, flail chest, SSRF, intramedullary fixation, and history. Studies were purposively selected if they described key fixation concepts, major shifts in management strategy, recurrent technical problems still seen in modern SSRF, or contemporary guideline context. Forty-seven articles were included and synthesized using a problem-based rather than purely chronological framework.

RESULTS: Across eras, rib fixation evolved around recurring challenges: (1) restoration of chest wall stability and thoracic continuity, (2) the tension between internal pneumatic stab […]

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### Patient-reported outcome measures within European cohorts of severely injured patients: a systematic review and meta-analysis.

*van den Berg L, van Ditshuizen JC, Leliefeld PHC, den Hartog D, Giannakópoulos GF, Verhofstad MHJ, de Jongh MAC*

PURPOSE: Severe injury affects multiple health-related domains, yet comprehensive European data on patient-reported outcomes remain limited. This systematic review and meta-analysis evaluates patient-reported outcome measures (PROMs) use and outcomes in severely injured European cohorts.

METHODS: A systematic search of four databases up to October 14, 2025, identified European studies from 2000 onward reporting PROMs in severely injured patients. Severe injury was defined as an Injury Severity Score ≥ 16, Glasgow Coma Scale ≤ 8, intensive care unit admission, spinal cord injury, traumatic amputations, or pelvic fractures. Two reviewers independently screened records, with disagreements resolved by a third reviewer. Meta-analysis was performed when ≥ 3 studies reported comparable PROMs at similar follow-up timepoints.

RESULTS: Of 2,479 studies, 119 were included. Most cohorts originated from the Netherlands (26%), Norway (18%), and Germany (16%). General severely injured cohorts were most frequently studied (61%), followed by traumatic brain injury (17%), and spinal cord injury (15%). In total, 94 PROMs were used across 277 follow-up timepoints. Health-related quality of life was assessed most frequently (63%), after that anxiety/depression (14%), post-traumatic stress (9%), and social functioning (6%). At one year follow-up, the pooled EuroQol-5D-3 L index score was 0.70 (95% […]

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### ABCB5■ mesenchymal stem cells enable high-ratio cellular augmentation of bone grafts without loss of osteogenic function.

*Hofmann J, Bewersdorf TN, Sommer U, Borcherding K, Thiel K, Ganss C, Gerstner M, Frank MH et al.*

Autologous bone grafting remains the standard for reconstruction of bone defects but is constrained by limited graft volume and loss of osteogenic capacity when expanded with acellular materials. We hypothesized that allogeneic ABCB5■ mesenchymal stem cells (MSCs) can serve as a biologic cell source to augment bone marrow-derived MSC (BM-MSC) populations without impairing osteogenesis. Primary human BM-MSCs from five donors were combined with GMP-manufactured ABCB5■ MSCs at graded replacement ratios (0-95%) and cultured under osteogenic conditions for 21 days. Osteogenic potential was evaluated using complementary assays of mineralization, biochemical activity, and matrix composition. Robust osteogenesis was maintained across all groups, with cultures containing 50% ABCB5■ MSCs demonstrating comparable functional osteogenic activity to BM-MSC controls, including similar 99mTc-HDP uptake (2.96 vs. 2.68 MBq) and calcium-phosphate ratios (1.2 vs. 1.1). Higher substitution levels showed a progressive decline in osteogenic output, consistent with dilution of BM-MSC-driven effects. These findings establish a quantitative threshold for biologic augmentation and demonstrate that substantial replacement of BM-MSCs is feasible without loss of function. This approach may enable cell-based expansion of bone grafts and supports further evaluation in translational in vivo models.

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### Surgical outcomes in an improvised underground operating room during missile attacks.

*Mitchnik IY, Essa A, Heinig O, Silberzweig JG, Ben Sira Y, Rabau O*

PURPOSE: Armed conflict disrupts healthcare when hospital infrastructure is vulnerable to missile attacks. During a 12-day missile strike period, our hospital established an improvised operating room (OR) in an underground parking lot. This study evaluated perioperative and short-term outcomes among patients treated in the improvised OR compared with those treated in the hospital's fortified standard OR.

METHODS: This retrospective cohort study included patients undergoing general or orthopaedic surgery during the 12-day conflict. Procedures identified via surgical logs were categorized by OR location (improvised or standard). Demographic, perioperative, and post-discharge data were collected from electronic records. Primary outcomes included operative time, estimated blood loss, transfusions, complications, hospital and ICU length of stay, ED visits, readmissions, and 3-month mortality.

RESULTS: Overall, 74 surgeries took place in the improvised OR and 77 in the standard OR. Patients in each group were comparable in age, sex, comorbidities, and anesthesia. Improvised OR surgery time was longer (mean 68.1 vs. 47.7 minutes, $p < 0.001$). There were no significant differences between groups in terms of blood loss, transfusions, postoperative complications, or hospital length of stay. After 3-month follow-up, there were no significant differences in readmissions or mortality rates […]

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## Repeated scoring with the adult appendicitis score improves the sensitivity and the specificity of appendicitis diagnosis in patients with early equivocal signs of appendicitis: a secondary analysis.

*Lastunen KS, Leppäniemi AK, Mentula PJ*

PURPOSE: The utilization of computed tomography in the early stage of acute appendicitis may result in overdiagnosis and unnecessarily expose patients to ionising radiation. The Adult Appendicitis Score (AAS) can be used to select patients for imaging. Observation and re-scoring in the DIAMOND trial reduced the need for imaging. Now, we wanted to determine if the change in AAS (ΔAAS) can serve as a diagnostic tool to select patients for imaging even more precisely.

METHODS: Eighty-eight patients with early equivocal appendicitis participated in the observation arm of the DIAMOND trial. The data for these patients were reanalysed, and ΔAAS during the observation was calculated. The baseline AAS, final AAS, and the change in C-reactive protein (ΔCRP) were selected as reference standards.

RESULTS: Eighty-three patients with complete data were included in the analysis. The AUROC (Area Under the Receiver Operating Characteristic) values are as follows: ΔAAS, 0.932 (95% CI 0.868-0.996); baseline AAS, 0.629 (95% CI 0.498-0.760); final AAS, 0.936 (95% CI 0.886-0.987); and ΔCRP, 0.796 (95% CI 0.696-0.897). Using receiver operating characteristic curves, we established the thresholds for low (AAS ≤ -2), intermediate (AAS -1 to 0), and high (AAS ≥ 1) probability of appendicitis. The negative predictive value for the low-probability group and the positive predictive value for the high-pr […]

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## Long-term patient-reported outcomes and quality of life after tibiofemoral knee dislocations in patients with open growth plates: a retrospective case series from a Level 1 trauma center.

*Hepp P, Youssef Y, Henkelmann R*

BACKGROUND: Tibiofemoral dislocations in children and adolescents are rare, and treatment guidelines are primarily based on adult data and algorithms. This case series of four pediatric and adolescent patients aims to add to the limited literature and highlights key aspects of diagnosis, treatment, and patient-reported outcomes.

METHODS: We conducted a retrospective-prospective single-center case series of pediatric and adolescent patients (< 18 years, open growth plates) treated for tibiofemoral knee dislocations at a Level 1 trauma center between 2017 and 2019. Clinical, radiographic, and intraoperative data were reviewed. In addition, long-term patient-reported outcomes covering both functional recovery and health-related quality of life were assessed using KOOS, IKDC, and EQ-55 D-L.

RESULTS: Four patients (2 male, 2 female), aged between 12 and 16 years, were included in this case series. The mean follow-up was 7.3 years (range 6 to 8 years). All patients sustained Schenck Type III-L dislocations. Early MRI and vascular imaging identified frequent associated injuries such as meniscal and chondral lesions, avulsion

fractures, in one case, peroneal nerve injury. A staged surgical approach was used in two of the four cases. No growth disturbances or deformities were observed in the patients based on the available clinical and radiological follow-up. Functional outcomes range […]

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### The need for, feasibility of, and barriers to implementation of a trauma traveling fellowship: a survey of ESTES members.

*Wohlgemut JM, Gupta S, Cioffi SPB, Llaquet-Bayo H, Wohlgemut HS, Hobby J, Mohseni S, Mariani D et al.*

PURPOSE: Many European surgeons lack access to high-volume trauma training. A Trauma Traveling Fellowship, supported by the European Society of Trauma and Emergency Surgery (ESTES), could broaden educational and clinical exposure to trauma care, but feasibility and implementation factors are unclear. The aims of this survey are to assess the aspirations, expectations, and perceived challenges of prospective fellows and host centres for an ESTES Trauma Traveling Fellowship.

METHODS: A cross-sectional, web-based survey was distributed to all ESTES members, capturing demographics, fellowship aspirations, perceived barriers, and host centre capacities. The survey incorporated Likert scales, free-text responses, and was guided by the Consolidated Framework for Implementation Research (CFIR) and the Reach, Effectiveness, Adoption, Implementation, and Maintenance / Practical Robust Implementation and Sustainability Model (RE-AIM/PRISM).

RESULTS: Of 903 invited, 210 responses (23.3%) were analysed. The median age was 38 years; 71% were consultants, and 29% were residents. Most (71%) would consider applying for a Fellowship, and 47% stated their centre could host Fellows. Only 23% currently access funded fellowships. Key aspirations included hands-on participation in trauma care, although hosts could more often only guarantee observational roles. Major barriers included funding and in […]

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### Antegrade transcystic balloon dilatation in emergency laparoscopic cholecystectomy with MRCP validation.

*Lipping E, Saar S, Kirss J, Aaspõllu H, Bahhir A, Gapejeva O, Kangur T, Rauk M et al.*

PURPOSE: Endoscopic retrograde cholangiopancreatography (ERCP) remains the predominant treatment of common bile duct stones but single-stage laparoscopic management via antegrade transcystic balloon dilatation (ATBD) offers a compelling alternative. All prior studies validate success solely by intraoperative fluoroscopic cholangiography, with no prospective trial employing postoperative magnetic resonance cholangiopancreatography (MRCP).

METHODS: This single-centre prospective study enrolled 100 consecutive patients with confirmed CBD stones requiring emergency laparoscopic cholecystectomy (LC). ATBD was attempted in all patients. Primary outcome was MRCP-confirmed CBD stone clearance. Secondary outcomes included operative time and postoperative complications.

RESULTS: One hundred consecutive patients were enrolled (median age 69 years; 38% male; 56% ASA III-IV). ATBD was performed in 88 patients (88%). Postoperative MRCP was performed in 86 of 88 patients (98%), with residual stones identified in 24 (28%; 95% CI 19.5-38.2%), corresponding to an MRCP-confirmed CBD clearance rate of 72% (95% CI 61.8-80.5%). The median operative time was 90 min (IQR 75-111.5). Among the 88 patients who underwent ATBD, postoperative complications occurred in 12 (14%), most commonly pancreatitis (4.5%) and deep surgical site infection (4.5%); one patient died following severe post-procedural panc […]

PubMed · DOI

### Closed reduction and percutaneous fixation versus open reduction and internal fixation for displaced medial malleolar fractures: a randomized controlled trial.

*Elkohail A, Elkasaby A, Shaheen AMZ, Teama M, Raza ML, Ibrahim IA*

OBJECTIVE: To evaluate the radiographic and functional outcomes of closed reduction and percutaneous fixation (CRPF) compared to open reduction and internal fixation (ORIF) in the treatment of isolated displaced medial malleolar fractures.

METHODS: This randomized controlled trial included 50 adult patients with isolated displaced medial malleolar fractures, Herscovici classification type B and C, across two hospitals. Participants were randomly assigned (1:1) to either CRPF (Group A, n = 25) or ORIF (Group B, n = 25). The principal objectives were radiological union and clinical function evaluated by the American Orthopedic Foot and Ankle Society (AOFAS) score at a 6-month follow-up. Secondary outcomes included the incidence of postoperative complications and correlations with patient demographics and mechanism of injury.

RESULTS: At 6 months, there was no significant difference between groups ($p > 0.05$); in Group A, 13 patients rated as "Good," 11 as "Fair," and 1 as "Poor," while in Group B, 15 were "Good," 9 "Fair," and 1 "Poor." AOFAS outcomes showed 13 excellent, 7 good, 4 fair, and 1 poor in Group A, versus 12 excellent, 9 good, 3 fair, and 1 poor in Group B, with no significant difference ($p > 0.05$). Complication rates were similar: wound infection (CRPF 4% vs. ORIF 12%), malunion/non-union (8% vs. 4%), hardware discomfort (12% vs. 20%), and deep venous thrombosis (4% […]

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### Long-term radiological and clinical outcomes after surgical treatment of non-union: a ten-year follow-up study.

*Böpple JC, Berger J, Ferbert T, Bewersdorf TN, Schmidmaier G, Findeisen S*

PURPOSE: Non-unions are a major challenge in trauma surgery. They are known to affect patients' lives for years after the initial trauma. Nevertheless, most studies only examine the first two years after treatment. The question what the long-term results look like ten years after surgery was posed in light of observations made in everyday clinical practice. To answer this, our study was designed to examine all patients who underwent non-union surgery in our department ten years ago.

METHODS: All patients who underwent surgery between July 2014 and October 2015 were invited to participate in the study. During a one-site appointment, an X-ray image was taken, which was then used to assess consolidation. The patients underwent a physical examination and answered a questionnaire, covering topics such as pain on a visual analogue scale (VAS) and overall quality of life with the short-form 12 (SF-12).

RESULTS: A total of 54 patients were included in the study. The median follow-up period was 122 months (IQR: 118-124). After 10 years 88.9% patients showed osseus consolidation with a median Lane-Sandhu-Score (LSS) of 4 (IQR: 3-4). The Wilcoxon signed-rank test revealed a significant difference compared to 24 months postoperatively ($Z = 4.716$, $p < 0.001$). The difference in the consolidation showed a marginally significant result in the McNemar test with $p = 0.063$. A total of 51 patien […]

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### Artificial intelligence for predicting 30-day mortality after emergency laparotomy.

*Bedrikovetski S, Murshed I, Seow W, Bunjo Z, De Fontgalland JS, Traeger L, Vather R*

PURPOSE: Accurate pre-operative risk prediction in emergency laparotomy (EL) is essential for appropriate resource allocation and improve patient outcomes. This study aimed to establish the accuracy of an artificial intelligence (AI) model in predicting 30-day mortality after EL using a national dataset.

METHODS: Data was extracted from the Australian and New Zealand Emergency Laparotomy Audit-Quality Improvement (ANZELA-QI) database from 1 July 2018 to 24 July 2023. AI models were created employing multilayer perceptron (MLP) and radial basis function (RBF) architectures. Extended AI models were developed using all available pre-operative variables and simplified AI models were developed using statistically significant variables identified through univariate analysis. We assessed the performance of AI models with that of a multivariate logistic regression (LR) and the UK-based National Emergency Laparotomy Audit (NELA) models using the area under the receiver operator characteristic curve (AUROC).

RESULTS: Data from 8293 ELs were included in the model and were randomly divided into a training dataset of 6619 (80%) and a testing dataset of 1674 (20%). There were 537 (6.5%) deaths 30 days after EL. In the testing dataset, the NELA model performed the best (AUROC 0.836), followed by the multivariate LR model (0.824), MLP simplified (0.817). MLP extended (0.802), RBF extended (0 […]

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## Predicting postoperative course in Fournier's gangrene surgery: evaluation of the E-PASS score in a single-center retrospective study.

*Gercek O, Senkol M, Ulucak MA, Yazar VM*

BACKGROUND: The E-PASS scoring system, which incorporates preoperative data and surgical stress factors, may provide a reliable risk classification for predicting mortality risk in patients with Fournier's gangrene. This study aimed to investigate the value of the E-PASS score in predicting mortality in patients with Fournier's gangrene.

METHODS: The study included 41 patients who were diagnosed with FG clinically and underwent surgery. Preoperative demographic and clinical data, laboratory values, FGSI, and E-PASS scores were recorded. The patients were divided into two groups: 8 (19.5%) patients who developed mortality and 33 (80.5%) patients who were discharged without mortality. Demographic, clinical, and laboratory data, as well as FGSI and E-PASS scores, were compared between the groups.

RESULTS: In the mortality group, CRP, serum creatinine levels, FGSI, E-PASS PRS, and E-PASS CRS scores were observed to be significantly higher, while serum albumin and hematocrit levels were lower compared to the other group (respectively, $p = 0.020$, $p = 0.001$, $p < 0.001$, $p < 0.001$, $p < 0.001$, $p = 0.001$, $p < 0.001$). FGSI, E-PASS CRS, CRP, and serum albumin values were found to have diagnostic value in predicting mortality. The cut-off values were determined as 7.5 for FGSI, 0.55 for E-PASS CRS, 186.25 for CRP, and 2.92 for albumin.

CONCLUSION: Our findings show that E-PASS can provide […]

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## Thoracoscopic internal fixation, open surgical internal fixation, and conservative treatment of rib fractures: differences in postoperative rehabilitation and chest volume (a randomized controlled trial).

*Li T, Zhao P, Li W, Liu B*

BACKGROUND: Currently, the standard treatment for multiple rib fractures is still under debate. This study explores the efficacy of three treatments (thoracoscopic internal fixation, open surgical internal fixation and conservative treatment) for multiple rib fractures and their influence on thoracic cavity volume and postoperative rehabilitation.

METHODS: Patients with multiple rib fractures admitted to Weifang People's Hospital of Shandong Province from March 2019 to March 2021 were enrolled in this study. The thoracic volume changes before and after treatment were measured by three-dimensional computed tomography (CT) reconstruction. Twenty-seven patients in the open operation group underwent internal fixation with the rib cage, and 24 patients in the non-internal fixation group underwent external fixation with the chest band. The recovery of multiple rib fractures was evaluated by comparing the difference in thoracic volume before and after treatment, the incidence of related complications and the Comprehensive Assessment of Generic Quality of Life Inventory-74 (GQOLI-74) score one month after the operation.

RESULTS: After 12 months of follow-up, thoracic volume was significantly different between the thoracoscopic and open fixation groups ($P < 0.001$). One month after the thoracoscopic internal fixation of rib fractures, the scores of physical function, psychological func […]

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## What is the rate and degree of deformity in femoral diaphyseal nonunion? A CT-based retrospective cohort study.

*Nabben J, Wagner RK, Groepenhoff F, Verwest TA, Bont Y, Kloen P, Janssen SJ*

BACKGROUND: This study investigated the rate and degree of deformity in femoral diaphyseal nonunion using preoperative bilateral CT-scans.

METHODS: A retrospective cohort study was conducted at a single Level 1 trauma center. Patients aged > 18 years who underwent operative treatment for femoral diaphyseal nonunion between 2015 and 2025 and had preoperative bilateral femur CT-scans were included. Axial, sagittal, coronal, and length differences between the femoral nonunion and the contralateral healthy femur were measured. Torsional deformity ≥ 15° and length discrepancy ≥ 20 mm were considered clinically relevant. Thresholds of ≥ 5° and ≥ 10° were used to identify

outliers in coronal and sagittal plane deformity, respectively.

RESULTS: Thirty-two patients met the inclusion criteria (median age 57 years, 63% male). Torsional deformity ≥ 15° was present in 10 patients (31%), including internal rotation in 4 (13%) and external rotation in 6 (19%). Sagittal plane deformity ≥ 10° was observed in 5 patients (16%), with apex anterior angulation in 1 (3%) and apex posterior angulation in 4 (13%). Mean sagittal plane angulation was 9.7° ± 6.7° in the affected limb and 12.5° ± 2.6° in the unaffected limb (p = 0.017). Coronal plane deformity ≥ 5° was present in 10 patients (31%), including varus in 8 (25%) and valgus in 2 (6%). Mean coronal angulation was 5.3° ± 6.0° in the affected li […]

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### The impact of pre- versus postoperative geriatric consultation in elderly patients with hip fractures.

*Aydin FB, Laane DWPM, van Rossum du Chattel A, DiStefano MT, King AH, Wignakumar T, Nwagbata A, van der Velde D et al.*

PURPOSE: To determine whether preoperative geriatric consultation was independently associated with lower incidence of delirium for elderly patients with hip fractures compared with postoperative geriatric consultation.

METHODS: This retrospective cohort study was conducted at two urban Level 1 trauma centers in the United States. Patients aged ≥ 70 years with isolated proximal femur fractures treated surgically between 2018 and 2024 were included. Patients with pathological or periprosthetic fractures, non-orthopedic admissions, and those without geriatric consultation were excluded. Patients were categorized based on timing of initial geriatric consultation: preoperative versus postoperative. Delirium was selected as the primary outcome because geriatric consultation most directly addresses modifiable delirium risk factors. Secondary outcomes included the incidence of one or more in-hospital complications, length of stay, 30-day all-cause readmissions, and mortality. Multivariable logistic and linear regression analyses were used to determine the independent association between timing of initial geriatric consultation and primary and secondary outcomes.

RESULTS: Of 1776 patients included, 1382 (78%) received initial geriatric consultation preoperatively and 394 (22%) postoperatively. Median age was 84 years (IQR 78-90), 70% were female, 73% ASA III, and 87% lived at home pr […]

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### Volume-dependent detection of free intraperitoneal fluid by FAST: a prospective in-vivo study with sequential fluid instillation.

*Güsgen C, Nada A, Kornek MT, Willms A, Schaaf S*

PURPOSE: To assess the volume-dependent sensitivity of the Focused Assessment with Sonography for Trauma (FAST) and to analyse regional detection performance under controlled in-vivo conditions.

METHODS: In this prospective single-centre study, sequential intraperitoneal instillation of 0.9% NaCl (250-1500 mL) was performed during elective laparoscopic procedures in 35 adults. Standardised FAST examinations were conducted by two experienced sonographers using transverse and longitudinal views of Morison's pouch, the Douglas recess, and Koller's space. Sensitivity was calculated for each instilled volume and anatomical region. Specificity could not be assessed due to the experimental design with mandatory fluid presence. Regional differences were tested using one-way ANOVA, and a Monte Carlo simulation (1000 resamples) evaluated detection stability across body-mass-index (BMI) groups.

RESULTS: FAST sensitivity increased with volume: 14.3% (95% CI 4.8-30.3) at 250 mL, 37.1% (21.5-55.1) at 500 mL, 74.3% (56.7-87.5) at 1000 mL, and 91.4% (76.9-98.2) at 1500 mL. The highest detection rate occurred in Morison's pouch, followed by the Douglas and Koller recesses. Longitudinal scanning of the Douglas pouch improved sensitivity by > 20% compared with the transverse view. ANOVA confirmed significant regional variability ($F(5, 153) = 4.37$, $p = 0.001$, $\eta^2 = 0.125$). Detection rates decline […]

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### When demography challenges policy: the rising burden of proximal femur fractures as a role model and driver for geriatric trauma care in Germany.

*Klug A, Münzberg M, Sibbel R*

PURPOSE: Germany's hospital reforms will substantially reshape service delivery and capacity planning. Proximal femur fractures (PFFs), the most frequent geriatric fractures, are highly sensitive to demographic aging and therefore serve as a model condition to assess future healthcare system requirements.

METHODS: Nationwide inpatient data from the German Federal Statistical Office (2000-2023) were analyzed for proximal femur fractures (ICD-10-GM S72.0-S72.2). Demographic projections up to 2050 were combined with Poisson regression, exponential smoothing, and autoregressive integrated moving average models to forecast future incidence and case numbers. Model performance was evaluated in a hold-out period (2016-2023) using mean absolute percentage error and root mean square error. Direct costs were estimated using the official German cost-of-illness framework. Staffing and hospital bed requirements were derived from case-based activity times, profession-specific full-time equivalent coefficients, and observed department-specific occupancy rates.

RESULTS: Between 2000 and 2023, annual proximal femur fracture cases in Germany increased from 118,243 to 182,447 (+ 54.3%). Forecasts project a rise to 317,877 cases by 2050 (95% CI 262,436-373,317), corresponding to an increase of approximately 75% compared with 2023. Patients aged ≥ 80 years are projected to account for 74% of all c […]

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## Orbital floor fractures: clinical characteristics, mechanisms of injury, and surgical outcomes in a 25-year single-center study.

*Kasap ■, Ayd■n MS, Özler G, Ni■anc■ M, Özvurmaz O*

PURPOSE: Orbital floor fractures are common injuries following blunt midfacial trauma, presenting as isolated defects or as part of complex craniofacial fractures. This study aimed to evaluate clinical characteristics, fracture patterns, management strategies, and postoperative outcomes in surgically treated patients.

METHODS: This retrospective single-center study included 56 patients who underwent surgical treatment for orbital floor fractures between 2000 and 2025. Demographic data, injury mechanisms, imaging findings, associated fractures, clinical presentation, surgical techniques, and outcomes were analyzed. Minimum follow-up was 3 months. Statistical analysis was performed using IBM SPSS Statistics version 25.0, with $p < 0.05$ considered significant.

RESULTS: Among 56 patients, 80.4% were male, with a mean age of 38.6 years (10-70). Assault (44.6%) was the most common cause, followed by in-vehicle accidents (21.4%) and falls (17.9%). Inferior wall fractures were most frequent (55.4%), followed by medial (28.6%) and lateral (16.1%) involvement; no superior fractures were observed. Common findings included periorbital edema/ecchymosis (75.0%), diplopia (26.8%), and gaze limitation (19.6%); infraorbital hypoesthesia occurred in 12.5%. Associated fractures involved the maxilla (60.7%) and zygoma (46.4%).Surgical treatment included exploration/elevation (46.4%), titanium mes […]

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## Epidemiology and healthcare burden of implant-associated complications in Germany: a nationwide population-based analysis.

*Anees H, Heiss C, El Khassawna T*

PURPOSE: Implant-associated complications remain a major challenge in orthopaedic and trauma surgery, particularly among elderly patients requiring revision procedures or infection management. This study analysed nationwide trends and healthcare burden of implant-associated complications in Germany.

METHODS: Nationwide German hospital discharge data from 2000 to 2023 were analysed and complemented by Diagnosis-Related Group (DRG) data from 2024. Cases were identified using ICD-10-GM code T84 and its subcategories. Outcomes included temporal trends, age-standardised incidence, demographic characteristics, in-hospital mortality, length of stay (LOS), and DRG-based healthcare resource utilisation.

RESULTS: Hospitalisations for implant-associated complications increased from 45,121 in 2000 to a peak of 95,154 in 2011, followed by a decline to 72,846 in 2023. Age-standardised incidence decreased by approximately 31% between 2010 and 2023. Segmented regression identified a temporal breakpoint in 2010.6, with annual percentage changes of + 6.21% before and - 3.55% after the breakpoint. Annual hospitalisations for periprosthetic joint infection increased from 5,367 to 11,215, while osteosynthesis-associated infections increased from 2,469 to 7,059. Mean LOS decreased from 20.8 to 11.8 days, whereas in-hospital mortality remained stable. Compared

with low-complexity cases, infection-r […]

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### Effect of periprosthetic greater trochanter fractures (Vancouver AG) on primary stability of the femoral stem in total hip arthroplasty: a biomechanical micromotion analysis.

*Bruder J, Goronzi M, Cavalcanti Kußmaul A, Wulf J, Pachmann F, Holzapfel BM, Böcker W, Kistler M*

PURPOSE: Primary stability is a crucial factor for the long-term success of cementless total hip arthroplasty (THA). Periprosthetic fractures represent an important cause of loss of primary implant stability. While Vancouver AG fractures rarely lead to stem loosening, their effect on the metaphyseal anchoring of the stems has not yet been investigated. The present study aimed to evaluate the influence of a Vancouver AG fracture on the primary stability of the CLS Spotorno stem.

METHODS: Six fourth-generation composite femora were implanted with cementless CLS Spotorno stems (Zimmer, Warsaw, USA). Primary stability was determined by three-dimensional digital image correlation under dynamic loading conditions (300-1700 N, 1 Hz) simulating physiological gait. Micromotion was recorded at six standardized measurement points (proximal, mid, distal, medial and ventral) before and after a simulated Vancouver AG fracture. Data was statistically analyzed using paired one-sided t-tests ($\alpha = 0.05$).

RESULTS: The results revealed no significant increase in micromotion at the primary anchorage zone of the stem. However, a non-significant increase in micromotion was observed at the prosthesis tip (DGventral and DGmedial), where the micromotion values increased from 164.4 µm (± 59.6 µm) and 140.4 µm (± 75.6 µm) to 193.1 µm (± 48.8 µm) and 186.9 µm (± 60.5 µm), respectively.

CONCLUSION: Vanco […]

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### Injury patterns, management and outcomes of hemodynamically unstable pelvic fractures at an Australian level 1 trauma centre: a 10-year registry analysis.

*Louw L, Raubenheimer K, Blakeney WG, Young S, Balogh ZJ, Weber DG*

INTRODUCTION/BACKGROUND: Pelvic ring injuries (PRI) in traumatic shock are associated with over 30% mortality even in high income countries' mature trauma systems, but recent Australian data showed no haemorrhage related mortality in this group. We aimed to describe the epidemiology, management and outcomes of PRI with haemodynamic instability managed in a state's only Level 1 adult trauma centre.

DESIGN: We conducted a retrospective cohort study of all trauma patients with major PRI (Abbreviated Injury Scale (AIS) > 3) and hemodynamic instability (defined as systolic blood pressure (SBP) <90 mmHg or a shock index (heart rate (HR)/SBP) > 0.7) presenting to the state adult trauma centre between 2013 and 2022. Data on demographics, injury patterns, management strategies, and outcomes were collected from the State Trauma Registry and hospital records.

RESULTS: Of 1369 patients with pelvic or acetabular fractures during the 10-year study period, 152 (11%) had major PRI with hemodynamic instability at presentation. Most were male (72%) with a median age of 35 years. Motor vehicle and motorbike crashes accounted for 52% of injuries. The median Injury Severity Score was 33. Massive transfusion protocols were activated in 45% of cases, while angioembolisation was utilised in 5.9%. Overall in-hospital mortality was 7.2% and mortality due to haemorrhage was 2%.

CONCLUSIONS: In our mat […]

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### Time to surgery and its impact on length of stay and mortality for neck of femur fractures at a major trauma centre.

*Tang LY, Ho SYS, Zhou AK, Geetala R, Krkovic M*

PURPOSE: Neck of femur (NOF) fractures are a major public health concern. Timely surgical intervention (current guidelines is within 36 h) is associated with improved survival, shorter hospital stays, and better functional outcomes. However, surgical delays remain common in hospitals, especially in major trauma centres, and their impact on length of stay (LOS) and mortality is not well defined across different NOF fracture subtypes.

METHODS: We conducted a retrospective cohort study of 4415 patients with NOF fractures, classified into intracapsular (n = 2507) and extracapsular (n = 1908). The primary outcome data gathered were LOS (from surgery to discharge) and in-hospital mortality. Regression analyses were performed to assess associations between LOS, and mortality, adjusting for age, sex, Charlson Comorbidity Index (CCI), American Society of Anesthesiologists (ASA) score, intra-operative blood transfusion, and surgical methods.

RESULTS: For intracapsular fractures, delays in time to surgery were significantly associated with increased LOS in patients expected to have short to median lengths of stay, after adjusting for other variables ($p < 0.05$). Delayed surgery was associated with increased mortality for both intracapsular and extracapsular fractures ($p < 0.0001$). Survival curves showed significant differences for the surgical delayed group. After controlling for time to […]

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## Pattern of blast injuries. A Systematic Review: Part 1-Improvised, multiple and unspecified explosive devices.

*Neubert A, Seidelmann M, Gaeth C, Bieler D*

PURPOSE: This systematic review aims to comprehensively identify and categorize the specific injury patterns associated with IEDs and other explosive devices across military and civilian populations to improve medical preparedness, resource planning strategies, and treatment approaches.

METHODS: A search on PubMed und Scopus was conducted on 26. March 2025 for observational studies reporting blast injuries in civilian and military populations of all ages. Studies were clustered by explosion type - Part 1 IED and multiple/unspecified explosives. A narrative data synthesis was performed. Reporting was assessed with the JBI tool for case series.

PROSPERO: CRD42023391736.

RESULTS: 33 studies with 4,373 victims (mainly military personnel) from 2002 - 2022 reported injuries caused by IEDs. The average age was 25.1 years. 97.2% were male. Injuries to the upper and lower extremities were predominant in all study populations followed by head injuries among military personnel, and thorax and abdomen injuries among civilians. 37 studies with 24,600 casualties (mean age 25.7, 85.8% males) were included in multiple/unspecified explosives. Extremities were predominant followed by facial injuries among military personnel, and thorax injuries among civilians.

CONCLUSION: The first part of this systematic review on the pattern of blast injuries shows that injuries to the extremities, includ […]

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## Clinical and laboratory predictors of mortality in Fournier's gangrene: prognostic role of the HALP score.

*Yavuz A, Ersöz V, Çamöz MS, Yurdakul ZN, ■■deci Tut G, Demirci BF, Koçsoy NS, Dönmez M*

OBJECTIVE: Fournier's gangrene (FG) is a rapidly progressive necrotizing soft-tissue infection associated with substantial mortality. This study evaluated the prognostic performance of the Hemoglobin-Albumin-Lymphocyte-Platelet (HALP) score and routinely available clinical and laboratory parameters in patients with FG.

METHODS: This retrospective cohort study included 236 consecutive patients who underwent surgical treatment for FG between February 2019 and May 2025. The primary outcome was in-hospital mortality. Prognostic factors were evaluated using univariate and multivariable logistic regression analyses, and discriminative performance was assessed by receiver operating characteristic (ROC) curve analysis.

RESULTS: The overall in-hospital mortality rate was 12.3%. Non-survivors had significantly lower HALP scores, hemoglobin, hematocrit, albumin, lymphocyte, and monocyte levels (all $p < 0.05$). In univariate analysis, younger age, disease confined to the perianal region, and higher albumin, monocyte, hemoglobin, hematocrit, and HALP values were associated with survival. In the primary multivariable model, advanced age (OR = 0.966, 95% CI: 0.936-0.997; $p = 0.030$) and monocyte count (OR = 5.243, 95% CI: 1.055-26.049; $p = 0.043$) remained independent predictors of survival. Hemoglobin showed the highest discriminative performance (AUC = 0.781), followed by albumin (0.767), he […]

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### Surgical stabilization of rib fractures is associated with lower adjusted odds of adverse pulmonary outcomes across RibScore-defined injury severity: a single-center cohort study.

*Muldiiarov V, Liu JL, Campbell A, Bouldoukian S, Daro A, Kemp SE, Matos M, Cantrell E et al.*

PURPOSE: Surgical stabilization of rib fractures (SSRF) is increasingly used for selected patients with severe chest wall injury, but its relationship with pulmonary outcomes after accounting for RibScore-defined injury severity remains unclear. This study evaluated adverse pulmonary outcomes after SSRF versus nonoperative management (NONOP) among adults with traumatic rib fractures.

METHODS: Single-center retrospective cohort study at an ACS-verified Level I trauma center from January 2016 to April 2023. Adults with CT-confirmed blunt traumatic rib fractures were categorized as SSRF or nonoperative management. RibScore was calculated from initial CT imaging. The primary outcome was a composite of pneumonia, invasive mechanical ventilation for more than 48 h, or tracheostomy. Multivariable logistic regression adjusted for demographics, RibScore, thoracic and extra-thoracic injury severity, calendar year, COPD, and current smoking status.

RESULTS: Among 3,066 patients, 444 underwent SSRF and 2,622 were managed nonoperatively. SSRF patients had greater radiographic chest wall injury severity, including a higher median RibScore than NONOP patients (2 vs. 0; $p < 0.001$). Composite adverse pulmonary outcome occurred in 14.6% of SSRF patients and 16.6% of NONOP patients. In adjusted analysis, SSRF was associated with lower odds of composite adverse pulmonary outcome (aOR 0.57, 95% C […]

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### Physician-staffed prehospital units for severely injured patients in the Stockholm trauma system - is there a survival benefit?

*Helmersson K, Lapidus O, Alm T, Bäckström D, Rubensson Wahlin R*

BACKGROUND: The role of prehospital physicians for patients with major trauma has previously been examined with mixed results. No studies have assessed the survival benefit of physician-staffed prehospital units in the context of the Stockholm trauma system.

AIM: To investigate the association between exposure to prehospital physicians and mortality for severely injured trauma patients in the Stockholm trauma system.

METHODS: A total of 11,063 trauma patients were obtained from the Swedish Trauma Registry and stratified according to physician exposure in the prehospital setting. Logistic regression was used to evaluate the association between prehospital physician exposure and 30-day mortality and Glasgow Outcome Score (GOS).

RESULTS: No significant reduction in 30-day mortality was observed in the P-EMS group compared to the EMS group (OR 1.06, $p = 0.71$). However, transport with helicopter emergency medical services (HEMS) was independently associated with greater odds of survival. The P-EMS group had significantly lower GOS at discharge compared to EMS (OR 1.72, $p < 0.001$).

CONCLUSION: The potential benefit of prehospital physician exposure for major trauma patients remains uncertain. However, HEMS transport was associated with improved survival. Further research is needed to determine which patients benefit from prehospital physicians and to evaluate the optimal allocati […]

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### Clinical impact and economic burden of electric scooter-related facial fractures in Barcelona.

*Lugilde-Guerbek A, López-Masramon B, García Sánchez V, Andreu-Sola V, Collado-Delfa JM, Pueyo Morillo C, Arteaga A, Grabosky-Elbaile A et al.*

PURPOSE: The landscape of facial trauma has evolved with the integration of micromobility devices into urban environments. This study aimed to characterize the clinical profile of electric scooter (e-scooter) accidents compared with traditional etiologies (traffic, assaults, falls) and evaluate the economic burden imposed on the public health system.

METHODS: A cohort of 119 patients who underwent surgery for facial fractures at a Spanish tertiary hospital between 2020 and 2024 was analyzed. To accurately assess economic impact, a specific subgroup of 51 "pure facial fractures"-those with isolated injuries-was utilized for cost analysis.

RESULTS: E-scooter users (10.1% of the total cohort) were significantly younger (mean age 27.9 years, p = 0.044), had shorter hospital stays (7.7 days, p = 0.007), and lower ICU requirements (16.7%) than the traditional group. Adverse outcomes accounted for 29.4% of cases overall, with a significantly higher rate in the e-scooter group compared to other etiologies (58.3% vs. 26.2%; p = 0.039). The median cost for an e-scooter injury was €8,092; though higher than other causes, this was not statistically significant (p = 0.33).

CONCLUSION: E-scooters generate complex facial injuries with clinical severity and complication rates comparable to high-energy trauma. The high economic cost and reconstructive complexity, suggest a need for stricter […]

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## Significant other reported outcome measures within European cohorts of severely injured patients: A systematic review.

*Van Ditshuizen JC, Berg LVD, Verhofstad MHJ, Leliefeld PHC, Jongh MAC, Hartog DD*

PURPOSE: Major trauma can have a long-term impact, leading to diminished health-related quality of life (HRQoL) for patients and their significant others (SOs). Patient reported outcome measures (PROMs) make this impact visible, providing expectation- and self-management. European studies assessing Significant Other Reported Outcome Measures (SOROMs) in cohorts of severely injured patients were reviewed.

METHODS: A systematic literature search was conducted. European studies from 2000 onwards assessing SOROMs in severely injured patient cohorts were included. Two reviewers independently screened all records, consensus was reached by a third reviewer.

RESULTS: 2,909 studies were screened, resulting in 30 inclusions. Studies originated from Western and Northern Europe, overrepresented by traumatic brain injury (n = 27, 90%). Forty-three different SOROMs were identified. Twenty (67%) studies reported subacute and long-term reduction in HRQoL, and high prevalence (28-56%) of depression, anxiety, and stress within SOs. Outcomes of SOs and patients run parallel and are associated with coping strategies. Sixteen (53%) studies reported around 50% of unmet family needs (information and emotional support), and a disproportionate burden of care associated with loneliness and life satisfaction.

CONCLUSION: SOs of severely injured patients reported reduced HRQoL, with high levels of depr […]

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## Early treatment goal uncertainty as a barrier to regional analgesia and respiratory deterioration in traumatic rib fractures: a propensity-weighted analysis.

*Yi SB, Wu JH, Zhang XN, Lin DC, Zheng YL*

PURPOSE: Patients with multiple rib fractures (≥ 3) admitted to the intensive care unit (ICU) are at high risk of early respiratory deterioration, for which timely and effective analgesia is essential. Although multimodal analgesic strategies are widely recommended, their real-world implementation may be influenced by variability in early clinical decision-making, including treatment goal uncertainty (TGU). We aimed to investigate the association between early TGU, analgesic strategy selection, and respiratory outcomes in critically injured patients with rib fractures.

METHODS: This single-center observational cohort study included 236 adult ICU patients with ≥ 3 radiologically confirmed rib fractures who did not require invasive mechanical ventilation at admission (2020-2024). TGU within the first 48 h was assessed using blinded electronic medical record review as a process-level marker of variability in early care prioritization. Analgesic strategies were categorized as regional-based analgesia (RA), systemic-dominant analgesia (SA), or multimodal conservative management. The primary outcome was respiratory deterioration, defined as a composite of escalation of respiratory support, endotracheal intubation, or pulmonary complications requiring intervention. Associations were evaluated using propensity score overlap weighting and adjusted regression models.

RESULTS: High TGU […]

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### Multifocal necrotizing fasciitis: what makes it different? A German trauma center experience.

*Keßler F, Frank J, Störmann P, Wagner N, Saman AE, Marzi I, Leiblein M*

BACKGROUND: Multifocal necrotizing fasciitis (MulNF) is a rare presentation of necrotizing fasciitis (NF) characterized by involvement of more than one non-contiguous anatomical site during the same disease episode. Comparative data between monofocal and multifocal NF remain limited.

METHODS: We performed a retrospective single-center cohort study of patients treated for NF between 2015 and 2026. Patients were classified as MNF (single site) or MulNF (multiple non-contiguous sites) based on chart review. Demographics, comorbidities, microbiology, laboratory values, organ failure, ICU course, surgical burden, and outcomes were compared.

RESULTS: Forty-seven patients were included, of whom five (10.6%) had multifocal NF. Four of these showed clearly synchronous non-contiguous involvement, whereas one developed a delayed second non-contiguous focus within the first four days of the same disease episode. Compared with MNF, MulNF was associated with higher ICU burden, more frequent mechanical ventilation, and a greater number of debridements and total operations. Higher rates of septic shock, hemodynamic failure, and renal replacement therapy were observed in MulNF. No deaths occurred in the small multifocal cohort, whereas 11/42 patients with monofocal NF died during hospitalization. Given the small multifocal sample size and multiple comparisons performed, all findings should be […]

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### Trauma-related retrobulbar hematoma: predictive risk factors and clinical outcome in 52 cases.

*Bicsák Á, ■ahin H, Brandt L, Hassfeld S, Bonitz L*

PURPOSE: Retrobulbar hematoma (RBH) is a vision-threatening emergency. This large-scale study integrates clinical data and 3D analysis to identify prognostic factors, refine diagnosis, and support guideline-based emergency management.

METHODS: This study standardizes terminology, applies AO fracture classification, and uses CT-based 3D segmentation to analyze anatomical patterns, treatment protocols, providing evidence-based guidance for diagnosis, surgical management, and postoperative care.

RESULTS: The Dortmund Maxillofacial Trauma Registry (2018-2025) recorded 52 RBHs among 7,671 trauma patients (0.7%). RBH occurred predominantly in middle-aged males and elderly females, with overproportionate incidence of anticoagulant therapy. Most fractures clustered around the orbital floor, medial wall, zygomatic surface, often near the Frankfurt horizontal plane. RBHs were mainly extraconal; poor outcome was predicted with intraconal location, optic nerve contact, orbital floor fracture, high RBH/orbital volume ratio, and aspirin use. Surgical decompression was performed in 15 cases (28.9%); 65,4% (34 of 52) retained vision. Neural network analysis confirmed fracture site and hematoma position as key prognostic factors over demographic or comorbidity variables.

CONCLUSION: In 7,671 midface fracture cases, RBHs occurred in 0.7% (n = 52), mainly after trauma. Falls predominated in ae […]

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### Patterns of negative trauma laparotomy worldwide: a sub-analysis of a global dataset.

*Laird WLA, Bath MF, Amoako J, Wohlgemut J, Nuño-Guzmán CM, Khajanchi M, Smith BG, Hobbs L et al.*

PURPOSE: Trauma remains a major contributor to global disability. While the trauma laparotomy can be a life-saving procedure in abdominal injury, a select proportion are performed with no pathology identified intra-operatively, potentially exposing patients to unnecessary morbidity. We aimed to assess the global prevalence of negative laparotomy and identify any influencing factors at both the patient- and system-level.

METHODS: This was a secondary analysis of the GOAL-Trauma study, a multicentre prospective international observational study on trauma laparotomy patients, conducted from April to December 2024. We defined a negative laparotomy as where no intra-abdominal injuries were identified during the index operation. A multivariable logistic regression was performed to identify predictive factors for a negative laparotomy.

RESULTS: Of the 1769 patients included, 128 patients (7.2%) underwent a negative laparotomy. Regression analysis demonstrated that a penetrating mechanism of injury (OR 2.37, CI:1.53-3.69, p < 0.001) and a lower grade of surgeon (surgical registrar relative to consultant - OR 2.91, CI:1.90-4.47, p < 0.001) were independent predictors for negative laparotomy. Neither human development index nor hospital resource level impacted negative laparotomy rates.

CONCLUSION: One in every fourteen trauma laparotomies performed in our global cohort was negative, […]

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### Preoperative stay under 48 h after admission for proximal femur fractures is linked to fewer complications and better outcomes.

*Moskal M, Feulner S, Grimm A, Müller-Heck R, Heep H, Godolias P*

PURPOSE: To evaluate the association between admission-to-surgery time (with a focus on < 48 h) and postoperative complications and discharge mobility in patients with proximal femur fractures (PFFs), using a large administrative dataset from routine inpatient care in Germany.

METHODS: In this retrospective database analysis, 128,092 patient records from external inpatient quality assurance (North Rhine-Westphalia, Germany, 2016-2020) were included. All patients underwent surgical treatment for PFFs and were stratified by time to surgery as early surgery (Group A; within 48 h of admission) or delayed surgery (Group B; more than 48 h after admission). Two cohorts were analysed separately: hip arthroplasty (HA: early 57,635; delayed 7,504) and osteosynthesis (OST: early 59,466; delayed 3,487). Associations between admission-to-surgery time, postoperative complications, pre-admission anticoagulant therapy and inability to walk at discharge were assessed using descriptive statistics and binary logistic regression. Patients unable to walk before the fracture were excluded from the mobility analysis. Reporting was guided by the STROBE statement.

RESULTS: In-hospital mortality was 4.7% after early versus 9.7% after delayed surgery (OST) and 5.8% versus 9.3% (HA). Delayed surgery was associated with higher odds of postoperative complications (OST: OR 1.21, 95% CI 1.11-1.34; HA: OR 1. […]

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### Impact of routine use of the GFAP and UCH-L1 TBI blood test for management of mild TBI on patient care in a European emergency department.

*Brüning-Wolter F, Wolff N, Schrader M, Musaelyan K, Chandran R, Hoffmann M, Marino JA, McQuiston B et al.*

OBJECTIVES: Blood biomarkers glial fibrillary acidic protein (GFAP) and ubiquitin C-terminal hydrolase L1 (UCH-L1) were recently introduced into clinical practice for mild traumatic brain injury (mTBI) assessment. However, real world data demonstrating TBI biomarkers' impact on the emergency department (ED) practice is lacking. This study aimed to describe the experience of a single tertiary hospital ED that introduced TBI biomarkers GFAP and UCH-L1 measurements using Alinity i® TBI test into routine mTBI assessment.

METHODS: Retrospective data were extracted from hospital records of patients who had TBI test ordered in the ED. Test results and subsequent clinical care pathway data (CT order, CT scan result, admission or discharge) were analysed.

RESULTS: Data was available for 460 mTBI patients (median age 43, 54% male). Of the 216 patients (47%) with a negative TBI test, 198 did not have a CT scan, and 18 had a CT scan with no abnormalities detected. Among 244 patients with a positive TBI test, 162 had CT scans, with 12 (6.6%) showing intracranial lesions. Overall, among all 180 scanned patients every patient with intracranial lesion findings had a positive TBI test result. The test could rule out CT scans in 47% of mild TBI patients. In practice, 42% of patients avoided a CT scan based on the TBI test results with additional 12% of patients ruled out based on clinical asse […]

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## Electric bicycle versus conventional bicycle crashes: comparative analysis of injury patterns, severity, and the modifying effect of body mass index.

*Bauer D, Maier JP, Holzfurtner L, Pietsch C, Wagner FC, Schmal H, Erdle B, Mühlenfeld N*

PURPOSE: E-bike trauma is often presumed to result in greater injury severity than conventional cycling, yet evidence remains inconsistent. Furthermore, the impact of factors such as Body Mass Index (BMI) on injury patterns is poorly understood. This study therefore evaluated whether e-bike trauma is associated with increased overall severity or distinct regional injury patterns.

METHODS: This retrospective single-center cohort study analyzed 261 adult trauma-bay patients (e-bike n = 37; conventional bicycle n = 224). Primary endpoints included Injury Severity Score (ISS) and regional maximum Abbreviated Injury Scale (AIS) scores. Secondary endpoints included physiological burden (SAPS II) and resource utilization (LOS, TISS-10). Multivariate logistic regression was used to identify independent predictors of intracranial hemorrhage, e-bike riding and extremity fractures. Event counts and events-per-variable (EPV) ratios are reported for each model.

RESULTS: E-bike riders had a significantly higher BMI than conventional cyclists (26.1 vs. 24.6 kg/m²; p = 0.029). However, the absolute difference of approximately 1.5 kg/m² is modest, and both group medians fall within the normal-to-overweight range. No statistically significant differences were observed in ISS, physiological burden, or resource utilization. Conventional cyclists had a higher proportion of patients with AIS Pelvi […]

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## Effect of management in intermediate care unit versus conventional hospitalization on long-term mortality and autonomy in patients with rib fractures aged 75 years and older: a retrospective study.

*Vallod E, Debord-Peguet S, Mikail N, Termoz A, Tripoz S, Abrard S, Gueret-du-Manoir M*

PURPOSE: Rib fractures in older adults are associated with substantial morbidity and mortality. Management in an intermediate care unit (IMCU) may improve outcomes, but its long-term benefit remains uncertain. We compared IMCU versus conventional management regarding 12-month mortality and autonomy in patients aged ≥ 75 years with rib fractures.

METHODS: We conducted a retrospective observational cohort study in two Lyon teaching hospitals (Hôpital Edouard Herriot and Hôpital Lyon Sud, Hospices Civils de Lyon, France) between January 2015 and December 2019. The primary outcome was a composite of death or institutionalization within 12 months of hospital admission. Secondary outcomes included discharge modality, opioid prescription at discharge, hospital length of stay, unplanned readmission within 6 months, and ADL/IADL scores. Logistic regression was adjusted for hospitalization unit, age, sex, pre-existing institutionalization, and associated injuries.

RESULTS: A total of 196 patients were included: 126 (64%) in the conventional group and 70 (36%) in the IMCU group. Death or institutionalization within 12 months occurred in 31 patients (26%) in the conventional group and 8 (12%) in the IMCU group (p = 0.026). However, after adjustment, conventional hospitalization was not significantly associated with the primary outcome (OR 2.25, 95% CI 0.90-6.09; p = 0.092). Institutional […]

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## The road to the Pearl-String Technique: lessons learned from early IMT modifications using RIA autograft and cancellous allograft granules.

*Hückstädt M, Scholl C, Langwald S, Klauke F, Mendel T, Kobbe P, Fischer C*

BACKGROUND: The induced membrane technique (IMT) is widely used for reconstruction of large bone defects; however, reported outcomes vary considerably among technical modifications. One potential mechanism impairing consolidation is graft sedimentation, particularly when low-density cancellous autograft harvested using the reamer-irrigator-aspirator (RIA) is applied. This study evaluates a standardized modification of the IMT and explores mechanical factors potentially affecting graft consolidation.

METHODS: In this retrospective single-center study, 28 patients were treated following eradication of osteomyelitis using a defined modification of the IMT. Mean follow-up was 7.4 years. Defect length ranged from 3.2 to 15.7 cm. During the second stage, RIA-harvested autologous cancellous bone was combined with lyophilized cancellous

bone allograft granules. Definitive fixation was achieved using intramedullary nailing or locking plate fixation; one patient was treated with a ring fixator.

RESULTS: Primary consolidation was achieved in 57.1% of cases. Aseptic nonunion occurred in 10.7%, with secondary consolidation achieved after revision in all affected patients. Recurrent infection was observed in 35.7%; infection control was achieved in 90% of these cases following additional procedures, and reconstruction was ultimately completed using alternative strategies. Median time to fu […]

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### Non-tourniquet strategy with high-position low-perfusion in distal humeral fracture surgery: reduces swelling without increasing blood loss.

*Wang Z, Ai X, Ren T, Wu B, Zhang K, Li Z, Xue H, Li M et al.*

INTRODUCTION: To evaluate the feasibility of non-tourniquet strategy combined with "High-position Low-Perfusion" positioning in distal humeral fractures ORIF (DHF-ORIF), and to investigate its effects on perioperative blood loss, postoperative rehabilitation, and functional outcomes.

METHODS: Prospective inclusion of patients aged 18-45 with distal humeral fractures, randomly divided into an experimental group without tourniquets (Group A) and a control group with traditional tourniquets (Group B). The Group A adopted a lateral position with high and low perfusion of the affected limb, and received intravenous injection of TXA (15 mg/kg) combined with continuous flushing with 4 ■ physiological saline during the operation; The control group received routine use of tourniquets.

PRIMARY OUTCOMES: perioperative blood loss, elbow ROM and degree of limb swelling.

SECONDARY OUTCOMES: Postoperative VAS score, Mayo elbow joint score and DASH score, and incidence of complications.

RESULTS: No significant difference was observed in total perioperative blood loss between the Group A and the Group B (240.78 mL vs. 228.53 mL, $p > 0.05$). However, the Group A demonstrated significantly reduced postoperative drainage volume and hidden blood loss ($p < 0.05$). The Group A exhibited significantly superior outcomes in terms of VAS scores and limb swelling, compared with the control group. Additi […]

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### Impacts of intracranial pressure monitoring on mortality and length of stay in severe traumatic brain injury: a systematic review and meta-analysis.

*Merakis M, Kim N, Velli G, Balogh ZJ*

PURPOSE: This systematic review and meta-analysis examined the association between intracranial pressure monitoring (ICPm) and mortality in patients with severe traumatic brain injury (sTBI), with additional focus on functional neurological outcomes and intensive care unit (ICU) and hospital length of stay (LOS).

METHODS: This review followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines and was registered with PROSPERO (CRD42025643607). MEDLINE, Embase, and the Cochrane Central Register of Controlled Trials were searched through August 1, 2025, for studies comparing ICP-monitored and non-ICP-monitored patients with traumatic brain injury. Eligible study designs included randomized controlled trials, prospective observational cohort studies, retrospective observational cohort studies, registry-based cohort studies, and case-control studies. Study quality was assessed using the Strengthening the Reporting of Observational Studies in Epidemiology checklist, Risk of Bias 2.0 for randomized trials, and the Oxford Centre for Evidence-Based Medicine Levels of Evidence framework.

RESULTS: Thirty-one studies met inclusion criteria. The mortality meta-analysis included 121,799 patients. Pooled crude analysis showed lower mortality in ICP-monitored patients than in non-monitored patients (OR 0.79, 95% CI 0.68-0.91, $p = 0.002$), with high heteroge […]

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### Prehospital point-of-care lactate for identification of severe injury in trauma team-activated patients: a prospective multicenter cohort study.

*Faul P, Hagebusch P, Bertram E, Naujoks F, Störmann P, Schweigkofler U, Koch D*

BACKGROUND: Accurate prehospital trauma triage remains challenging despite structured activation criteria, particularly in patients without overt physiologic instability. Prehospital point-of-care lactate has been proposed as a physiologic adjunct to support early identification of severely injured patients. This prospective multicenter cohort study evaluated the association between prehospital lactate and injury severity in trauma team-activated patients within a structured guideline-based trauma system.

METHODS: This prospective observational multicenter cohort study included adult trauma patients transported by emergency medical services to three Level I trauma centers and managed in the trauma room following trauma team activation (TTA). Prehospital lactate was measured by capillary earlobe sampling before or during transport and was not used for clinical decision-making. The primary outcome was severe injury, defined as Injury Severity Score (ISS) ≥16. Secondary outcomes were Abbreviated Injury Scale (AIS) ≥4 in at least one body region and intensive care unit (ICU) admission > 24 hours. Diagnostic performance was assessed using receiver operating characteristic (ROC) analysis and predefined lactate thresholds of ≥2 mmol/L and ≥4 mmol/L.

RESULTS: A total of 108 patients were included, of whom 26 (24.1%) had ISS ≥ 16. Prehospital lactate levels were higher in severely inj […]

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## Surgical timing and mortality in acute traumatic subdural hematoma: is the four-hour threshold confounded by injury severity?

*Rakip U, Y■ld■zhan S, Canbek ■, Boyac■ MG, Korkmaz S, Öztürk KH, Aslan MZ, Aslan A*

PURPOSE: Early surgical evacuation within four hours has long been advocated for acute traumatic subdural hematoma (ASDH), yet the supporting evidence is largely observational and unadjusted for injury severity. We examined whether surgical timing is independently associated with in-hospital mortality after accounting for baseline severity.

METHODS: We retrospectively analyzed 91 consecutive patients who underwent surgical evacuation for ASDH between January 2014 and December 2024. The primary outcome was in-hospital mortality; the secondary outcome was functional status at discharge measured by the Glasgow Outcome Scale (GOS). Surgical timing (≤ 4 vs. >4 h) was assessed through unadjusted comparison, stratification by Glasgow Coma Scale (GCS) score, multivariable Firth penalized logistic regression with 1000-iteration bootstrap internal validation, and sensitivity analyses including alternative timing thresholds (1-12 h), restricted cubic spline modeling, and timing-by-severity interaction testing.

RESULTS: In-hospital mortality was 33.0% (30/91; 95% Wilson CI 24.2-43.1%). Crude analysis showed higher mortality with early surgery (50.0% vs. 15.6%, p = 0.001), but these patients had lower admission GCS and higher rates of CT-defined traumatic axonal injury (TAI) and cerebral contusion. After stratification, no statistically detectable mortality difference was observed within […]

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## Association between time to transfusion and outcomes in patients with traumatic haemorrhagic shock: an analysis of a nationwide trauma registry in Japan.

*Shibahashi K, Inoue K, Kato T, Sugiyama K*

PURPOSE: To investigate the association of in-hospital time to transfusion on outcomes of patients with traumatic haemorrhagic shock.

METHODS: Using data from the Japan Trauma Data Bank (January 2019-December 2023), we identified adult patients (aged ≥ 18 years) with haemorrhagic shock following trauma (systolic blood pressure < 90 mmHg; shock index ≥ 1.0 upon hospital arrival) and received ≥ 10 units red blood cells within 24 h of injury. We excluded patients transferred from or to other hospitals after emergency treatment, with burn injuries or cardiac arrest, without transfusion time records, and with first transfusion > 60 min after hospital arrival. Mixed-effect multivariable logistic regression estimated the association between time to transfusion for in-hospital mortality, adjusting for patient characteristics, comorbidities, and injury severity. Heterogeneity in the association between transfusion timing and mortality was assessed through nine subgroup analyses.

RESULTS: Overall, 698 patients (median age 59 years, 66.2% male) from 132 hospitals were analysed, of whom 265 (38.0%) died during hospitalisation. Mean time to transfusion varied from 4 to 58 min across hospitals. A

positive association was observed between time to transfusion and in-hospital mortality. Each 1-min delay in transfusion initiation was associated with higher in-hospital mortality (adjusted odds […]

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### Neutrophil phenotyping reflects injury severity when base excess and lactate remain normal.

*van Eerten FJC, Duebel LV, de Fraiture EJ, Brands SR, van Wessem KJP, Vrisekoop N, Venhuizen JH, de Groot MCH et al.*

INTRODUCTION: Accurate assessment of injury severity is essential for optimal trauma care, as it dictates timing and intensity of interventional strategy. Conventional markers, such as base excess and lactate levels are usually borderline deranged in patients with midrange injury severity. This study explored whether determining circulating neutrophil phenotypes in these patients (ISS 9-35) might aid in the assessment of injury severity.

METHODS: A descriptive single-center cohort study of trauma patients ≥ 16 years between 2013 and 2023 was conducted. Data was collected from the Dutch National Trauma Registry and the Utrecht Patient-Oriented Database (< 10% missing values). Base excess, lactate levels and neutrophil phenotypes were routinely measured in emergency department blood samples. The dynamics of these parameters was analyzed in the context of increasing Injury Severity Scores (ISS).

RESULTS: 807 patients were included, of whom 591 were classified as midrange severely injured. In these patients, banded neutrophils showed a greater fold increase between ISS 9 and 35 compared with base excess and lactate (6.7, 2.7, and 1.5, respectively). Furthermore, the number of banded neutrophils exceeded the upper interquartile range of controls at lower trauma severities compared to base excess or lactate (ISS 17 vs. 30 and 39, respectively).

DISCUSSION: Although base excess and […]

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### Limb occlusion pressure-based individualized tourniquet pressure management in patients undergoing internal fixation for extremity fractures: a prospective comparative feasibility study.

*Yang J, Hu P, Wang L*

OBJECTIVE: This study aimed to evaluate the feasibility of limb occlusion pressure (LOP)-based individualized tourniquet pressure management and to quantify the reduction in applied tourniquet pressure compared with a traditional fixed-pressure protocol in patients undergoing internal fixation for extremity fractures.

METHODS: A prospective comparative study was conducted at Civil Aviation Shanghai Hospital from January to September 2025. Patients undergoing internal fixation for extremity fractures were enrolled and allocated to either the LOP-based group (n = 171) or the traditional fixed-pressure group (n = 235) according to the attending surgical team and operative scheduling, without randomization. The LOP-based group received individualized tourniquet pressure based on preoperative LOP measurement plus a safety margin, while the control group received standardized pressures (40-50 kPa; 300-375 mmHg). The primary outcome was applied tourniquet inflation pressure. Secondary outcomes included demographic characteristics, extremity location, operative duration, and subgroup analyses stratified by extremity location, age, and sex; multivariable linear regression was performed to adjust for potential confounders.

RESULTS: The LOP-based group demonstrated significantly lower mean tourniquet pressure compared with the traditional group (32.82 +/- 3.63 kPa [246 +/- 27 mmHg] vers […]

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### Impairment in activities of daily living and textbook outcome after major emergency abdominal surgery.

*Mima K, Nitta H, Shikiji Y, Nakashima R, Uemura S, Morinaga T, Itoyama R, Umezaki N et al.*

PURPOSE: Textbook Outcome (TO) has been a useful tool to assess benchmark patient-centred surgical outcomes in elective abdominal surgery. This study aimed to evaluate the association of preoperative activities of

daily living (ADL) status with TOs following major emergency abdominal surgery.

METHODS: This retrospective study included 301 consecutive patients who underwent major emergency abdominal surgery for high-risk acute abdominal conditions, including perforated peptic ulcer, intestinal ischemia, bowel obstruction, bowel perforation, and severe intraabdominal bleeding or infection, at Kumamoto Regional Medical Center between April 2019 and March 2025. Preoperative ADL was evaluated using the Barthel Index (range, 0 to 100), with higher scores indicating greater independence. Barthel Index ≤ 85 was defined as ADL impairment. According to a recent international consensus, TO was defined as discharge to "home" without major postoperative complications (Clavien-Dindo grade ≥ III). Multivariable logistic regression analysis was conducted to assess the association of preoperative ADL with TO achievement, adjusting for clinical features and surgery-related factors.

RESULTS: Among the 301 patients, 90 (30%) had preoperative ADL impairment (Barthel Index ≤ 85) and 211 (70%) achieved TO. Multivariable analysis revealed that preoperative ADL impairment was independently associated […]

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### Early calcium replacement after minimal blood transfusion is associated with adverse outcomes.

*Hewgley WP, Griffin RL, Baird E, Hashmi ZG, Baker SJ, Jansen JO, Kerby J, Holcomb JB et al.*

PURPOSE: Calcium disturbances are common in patients with traumatic injuries, exacerbated by shock and blood product resuscitation. Recently, empiric calcium supplementation has been recommended with the first unit of blood transfusion, despite a lack of data supporting this practice. We hypothesize that there are risks associated with calcium administration in patients receiving small volume transfusion.

METHODS: This is a retrospective review of patients admitted to our Level 1 trauma center from 2018 to 2024 who received 1-3 units of red blood cells (RBCs). Patients were stratified by calcium levels and whether they received early calcium supplementation. A logistic regression was used to estimate odds ratios for the association between calcium levels, calcium supplementation, and outcomes. The primary outcome was in-hospital mortality. Secondary outcomes included acute kidney injury, myocardial infarction, acute respiratory distress syndrome, and venous thromboembolism. The model was adjusted for age, sex, ISS and shock index.

RESULTS: A total of 2,680 patients were included. When stratified by calcium levels and supplementation status, patients were similar in terms of age, sex, mechanism and injury severity score, but shock index was higher among patients receiving calcium supplementation. After adjustment, hypocalcemic patients who received calcium supplementation had […]

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### Peripheral perfusion index as a predictor of severe shock in trauma patients: a prospective observational study.

*Sundar H, Purushothaman V, Mathuram SR, Chandy GM, Sharma SL, Nayak S, James JD*

BACKGROUND: Early recognition of haemorrhagic shock is critical in trauma care, particularly in low- and middle-income countries where delayed referral is common. Conventional vital signs may remain normal despite ongoing hypoperfusion. The Perfusion Index (PI), derived from pulse oximetry, is a non-invasive and continuously available marker of peripheral perfusion. This study evaluated the ability of PI to identify severe haemorrhagic shock in trauma patients and compared its performance with the Shock Index (SI) and Modified Shock Index (mSI).

METHODS: This prospective, single-centre study was conducted at a Level 1 trauma centre between January and May 2024. Adult Priority 1 trauma patients were enrolled and classified as having no/mild shock (ATLS grades 1-2) or severe shock (grades 3-4). PI was recorded during initial assessment. SI, mSI, lactate, injury severity, transfusion requirement, ICU admission, and 24-hour mortality were analysed. Receiver operating characteristic (ROC) analysis and multivariable logistic regression were performed.

RESULTS: A total of 120 patients were included (mean age 40.4 ± 17.2 years; 90% male), predominantly with blunt trauma (85% road traffic injuries). Severe shock was present in 35 patients (29.2%). Median PI was significantly lower in severe shock compared with no/mild shock (0.54 [IQR 0.28-1.31] vs. 1.33 [0.71-2.16], p = 0.008). Mean […]

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### A prospective study on abdominal trauma: patterns of injury, clinical presentation, organ involvement, associated injuries, and outcomes at a tertiary hospital in Mogadishu, Somalia.

*Mohamud AA, Hassan AI, Ali AK, Jubur AA, Ahmed MR, Mohamed SS, Mohamed MM, Bashir AM*

BACKGROUND: Trauma is a major global public health problem and one of the leading causes of death and disability worldwide, particularly among individuals younger than 44 years of age. Approximately 6 million people die each year as a result of traumatic injuries, with nearly 90% of these deaths occurring in low- and middle-income countries. Previous epidemiological studies have reported postoperative complication rates ranging from 18% to 42% and mortality rates between 8.5% and 13%.

OBJECTIVE: To evaluate the patterns of injury, clinical presentation, organ involvement, associated injuries, management strategies, and outcomes among adult patients with abdominal trauma in Mogadishu, Somalia.

METHODS: A prospective observational study was conducted from August 2023 to July 2024 at Mogadishu Somali-Türkiye Recep Tayyip Erdo■an Training and Research Hospital. A total of 146 adult patients with abdominal trauma were enrolled. Data were collected using a pretested structured questionnaire and analyzed using Stata version 16. Univariable and multivariable logistic regression analyses were performed to identify factors independently associated with in-hospital mortality.

RESULTS: The study population consisted predominantly of young adult males, with a mean age of 30.18 ± 9.2 years, and males accounted for 97.3% of all patients. Penetrating trauma was the predominant mechanism of […]

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### Comparison of fluoroscopic versus electromagnetic distal locking in humeral shaft osteosynthesis using intramedullary nail: a retrospective study.

*Madeja R, Szeliga J, Voves J, Pometlova J, Konde A, Janosek J, Dousa P*

PURPOSE: Intramedullary nailing is a standard method for osteosynthesis of diaphyseal humeral fractures. Distal locking, however, typically requires fluoroscopic guidance, which increases radiation exposure. This study aimed to compare standard fluoroscopy-guided distal locking with the Ezy-Aim system (Austofix, Australia) based on electromagnetic pulse detection from the perspectives of surgical times, radiation exposure, and complications.

METHODS: A retrospective study included 58 patients treated for diaphyseal humeral fractures (AO/OTA 12A, 12B) between 2019 and 2023. Group A (n = 28) received osteosynthesis using the Ezy-Aim system; Group B (n = 30) underwent fluoroscopy-guided distal locking. Data on operative time, fluoroscopy time, radiation dose, and complications were analysed.

RESULTS: In all, distal locking was not successful at the first attempt in 14.3% (4/28) of screws placed with Ezy-Aim and 20.0% (6/30) when fluoroscopy was used. Time and dose analyses were performed on the remaining 48 patients. The total surgical time was significantly longer with the Ezy-Aim system (median 58 min) compared to fluoroscopy (median 52 min, p = 0.034), primarily due to the time required for the device assembly. However, distal screw placement was faster with Ezy-Aim (median 12 min vs. 14 min, p = 0.019). Radiation exposure was significantly lower in the Ezy-Aim group (median […]

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### Patterns of patient worry following major emergency abdominal surgery: a 180-day follow-up study.

*Parner MK, Jensen LR, Gamst-Jensen H, Burcharth J, Kokotovic D*

PURPOSE: Major emergency abdominal surgery is associated with high postoperative morbidity and mortality. Beyond physical complications, patients experience psychological distress. This study aimed to describe the distribution of patient-reported worry across independent cross-sectional samples and its association with Days Alive and out of Hospital (DAOH) over 180 days postoperatively.

METHODS: Prospective cohort at Copenhagen University Hospital Herlev (March 30, 2023-March 31, 2024). Degree of Worry (DOW) was surveyed at discharge and postoperative day (POD) 30, 90, and 180, categorized as

physical, mental, and physical-generic domains of worry. DAOH was computed cumulatively and incrementally for 0-30, 30-90, and 90-180 days. Associations between high DOW (hDOW) and subsequent DAOH were tested with Mann-Whitney U.

RESULTS: hDOW was reported by 36.1% at discharge, 22.4% at POD 30, 45.0% at POD 90, and 32.0% at POD 180. Across 180 days, the most frequent worries were relapse (21.3%), disease (14.1%), functional level (12.0%) and daily living (9.1%). The proportion of physical worries was significantly higher at POD 30 than at discharge and significantly lower at POD 180; mental worries varied between time points without statistically significant differences. hDOW at POD 30 was associated with fewer cumulative DAOH by POD 90 (p = 0.008).

CONCLUSION: Patient-reported worry f […]

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### Outcomes and complications of metallic vs. bioabsorbable screw fixation for displaced medial epicondylar fractures in pediatric and adolescent patients.

*Jhala T, Scherer S, Esser M, Lieber J, Fuchs J, Dietzel M*

PURPOSE: The use of bioabsorbable screws in orthopedics and trauma surgery is increasing. The major advantage of not having to remove the implant is of particular interest in pediatric surgery. Due to drawbacks, most notably implant fracture, bioabsorbable screw osteosynthesis has not yet surpassed the gold standard of metallic screw osteosynthesis. This single-centre study compares metallic screw and bioabsorbable screw osteosynthesis of medial humeral epicondyle fractures in children and adolescents over a two-year period.

MATERIALS AND METHODS: Retrospective analysis of surgically treated (screw fixation with either bioabsorbable magnesium-based or metallic screw) patients with a medial humeral epicondyle fracture between 2020 and 2022 at the author's institution.

RESULTS: Seventeen patients with medial epicondyle fractures underwent surgical treatment using either bioabsorbable screws (n = 8) or metallic screws (n = 9). Baseline characteristics, surgical approach, and postoperative care were similar between groups. Two complications (Clavien-Dindo I) occurred in the bioabsorbable group due to implant failure (screw breakage), though without clinical sequelae. These implant failures led to a significantly increased number of postoperative radiographs in the bioabsorbable group (median 4.8 vs. 2.5; p = 0.014). Functional outcomes (LOM, MEPI) and follow-up durations were com […]

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### Characteristics and temporal trends of mobility scooter accidents in the Netherlands: a multicenter cohort study.

*Stensen Z, Peters A, Keereweer D, Krul I, van Osch F, de Groot B, Barten D*

PURPOSE: Mobility scooters are increasingly used by older individuals. The aim of this study was to assess the characteristics and temporal trends of mobility scooter accidents (MSAs) in the Netherlands over a 10-year period using a representative dataset derived from fourteen Emergency Departments (EDs).

METHODS: In this retrospective multicenter cohort study, we analyzed MSA-related ED visits between January 2014 and December 2023, using the Dutch Injury Surveillance System (DISS). Information about age, sex, hospital admission, type and location of injury, mechanism of injury, and operational errors was analyzed. Injury severity was classified using the maximum abbreviated injury score (MAIS). Trend analyses were only performed for ED visits of patients with MAIS2+ injuries.

RESULTS: In total, 2,702 MSA-related ED visits were identified. Forty-nine percent of patients was male and the median age was 75 years (IQR 64.0-83.0). The most common mechanism of injury was a mobility scooter rollover (34.2%). From 2014 to 2023, the number of MSA-related ED visits of patients with MAIS2+ injuries increased by 38.0%. In the 70-79 age group, this rise was 91%. The total number of traumatic brain injuries (TBI) rose steeply bij 103.0%, which includes a modest (+34.0%) increase in moderate-severe TBI. The hospital admission rate was 30.1%, with a median length of stay of 7 days (IQR 3.0 […]

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### "Is the effect of intraoperative 3D imaging beneficial in plate fixation of distal radius fractures: a matched-pair analysis".

*Zumsteg JS, Pieringer A, Hinkelmann MA, Isenmann K, Pape HC, Allemann F, Barthel C*

BACKGROUND: Distal radius fractures (DRFs) are among the most common adult fractures and are frequently treated with volar plate fixation. While intraoperative 3D imaging has been proposed to improve detection of malreduction and implant malposition, its impact on clinical and radiological outcomes remains unclear. This study aimed to evaluate whether intraoperative 3D imaging provides advantages over conventional 2D fluoroscopy. Outcome criteria were accuracy of fracture reduction, plate positioning relative to the watershed line, complication rates, operative time, and functional outcomes following volar plate fixation of distal radius fractures.

METHODS: This retrospective single-centre study included patients aged ≥ 16 years who underwent volar plate fixation for distal radius fractures between 2016 and 2024. Patients were assigned to a 3D imaging group or a conventional 2D fluoroscopy group. To ensure comparability, patients were matched 1:1 using exact AO/OTA subgroup matching and propensity score matching for age and sex. Outcomes included operative duration, radiographic reduction quality, plate positioning, complications, and wrist range of motion. Matched-pair comparisons were performed using appropriate paired statistical tests.

RESULTS: Of 1,259 patients screened, 418 patients (209 matched pairs) were included after exact AO/OTA subgroup matching and propensity sc […]

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## High-velocity gunshot chest trauma: tailoring patient care through insights from a mass casualty experience.

*Azulay AA, Stein M, Tabo LY, Lebenthal A, Ruderman L, Refaely Y*

OBJECTIVES: This study describes the management strategies and clinical outcomes of a cohort of patients sustaining high-velocity military rifle gunshot wounds to the thorax during a mass casualty incident (MCI). The analysis focuses on damage control surgical techniques, the incidence and timeline of venous thromboembolism (VTE), regional pain management, and early psychiatric protocols.

METHODS: A descriptive retrospective analysis was conducted on patients presenting with a single penetrating high-velocity ballistic wound to the thorax on October 7, 2023. Inclusion criteria were limited to patients with a projectile trajectory traversing the lung parenchyma, managed by the cardiothoracic surgery department. Injuries to immediately adjacent structures (subclavian vessels, brachial plexus, clavicle, and scapula) were recorded as associated injuries. Patients presenting with multi-system trauma involving entirely separate, distant anatomical regions were excluded.

RESULTS: The cohort comprised 11 male patients (age range: 19-49 years; 9 soldiers, 2 civilians) injured by 7.62 mm assault rifle rounds. Six patients (54.5%) required emergency thoracotomy, sternotomy, or VATS within the first 24 h due to ongoing hemorrhage; primary hemorrhage control consisted of deep parenchymal lung suturing (n = 5) and temporary chest packing (n = 1). Four patients (36.4%) underwent delayed, el […]

PubMed · DOI

## Elevated risk of wound and systemic complications in MRSA-colonized patients undergoing lower extremity orthopedic trauma surgery.

*Areti AS, Hafeez H, Sangineni PV, Argao L, Deveza L, Dawson J*

PURPOSE: Preoperative colonization with methicillin-resistant Staphylococcus aureus (MRSA) is a known risk factor for adverse outcomes in arthroplasty but remains under-investigated in orthopedic trauma. This study evaluates the impact of preoperative MRSA colonization on postoperative complication rates in patients undergoing lower extremity orthopedic trauma surgery.

METHODS: We conducted a retrospective cohort study using the TriNetX Research Network, identifying adult patients who underwent lower extremity orthopedic trauma procedures over a 20-year period. Patients with a documented MRSA diagnosis within three months before surgery were matched 1:1 to MRSA-negative controls based on demographics and comorbidities. Postoperative outcomes were assessed within three months of surgery, including surgical site infections (SSIs), wound dehiscence, organ dysfunction, thromboembolic events, and healthcare utilization. Odds ratios (ORs) and p-values were calculated, with statistical significance set at $p < 0.05$.

RESULTS: A total of 1,868 MRSA+ and 629,342 control patients were identified. After matching, MRSA+ patients remained at significantly increased risk for wound dehiscence (OR = 1.80, p = 0.005), pneumonia (OR = 2.33, p < 0.001), sepsis (OR = 3.71, p < 0.001), acute renal failure (OR = 1.45, p < 0.001), blood transfusion (OR = 1.74, p < 0.001), deep vein thrombosis (OR = 2 […]

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### A global contrast shortage: a descriptive study exploring IV contrast utilization and outcomes in trauma patients at a level I trauma center.

*Zakaria H, Johansen C, Waldrop C, Wan B, Graham C, Fetzer D, Minei J, Bruns B et al.*

PURPOSE: Computed Tomography (CT) with intravenous (IV) contrast is critical to the evaluation of the trauma patient. A pandemic-related lockdown occurred in Spring 2022 in the city where iohexol iodinated contrast media) is manufactured, resulting in a global shortage. We explored the potential impact of this shortage on contrast utilization and patient outcomes.

METHODS: A retrospective study was performed of all trauma patients who underwent CT between January and December 2022 at a level I trauma center. Pre-contrast conservation (PRE) period was defined as January to April 2022, and post-conservation (POST) as May to December 2022. Demographics, utilization rates (bolus/patient) and outcomes were evaluated. Chi-square tests were performed for categorical variables and Mann-Whitney U tests for continuous variables. Statistical significance was set at p < 0.05.

RESULTS: 1,097 patients were included; 509 in the pre-shortage period, 588 post-shortage. 857 CTs with contrast (1.68 contrast bolus/patient) were performed pre-conservation and 979 post-conservation (1.66 contrast bolus/patient) (p = 0.04). Maximum Abbreviated Injury Scale (AIS) head, head/neck, and extremity were higher post-shortage (p < 0.05); however, mortality was higher in the pre-conservation group (p < 0.0012). There were 9 cases of Acute Kidney Injury (AKI, 8 PRE, 1 POST), 22 delayed diagnoses (14 PRE, 8 P […]

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### Trauma stabilization points in armed conflict: operational evolution and strategic reflections from mosul to Gaza.

*Marta C, Luca P, Kai-Hsun H, Areej AF, Athanasios G, Wessam EG, Jonas Z, Johan VS et al.*

BACKGROUND: Forward trauma care remains a critical gap in emergency responses to armed conflict, particularly where conventional medical evacuation is disrupted. Trauma Stabilization Points (TSPs) were introduced in previous conflicts to bring life-saving care closer to the point of injury. To improve standards and adaptability, the World Health Organization convened a Technical Working Group to develop operational guidance for TSPs, drawing on diverse field experiences, including the ongoing Gaza deployments.

CASE PRESENTATION: Between February and July 2024, three TSPs were established in Gaza, managing over 4,000 consultations. Initially focused on trauma stabilization and referral, these sites quickly adapted to minor injuries and non-traumatic conditions, reflecting population needs and access challenges. Referral rates varied across sites due to hospital proximity, ambulance availability, and shifting frontlines. Security threats limited forward deployment and safe patient access, requiring high mobility and rapid relocation. Experience from Gaza highlighted key operational principles: locating TSPs near the point of injury; integrating within a functioning trauma referral pathway supported by evacuation capacity and hospital readiness; maintaining clear clinical functions and staffing standards; and using standardized documentation for quality assurance and continuity. […]

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### Acute-on-chronic versus isolated acute subdural hematoma in complicated mild traumatic brain injury: association with radiological mass effect markers in a multicenter cohort.

*Crombé A, Benhamed A, Seux M, Frassin L, Matichard R, L'Huillier R, Millon D, Gorincour G*

PURPOSE: The clinical impact of acute-on-chronic subdural hematoma (acSDH) in mild traumatic brain injury (mTBI) remains incompletely characterized. We assessed whether acSDH, compared with isolated acute SDH (aSDH), is associated with worse neurological presentation and increased radiological markers of mass effect.

METHODS: We conducted a retrospective multicenter cohort study using the IMADIS teleradiology head trauma workflow (103 emergency departments in France, January 2020 to December 2022). Adult patients with

complicated mTBI (i.e., GCS 13-15 plus intracranial hemorrhage or skull fracture) and acute SDH on non-contrast head CT were eligible. Exposure was acSDH versus aSDH based on structured radiology reports. Outcomes were (1) GCS < 15 at presentation, (2) SDH maximal thickness ≥ 7 mm, (3) radiological brain herniation, and (4) intermediate or high QueBIC risk category. Univariable and multivariable logistic regression models were used to estimate associations.

RESULTS: Among 935 included patients, 107 (11.4%) had acSDH. Patients with acSDH were older (≥ 75 years: 57.0% vs. 39.7%, p < .0001) and more frequently had dementia (8.4% vs. 3.7%, p = .0465). Compared with aSDH, acSDH was associated with GCS < 15 (37.4% vs. 21.7%, P = .0010) and confusion (41.0% vs. 25.8%, p = .0016). On CT, acSDH was associated with SDH thickness ≥ 7 mm (54.2% vs. 20.4%, OR 4.50, 95%CI 2.9 […]

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### Persistent neuropathic pain symptoms after FPV drone-related thoracic blast injury without rib fractures: a pilot observational study from the Ukraine War.

*Battle C, Baker E, Giamello JD, Melchio R, Dmytriiev D*

PURPOSE: The aim of this exploratory pilot study was to describe First Person View drone (FPV) related, blast-mechanism thoracic trauma with no rib fractures in Ukrainian military patients sustained during the Russia-Ukraine Conflict, and report the prevalence of persistent neuropathic pain symptoms at three months among patients admitted to a rehabilitation and pain medicine service.

METHODS: A retrospective, single-centre observational, exploratory study design was used. Military patients (≥ 16 years old) with blast-mechanism thoracic injuries with no rib fractures, admitted to a hospital's rehabilitation and pain medicine service in Ukraine for ≥ 24 h were included. The PainDETECT survey was completed at three months post-injury. Participant characteristics were summarised using descriptive statistics. Exploratory univariate and multivariate analyses were performed to explore possible associations between variables and the development of persistent neuropathic pain symptoms.

RESULTS: 217 patients were included in the final analysis. Median age was 39 years (IQR: 32-47.5). 131 (60.4%) patients were admitted to critical care with 91 (41.9%) patients requiring mechanical ventilation. The median hospital length of stay was 22 days (IQR: 16-25) and critical care length of stay was 4 days (IQR: 2-5). Persistent neuropathic pain symptoms were reported in 173 out of 217 patients ( […]

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### The association between pancreatic enzyme serum concentrations and blunt pancreatic trauma: a systematic review.

*Sykes B, Gracias D, Bendinelli C, Balogh ZJ*

PURPOSE: Blunt pancreatic trauma is uncommon. When suspected, serum amylase and lipase are measured, but their diagnostic value remains uncertain. This systematic review evaluated the diagnostic performance of serum amylase and serum lipase in detecting blunt pancreatic trauma in adults.

METHODS: MEDLINE, Embase, Scopus and Cochrane Central were searched for studies evaluating serum amylase and/or lipase in adults with blunt pancreatic trauma. Observational studies reporting diagnostic accuracy or clinical outcomes were included. Two reviewers independently screened studies and assessed risk of bias using the Newcastle-Ottawa Scale.

RESULTS: Four observational studies met inclusion criteria. A total of 4937 patients with blunt abdominal trauma, including 150 with pancreatic injury were investigated. Two studies evaluated serum lipase alone and two compared amylase and lipase. Pancreatic injuries were graded according to the American or Japanese Association for the Surgery of Trauma. Early serum amylase measurement (< 90 min post-injury) demonstrated low sensitivity. In contrast, testing performed more than 3 h after injury has shown improved diagnostic accuracy with one study suggesting combined serum amylase and lipase testing can achieve a sensitivity of 85% and specificity of 100%.

CONCLUSION: Early serum pancreatic enzymes measurement does not appear to aid initial manag […]

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## Extension of the fluidics NTF-Biobank of polytraumatized patients with extracellular vesicles: improved vesicle preservation through local processing.

*Weber B, Leppik L, Han J, Becker N, Wang T, Groven RVM, Balmayor ER, Zhao Q et al.*

PURPOSE: Extracellular vesicles (EVs) show diagnostic relevance in trauma, prompting the development of the NTF■EV■Biobank as an extension of the established Network Trauma Research (NTF) plasma/serum biobank. After optimizing the EV isolation and characterization standard operating procedure (SOP) at a single center, this multicenter study evaluates its transferability and examines how sample transport affects EV quality before large■scale implementation in a nationwide biobank setting.

METHODS: Plasma and serum from polytrauma patients and healthy controls (n=10) were processed under the standardized NTF■Biobank protocol and distributed to three centers. EVs were isolated using the unified NTF■EV■Biobank workflow, exchanged between centers, and analyzed for protein markers, particle size and concentration (NTA), total protein content, and morphology (TEM) to compare EV quality across sites.

RESULTS: Implementation of the EV biobank protocol across participating centers, as well as inter-center transport on dry ice, was successfully performed without complications. All samples arrived at the analyzing centers in a frozen state. EVs exhibited typical vesicular morphology as confirmed by transmission electron microscopy and remained positive for CD9 and CD81, indicating that transportation did not affect morphology or surface marker expression. However, transportation did redu […]

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## Whole blood first resuscitation and association of blood product utilization based on mechanism of injury.

*Hadley J, Conner J, Thomas C, Llorente K, Gilani A, White B, Miller Iii P, Nunn A et al.*

INTRODUCTION: Studies evaluating the effect of whole blood (WB) vs. component-based resuscitation accounting for mechanism of injury are lacking. We sought to evaluate the effects of a WB first resuscitation strategy regarding mechanism of injury.

METHODS: This is a retrospective cohort study of trauma patients who received any blood product during resuscitation from January 2016 to November 2021. The primary outcomes were total blood volume utilized within 24 h and 30-day mortality. Multivariable gamma regression with a log link was used for blood volume and logistic regression for mortality. A likelihood ratio test evaluated whether injury mechanism modified the treatment effect.

RESULTS: Between January 2016 and November 2021, 1,013 injured patients received blood products (785 WB-first). We did not detect statistically significant effect modification by injury mechanism for either outcome (interaction p = 0.44 for blood volume, p = 0.25 for mortality). In multivariable gamma regression, WB-first resuscitation was associated with a ratio of mean total blood volume of 0.71 (95% CI 0.55 to 0.90, p = 0.005) relative to components, representing approximately 29% lower utilization. WB-first was not associated with 30-day mortality (OR 0.90, 95% CI 0.48 to 1.71, p = 0.73).

CONCLUSIONS: A WB-first resuscitation strategy is associated with lower blood volume utilized within the f […]

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## Reported complications after spinopelvic fixation for U-type sacral fractures in older adults: a systematic review with narrative synthesis.

*Giordano V, Pinheiro HRS, Pires RE, Varga P, Freitas A, Giannoudis PV*

BACKGROUND: Spinopelvic fixation (SPF), alone or combined with iliosacral or transiliac-transsacral fixation, is often used for highly unstable posterior pelvic ring injuries and some fragility fracture patterns in older adults. However, the evidence specifically describing complications after SPF in older patients with U-type sacral fractures is limited.

METHODS: This systematic review was conducted according to the Cochrane Handbook and PRISMA recommendations. PubMed, Embase, and grey-literature sources were searched from January 1, 2000 to July 31, 2025. Eligible studies included adults aged 60 years or older with U-type sacral fractures or closely related spinopelvic dissociation/FFP IVb patterns treated with SPF, and reporting at least one complication. Because of the small number of eligible studies, probable cohort overlap, heterogeneity in fracture definitions, treatment

indications, fixation constructs, and outcome reporting, findings were synthesized narratively rather than pooled quantitatively.

RESULTS: Three studies met the eligibility criteria, comprising 157 patients overall; however, only a subset underwent SPF, and two studies were produced by the same group with probable cohort overlap. The available cohorts mixed bilateral sacral fragility fractures, FFP IVb patterns, and comparative SPF-versus-trans-sacral constructs. Reported adverse events included wound […]

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### Influence of allograft augmentation on surgical outcomes and complications in proximal humerus fractures: a matched-pair analysis.

*Barthel C, Rudolph F, Kalbas Y, Russek R, Pape HC, Allemann F*

BACKGROUND: Proximal humerus fractures predominantly affect elderly patients and pose a growing clinical burden. This study evaluates the impact of allograft augmentation on operative time, fracture reduction quality, and complication rates compared with standard fixation across different fracture morphologies.

METHODS: This retrospective matched-pair study included patients surgically treated for proximal humerus fractures with locking plate fixation at a single tertiary trauma center between 01.01.2014 and 31.12.2024. Patients treated with allograft augmentation (Group A) were matched 1:1 to patients without allograft (Group NA) using exact matching for Mayo FJD fracture morphology and sex, and propensity score matching for age. Radiological outcomes were assessed using pre- and postoperative neck-shaft angles, with normalization defined as correction into the physiological range (120°-150°). Complications and operative parameters were recorded. Paired statistical analyses were applied.

RESULTS: A total of 1,012 patients were treated during the study period; after selection and matching, 102 matched patient pairs (204 patients) were analyzed. Operative time was longer with allograft augmentation (141.3 ± 40.3 vs. 126.9 ± 50.0 min; p = 0.08), reaching statistical significance only in varus posteromedial fractures (144.4 ± 37.8 vs. 122.2 ± 39.6 min; p = 0.007). Although overa […]

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### Impact of a comprehensive trauma pathway on workflow and clinical processes in a level III trauma center.

*Neitzert L, Druetto C, Forno D, Molino P, Vitale D, Grio M*

PURPOSE: Organized trauma systems improve coordination of care and may enhance the efficiency of trauma management, potentially improving outcomes and reducing hospital length of stay and costs. The aim of the study was to evaluate key process-of-care indicators and clinical outcomes before and after the introduction of a comprehensive trauma pathway.

METHODS: We conducted a single-center retrospective-prospective cohort study at Rivoli Hospital, Italy. Trauma patients admitted to the Emergency Department (ED) between 2024 and 2025 who met the criteria for trauma team activation were prospectively included. Outcomes were compared with a retrospective cohort from the period prior to the implementation of the dedicated trauma pathway (2023-2024).

RESULTS: A total of 465 patients were included. Implementation of a comprehensive trauma pathway was associated with significant improvements in several time-dependent quality indicators, including ED boarding time (888 vs. 1021 min; p = 0.02), time to imaging (48 vs. 78 min; p < 0.01), time to surgery (183 vs. 237 min; p = 0.03), time to ICU admission (226 vs. 343 min; p = 0.02), and time to interhospital transfer (238 vs. 305 min; p = 0.04). We observed a trend toward a reduction in early mortality (2% vs. 3.6%; p = 0.41) and 30-day mortality (3.7% vs. 5.9%; p = 0.62).

CONCLUSIONS: The adoption of a structured multidisciplinary trau […]

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### The DRoPH score: development of a tool for predicting reduction difficulty in proximal humerus fracture osteosynthesis.

*Barthel C, Rudolph F, Kraus M, Pape HC, Allemann F*

BACKGROUND: Proximal humerus fractures (PHFs) are among the most common and challenging fractures due to complex anatomy and variable fracture patterns. While anatomical reduction is critical for successful outcomes, no tool currently predicts the intraoperative difficulty of achieving it. Despite advances in surgical techniques, complications persist. Existing classification systems, such as Codman's, Neer's, AO/OTA, and Mayo-FJD, focus on fracture morphology and displacement but do not address intraoperative complexity. This study aims to develop a new score for the early prediction of intraoperative reduction difficulty in proximal humerus fracture surgery, the DRoPH score (Difficulty of Reduction of Proximal Humerus Fractures).

METHODS: An online survey was conducted among experienced upper extremity surgeons in Switzerland to identify factors influencing the difficulty of reducing and retaining proximal humerus fractures. Inclusion criteria for surgeon grading comprised experience with more than 100 surgically treated proximal humerus fractures. Eligible surgeons rated the relevance of factors influencing reduction difficulty using a five-point Likert scale. A rank-based weighting system derived from expert ratings was applied, assigning two points to the three highest-ranked factors. The resulting DRoPH-Score ranged from 0 to 10, with higher values indicating greater sur […]

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## Interface stability of intramedullary nail locking configurations under combined axial and torsional loading.

*Marianne H, Martin W, Claus G, Anja B, Peter A*

The biomechanical performance of intramedullary nails in metaphyseal regions strongly depends on distal locking configuration. This study aimed to evaluate screw-bone interface stability under combined axial and torsional cyclic loading for different distal locking configurations of intramedullary nails, focusing on the effects of screw number, spatial distribution, interscrew distance, and multiplanar versus coplanar fixation using a standardized synthetic bone model. Six distal locking configurations (n = 5 per group) were tested under stepwise cyclic axial compression, superimposed with a constant torsional moment (± 2 Nm, 2 Hz). Primary endpoints were cycles to failure, cycles to 1 mm axial displacement, and cycles to 0.5° rotation. All constructs failed by progressive screw cut-through within the synthetic bone. The three-screw multiplanar configuration exhibited the highest endurance (≈ 36,000 cycles to failure) and lowest micromotion (≈ 34,000 cycles to 1 mm displacement), significantly outperforming the two-screw configurations overall (p = 0.002). Increasing interscrew distance from 10 mm to 34 mm improved migration resistance by approximately 25%, whereas the short 10 mm spacing led to premature failure. Blocked and free-hand configurations showed comparable behavior to the standard setup, indicating no added benefit from toggle elimination alone. Locking configuratio […]

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